



AFAQ Iqbal Series

KEYBOOK Mathematics

GRADE

6



Based on National Curriculum 2022

1.1: Let's Practice

1. Find the factors of the following numbers.

To find the factors of a number, we first write down all its divisors. i.e.

a) 14

Solution:

Divisors of 14:

$$1 \times 14 = 14$$

$$2 \times 7 = 14$$

So, the factors of 14 are: 1, 2, 7, and 17

b) 98

Solution:

Divisors of 98:

$$1 \times 98 = 98$$

$$2 \times 49 = 98$$

$$7 \times 14 = 98$$

So, the factors of 98 are: 1, 2, 7, 14, 49 and 98.

c) 112

Solution:

Divisors of 112:

$$1 \times 112 = 112$$

$$2 \times 56 = 112$$

$$4 \times 28 = 112$$

$$7 \times 16 = 112$$

$$8 \times 14 = 112$$

So, the factors of 112 are: 1, 2, 4, 7, 8, 14, 16, 28, 56 and 112.

d) 180

Solution:

Divisors of 180:

$$1 \times 180 = 180$$

$$2 \times 90 = 180$$

$$3 \times 60 = 180$$

$$4 \times 45 = 180$$

$$5 \times 36 = 180$$

$$6 \times 30 = 180$$

$$9 \times 20 = 180$$

$$10 \times 18 = 180$$

$$12 \times 15 = 180$$

So, the factors of 180 are: 1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90 and 180.

e) 256

Solution:

Divisors of 256:

$$1 \times 256 = 256 \quad 2 \times 128 = 256$$

$$4 \times 64 = 256 \quad 8 \times 32 = 256$$

$$16 \times 16 = 256$$

Therefore, the factors of 256 are: 1, 2, 4, 8, 16, 32, 64, 128 and 256.

f) 340

Solution:

Divisors of 340:

$$1 \times 340 = 340 \quad 2 \times 170 = 340$$

$$4 \times 85 = 340 \quad 5 \times 68 = 340$$

$$10 \times 34 = 340 \quad 17 \times 20 = 340$$

Therefore, the factors of 340 are: 1, 2, 4, 5, 10, 17, 20, 34, 68, 85, 170 and 340

g) 65

Solution:

Divisors of 65:

$$1 \times 65 = 65$$

$$5 \times 13 = 65$$

So, the factors of 65 are: 1, 5, 13 and 65.

h) 200

Solution:

Divisors of 200:

$$1 \times 200 = 200 \quad 2 \times 100 = 200$$

$$4 \times 50 = 200 \quad 5 \times 40 = 200$$

$$8 \times 25 = 200 \quad 10 \times 20 = 200$$

So, the factors of 200 are: 1, 2, 4, 5, 8, 10, 20, 25, 40, 50, 100 and 200

2. Check the see if 7 and 16 are factors of 240.

Solution: To check if a number is a factor of another bigger number we divide the bigger number by the given number. If it divides the bigger number completely, it is a factor of the bigger number.

To check if 7 is a factor of 240, we divide 240 by 7.

As 240 is not divided completely by 7, so, 7 is not a factor of 240.

Similarly, to check if 16 is a factor of 240, we divide 240 by 16.

As 240 is divided completely by 16, so 16 is a factor of 240.

$$\begin{array}{r} 34 \\ 7 \overline{) 240} \\ \underline{- 21} \\ 30 \\ \underline{- 28} \\ 2 \end{array} \rightarrow \text{remainder}$$

$$\begin{array}{r} 15 \\ 16 \overline{) 240} \\ \underline{- 16} \\ 80 \\ \underline{- 80} \\ 0 \end{array} \rightarrow \text{remainder}$$

3. Find the first 6 multiples of the given numbers.

a)

6

Solution:

The first 6 multiples of 6:

$$\begin{array}{ll} 1 \times 6 = 6 & 2 \times 6 = 12 \\ 3 \times 6 = 18 & 4 \times 6 = 24 \\ 5 \times 6 = 30 & 6 \times 6 = 36 \end{array}$$

So, the first 6 multiples of 6 are: 6, 12, 18, 24, 30 and 36.

c)

32

Solution:

The first 6 multiples of 32:

$$\begin{array}{ll} 1 \times 32 = 32 & 2 \times 32 = 64 \\ 3 \times 32 = 96 & 4 \times 32 = 128 \\ 5 \times 32 = 160 & 6 \times 32 = 192 \end{array}$$

So, the first 6 multiples of 32 are: 32, 64, 96, 128, 160 and 192

e)

54

Solution:

The first 6 multiples of 54:

$$\begin{array}{ll} 1 \times 54 = 54 & 2 \times 54 = 108 \\ 3 \times 54 = 162 & 4 \times 54 = 216 \\ 5 \times 54 = 270 & 6 \times 54 = 324 \end{array}$$

So, the first 6 multiples of 54 are: 54, 108, 162, 216, 270 and 324.

b)

15

Solution:

The first 6 multiples of 15:

$$\begin{array}{ll} 1 \times 15 = 15 & 2 \times 15 = 30 \\ 3 \times 15 = 45 & 4 \times 15 = 60 \\ 5 \times 15 = 75 & 6 \times 15 = 90 \end{array}$$

So, the first 6 multiples of 15 are: 15, 30, 45, 60, 75 and 90.

d)

39

Solution:

The first 6 multiples of 39:

$$\begin{array}{ll} 1 \times 39 = 39 & 2 \times 39 = 78 \\ 3 \times 39 = 117 & 4 \times 39 = 156 \\ 5 \times 39 = 195 & 6 \times 39 = 234 \end{array}$$

So, the first 6 multiples of 39 are: 39, 78, 117, 156, 195 and 234.

f)

63

Solution:

The first 6 multiples of 63:

$$\begin{array}{ll} 1 \times 63 = 63 & 2 \times 63 = 126 \\ 3 \times 63 = 189 & 4 \times 63 = 252 \\ 5 \times 63 = 315 & 6 \times 63 = 378 \end{array}$$

So, the first 6 multiples of 63 are: 63, 126, 189, 252, 315 and 378.

g) 79

Solution:

The first 6 multiples of 79:

$$1 \times 79 = 79$$

$$2 \times 79 = 158$$

$$3 \times 79 = 237$$

$$4 \times 79 = 316$$

$$5 \times 79 = 395$$

$$6 \times 79 = 474$$

So, the first 6 multiples of 79 are: 79, 158, 237, 316, 395 and 474.

h) 84

Solution:

The first 6 multiples of 84:

$$1 \times 84 = 84$$

$$2 \times 84 = 168$$

$$3 \times 84 = 252$$

$$4 \times 84 = 336$$

$$5 \times 84 = 420$$

$$6 \times 84 = 504$$

So, the first 6 multiples of 84 are: 84, 168, 252, 336, 420 and 504

4. Find all the possible groups that can be made of students in a computer class if there are 32 students (e.g., 16 students in 2 groups).

Solution: To find all the possible groups that can be made of 32 students, we find all the factors pairs for the number 32.

All factor pairs of 32:

$$1 \times 32 = 32 \text{ (we all skip this pair as we need to make groups)}$$

$$2 \times 16 = 32 \qquad 4 \times 8 = 32$$

So, the factor pairs of 32 are, (4, 8)

2 and 16 (2, 16), 4 and 8

So there are the following four options

We can make 2 groups of 16 students each or 16 groups of 2 students each.

Also, 4 groups of 8 students each or 8 groups of 4 students each.

5. In how many ways can a baker divide 105 cupcakes into trays so that each tray has an equal number of cupcakes?

Solution: To find the solution we find the factor pairs of 105, except 1 and 105.

Factor pairs of 105:

$$3 \times 35 = 105$$

$$5 \times 21 = 105$$

$$7 \times 15 = 105$$

So the factor pairs of 105 are: (3, 35), (5, 21), (7, 15)

Therefore, the baker can divide the cups as follows:

3 trays having 35 cupcakes each

5 trays having 21 cupcakes each

7 trays having 15 cupcakes each

6. Find all the multiples of 49 less than 300 but greater than 180.

Solution: To find all the multiples of 49 less than 300 and greater than 180, we multiply 49 by 1, 2, 3, 4, And so on until we get a multiple less than 300.

$$49 \times 1 = 49$$

$$49 \times 2 = 98$$

$$49 \times 3 = 147$$

$$49 \times 4 = 196$$

$$49 \times 5 = 245$$

$$49 \times 6 = 294$$

$$49 \times 7 = 343 \rightarrow \text{(this will not be included.)}$$

So the multiples of 49 greater than 180 and less than 300 are: 196, 245 and 294.

7. Find the first 8 multiples of the given numbers.

a) 21

Solution:

$$1 \times 21 = 21$$

$$2 \times 21 = 42$$

$$3 \times 21 = 63$$

$$4 \times 21 = 84$$

$$5 \times 21 = 105$$

$$6 \times 21 = 126$$

$$7 \times 21 = 147$$

$$8 \times 21 = 168$$

So, the first 8 multiples of 21 are: 21, 42, 63, 84, 105, 126, 147 and 168.

b) 65

Solution:

$$1 \times 65 = 65$$

$$2 \times 65 = 130$$

$$3 \times 65 = 195$$

$$4 \times 65 = 260$$

$$5 \times 65 = 325$$

$$6 \times 65 = 390$$

$$7 \times 65 = 455$$

$$8 \times 65 = 520$$

So, the first 8 multiples of 65 are: 65, 130, 195, 260, 325, 390, 455 and 520.

c) 40

Solution:

$$1 \times 40 = 40$$

$$2 \times 40 = 80$$

$$3 \times 40 = 120$$

$$4 \times 40 = 160$$

$$5 \times 40 = 200$$

$$6 \times 40 = 240$$

$$7 \times 40 = 280$$

$$8 \times 40 = 320$$

So, the first 8 multiples of 40 are: 40, 80, 120, 160, 200, 240, 280 and 320.

d) 77

Solution:

$$1 \times 77 = 77$$

$$2 \times 77 = 154$$

$$3 \times 77 = 231$$

$$4 \times 77 = 308$$

$$5 \times 77 = 385$$

$$6 \times 77 = 462$$

$$7 \times 77 = 539$$

$$8 \times 77 = 616$$

So, the first 8 multiples of 77 are: 77, 154, 231, 308, 385, 462, 539 and 616.

e) 82

Solution:

$$1 \times 82 = 82$$

$$2 \times 82 = 164$$

$$3 \times 82 = 246$$

$$4 \times 82 = 328$$

$$5 \times 82 = 410$$

$$6 \times 82 = 492$$

$$7 \times 82 = 574$$

$$8 \times 82 = 656$$

So, the first 8 multiples of 82 are: 82, 164, 246, 328, 410, 492, 574 and 656.

f) 9

Solution:

$$1 \times 9 = 9$$

$$2 \times 9 = 18$$

$$3 \times 9 = 27$$

$$4 \times 9 = 36$$

$$5 \times 9 = 45$$

$$6 \times 9 = 54$$

$$7 \times 9 = 63$$

$$8 \times 9 = 72$$

So, the first 8 multiples of 9 are: 9, 18, 27, 36, 45, 54, 63 and 72.

1.2: Let's Practice

1. Factorize the following using prime factorization and write them in index form.

a) 56

Solution: To find prime factors of 56, we divide 56 into its factors until we get all its prime factors.

Prime factorization of 56:

Prime factors of 56 = 2, 2, 2, 7

Prime factorization of 56 = $2 \times 2 \times 2 \times 7$

Index form: As 2 is multiplied three times by itself.

So we write 2 to the power 3. 7 appears once so we write it as it is:

So in index form: $56 = \overline{2 \times 2 \times 2} \times 7$
 $= 2^3 \times 7$

2	56
2	28
2	14
7	7
	1

b) 75

Solution: To find prime factors of 75, we divide 75 into its factors until we get all its prime factors.

Prime factorization of 75:

Prime factors of 75 = 5, 5, 3

Prime factorization of 75 = $5 \times 5 \times 3$

Index form: As 5 is multiplied two times by itself, so we write 5 to the power 2. 3 appears only once so we write it as it is.

So in index form: $75 = \overline{5 \times 5} \times 3$
 $= 5^2 \times 3$

5	75
5	15
3	3
	1

c) 396

Solution: To find prime factors of 396, we divide 396 into its factors until we get all its prime factors.

Prime factorization of 396:

Prime factors of 396 = 2, 2, 3, 3, 11

Prime factorization of 396 = $2 \times 2 \times 3 \times 3 \times 11$

Index form: As 2 is multiplied 2 times by itself, so we write 2 to the power 2. Also 3 is multiplied 2 times by itself so we write 3 to the power 2. 11 appears once, so we write it as it is.

So in index form: $396 = \overline{2 \times 2} \times \overline{3 \times 3} \times 11$
 $= 2^2 \times 3^2 \times 11$

2	396
2	198
3	99
3	33
11	11
	1

d) 125

Solution:

Prime factorization of 125:

Prime factors of 125 = 5, 5, 5

Prime factorization of 125 = $5 \times 5 \times 5$

Index form: $125 = \overline{5 \times 5 \times 5}$
 $= 5^3$

5	125
5	25
5	5
	1

e) 426

Solution:

Prime factorization of 426

Prime factors of 426 = 2, 3, 71

Prime factorization of 426 = $2 \times 3 \times 71$

Index form: As all factors appear once, they will be written as it is in index form.

So in index form: $426 = 2 \times 3 \times 71$

2	426
3	213
71	71
	1

f) 7938

Solution:

Prime factorization of 7938

Prime factors of 7938 = 2, 3, 3, 3, 3, 7, 7

Prime factorization of 7938 = $2 \times 3 \times 3 \times 3 \times 3 \times 7 \times 7$

In Index form: $7938 = 2 \times \overline{3 \times 3 \times 3 \times 3} \times \overline{7 \times 7}$
 $= 2 \times 3^4 \times 7^2$

2	7938
3	3969
3	1323
3	441
3	147
7	49
7	7
	1

g) 2025

Solution:

Prime factorization of 2025.

Prime factors of 2025 = 3, 3, 3, 3, 5, 5

Prime factorization of 2025 = $3 \times 3 \times 3 \times 3 \times 5 \times 5$

In index form: $2025 = \overline{3 \times 3 \times 3 \times 3} \times \overline{5 \times 5}$
 $= 3^4 \times 5^2$

3	2025
3	675
3	225
3	75
5	25
5	5
	1

h) 2401

Solution:

Prime factorization of 2401

Prime factors of 2401 = 7, 7, 7, 7

Prime factorization of 2401 = $7 \times 7 \times 7 \times 7$

In Index form: $2401 = \overline{7 \times 7 \times 7 \times 7} = 7^4$

7	2401
7	343
7	49
7	7
	1

2. Find the prime factors of the given numbers by factor tree method and write them in index form.

a) 25

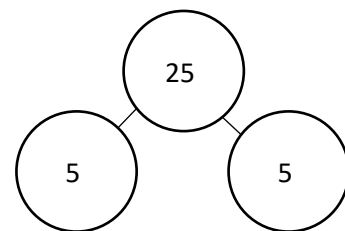
Solution:

To find prime factors using factor tree method, we divide the given number into its factors until we get all its prime factors.

Prime factors of 25 = 5, 5

Prime factorization of 25 = 5×5

In index form: $25 = \overline{5 \times 5}$
 $= 5^2$



b) 712

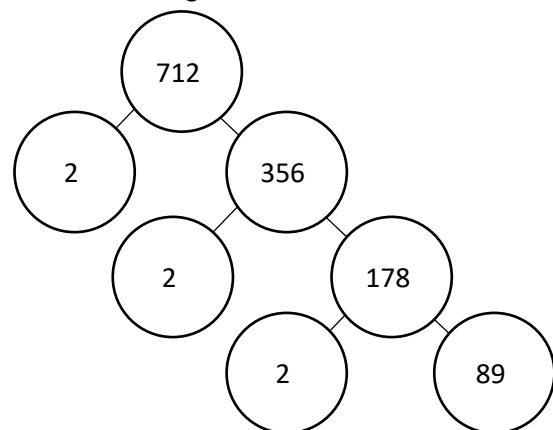
Solution:

To find prime factors using factor tree method, we divide the given number into its factors until we get all its prime factors.

Prime factors of 712 = 2, 2, 2, 89

Prime factorization of 712 = $2 \times 2 \times 2 \times 89$

In index form: $712 = \overline{2 \times 2 \times 2} \times 89$
 $= 2^3 \times 89$



c) 496

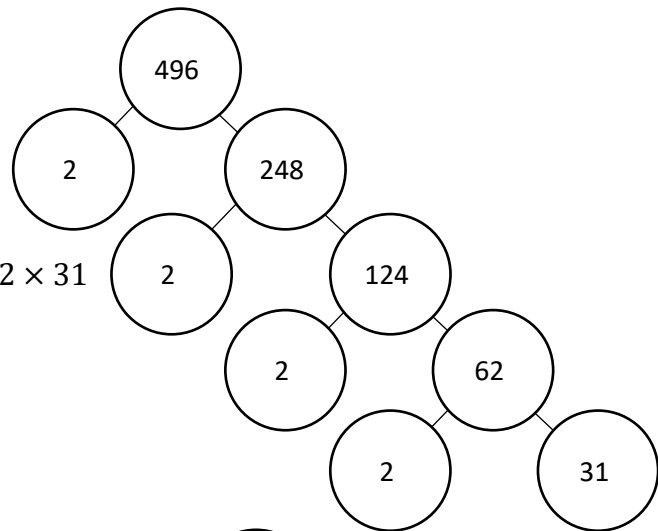
Solution:

Factor tree for 496:

Prime factors of 496 = 2, 2, 2, 2, 31

Prime factorization of 496 = $2 \times 2 \times 2 \times 2 \times 31$

In index form: $496 = \overline{2 \times 2 \times 2 \times 2} \times 31$
 $= 2^4 \times 31$



d) 3484

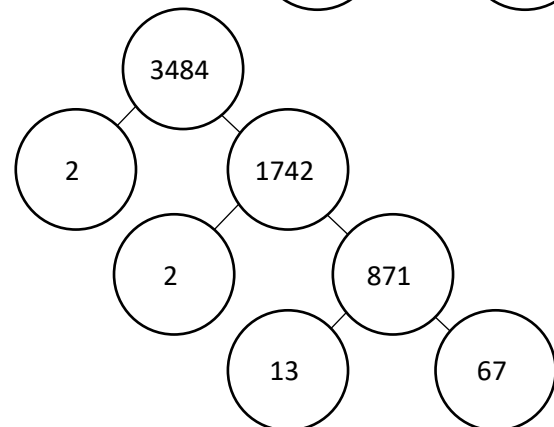
Solution:

Factor tree for 3484:

Prime factors of 3484 = 2, 2, 13, 67

Prime factorization of 3484 = $2 \times 2 \times 13 \times 67$

In index form: $3484 = \overline{2 \times 2} \times 13 \times 67$
 $= 2^2 \times 13 \times 67$



e) 6745

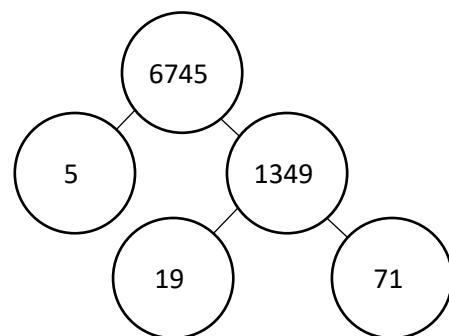
Solution:

Factor tree for 6745

Prime factors of 6745 = 5, 19, 71

Prime factorization of 6745 = $5 \times 19 \times 71$

In index form: $6745 = 5 \times 19 \times 71$



f) 1268

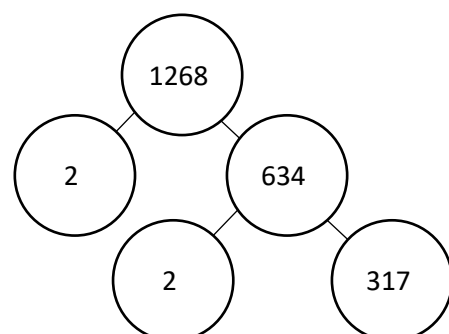
Solution:

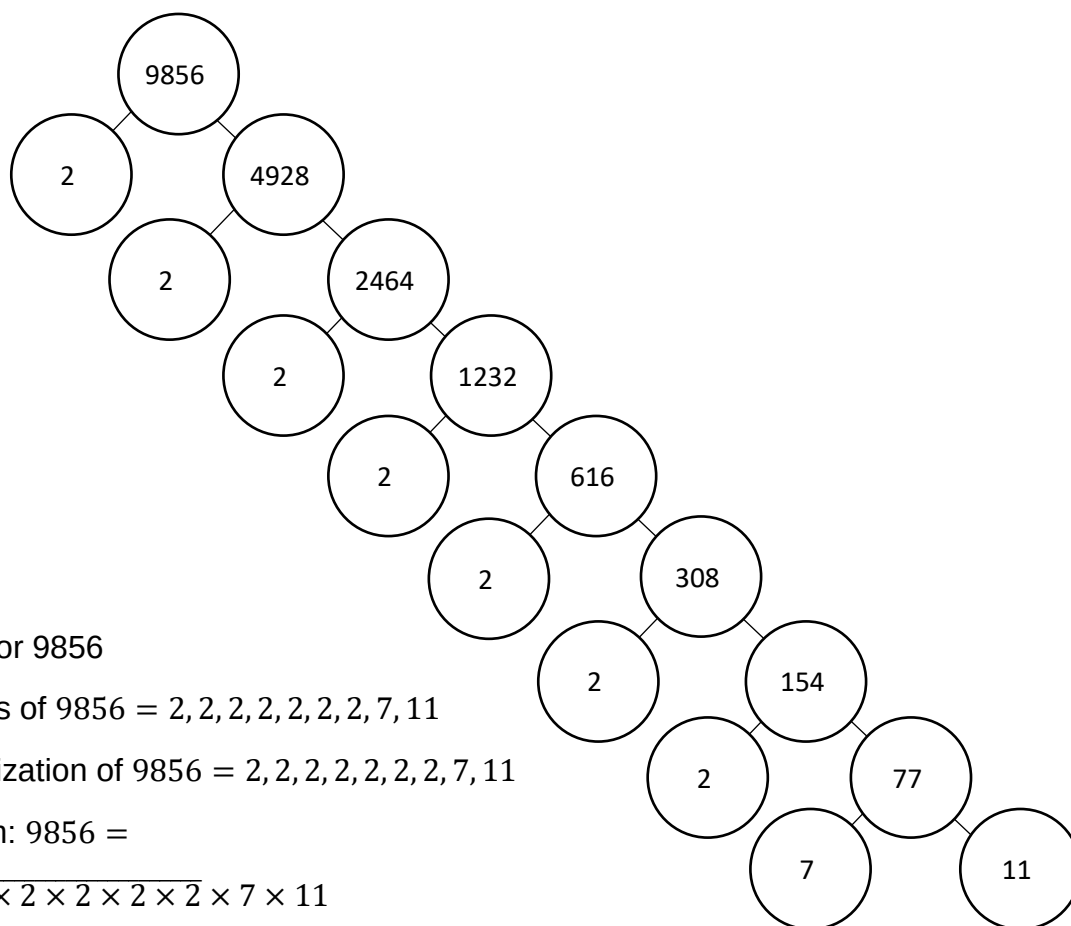
Factor tree for 1268

Prime factors of 1268 = 2, 2, 317

Prime factorization of 1268 = $2 \times 2 \times 317$

In index form: $1268 = \overline{2 \times 2} \times 317$
 $= 2^2 \times 317$





g) 9856

Solution:

Factor tree for 9856

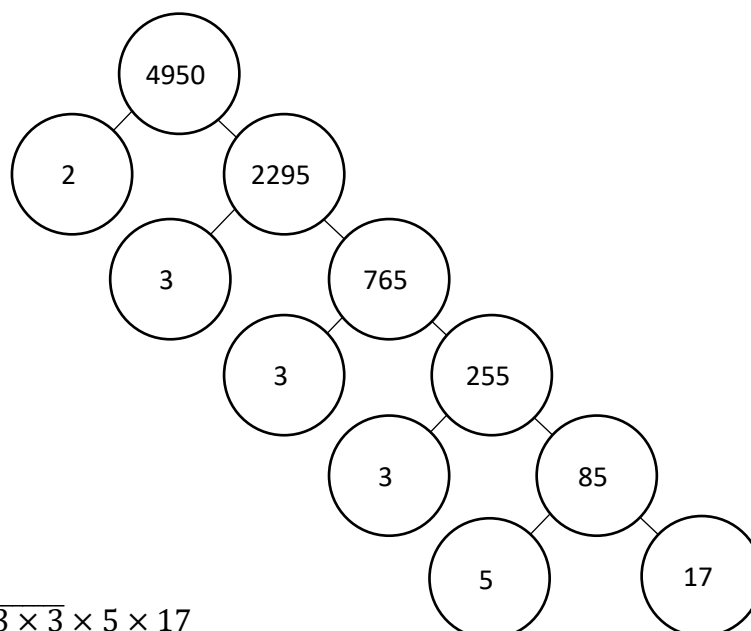
Prime factors of 9856 = 2, 2, 2, 2, 2, 2, 2, 7, 11

Prime factorization of 9856 = 2, 2, 2, 2, 2, 2, 2, 7, 11

In index form: 9856 =

$$= \overline{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2} \times 7 \times 11$$

$$= 2^7 \times 7 \times 11$$



h) 4950

Solution:

Factor tree for 4950

Prime factors of 4950 =

= 2, 3, 3, 3, 5, 17

Prime factorization of 4950 =

= 2 × 3 × 3 × 3 × 5 × 17

In index form: 4950 = 2 × $\overline{3 \times 3 \times 3}$ × 5 × 17

= 2 × 3³ × 5 × 17

3. Write the given numbers in expanded form.

a) 3^6

Solution:

3 raised to the power 6, i.e. 3^6 means 3 is being multiplied 6 times by itself.

So, in expanded form: $3^6 = 3 \times 3 \times 3 \times 3 \times 3 \times 3$

b) $3^2 \times 5^2$

Solution:

Here 3^2 means 3 is multiplied two times by itself and 5^2 means 5 is multiplied two times by itself.

So, in expanded form: $3^2 \times 5^2 = 3 \times 3 \times 5 \times 5$

c) $2^4 \times 7^2 \times 13^4$

Solution:

Here 2^4 means 2 is multiplied 4 times by itself, 7^2 means 7 is multiplied 2 times by itself and 13^4 means 13 is multiplied 4 times by itself.

So, in expanded form: $2^4 \times 7^2 \times 13^4 = 2 \times 2 \times 2 \times 2 \times 7 \times 7 \times 13 \times 13 \times 13 \times 13$

d) $3^2 \times 5^3$

Solution:

Here 3^2 means 3 is multiplied 2 times by itself and 5^3 means 5 is multiplied 3 times by itself.

So, in expanded form: $3^2 \times 5^3 = 3 \times 3 \times 5 \times 5 \times 5$

e) $11^2 \times 13^3$

Solution:

Here 11^2 means 11 is multiplied 2 times by itself and 13^3 means 13 is multiplied 3 times by itself.

So, in expanded form: $11^2 \times 13^3 = 11 \times 11 \times 13 \times 13 \times 13$

f) 2×5^5

Solution:

Here 2 means it is appearing in this factorization once and 5^5 means 5 is multiplied 5 times by itself.

So, in expanded form: $2 \times 5^5 = 2 \times 5 \times 5 \times 5 \times 5 \times 5$

4. Express the following in index notation.

a) $5 \times 5 \times 5$

In index form $= \overline{5 \times 5 \times 5} = 5^3$

b) $2 \times 2 \times 5 \times 5 \times 5 \times 5$

In index form $= \overline{2 \times 2} \times \overline{5 \times 5 \times 5 \times 5} = 2^2 \times 5^4$

c) $2 \times 2 \times 7 \times 7 \times 7$

In index form $= \overline{2 \times 2} \times \overline{7 \times 7 \times 7} = 2^2 \times 7^3$

d) $3 \times 3 \times 3 \times 17 \times 17 \times 23$

In index form $= \overline{3 \times 3 \times 3} \times \overline{17 \times 17} \times 23 = 3^3 \times 17^2 \times 23$

1.3 Let's Practice

1. Find the HCF and LCM of the following numbers using prime factorization method.

a) 30, 50, 80

Solution:

First find the prime factors of 170, 156 and 190.

Prime factorization of 30, 50 and 80

$$\begin{array}{r|l} 2 & 30 \\ \hline 3 & 15 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 50 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 80 \\ \hline 2 & 40 \\ \hline 2 & 20 \\ \hline 2 & 10 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 30 = $2 \times 3 \times 5$

Prime factorization of 50 = $2 \times 5 \times 5$

Prime factorization of 80 = $2 \times 2 \times 2 \times 2 \times 5$

Common prime factors = 2 and 5

Product of common prime factors = $2 \times 5 = 10$

So, the HCF of 30, 50 and 80 = 10

LCM

Prime factorization of 30 = $2 \times 3 \times 5$

Prime factorization of 50 = $2 \times 5 \times 5$

Prime factorization of 80 = $2 \times 2 \times 2 \times 2 \times 5$

Common prime factors of 30, 50 and 80 = 2 and 5

Non-common factors of 30, 50 and 80 = 3, 5, 2, 2 and 2

LCM = Product of common prime factors of two or more numbers $\times$ Product of non-common prime factors

$$= 2 \times 5 \times 3 \times 5 \times 2 \times 2 \times 2 = 1200$$

So, the LCM of 30, 50 and 800 = 1200

b) 170, 156, 190

Solution:

First find the prime factors of 170, 156 and 190.

Prime factorization of 170, 156 and 190

$$\begin{array}{r|l} 2 & 170 \\ \hline 5 & 85 \\ \hline 17 & 17 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 156 \\ \hline 2 & 78 \\ \hline 3 & 39 \\ \hline 13 & 13 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 190 \\ \hline 5 & 95 \\ \hline 19 & 19 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 170 = $2 \times 5 \times 17$

Prime factorization of 156 = $2 \times 2 \times 3 \times 13$

Prime factorization of 190 = $2 \times 5 \times 19$

Common prime factors of 170, 156 and 190 = 2

So, the HCF of 170, 156 and 190 = 2

LCM

Prime factorization of 170 = $2 \times 5 \times 17$

Prime factorization of 156 = $2 \times 2 \times 3 \times 13$

Prime factorization of 190 = $2 \times 5 \times 19$

Common prime factors of 170, 156 and 190 = 2, 5

non-common prime factors of 170, 156 and 190 = 17, 2, 3, 13, 19

LCM = Product of common prime factors $\times$ Product of non-common prime factors

$$= 2 \times 5 \times 17 \times 2 \times 3 \times 13 \times 19 = 251940$$

So, the LCM of 170, 156 and 190 = 251940

c) 400, 200, 300

Solution:

First find the prime factors of 400, 200 and 300.

Prime factorization of 400, 200 and 300

$$\begin{array}{r|l} 2 & 400 \\ \hline 2 & 200 \\ \hline 2 & 100 \\ \hline 2 & 50 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 200 \\ \hline 2 & 100 \\ \hline 2 & 50 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 300 \\ \hline 2 & 150 \\ \hline 3 & 75 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 400 = $2 \times 2 \times 2 \times 2 \times 5 \times 5$

Prime factorization of 200 = $2 \times 2 \times 2 \times 5 \times 5$

Prime factorization of 300 = $2 \times 2 \times 3 \times 5 \times 5$

Common prime factors of 400, 300 and 200 = 2, 2, 5, 5

Product of common prime factors = $2 \times 2 \times 5 \times 5 = 100$

So, the HCF of 400, 200 and 300 = 100

LCM

Prime factorization of 400 = $2 \times 2 \times 2 \times 2 \times 5 \times 5$

Prime factorization of 200 = $2 \times 2 \times 2 \times 5 \times 5$

Prime factorization of 300 = $2 \times 2 \times 3 \times 5 \times 5$

Common prime factors of 400, 200 and 300 = 2, 2, 2, 5, 5

non-common prime factors of 400, 200 and 300 = 2, 3

LCM = Product of common prime factors of two or more numbers $\times$ Product of non-common prime factors

$$= 2 \times 2 \times 2 \times 5 \times 5 \times 2 \times 3 = 1200$$

So, the LCM of 400, 200 and 300 = 1200

d) 650, 450, 750

Solution:

First find the prime factors of 650, 450 and 750.

Prime factorization of 650, 450 and 750

$$\begin{array}{r|l} 2 & 650 \\ \hline 5 & 325 \\ \hline 5 & 65 \\ \hline 13 & 13 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 450 \\ \hline 3 & 225 \\ \hline 3 & 75 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 750 \\ \hline 3 & 375 \\ \hline 5 & 125 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 650 = $2 \times 5 \times 5 \times 13$

Prime factorization of 450 = $2 \times 3 \times 3 \times 5 \times 5$

Prime factorization of 750 = $2 \times 3 \times 5 \times 5 \times 5$

Common prime factors of 400, 300 and 200 = 2, 5, 5

Product of common prime factors = $2 \times 5 \times 5 = 50$

So, the HCF of 650, 450 and 750 = 50

LCM

Prime factorization of 650 = $2 \times 5 \times 5 \times 13$

Prime factorization of 450 = $2 \times 3 \times 3 \times 5 \times 5$

Prime factorization of 750 = $2 \times 3 \times 5 \times 5 \times 5$

Common prime factors of 400, 300 and 200 = 2, 5, 5, 3

non-common prime factors of 400, 300 and 200 = 3, 13, 5

LCM = Product of common prime factors of two or more numbers $\times$ Product of non-common prime factors

$$= 2 \times 5 \times 5 \times 3 \times 3 \times 13 \times 5 = 29250$$

So, the LCM of 650, 450 and 750 = 29250

e) 786, 566, 824

Solution:

First find the prime factors of 786, 566 and 824.

Prime factorization of 786, 566 and 824

$$\begin{array}{r|l} 2 & 786 \\ \hline 3 & 393 \\ \hline 131 & 131 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 566 \\ \hline 283 & 283 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 824 \\ \hline 2 & 412 \\ \hline 2 & 206 \\ \hline 103 & 103 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 786 = $2 \times 3 \times 131$

Prime factorization of 566 = 2×283

Prime factorization of 824 = $2 \times 2 \times 2 \times 103$

Common prime factors of 786, 566 and 824 = 2

So, the HCF of 786, 566 and 824 = 2

LCM

Prime factorization of 786 = $2 \times 3 \times 131$

Prime factorization of 566 = 2×283

Prime factorization of 824 = $2 \times 2 \times 2 \times 103$

Common prime factors of 786, 566 and 824 = 2

non-common prime factors of 786, 566 and 824 = 3, 131, 283, 2, 2, 103

LCM = Product of common prime factors $\times$ Product of non-common prime factors
of two or more numbers
= $2 \times 3 \times 131 \times 283 \times 2 \times 2 \times 103 = 91644456$

So, the LCM of 786, 566 and 824 = 91644456

e) 145, 215, 325

Solution:

First find the prime factors of 145, 215 and 325.

Prime factorization of 145, 215 and 325

$$\begin{array}{r|l} 5 & 145 \\ \hline 129 & 29 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 5 & 215 \\ \hline 43 & 43 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 5 & 325 \\ \hline 5 & 65 \\ \hline 13 & 13 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 145 = 5×29

Prime factorization of 215 = 5×43

Prime factorization of 325 = $5 \times 5 \times 13$

Common prime factors = 5

So, the HCF of 145, 215 and 325 = 5

LCM

Prime factorization of 145 = 5×29

Prime factorization of 215 = 5×43

Prime factorization of 325 = $5 \times 5 \times 13$

Common prime factors 145, 215 and 325 = 5

non-common prime factors 145, 215 and 325 = 5, 29, 43, 13

LCM = Product of common prime factors $\times$ Product of non-common prime factors
of two or more numbers
= $5 \times 5 \times 29 \times 43 \times 13 = 405275$

So, the LCM of 145, 215 and 325 = 405275

2. Find the HCF and LCM of the following numbers using division method.

a) 50, 120, 150

Solution:

HCF

Step 1: Divide 150 by 120

$$\begin{array}{r} 1 \\ 120 \overline{) 150} \\ \underline{-120} \quad 4 \\ \text{HCF of 120 and 150} \longrightarrow 30 \overline{) 120} \\ \underline{-120} \\ 0 \end{array}$$

Step 2: Divide the third number i.e. 50 by the last divisor of step 1.

$$\begin{array}{r} 1 \\ 30 \overline{) 50} \\ \underline{-30} \quad 1 \\ 20 \overline{) 30} \\ \underline{-20} \quad 2 \\ \text{HCF of 150, 120} \longrightarrow 10 \overline{) 20} \\ \underline{-20} \\ 0 \end{array}$$

and 50

So, the HCF of 50, 120 and 150 = 10

LCM: Divide by a prime number at each step.

2	50, 120, 150	
2	25, 60, 75	← Dividing all by 2.
2	25, 30, 75	← Dividing 60 by 2.
3	25, 15, 75	← Dividing 30 by 2.
5	25, 5, 25	← Dividing 15 and 75 by 3.
5	5, 5, 5	← Dividing all by 5.
	1, 1, 1	← Dividing 5 by 5.

LCM = Product of prime factors

$$= 2 \times 2 \times 2 \times 3 \times 5 \times 5 = 600$$

b) 740, 390

Solution:

HCF

Step 1: Divide 740 by 390

$$\begin{array}{r} 1 \\ 390 \overline{) 740} \\ \underline{-390} \quad 1 \\ 350 \overline{) 390} \\ \underline{-350} \quad 8 \\ 40 \overline{) 350} \\ \underline{-320} \quad 1 \\ 30 \overline{) 40} \\ \underline{-30} \quad 3 \\ \text{HCF} \longrightarrow 10 \overline{) 30} \\ \underline{-30} \\ 0 \end{array}$$

So, the HCF of 390 and 740 is 10.

LCM: Divide by a prime number at each step.

2	740, 390	
2	370, 193	← Divding all by 2.
5	185, 195	← Divding 370 by 2.
37	37, 39	← Divding both by 5.
13	1, 39	← Divding 37 by 37.
3	1, 3	← Divding 39 by 13.
	1, 1	← Divding 3 by 3.

LCM = Product of prime factors

$$= 2 \times 2 \times 5 \times 37 \times 13 \times 3 = 28860$$

c) 170, 165, 155

Solution:

HCF

Step 1: Divide 170 by 165

HCF of 165
and 170

$$\begin{array}{r} 1 \\ 165 \overline{) 170} \\ \underline{-165} 33 \\ 5 \overline{) 165} \\ \underline{-165} \\ 0 \end{array}$$

Step 2: Divide 155 by the last divisor of step 1.

HCF of 170,
165 and 155

$$\begin{array}{r} 31 \\ 5 \overline{) 155} \\ \underline{-155} \\ 0 \end{array}$$

So, the HCF of 170, 165 and 155 = 5

LCM: Divide 170, 165 and 155 by prime factors.

2	170, 165, 155	
5	85, 165, 155	← Divding 170 by 2.
11	17, 33, 31	← Divding all by 5.
3	17, 3, 31	← Divding 33 by 11.
17	17, 1, 31	← Divding 3 by 3.
31	1, 1, 31	← Divding 17 by 17.
	1, 1, 1	← Divding 31 by 31.

LCM = Product of prime factors

$$= 2 \times 5 \times 11 \times 3 \times 17 \times 31$$

$$= 173910$$

d) 660, 760, 460

Solution:

HCF

Step 1: Divide 760 by 660

$$\begin{array}{r}
 1 \\
 660 \overline{) 760} \\
 \underline{-660} 6 \\
 100 \overline{) 660} \\
 \underline{-600} 1 \\
 60 \overline{) 100} \\
 \underline{-60} 1 \\
 40 \overline{) 60} \\
 \underline{-40} 2 \\
 20 \overline{) 40} \\
 \underline{-40} \\
 0
 \end{array}$$

HCF of 760 and 660 $\longrightarrow$

Step 2: Divide 460 by the last divisor of step 1.

$$\begin{array}{r}
 23 \\
 20 \overline{) 460} \\
 \underline{-460} \\
 0
 \end{array}$$

HCF of 760, 660 and 460 $\longrightarrow$

So, the HCF of 760, 660 and 460 is 20.

LCM: Divide 660, 760 and 460 by prime factors.

2	660, 760, 460	
2	330, 380, 230	← Dividing all by 2.
2	165, 190, 115	← Dividing all by 2.
3	165, 95, 115	← Dividing 190 by 2.
5	55, 95, 115	← Dividing 165 by 3
11	11, 19, 23	← Dividing all by 5.
19	1, 19, 23	← Dividing 11 by 11.
23	1, 1, 23	← Dividing 19 by 19.
	1, 1, 1	← Dividing 23 by 23.

LCM = Product of prime factors

$$= 2 \times 2 \times 2 \times 3 \times 5 \times 11 \times 19 \times 23$$

$$= 576840$$

e) 492, 578, 890

Solution:

HCF

Step 1: Divide 890 by 578

$$\begin{array}{r}
 1 \\
 578 \overline{) 890} \\
 \underline{-578} \quad 1 \\
 312 \overline{) 578} \\
 \underline{-312} \quad 1 \\
 266 \overline{) 312} \\
 \underline{-266} \quad 5 \\
 46 \overline{) 266} \\
 \underline{-230} \quad 1 \\
 36 \overline{) 46} \\
 \underline{-36} \quad 3 \\
 10 \overline{) 36} \\
 \underline{-30} \quad 1 \\
 6 \overline{) 10} \\
 \underline{-6} \quad 1 \\
 4 \overline{) 6} \\
 \underline{-4} \quad 2 \\
 2 \overline{) 4} \\
 \underline{-4} \\
 0
 \end{array}$$

HCF of 890 and 578 →

Step 2: Divide 492 by the last divisor of step 1.

$$\begin{array}{r}
 246 \\
 2 \overline{) 492} \\
 \underline{-492} \\
 0
 \end{array}$$

So, the HCF of 492, 578 and 890 is 2.

LCM: Divide all numbers by prime factors.

2	492, 578, 890	
2	246, 289, 445	← Dividing all by 2.
3	123, 289, 445	← Dividing 246 by 2.
5	41, 289, 445	← Dividing 123 by 3.
17	41, 289, 89	← Dividing 445 by 5.
17	41, 17, 89	← Dividing 289 by 17.
41	41, 1, 89	← Dividing 17 by 17.
89	1, 1, 89	← Dividing 41 by 41.
	1, 1, 1	← Dividing 89 by 89.

LCM = Product of prime factors

$$= 2 \times 2 \times 3 \times 5 \times 17 \times 17 \times 41 \times 89 = 63273660$$

f) 220, 250, 280

Solution:

HCF

Step 1: Divide 280 by 250

$$\begin{array}{r}
 1 \\
 250 \overline{) 280} \\
 \underline{-250} \quad 8 \\
 30 \overline{) 250} \\
 \underline{-240} \quad 3 \\
 \text{HCF of 280 and 250} \longrightarrow 10 \overline{) 30} \\
 \underline{-30} \\
 0
 \end{array}$$

Step 2: Divide 492 by the last divisor of step 1.

$$\begin{array}{r}
 10 \\
 10 \overline{) 220} \\
 \underline{-220} \\
 0
 \end{array}$$

So, the HCF of 220, 250 and 280 is 10.

LCM: Divide 220, 250 and 280 by prime factors.

2	220, 250, 280	
2	110, 125, 140	← Dividing all by 2.
2	55, 125, 70	← Dividing 110 and 140 by 2.
5	55, 125, 35	← Dividing 70 by 2.
5	11, 25, 7	← Dividing all by 5.
5	11, 5, 7	← Dividing 11 and 25 by 5.
11	11, 1, 7	← Dividing 5 by 5.
7	1, 1, 7	← Dividing 11 by 11.
	1, 1, 1	← Dividing 7 by 7.

LCM = Product of prime factors

$$= 2 \times 2 \times 2 \times 5 \times 5 \times 5 \times 11 \times 7 = 77000$$

3. Find the longest length of measuring tape that can measure the length of three ribbons of 145 cm, 150 cm, and 175 cm completely.

Solution:

To find the longest of measuring tape, we have to find the HCF of the three values.

Given:

Length of first ribbon = 145 cm

Length of second ribbon = 150 cm

Length of third ribbon = 175 cm

Prime factorization of 145, 150 and 175.

$$\begin{array}{r|l} 5 & 145 \\ 29 & 29 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 150 \\ 3 & 75 \\ 5 & 25 \\ 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 5 & 175 \\ 5 & 35 \\ 7 & 7 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 145 = 5×29

Prime factorization of 150 = $2 \times 3 \times 5 \times 5$

Prime factorization of 175 = $5 \times 5 \times 7$

Common prime factors = 5

So, the HCF of 145, 150 and 175 = 5

Therefore the longest length of measuring tape = 5 cm.

- 4. Find the length of the shortest rope that can be exactly cut into pieces of 15 cm, 25 cm and 90 cm, respectively, without leaving any remainder.**
Solution:

To find the shortest of length, we have to find the LCM of the three values.

Given:

Length of one rope = 15 cm

Length of second rope = 25 cm

Length of third rope = 90 cm

Prime factorization of the three values.

$$\begin{array}{r|l} 2 & 15, 25, 90 \\ 3 & 15, 25, 45 \\ 3 & 5, 25, 15 \\ 5 & 1, 5, 3 \\ 5 & 1, 1, 3 \\ \hline & 1, 1, 1 \end{array}$$

LCM = Product of prime factors

$$= 2 \times 3 \times 3 \times 5 \times 5 = 450$$

So, the length of the shortest rope = 450 cm.

- 5. Ali, Ahmad and Hassan walk around a circular walking track. They complete one round in 100 sec, 120 sec and 180 sec, respectively. After how many seconds will they meet again at the starting point?**

Solution:

To find after how many seconds they will meet again, we have to find the LCM of the three value.

Ali completes one round in = 100 cm

Ahmed completes one round in = 120 cm

Hassan completes one round in = 180 cm

Prime factorization of the three values.

$$\begin{array}{r|l} 2 & 100, 120, 180 \\ 2 & 50, 60, 90 \\ 2 & 25, 30, 45 \\ 3 & 25, 15, 45 \\ 3 & 25, 5, 15 \\ 5 & 25, 5, 5 \\ 5 & 5, 1, 1 \\ \hline & 1, 1, 1 \end{array}$$

LCM = Product of prime factors

$$= 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 = 1800$$

6. Find the greatest number that can divide 430, 170 and 550 completely.

Solution:

To find the greatest number we find the HCF of the given values.

Prime factorization of the given values.

$$\begin{array}{r|l} 2 & 430 \\ \hline 5 & 215 \\ \hline 43 & 43 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 170 \\ \hline 5 & 85 \\ \hline 17 & 17 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 550 \\ \hline 5 & 275 \\ \hline 5 & 55 \\ \hline 11 & 11 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 430 = $2 \times 5 \times 43$

Prime factorization of 170 = $2 \times 5 \times 17$

Prime factorization of 550 = $2 \times 5 \times 5 \times 11$

Common prime factors = 2, 5

Product of common factors = $2 \times 5 = 10$

So, the HCF = 10

Therefore the greatest number that can divide 430, 170 and 550 completely is 10.

7. Find the smallest number that is exactly divisible by 860, 448 and 568 without any remainder.

Solution:

To find the smallest number we have to find the LCM of the given values.

Prime factorization of 860, 448 and 568

$$\begin{array}{r|l} 2 & 860, 448, 568 \\ \hline 2 & 430, 224, 284 \\ \hline 2 & 215, 112, 142 \\ \hline 2 & 215, 56, 71 \\ \hline 2 & 215, 28, 71 \\ \hline 2 & 215, 14, 71 \\ \hline 5 & 215, 7, 71 \\ \hline 7 & 43, 7, 71 \\ \hline 43 & 43, 1, 71 \\ \hline 71 & 1, 1, 71 \\ \hline & 1, 1, 1 \end{array}$$

LCM = Product of prime factors

$$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 7 \times 43 \times 71 = 6838720$$

The smallest number that is exactly divisible by 860, 448 and 568 is 6838720.

1.4 Let's Practice

1. Find the square of the following.

a) 12

Solution:

When a number is multiplied it self, the product is called the square of that number. So, we multiple by itself to get its square.

$$\begin{aligned}\text{Square of } 12 &= 12 \times 12 \\ &= 12^2 = 144 \text{ Answer}\end{aligned}$$

d) 19

Solution:

$$\begin{aligned}\text{Square of } 19 &= 19 \times 19 \\ &= 19^2 = 361 \text{ Answer}\end{aligned}$$

f) 69

Solution:

$$\begin{aligned}\text{Square of } 69 &= 69 \times 69 \\ &= 69^2 = 4761 \text{ Answer}\end{aligned}$$

h) 91

Solution:

$$\begin{aligned}\text{Square of } 91 &= 91 \times 91 \\ &= 91^2 = 8281 \text{ Answer}\end{aligned}$$

j) 13

Solution:

$$\begin{aligned}\text{Square of } 13 &= 13 \times 13 \\ &= 13^2 = 169 \text{ Answer}\end{aligned}$$

l) 40

Solution:

$$\begin{aligned}\text{Square of } 40 &= 40 \times 40 \\ &= 40^2 = 1600 \text{ Answer}\end{aligned}$$

b) 35

Solution:

$$\begin{aligned}\text{Square of } 55 &= 55 \times 55 \\ &= 55^2 = 3025 \text{ Answer}\end{aligned}$$

c) 42

Solution:

$$\begin{aligned}\text{Square of } 42 &= 42 \times 42 \\ &= 42^2 = 1764 \text{ Answer}\end{aligned}$$

e) 78

Solution:

$$\begin{aligned}\text{Square of } 78 &= 78 \times 78 \\ &= 78^2 = 6084 \text{ Answer}\end{aligned}$$

g) 21

Solution:

$$\begin{aligned}\text{Square of } 21 &= 21 \times 21 \\ &= 21^2 = 441 \text{ Answer}\end{aligned}$$

i) 86

Solution:

$$\begin{aligned}\text{Square of } 86 &= 86 \times 86 \\ &= 86^2 = 7396 \text{ Answer}\end{aligned}$$

k) 25

Solution:

$$\begin{aligned}\text{Square of } 25 &= 25 \times 25 \\ &= 25^2 = 625 \text{ Answer}\end{aligned}$$

2. A square-shaped ground has a side length of 10 metres. Find the area of the ground.

Solution:

Given:

Length of a side of square-shaped ground = 10 m

Area of the ground = ?

Area of the square-shaped ground = length $\times$ length

$$= 10 \text{ m} \times 10 \text{ m}$$

$$= 100 \text{ m}^2$$

as meter is multiplied by metre so it is written as m raised to the power 2, $\text{m} \times \text{m} = \text{m}^2$, or metre square

Unit Evaluation

1. Choose the correct option.

a) (ii) factor

b) (iii) 65

$$\begin{array}{r} \leftarrow 13 \\ \times 5 \\ \hline 65 \end{array}$$

c) (iii) Prime factorization

d) (iii) $2 \times 2 \times 2 \times 2$

e) (iii) 65

f) (iv) square

2. Find the HCF and LCM of the following numbers using division method.

a) 124

Solution:

Divisors of 124:

$$1 \times 124 = 124, \quad 2 \times 62 = 124, \quad 4 \times 31 = 124$$

So, the factors of 124 are: 1, 2, 4, 31, 62, 124.

b) 69

Solution:

Divisors of 69:

$$1 \times 69 = 69, \quad 3 \times 23 = 69$$

So, the factors of 69 are: 1, 3, 23, 69.

c) 154

Solution:

Divisors of 154:

$$1 \times 154 = 154, \quad 2 \times 77 = 154, \quad 7 \times 22 = 154, \quad 11 \times 14 = 154$$

So, the factors of 154 are: 1, 2, 7, 11, 14, 22, 77, 154.

d) 988

Solution:

Divisors of 988:

$$1 \times 988 = 988, \quad 2 \times 494 = 988, \quad 4 \times 247 = 988, \quad 13 \times 76 = 988, \quad 19 \times 52 = 988$$

$$26 \times 38 = 988$$

So, the factors of 154 are: 1, 2, 4, 13, 19, 26, 38, 52, 76, 247, 494, 988.

3. Use prime factorization to find the factors of the given numbers and write them in index form.

a) 278

Solution:

Prime factorization of 278:

$$\begin{array}{r|l} 2 & 278 \\ \hline 139 & 139 \\ \hline & 1 \end{array}$$

Prime factors of 278 = 2, 139

Prime factorization of 278 = 2×139

Index form: $278 = 2 \times 139$ (as not term is repeated and appears only one)

b) 8088

Solution:

Prime factorization of 8088:

$$\begin{array}{r|l} 2 & 8088 \\ \hline 2 & 4044 \\ \hline 2 & 2022 \\ \hline 3 & 1011 \\ \hline 337 & 337 \\ \hline & 1 \end{array}$$

Prime factors of 8088 = 2, 2, 2, 3, 337

Prime factorization of 8088 = $2 \times 2 \times 2 \times 3 \times 337$

Index form: $8088 = 2 \times 2 \times 2 \times 3 \times 337$
 $= 2^3 \times 3 \times 337$

c) 300

Solution:

Prime factorization of 300:

$$\begin{array}{r|l} 2 & 300 \\ \hline 2 & 150 \\ \hline 3 & 75 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

Prime factors of 300 = 2, 2, 3, 5, 5

Prime factorization of 300 = $2 \times 2 \times 3 \times 5 \times 5$

Index form: $300 = 2 \times 2 \times 3 \times 5 \times 5$
 $= 2^2 \times 3 \times 5^2$

d) 5432

Solution:

Prime factorization of 5432:

$$\begin{array}{r|l} 2 & 5432 \\ \hline 2 & 2716 \\ \hline 2 & 1358 \\ \hline 7 & 679 \\ \hline 97 & 97 \\ \hline & 1 \end{array}$$

Prime factors of 5432 = 2, 2, 2, 7, 97

Prime factorization of 5432 = $2 \times 2 \times 2 \times 7 \times 97$

Index form: $5432 = 2 \times 2 \times 2 \times 7 \times 97$
 $= 2^3 \times 7 \times 97$

4. Find the HCF and LCM of the following numbers using the prime factorization method.

a) 286, 78

Solution:

First we find the prime factors of 286 and 78

$$\begin{array}{r|l} 2 & 286 \\ \hline 11 & 143 \\ \hline 13 & 13 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 78 \\ \hline 3 & 39 \\ \hline 13 & 13 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 286 = $(2) \times 11 \times (13)$

Prime factorization of 78 = $(2) \times 3 \times (13)$

Common prime factors = 2, 13

Product of common prime factors = $2 \times 13 = 26$

So, the HCF of 286 and 78 = 26

LCM

Prime factorization of 286 = $(2) \times 11 \times (13)$

Prime factorization of 78 = $(2) \times 3 \times (13)$

Common prime factors of 286 and 78 = 2, 13

Non-common prime factors of 286 and 78 = 11, 3

LCM = Product of common prime factors $\times$ Product of non-common prime factors

= $2 \times 13 \times 11 \times 3 = 858$

So, the LCM of 286 and 78 = 858

b) 788, 542, 128

Solution:

Prime factorization of 788, 542 and 128

$$\begin{array}{r|l} 2 & 788 \\ \hline 2 & 394 \\ \hline 197 & 197 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 542 \\ \hline 271 & 271 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 128 \\ \hline 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 788 = $(2) \times 2 \times 197$

Prime factorization of 542 = $(2) \times 271$

Prime factorization of 128 = $(2) \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$

Common prime factors = 2

So, the HCF of 788, 542 and 128 = 2

LCM

$$\text{Prime factorization of 788} = 2 \times 2 \times 197$$

$$\text{Prime factorization of 524} = 2 \times 2 \times 271$$

$$\text{Prime factorization of 128} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$\text{Common prime factors} = 2, 2$$

$$\text{Non-common prime factors} = 197, 271, 2, 2, 2, 2, 2$$

LCM = Product of common prime factors of two or more numbers $\times$ Product of non-common prime factors

$$= 2 \times 2 \times 197 \times 271 \times 2 \times 2 \times 2 \times 2 \times 2 = 6833536$$

So, the LCM of 788, 542 and 128 = 6833536

c) 560, 440, 220

Solution:

Prime factorization of 560, 440 and 220

2	560
2	280
2	140
2	70
5	35
7	7
	1

2	440
2	220
2	110
5	55
11	11
	1

2	220
2	110
5	55
11	11
	1

HCF

$$\text{Prime factorization of 560} = 2 \times 2 \times 2 \times 2 \times 5 \times 7$$

$$\text{Prime factorization of 440} = 2 \times 2 \times 2 \times 5 \times 11$$

$$\text{Prime factorization of 220} = 2 \times 2 \times 5 \times 11$$

$$\text{Common prime factors} = 2, 2, 5$$

$$\text{Product of common prime factors} = 2 \times 2 \times 5 = 20$$

So, the HCF of 560, 44 and 220 = 20

LCM

$$\text{Prime factorization of 560} = 2 \times 2 \times 2 \times 2 \times 5 \times 7$$

$$\text{Prime factorization of 440} = 2 \times 2 \times 2 \times 5 \times 11$$

$$\text{Prime factorization of 220} = 2 \times 2 \times 5 \times 11$$

$$\text{Common prime factors} = 2, 2, 2, 5, 11$$

$$\text{Non-common prime factors} = 2, 7$$

LCM = Product of common prime factors of two or more numbers $\times$ Product of non-common prime factors

$$= 2 \times 2 \times 2 \times 5 \times 11 \times 2 \times 7 = 6160$$

So, the LCM of 560, 440, 220 = 6160

d) 160, 180, 200

Solution:

Prime factorization of 160, 180 and 200

$$\begin{array}{r|l} 2 & 160 \\ \hline 2 & 80 \\ \hline 2 & 40 \\ \hline 2 & 20 \\ \hline 2 & 10 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 180 \\ \hline 2 & 90 \\ \hline 3 & 45 \\ \hline 3 & 15 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 200 \\ \hline 2 & 100 \\ \hline 2 & 50 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

HCF

Prime factorization of 160 = $2 \times 2 \times 2 \times 2 \times 2 \times 5$

Prime factorization of 180 = $2 \times 2 \times 3 \times 3 \times 5$

Prime factorization of 200 = $2 \times 2 \times 2 \times 5 \times 5$

Common prime factors = 2, 2, 5

Product of common prime factors = $2 \times 2 \times 5 = 20$

So, the HCF of 160, 180 and 200 = 20

LCM

Prime factorization of 160 = $2 \times 2 \times 2 \times 2 \times 2 \times 5$

Prime factorization of 180 = $2 \times 2 \times 3 \times 3 \times 5$

Prime factorization of 200 = $2 \times 2 \times 2 \times 5 \times 5$

Common prime factors = 2, 2, 2, 5

Non-common prime factors = 2, 2, 3, 3, 5

LCM = Product of common prime factors $\times$ Product of non-common prime factors

= $2 \times 2 \times 2 \times 5 \times 2 \times 2 \times 3 \times 3 \times 5 = 7200$

So, the LCM of 160, 180, 200 = 7200

5. Find the HCF and LCM of the following numbers by division method.

a) 44, 62

Solution:

HCF

Step 1: Divide 62 by 44

$$\begin{array}{r} 1 \\ 44 \overline{) 62} \\ \underline{-44} \quad 2 \\ 18 \overline{) 44} \\ \underline{-36} \quad 2 \\ 8 \overline{) 18} \\ \underline{-16} \quad 4 \\ 2 \overline{) 8} \\ \underline{-8} \quad 0 \end{array}$$

HCF of 44 and 62 $\longrightarrow$ 2

Step 2: Divide 24 by the last divisor of 44 and 62.

$$\begin{array}{r} \text{HCF of all three numbers} \longrightarrow 2 \overline{) 24} \\ \underline{- 24} \\ 0 \end{array}$$

So, the HCF of 44, 24 and 62 = 2

LCM: Divide by a prime number at each step.

$$\begin{array}{l} \begin{array}{l} 2 \overline{) 44, 24, 62} \\ 2 \overline{) 22, 12, 31} \\ 2 \overline{) 11, 6, 31} \\ 3 \overline{) 11, 3, 31} \\ 11 \overline{) 11, 1, 31} \\ 31 \overline{) 1, 1, 31} \\ 1, 1, 1 \end{array} \begin{array}{l} \longleftarrow \text{Dividing all by 2.} \\ \longleftarrow \text{Dividing 22 and 12 by 2.} \\ \longleftarrow \text{Dividing 6 by 2.} \\ \longleftarrow \text{Dividing 3 by 3.} \\ \longleftarrow \text{Dividing 11 by 11.} \\ \longleftarrow \text{Dividing 31 by 31.} \end{array} \end{array}$$

LCM = Product of prime factors

$$= 2 \times 2 \times 2 \times 3 \times 11 \times 31$$

$$= 8184$$

b) 990, 690, 190

Solution:

HCF

Step 1: Divide 990 by 690

$$\begin{array}{r} \begin{array}{r} 1 \\ 690 \overline{) 990} \\ \underline{- 690} \quad 2 \\ 300 \overline{) 690} \\ \underline{- 600} \quad 3 \\ 90 \overline{) 300} \\ \underline{- 270} \quad 3 \\ 30 \overline{) 90} \\ \underline{- 90} \\ 0 \end{array} \\ \text{HCF of 690 and 990} \longrightarrow \end{array}$$

Step 2: Divide 190 by the last divisor of step 1.

$$\begin{array}{r} \begin{array}{r} 6 \\ 30 \overline{) 190} \\ \underline{- 180} \quad 3 \\ 10 \overline{) 30} \\ \underline{- 30} \\ 0 \end{array} \\ \text{HCF of all the three numbers} \longrightarrow \end{array}$$

So, the HCF of 990, 690, and 190 = 10

LCM: Divide by a prime number at each step.

2	990, 690, 190	
3	495, 345, 95	← Dividing all by 2.
3	165, 115, 95	← Dividing 995 and 345 by 3.
5	55, 115, 95	← Dividing 165 by 3.
11	11, 23, 19	← Dividing all by 5.
23	1, 23, 19	← Dividing 11 by 11.
19	1, 1, 19	← Dividing 23 by 23.
	1, 1, 1	← Dividing 19 by 19.

LCM = Product of prime factors

$$= 2 \times 3 \times 3 \times 5 \times 11 \times 23 \times 19 = 432630$$

c) 452, 786, 121

Solution:

HCF

Step 1: Divide 786 by 452

$$\begin{array}{r}
 1 \\
 452 \overline{) 786} \\
 \underline{-452} \quad 1 \\
 334 \overline{) 452} \\
 \underline{-334} \quad 2 \\
 118 \overline{) 334} \\
 \underline{-236} \quad 1 \\
 98 \overline{) 118} \\
 \underline{-98} \quad 4 \\
 20 \overline{) 98} \\
 \underline{-80} \quad 1 \\
 18 \overline{) 20} \\
 \underline{-18} \quad 9 \\
 2 \overline{) 18} \\
 \underline{-18} \\
 0
 \end{array}$$

Step 2: Divide 121 by the last divisor of step 1.

$$\begin{array}{r} 60 \\ 30 \overline{) 121} \\ \underline{-12} \\ 10 \\ \underline{-0} \\ 1 \\ \underline{-2} \\ 0 \end{array}$$

HCF of all three numbers $\longrightarrow 1$

So, the HCF of 452, 786, and 121 = 1

LCM: Divide by a prime number at each step.

2	452, 786, 121	
2	226, 393, 121	← Dividing 452 and 786 by 2.
3	113, 393, 121	← Dividing 226 by 2.
11	113, 131, 121	← Dividing 393 by 3.
11	113, 131, 11	← Dividing 121 by 11.
113	113, 131, 1	← Dividing 11 by 11.
131	1, 131, 1	← Dividing 113 by 113.
	1, 1, 1	← Dividing 131 by 131.

LCM = Product of prime factors

$$= 2 \times 2 \times 3 \times 11 \times 11 \times 113 \times 131 = 21493956$$

d) 414, 512, 616

Solution:

HCF

Step 1: Divide 616 by 512

$$\begin{array}{r}
 1 \\
 512 \overline{) 616} \\
 \underline{-512} \quad 4 \\
 104 \overline{) 512} \\
 \underline{-416} \quad 1 \\
 96 \overline{) 104} \\
 \underline{-96} \quad 12 \\
 8 \overline{) 96} \\
 \underline{-96} \\
 0
 \end{array}$$

HCF of 616 and 512 $\longrightarrow$

Step 2: Now, divide 414 by the last divisor of step 1.

$$\begin{array}{r}
 51 \\
 8 \overline{) 414} \\
 \underline{-408} \quad 1 \\
 6 \overline{) 8} \\
 \underline{-6} \quad 3 \\
 2 \overline{) 6} \\
 \underline{-6} \\
 0
 \end{array}$$

HCF of all three numbers $\longrightarrow$

So, the HCF of 414, 512, and 616 = 2

LCM: Divide by a prime number at each step.

2	414, 512, 616	
2	207, 256, 308	← Dividing all by 2.
2	207, 128, 154	← Dividing 256 and 308 by 2.
2	207, 64, 77	← Dividing 128 and 154 by 2.
2	207, 32, 77	← Dividing 64 by 2.
2	207, 16, 77	← Dividing 32 by 2.
2	207, 8, 77	← Dividing 16 by 2.
2	207, 4, 77	← Dividing 8 by 2.
2	207, 2, 77	← Dividing 4 by 2.
3	207, 1, 77	← Dividing 2 by 2.
3	69, 1, 77	← Dividing 207 by 3.
7	23, 1, 77	← Dividing 69 by 3.
11	23, 1, 11	← Dividing 77 by 7.
23	23, 1, 1	← Dividing 11 by 11.
	1, 1, 1	← Dividing 23 by 23.

LCM = Product of prime factors

$$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 11 \times 23 = 8160768$$

6. Find the square of the following numbers.

a) 16

Solution:

$$\begin{aligned}\text{Square of } 16 &= 16 \times 16 \\ &= 16^2 = 256 \text{ Answer}\end{aligned}$$

c) 44

Solution:

$$\begin{aligned}\text{Square of } 44 &= 44 \times 44 \\ &= 44^2 = 1936 \text{ Answer}\end{aligned}$$

b) 18

Solution:

$$\begin{aligned}\text{Square of } 18 &= 18 \times 18 \\ &= 18^2 = 324 \text{ Answer}\end{aligned}$$

d) 62

Solution:

$$\begin{aligned}\text{Square of } 62 &= 62 \times 62 \\ &= 62^2 = 3844 \text{ Answer}\end{aligned}$$

7. The lengths of three halls are 55 m, 64 m and 72 m, respectively. Find the longest length of a measuring tape that can exactly measure the length of the three halls.

Solution:

Given:

$$\text{Length of one hall} = 55 \text{ m}$$

$$\text{Length of second hall} = 64 \text{ m}$$

$$\text{Length of third hall} = 72 \text{ m}$$

To find the longest length of measuring tape that exactly measures the length of three halls we find the HCF of three values.

Prime factorization of the three values.

$$\begin{array}{r|l} 5 & 55 \\ \hline 11 & 11 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 72 \\ \hline 2 & 36 \\ \hline 2 & 18 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

HCF

$$\text{Prime factorization of } 55 = 5 \times 11 \times \textcircled{1}$$

$$\text{Prime factorization of } 64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times \textcircled{1}$$

$$\text{Prime factorization of } 72 = 2 \times 2 \times 2 \times 3 \times 3 \times \textcircled{1}$$

$$\text{Common prime factors} = 1$$

So, the HCF = 1

Therefore the longest length of measuring tape is 1 m.

8. Find the least volume of oil that can be put into cans with capacities of 12 litres, 42 litres and 72 litres exactly.

Solution:

Given:

$$\text{Capacity of first can} = 12 \text{ litres}$$

$$\text{Capacity of second can} = 42 \text{ litres}$$

$$\text{Capacity of third can} = 72 \text{ litres}$$

To find the least volume of oil we have to find the LCM of the given values.

Prime factorization of 12, 42, 72.

2	12, 42, 72	
2	6, 21, 36	← Dividing all by 2.
2	3, 21, 18	← Dividing 6 and 36 by 2.
3	3, 21, 9	← Dividing 18 by 2.
3	1, 7, 3	← Dividing all by 3.
7	1, 7, 1	← Dividing 3 by 3.
	1, 1, 1	← Dividing 7 by 7.

$$\begin{aligned} \text{LCM} &= \text{Product of prime factors} \\ &= 2 \times 2 \times 2 \times 3 \times 3 \times 7 = 504 \end{aligned}$$

2.1 Let's Practice

1. Separate the positive integers and negative integers.

Positive Integers:

 $+4, +17, +32, +1, +12$

Negative Integers:

 $-9, -13, -23, -1, -15, -19$

2. Write the first 15 integers starting from -4 .

Solution:

First 15 integers starting from -4 are:

 $-4, -13, -1, 0, +1, +2, +3, +4, +5, +6, +7, +8, +9, +10.$

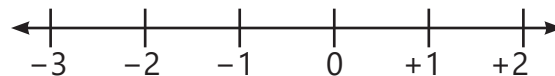
3. Represent the following sets of integers on a number line.

Solution:

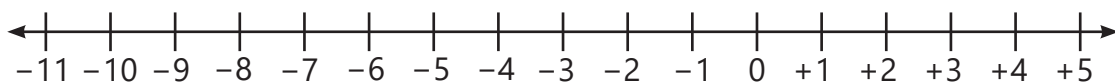
To represent any digit on a number line, first draw a line. Mark a point zero "0" in the middle of the line. Mark some points on the left and right of 0, at equal distance from each other. To mark negative integers, count exact number of points as the given negative integer, to the left of zero. For positive integers, count exact number of points as the given positives integer to the right of zero.

a) -3 to $+2$

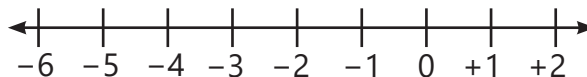
Solution:


b) -11 to $+5$

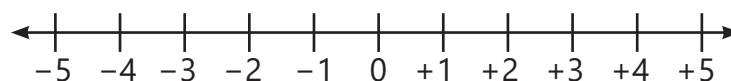
Solution:


c) -6 to $+2$

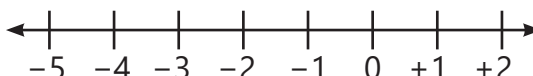
Solution:


d) -5 to $+5$

Solution:

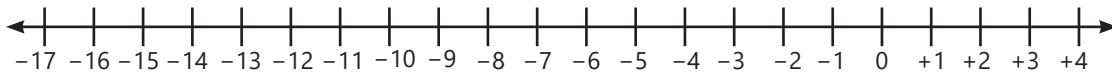

e) -5 to $+2$

Solution:



f) -17 to $+4$

Solution:



4. Separate the positive integers and negative integers.

a) -13

Solution:

-13

absolute value of $-13 = |-13| = 13$

b) $+15$

Solution:

$+15$

absolute value of $+15 = |+15| = 15$

c) -21

Solution:

-21

absolute value of $-21 = |-21| = 21$

d) $+32$

Solution:

$+32$

absolute value of $+32 = |+32| = 32$

e) -7

Solution:

-7

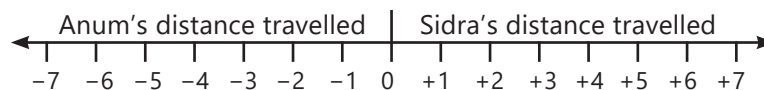
absolute value of $-7 = |-7| = 7$

The absolute values of an integer is the numerical value of that integer without “ $-$ ” or “ $+$ ” sign.

5. Write the absolute value of the given integers.

Solution:

If sidra and Anum started from the same point and travelled in opposite direction, we can represent their positions using a number line. Let’s suppose, the starting point is 0 km on the number line. If Sidra travelled 1 km towards the right, we can represent her position by counting 7 points to the right of 0. If Anum travelled 1km to the left, we can represent her position by counting 7 points to the left.



So, Sidra’s destination point in terms of integers is $+7$ and Anum’s destination point in terms of integers is -7 .

2.2 Let's Practice

1. Compare and circle the greater integer.

a) 300, -300

Solution:

A negative integer is always smaller than a positive integer. So -300 is the smaller integer and 300 is the greater integer.

300, -300

b) -100, -310

Solution:

On a number line -310 would lie to the left of -100. So -310 is the smaller integer and -100 is the greater integer.

-100, -310

c) -114, 99

Solution:

A negative integer is always smaller than a positive integer. So -114 is the smaller integer and 99 is the greater integer.

-114, 99

d) 12, -10

Solution:

A negative integer is always smaller than a positive integer. So -10 is the smaller integer and 12 is the greater integer.

12, -10

2. Arrange the following integers in ascending order and descending order

Solution:

All positive integers are greater than negative integers and greater integers with a negative sign are smaller than the smaller integers with a negative sign.

a) -10, +5, -17, +1, +15, -21, -24

Ascending Order:

Starting from greater integers with a negative sign and ending at greater integers with a positive sign.

-24, -21, -17, -10, +1, +5, +15

Descending Order:

Starting from greater integers with a positive sign and ending at greater integer with a negative sign.

+15, +5, +1, -10, -17, -21, -24

b) -76, -25, +7, -23, +12, +44

Solution:

Ascending Order:

-76, -25, +7, -23, +12, +44

Descending Order:

+44, +12, +7, -23, -25, -76

c) +13, -6, -15, +9, -14, -35, +35

Solution:

Ascending Order: -35, -15, -14, -6, +9, +13, +35

Descending ORder: +35, +13, +9, -6, -14, -15, -35

d) -4, -4, 0, +6, +18, -19, -11

Solution:

Ascending Order: -19, -11, -4, -4, 0, +16, +18

Descending Order: +18, +6, 0, -4, -4, -11, -19

3. Arrange the absolute value of the following integers in ascending order and descending order.

Solution:

The absolute value of each integer is positive. We write the given digits without their signs and arrange them, starting without their signs and arrange them, starting from greater integer and ending with the smaller integer for desending order, and starting from smaller integer and ending with the greater integer for ascending order.

a) -4, +5, -16, +12, +19

Solution:

Absolute values = $|-4| = 4, | +5| = 5, |-16| = 16, | +12| = 12, | +19| = 19$

Ascending Order: 4, 5, 12, 16, 19

Descending Order: 19, 16, 12, 5, 4

b) -11, -18, +12, +17, -22, -6

Solution:

Absolute values = $|-11| = 11, |-18| = 18, | +12| = 12, | +17| = 17, |-22| = 22, |-6| = 6$

Ascending Order: 6, 11, 12, 17, 18, 22

Descending Order: 22, 18, 17, 12, 11, 6

c) -33, +44, -23, +26, -37

Solution:

Absolute values: $|-33| = 33, | +44| = 44, |-23| = 23, | +26| = 26, |-37| = 37$

Ascending Order: 23, 26, 33, 37, 44

Descending Order: 44, 37, 33, 26, 23

d) -9, -8, -5, +12, +17, -13

Solution:

Abdolute values = $|-9| = 9, |-8| = 8, |-5| = 5, | +12| = 12, | +17| = 17, |-13| = 13$

Ascending order: 5, 8, 9, 12, 13, 17

Descending order: 17, 13, 12, 9, 8, 5

4. In a town, the lowest temperature on Monday was -5°C and on Tuesday it was -2°C . Which day was colder?

Solution:

As we know, the greater integers with a negative sign are smaller than the smaller integers with a negative sign.

On Monday the temperature was -5°C and on Tuesday it was -2°C . The temperature of -5°C is lower than the temperature of -2°C . So the temperature on Tuesday i.e., -2°C is higher than on Monday. Therefore, Monday was colder than Tuesday.

2.3 Let's Practice

1. Add the given integers.

- a) -7 and $+25$

Solution:

To add -7 and $+$, first take absolute values of both.

$$|-7| = 7, |+25| = 25$$

Subtract these absolute values as both have opposite symbols.

$$25 - 7 = 18$$

Now put the symbol of the greater integers.

$$\text{So, } (-7) + (+25) = +18$$

- b) -6 and -14

Solution:

Take absolute values of both

$$|-6| = 6, |-14| = 14$$

add these absolute values as both have the same symbols.

$$6 + 14 = 20$$

since both integers are negative, put $-$ symbol with the result.

$$(-6) + (-14) = -20$$

- c) $+16$ and $+21$

Solution:

Take absolute values of both.

$$|+16| = 16, |+21| = 21$$

add these absolute values as both have same symbols.

$$16 + 21 = 37$$

As both integers are positive, put the $+$ symbol with answer.

$$(+16) + (+21) = +37$$

- d) $+5$ and -34

Solution:

Take absolute values of both.

$$|+5| = 5, |-34| = 34$$

Subtract these absolute values as both have opposite symbols.

$$34 - 5 = 29$$

put the symbol of the greater integer

$$(+5) - (-34) = -29$$

e) +4 and +7

Solution:

take absolute values of both

$$|+4| = 4, |+7| = 7$$

add these absolute values as both have same symbols.

$$4 + 7 = 11$$

as both integers are positive put the + symbol with answer.

$$+4 + (+7) = + 11$$

f) -18 and +1

Solution:

take absolute values of both.

$$|-18| = 18, |+1| = 1$$

subtract these values as both have opposite symbol

$$18 - 1 = 17$$

put the symbol of greater integer.

$$(-18) + (+1) = -17$$

g) -5 and -13

Solution:

take absolute values of both.

$$|-5| = 5, |-13| = 13$$

add these absolute values as both have the same symbols.

$$5 + 13 = 18$$

as both integers are negative, put - sign with the answer.

$$(-5) + (-13) = -18$$

h) + 6 and -2

Solution:

take absolute values of both.

$$|+6| = 6, |-2| = 2$$

subtract these absolute values as both have opposite symbols.

$$6 - 2 = 4$$

put the sign of the greater integer.

$$(+6) + (-2) = +4$$

2. Subtract the first integer from the second integer.

Solution:

We have to subtract + 2 from + 5 i.e. $(+5) - (+2)$.

To subtract, first change the operation from subtraction to addition and change the

symbol of +2 to -2.

$$+5 - (+2) = +5 + (-2)$$

take absolute values of both

$$|+5| = 5, |-2| = 2$$

Subtract these values as both have opposite symbols.

$$5 - 2 = 3$$

put the symbol of greater integer.

$$(+5) - (+2) = +3$$

b) +7 and -8

Solution:

We have to subtract +7 from -8 i.e. $-8 - (+7)$

to do so first we change the operation from subtraction to addition and change the symbol of +7 to -7.

$$-8 - (+7) = -8 + (-7)$$

take absolute values of both

$$|-8| = 8, |-7| = 7$$

add these values as both have same symbols.

$$8 + 7 = 15$$

put the symbol of the greater number

$$-8 - (+7) = -15$$

c) -7 and -23

Solution:

We have to subtract -7 from -23, i.e. $-23 - (-7)$

To do so first we change the operation from subtraction to addition and change the symbol of -7 to +7.

$$-23 - (-7) = -23 + (+7)$$

take absolute values of both.

$$|-23| = 23, |+7| = 7$$

subtract these absolute values as both have opposite symbol,

$$23 - 7 = 16$$

As both integers have opposite symbols, put the symbol of the greater integer.

$$\text{so, } -23 - (-7) = -16$$

d) -17 and +16

Solution:

We have to subtract -17 from +16 i.e. $+16 - (-17)$. To do so first change the operation from subtraction to addition and change the symbol of -17 to +17.

$$+16 + (+17)$$

take absolute values of both

$$|+16| = 16, |+17| = 17$$

Add these values as both have the same signs

$$16 + 17 = 33$$

As both integers have positive symbols, put + symbol with the answer.

$$\text{So, } +16 - (-17) = +33$$

e) -18 and $+14$

Solution:

We have to subtract -18 from $+14$ i.e. $+14 - (-18)$. To do so first change the operation from subtraction to addition and change the symbol of -18 to $+18$.

$$+14 + (+18)$$

Take absolute values of both

$$|+14| = 14, |+18| = 18$$

Add these values as both have positive symbols

$$14 + 18 = 32$$

Put + symbol with the answer, as both integers are positive.

$$+14 - (-18) = +32$$

f) $+20$ and $+44$

Solution:

We have to subtract $+20$ from $+44$ i.e. $+44 - (+20)$.

To do so first we change the operation from subtraction to addition and change the symbol of $+20$ to -20 .

$$+44 - (+20) = +44 + (-20)$$

Take absolute values of both.

$$|-20| = 20, |+44| = 44$$

As both integers have opposite symbols, so subtract them.

$$44 - 20 = 24$$

Put the sign of greater integer.

$$+44 + (-20) = +24$$

g) -2 and $+21$

Solution:

We have to subtract -2 from $+21$, i.e. $+21 - (-2)$

To do so we change the operation from subtraction to addition and change the the symbol of -2 to $+2$.

$$+21 - (-2) = +21 + (+2)$$

take absolute values of both.

$$|+21| = 21, |+2| = 2$$

add these values as both have positive symbols.

$$21 + 2 = 23$$

as both values are positive so put positive symbol with the answer.

$$+21 - (-2) = +23$$

h) -45 and -54

Solution:

We have to subtract -45 from -54 i.e. $-54 - (-45)$

To do so first change the operation from subtraction to addition and change the symbol of -45 to $+45$.

$$-45 - (-45) = -54 + (+45)$$

Take absolute values of both

$$|-54| = 54, |+45| = 45$$

Subtract these absolute values as both have opposite symbols.

$$54 - 45 = 9$$

Put the sign of the greater value

$$-54 - (-45) = -9$$

3. Find the missing number.

a) $\underline{\hspace{1cm}} - (-43) = 145$

Solution:

$$-(-43) = 145$$

To solve this we add -43 to 145

taking absolute values:

$$|+145| = 145, |-43| = 43$$

the absolute value of subtract a both integers as they have opposite sign.

$$145 - 43 = 102$$

put the sign of the greater integer

$$\text{so } 145 + (-43) = 102$$

so 102 is the missing integer

$$102 - (-43) = 145$$

b) $54 - (\underline{\hspace{1cm}}) = -13$

Solution:

$$54 - (\underline{\hspace{1cm}}) = -13$$

to solve this we first bring 54 to the other side of equation and subtract it.

$$-(\underline{\hspace{1cm}}) = -13 - (+54)$$

Then according to subtraction of integers, we change the process from subtraction to addition and change the symbol of $+54$ to -54 .

$$-(\underline{\hspace{1cm}}) = -13 + (-54)$$

taking absolute values

$$|-13| = 13, |-54| = 54$$

adding both values as both integers have same sign.

$$13 + 54 = 67$$

put $-$ symbol with the answer.

$$\text{so, } -(\underline{\hspace{1cm}}) = -13 + (-54)$$

$$-(\underline{\hspace{1cm}}) = -67$$

The number subtracted on one side of equation is added to the other side and the number added to one side of equation is subtracted from the other side.

Now to remove $-$ symbol we multiply both sides by -1

$$-1 \times - (\underline{\quad}) = -1 \times (-54)$$

$$- (\quad) = -67$$

Therefore the missing number is 67 and we write the given equation as:

$$54 - (67) = -13$$

$$c) \underline{\quad} + (-31) = -76$$

Solution:

$$\underline{\quad} + (-31) = -76$$

subtract (-31) from -76 . To subtract first change the operation from subtraction to addition and change the symbol of -31 to $+31$

$$\underline{\quad} = -76 - (-31)$$

$$= -76 + (+31)$$

Take absolute values

$$|-76| = 76, |+31| = 31$$

subtract these values as the integers have opposite symbols.

$$76 - 31 = 45$$

put the symbol of greater integer.

$$76 - 31 = -45$$

$$\text{so } -45 + (-31) = -76$$

so the missing number is 45.

- 4. A diver is 8 feet below sea level. After a while, he moves 4 feet in an upward direction. What distance from sea level is the diver now?**

Solution:

If we consider being below sea level is negative and being above sea level as positive, we can write it as:

Initial distance from sea level = -8 feet

Upward movement = $+4$ feet

New distance from sea level = -8 feet

$$\text{or } = -8 + (+4)$$

to add we take absolute values of both

$$|-8| = 8, |+4| = 4$$

as they have opposite sign, we subtract the absolute value

$$8 - 4 = 4$$

Put the sign of the greater integer

$$-8 + (+4) = -4$$

so his new distance from sea level is -4 feet.

The diver is now 4 feet below sea level.

5. The sum of two numbers is -23 . If the smaller number is -45 , find the greater one.

Solution:

$$\underline{\hspace{1cm}} + (-45) = -23$$

To solve subtract -45 from -23 . First change the operation from subtraction to addition and change the symbol of -45 to $+45$.

$$-23 - (-45) = -23 + (+45)$$

take absolute values

$$|-23| = 23 \quad | +45| = 45$$

subtract them as the integers have opposite sign.

$$45 - 23 = 22$$

put the symbol of the greater integer.

$$-23 + (+45) = +22$$

so the other number is $+22$

$$\text{or } +22 + (-45) = -23$$

6. Asif got a loan of Rs 500. He took another Rs 750 as a loan from his friend. How much debt does he now have?

Solution:

If we consider debt as negative amount we can express Asif's initial loan and the loan he took from his friend as integers

$$\text{Asif's initial loan} = \text{Rs. } 500 = -500$$

$$\text{Loan taken from friend} = \text{Rs. } 750 = -750$$

To find the total debt we add these two amounts.

$$(-500) + (-750)$$

Taking absolute values

$$|-500| = 500, \quad |-750| = 750$$

Add both values as they have the same symbols.

$$500 + 750 = 1250$$

As both have $-$ symbol, write $-$ symbol with the answer.

$$(-500) + (-750) = -1250$$

So now Asif's total debt is Rs. 1250

2.4 Let's Practice

1. Observe and write the properties of integers in the given boxes.

Example

Property name

$$\text{a) } 2 + (-2) = 0$$

Additive inverse

$$\text{b) } -7 + [(-4)]$$

Associative law

$$\text{c) } -6 + 0 = -6$$

Additive identity

$$\text{d) } -5 + (-8)$$

Commutative law

$$\text{e) } -3 + (-4)$$

Closure law

2. Verify the commutative property of addition for the following.

Solution:

The commutative property of addition states that changing the order of the integers does not affect the result, when we add two integers. Now let's verify this property for the given integers.

a) +4 and -6

Solution:

Changing the order of integers and adding:

$$(-6) + (+4) = -2$$

So changing the order of integers does not affect the result. Hence proved.

b) -5 and -16

Solution:

Changing the order of integers and adding:

$$-16 + (-5) = -21$$

So, changing the order of integers does not affect the result. Hence proved.

c) +13 and +17

Solution:

$$+13 + (+17) = 30$$

Changing the order of integers and adding.

$$+17 + (+13) = 30$$

So, changing the order of integers does not affect the result. Hence proved.

d) -21 and +15

Solution:

$$-21 + (+15) = -6$$

changing the order of integers and adding.

$$+15 + (-21) = -6$$

So changing the order of integers does not affect the result. Hence proved.

3. Verify the associative property of addition for the following.

Solution:

According to the associative law, when we add three integers, we add two numbers at a time and then add the third number to it. The order of grouping in addition does not affect the result. Let us verify this property for the following sets of integers.

a) +2 , -7 and -11

Solution:

According to associative law:

$$[+2 + (-7)] + (-11) = +2 + [(-7) + (-11)]$$

LHS

$$= [+2 + (-7)] + (-11)$$

$$= [+2 - 7] + (-11)$$

$$= (-5) + (-11)$$

$$= -16$$

LHS = RHS. Hence proved

RHS

$$= +2 + [(-7) + (-11)]$$

$$= +2 + [-7 - 11]$$

$$= +2 + [-18]$$

$$= -16$$

b) -1, -6 and -8

Solution:

According to associative law:

$$[-1 + (-6)] + (-8) = -1 + [(-6) + (-8)]$$

LHS

$$= [-1 + (-6)] + (-8)$$

$$= [-1 - 6] + (-8)$$

$$= (-7) + (-8)$$

$$= -15$$

LHS = RHS. Hence proved

RHS

$$= -1 + [(-6) + (-8)]$$

$$= -1 + [-6 - 8]$$

$$= -1 + [-14]$$

$$= -15$$

c) +4 , +14 and -31

Solution:

According to associative law:

$$[+4 + 14] + (-31) = +4 + [+14 + (-31)]$$

LHS

$$= [+4 + 14] + (-31)$$

$$= +18 + (-31)$$

$$= -13$$

LHS = RHS. Hence proved

RHS

$$= +4 + [+14 - 31]$$

$$= +4 - 17]$$

$$= -13$$

d) -4 , +5 and -17

Solution:

According to associative law:

$$[-4 + 5] + (-17) = -4 + [+5 + (-17)]$$

LHS

$$= [-4 + 5] + (-17)$$

$$= +1 + (-17)$$

$$= -16$$

LHS = RHS. Hence proved

RHS

$$= -4 + [+5 + (-17)]$$

$$= -4 + [-12]$$

$$= -16$$

e) +3 , -15 and -10

Solution:

According to associative law:

$$[+3 + (-15)] + (-10) = +3 + [(-15) + (-10)]$$

$$\begin{aligned} \text{LHS} \\ &= [+3 + (-15)] + (-10) \\ &= (-12) + (-10) \\ &= -22 \end{aligned}$$

$$\begin{aligned} \text{RHS} \\ &= +3 + [(-15) + (-10)] \\ &= +3 + [-25] \\ &= -22 \end{aligned}$$

LHS = RHS. Hence proved

f) -18 , -20 and +14

Solution:

According to associative law:

$$[-18 + (-20)] + 14 = -18 + [(-20) + 14]$$

$$\begin{aligned} \text{LHS} \\ &= [-18 + (-20)] + 14 \\ &= (-38) + 14 \\ &= -24 \end{aligned}$$

$$\begin{aligned} \text{RHS} \\ &= -18 + [(-20) + 14] \\ &= -18 + (-6) \\ &= -24 \end{aligned}$$

LHS = RHS. Hence proved

4. Verify the closure property of addition for the following.

Solution:

According to closure law of addition, when we add two integers, the result is always an integer.

a) +9 and +3

Solution:

$$\begin{aligned} &+9 + 3 \\ &= + 12 \\ &+ 12 \text{ is also an integer. Hence proved.} \end{aligned}$$

b) -7 and + 19

Solution:

$$\begin{aligned} &-7 + 19 \\ &= + 12 \\ &+12 \text{ is also an integer. Hence proved.} \end{aligned}$$

c) +45 and -54

Solution:

$$\begin{aligned} &+45 + (-54) \\ &= -9 \\ &-9 \text{ is also an integer. Hence proved.} \end{aligned}$$

d) -21 and -16

Solution:

$$-21 + (-16)$$

$$= -37$$

-37 is also an integer. Hence proved.

5. Write the additive inverse of each of the following.

a) $+1$

Solution:

We can find the additive inverse by changing the sign of the number. So the additive inverse of $+1$ is -1 . Let's check.

b) -66

Solution:

We can find the additive inverse by changing the sign of the number. So the additive inverse of -66 is $+66$. Let's check.

$$-66 + 66 = 0$$

Hence $+66$ is the additive inverse of -66 .

c) $+10$

Solution:

We can find the additive inverse of a number by changing the sign of the number. So the additive inverse of $+10$ is -10 . Let's check.

$$+10 + (-10) = 0$$

Hence the additive inverse of $+10$ is -10 .

d) -17

Solution:

We can find the additive inverse of a number by changing the sign of the number. So the additive inverse of -17 is $+17$. Let's check.

$$-17 + 17 = 0$$

Hence $+17$ is the additive inverse of -17 .

2.5 Let's Practice

1. Multiply the following integers.

a) $+11$ and -44

Solution:

As both integers have opposite signs, the result of their product will be a negative integer.

$$(+11) \times (-44)$$

$$\begin{array}{r} 11 \\ \times 44 \\ \hline 44 \\ + 440 \\ \hline 484 \end{array}$$

$$\text{So, } (+11) \times (-44) = -484$$

c) -16 and -18

Solution:

As both integers have same signs, their result will be a positive integer.

$$(-16) \times (-18)$$

$$\begin{array}{r} 16 \\ \times 18 \\ \hline 128 \\ + 160 \\ \hline 288 \end{array}$$

$$\text{So, } (-16) \times (-18) = +288$$

e) -19 and $+23$

Solution:

As both integers have opposite signs, their product will be a negative integer.

$$(-19) \times (+23)$$

$$\begin{array}{r} 19 \\ \times 23 \\ \hline \textcircled{1} 57 \\ + 380 \\ \hline 437 \end{array}$$

$$\text{So, } (-19) \times (+23) = -437$$

b) -21 and $+37$

Solution:

As both integers have opposite signs, the result of their product will be a negative integer.

$$(-21) \times (+37)$$

$$\begin{array}{r} 21 \\ \times 37 \\ \hline 147 \\ + 630 \\ \hline 777 \end{array}$$

$$\text{So, } (-21) \times (+37) = -777$$

d) $+12$ and $+24$

Solution:

As both integers have same sign, the result of their product will be a positive sign.

$$(+12) \times (+24)$$

$$\begin{array}{r} 12 \\ \times 24 \\ \hline 48 \\ + 240 \\ \hline 288 \end{array}$$

$$\text{So, } (+12) \times (+24) = +288$$

f) -5 and -13

Solution:

As both integers are of same sign, their product will be positive.

$$(-5) \times (-13)$$

$$\begin{array}{r} 13 \\ \times 5 \\ \hline 65 \\ \hline 65 \end{array}$$

$$\text{So, } (-5) \times (-13) = +65$$

g) +52 and +67

Solution:

As both integers have have sign,
the result of thier product will be a
postive sign.

$$(+52) \times (+67)$$

$$\begin{array}{r} 52 \\ \times 67 \\ \hline 364 \\ + 3120 \\ \hline 3484 \end{array}$$

$$\text{So, } (+52) \times (+67) = +3484$$

d) -32 and -21

Solution:

As both integers have have sign,
their product will be a positive
integer.

$$(-32) \times (-21)$$

$$\begin{array}{r} 32 \\ \times 21 \\ \hline 32 \\ + 640 \\ \hline 672 \end{array}$$

$$\text{So, } (-32) \times (-21) = +672$$

2. Divide the following.

a) +20 by -5

Solution:

The result of division of integers
having opposite signs is negative.

$$(+20) \div (-5)$$

$$\begin{array}{r} 4 \\ 5 \overline{) 20} \\ - 20 \\ \hline 0 \end{array}$$

$$\text{So, } (+20) \div (-5) = -4$$

b) -38 by +2

Solution:

The result of division of integers
having opposite signs is negative.

$$(-38) \div (+2)$$

$$\begin{array}{r} 19 \\ 2 \overline{) 38} \\ - 2 \downarrow \\ \hline 18 \\ - 18 \\ \hline 0 \end{array}$$

$$\text{So, } (-99) \div (+9) = -11$$

c) -60 by -12

Solution:

The result of division of integers
having same signs is positive.

$$(-60) \div (-12)$$

$$\begin{array}{r} 5 \\ 12 \overline{) 60} \\ - 60 \\ \hline 0 \end{array}$$

$$\text{So, } (-60) \div (-12) = +5$$

d) -99 by +9

Solution:

The result of division of integers
having opposite signs is negative.

$$(-99) \div (+9)$$

$$\begin{array}{r} 11 \\ 9 \overline{) 99} \\ - 99 \\ \hline 0 \end{array}$$

$$\text{So, } (-99) \div (+9) = -11$$

e) +72 by +8

Solution:

The result of division of integers having same signs is positive.

$$(+72) \div (+8)$$

$$\begin{array}{r} 9 \\ 8 \overline{) 72} \\ \underline{- 72} \\ 0 \end{array}$$

$$\text{So, } (+72) \div (+8) = +9$$

g) +66 by -6

Solution:

The result of division of integers having opposite signs is negative.

$$(+66) \div (-6)$$

$$\begin{array}{r} 11 \\ 6 \overline{) 66} \\ \underline{- 66} \\ 0 \end{array}$$

$$\text{So, } (+66) \div (-6) = -11$$

f) -45 by -3

Solution:

The result of division of integers having same signs is positive.

$$(-45) \div (-3)$$

$$\begin{array}{r} 15 \\ 3 \overline{) 45} \\ \underline{- 3} \downarrow \\ 15 \\ \underline{- 15} \\ 0 \end{array}$$

$$\text{So, } (-45) \div (-3) = +15$$

h) -78 by -13

Solution:

The result of division of integers having same signs is positive.

$$(-78) \div (-13)$$

$$\begin{array}{r} 6 \\ 13 \overline{) 78} \\ \underline{- 78} \\ 0 \end{array}$$

$$\text{So, } (-78) \div (-13) = +6$$

3. Ahmad participates in a game show. For every correct answer, he gains 3 marks and for every incorrect answer, he loses 2 marks. If he gets the answers to 8 questions right and 3 questions wrong, how many marks does he get in all?

Solution:

For every correct answer, Ahmad gains 3 marks. We can write it as +3. Similarly for every incorrect answer, he loses 2 marks. So we can write it as -2.

If he get the answers to 8 questions right, we can write it as:

$$8 \times +3 \text{ or } (+8) \times (+3) = +24.$$

Similarly for his 3 incorrect answers, we can write it as: 3×-2 or $+3 \times -2 = -6$

Ahmad's total score can be found by adding both these values. i.e. marks obtained + marks lost.

$$\text{Ahmad's total score} = (+24) + (-6)$$

$$\text{Taking absolute values } |+24| = 24, |-6| = 6$$

Subtracting the values as both integers have opposite signs. $24 - 6 = 18$ putting the sign of the greater integer.

$$\text{So, } (+24) + (-6) = +18$$

So, Ahmad gets a total of +18 marks.

4. The temperature of the Siachen Glacier is -4°C . At night, the temperature decreases by 2. What is the temperature at night?

Solution:

If the temperature of the Siachen glacier is -4°C and it decreases by 2 at night, we can represent the decrease as -2°C . Then, according to the given condition the temperature at night can be found using the following method.

Temp. at night = Temp. of Siachen Glacier – Decrease in temp. at night

$$= -4^{\circ}\text{C} - (-2^{\circ}\text{C}) \text{ or } = -4 - (-2)$$

$$\text{or } = -4 + (+2) \text{ (according to the rule for subtraction of integers)}$$

Taking absolute values $|-4| = 4$, $|+2| = 2$

Subtracting both values as both integers have opposite sign.

$$4 - 2 = 2$$

Put $-$ sign with the answer, as the greater integer has negative sign.

$$\text{So } (-4^{\circ}\text{C}) - (-2^{\circ}\text{C}) = -2^{\circ}\text{C}.$$

So the temperature of Siachen Glacier at night is -2°C .

2.6 Let's Practice

1. Verify the closure law for the following integers under multiplication.

Solution:

According to the closure law, when we multiply two integers, the product is always an integer.

a) -4 and -3

Solution:

$$(-4) \times (-3) = +12$$

$+12$ is also an integer so, closure law proved.

b) -7 and $+9$

Solution:

$$(-7) \times (+9) = -63$$

-63 is also an integer so, closure property is proved.

c) $+5$ and $+6$

Solution:

$$(+5) \times (+6) = +30$$

$+30$ is also an integer so, closure property is proved.

d) $+8$ and -15

Solution:

$$(+8) \times (-15) = -120$$

-120 is also an integer so, closure property is proved.

2. Verify the commutative law for the following integers under multiplication.

Solution:

According to the commutative law of multiplication, when two integers are multiplied, the result will be the same regardless of the order of the two integers.

a) +18 and -19

Solution:

First multiply +16 by -19

$$(+16) \times (-19)$$

$$\begin{array}{r} 16 \\ \times 19 \\ \hline 144 \\ + 160 \\ \hline 304 \end{array}$$

Now multiply -19 by +16

$$(-19) \times (+16)$$

$$\begin{array}{r} 19 \\ \times 16 \\ \hline 114 \\ + 190 \\ \hline 304 \end{array}$$

$$\text{So, } (+16) \times (-19) = -304$$

$$\text{also } (-19) \times (+16) = -304$$

Hence changing the order of the integers does not change the result. Hence proved.

c) +46 and -32

Solution:

First multiply +46 by -32

$$(+46) \times (-32)$$

$$\begin{array}{r} 46 \\ \times 32 \\ \hline 92 \\ + 1380 \\ \hline 1472 \end{array}$$

Now multiply -32 by +46

$$(-32) \times (+46)$$

$$\begin{array}{r} 32 \\ \times 46 \\ \hline 192 \\ + 1280 \\ \hline 1472 \end{array}$$

$$\text{So, } (+46) \times (-32) = -1472$$

$$\text{also } (-32) \times (+46) = -1472$$

So the result is not changed, when the order is changed.

Hence commulative law proved.

b) +21 and -52

Solution:

First multiply +21 by -52

$$(+21) \times (-52)$$

$$\begin{array}{r} 21 \\ \times 52 \\ \hline 42 \\ + 1050 \\ \hline 1092 \end{array}$$

Now multiply -52 by +21

$$(-52) \times (+21)$$

$$\begin{array}{r} 52 \\ \times 21 \\ \hline 52 \\ + 1040 \\ \hline 1092 \end{array}$$

$$\text{So, } (+21) \times (-52) = -1092$$

$$\text{also } (-52) \times (+21) = -1092$$

Hence changing the order does not change the result. Hence commulative law proved.

d) -17 and -40

Solution:

First multiply -17 by -40

$$(-17) \times (-40)$$

$$\begin{array}{r} 17 \\ \times 40 \\ \hline 00 \\ + 680 \\ \hline 680 \end{array}$$

Now multiply -40 by -70

$$(-40) \times (-70)$$

$$\begin{array}{r} 40 \\ \times 17 \\ \hline 280 \\ + 480 \\ \hline 680 \end{array}$$

$$\text{So, } (-17) \times (-40) = +680$$

$$\text{also } (-40) \times (-17) = +680$$

So changing the order does not change the result. Hence commulative law proved.

3. Verify the associative law for the following integers under multiplication.

a) +5, -9 and -28

Solution:

According to associative law

$$[+5 \times (-9)] \times (-28) = +5 \times [(-9) \times (-28)]$$

LHS

$$= [+5 \times (-9)] \times (-28)$$

$$= (-45) \times (-28)$$

$$= +1260$$

LHS = RHS, Hence proved.

RHS

$$= +5 \times [(-9) \times (-28)]$$

$$= (+5) \times (+252)$$

$$= +1260$$

b) -11, +16 and +15

Solution:

According to associative law

$$[(-11) \times (+16)] \times (+15) = (-11) \times [(+16) \times (+15)]$$

LHS

$$= [(-11) \times (+16)] \times (+15)$$

$$= (-176) \times (+15)$$

$$= -2640$$

LHS = RHS, Hence proved.

RHS

$$= (-11) \times [(+16) \times (+15)]$$

$$= (-11) \times (+240)$$

$$= -2640$$

c) +21, +34 and +13

Solution:

According to associative law

$$[(+21) \times (+34)] \times (+13) = (+21) \times [(+34) \times (+13)]$$

LHS

$$= [(+21) \times (+34)] \times (+13)$$

$$= (+714) \times (+13)$$

$$= +9282$$

LHS = RHS, Hence proved.

RHS

$$= (+21) \times [(+34) \times (+13)]$$

$$= (+21) \times (442)$$

$$= +9282$$

d) -7, -5 and -14

Solution:

According to associative law

$$[(-7) \times (-5)] \times (-14) = (-7) \times [(-5) \times (-14)]$$

LHS

$$= [(-7) \times (-5)] \times (-14)$$

$$= (+35) \times (-14)$$

$$= -490$$

LHS = RHS, Hence proved.

RHS

$$= (-7) \times [(-5) \times (-14)]$$

$$= (-7) \times (+70)$$

$$= -490$$

4. Verify the distributive law of multiplication over addition and subtraction for the following integers.

Solution:

a) +45, -21 and -17

Solution:

We have to prove that:

i) $+45 \times [(-21) \times (-17)] = [(+45) \times (-21)] + [(+45) \times (-17)]$

ii) $+45 \times [(-21) - (-17)] = [(+45) \times (-21)] - [(+45) \times (-17)]$

i) $+45 \times [(-21) \times (-17)] = [(+45) \times (-21)] + [(+45) \times (-17)]$

LHS

$$= +45 \times [(-21) + (-17)]$$

$$= +45 \times (-38)$$

$$= -1710$$

RHS

$$= [(+45) \times (-21)] + [(+45) \times (-17)]$$

$$= (-945) + (-765)$$

$$= -1710$$

as LHS = RHS, Hence proved.

ii) $+45 \times [(-21) - (-17)] = [(+45) \times (-21)] - [(+45) \times (-17)]$

LHS

$$= +45 \times [(-21) - (-17)]$$

$$= +45 \times [(-21) + (+17)]$$

$$= +45 \times (-4)$$

$$= -180$$

RHS

$$= [(+45) \times (-21)] - [(+45) \times (-17)]$$

$$= (-945) + (-765)$$

$$= (-945) + (+765)$$

$$= -180$$

as LHS = RHS, Hence proved.

b) 34, -56, 19

Solution:

We have to prove that:

i) $34 \times [(-56) + 19] = [34 \times (-56)] + [34 \times 19]$

ii) $34 \times [(-56) - 19] = [34 \times (-56)] - [34 \times 19]$

i) $34 \times [(-56) + 19] = [34 \times (-56)] + [34 \times 19]$

LHS

$$= 34 \times [(-56) + 19]$$

$$= 34 \times (-37)$$

$$= -1258$$

RHS

$$= [34 \times (-56)] + [34 \times 19]$$

$$= (-1904) + (646)$$

$$= -1258$$

as LHS = RHS, Hence proved.

ii) $34 \times [(-56) - 19] = [34 \times (-56)] - [34 \times 19]$

LHS

$$= 34 \times [(-56) - 19]$$

$$= 34 \times (-75)$$

$$= -2550$$

RHS

$$= [34 \times (-56)] - [34 \times 19]$$

$$= -1904 - 646$$

$$= -2550$$

as LHS = RHS, Hence proved.

c) $-9, -19, -29$

Solution:

We have to prove that:

$$\text{i) } (-9) \times [(-19) + (-29)] = [(-9) \times (-19)] + [(-9) \times (-29)]$$

$$\text{ii) } (-9) \times [(-19) - (-29)] = [(-9) \times (-19)] - [(-9) \times (-29)]$$

$$\text{i) } (-9) \times [(-19) + (-29)] = [(-9) \times (-19)] + [(-9) \times (-29)]$$

LHS

$$= (-9) \times [(-19) + (-29)]$$

$$= (-9) \times (-48)$$

$$= +432$$

RHS

$$= [(-9) \times (-19)] + [(-9) \times (-29)]$$

$$= (+171) + (+261)$$

$$= +432$$

as LHS = RHS, Hence proved.

$$\text{ii) } (-9) \times [(-19) - (-29)] = [(-9) \times (-19)] - [(-9) \times (-29)]$$

LHS

$$= (-9) \times [(-19) + (+29)]$$

$$= (-9) \times (10)$$

$$= -90$$

RHS

$$= (+171) - (+261)$$

$$= +171 + (-261)$$

$$= -90$$

as LHS = RHS, Hence proved.

d) $+14, -28$ and 81

Solution:

We have to prove that:

$$\text{i) } +14 \times [(-28) + (+81)] = [+14 \times (-28)] + [+14 \times (+81)]$$

$$\text{ii) } +14 \times [(-28) - (+81)] = [+14 \times (-28)] - [+14 \times (+81)]$$

$$\text{i) } +14 \times [(-28) + (+81)] = [+14 \times (-28)] + [+14 \times (+81)]$$

LHS

$$= +14 \times [(-28) + (+81)]$$

$$= +14 \times (+53)$$

$$= +742$$

RHS

$$= [+14 \times (-28)] + [+14 \times (+81)]$$

$$= (-392) + 1134$$

$$= +742$$

as LHS = RHS, Hence proved.

$$\text{ii) } +14 \times [(-28) - (+81)] = [+14 \times (-28)] - [+14 \times (+81)]$$

LHS

$$= +14 \times [(-28) - (+81)]$$

$$= +14 \times [(-28) + (-81)]$$

$$= +14 \times [-109]$$

$$= -1526$$

RHS

$$= [+14 \times (-28)] - [+14 \times (+81)]$$

$$= -392 - (+1134)$$

$$= -392 + (-1134)$$

$$= -1526$$

as LHS = RHS, Hence proved.

5. Write the inverse of each integer.

Solution:

When we multiply two integers and the result is 1, then both integers are called the multiplicative inverse of each other.

a) -4

As $-4 \times \frac{1}{-4} = 1$. When we multiply -4 by $\frac{1}{-4}$ we get 1 as their product. So, the reciprocal of -4 is $\frac{1}{-4}$.

b) $+21$

As $+21 \times \frac{1}{+21} = 1$. When we multiply $+21$ by $\frac{1}{+21}$ we get 1 as their product. So, the reciprocal of $+21$ is $\frac{1}{+21}$.

c) -15

As $-15 \times \frac{1}{-15} = 1$. When we multiply -15 by $\frac{1}{-15}$ we get 1 as their product. So, the reciprocal of -15 is $\frac{1}{-15}$.

d) $+39$

As $+39 \times \frac{1}{+39} = 1$. When we multiply $+39$ by $\frac{1}{+39}$ we get 1 as their product. So, the reciprocal of $+39$ is $\frac{1}{+39}$.

2.7 Let's Practice

1. Simplify the following.

a) $9 \times 56 \div 4 - 3$

Solution:

We solve it according to the BODMAS Rule

$$\begin{aligned} 9 \times 56 \div 4 - 3 & \leftarrow \text{Dividing 56 by 4} \\ = 9 \times 14 - 3 & \leftarrow \text{Multiplying 9 by 14} \\ = 126 - 3 & \leftarrow \text{Subtracting 3 from 123} \\ = 123 \end{aligned}$$

c) $65 + \{23 - (44 - 32 \div 8)\}$

Solution:

$$\begin{aligned} 65 + \{23 - (44 - 32 \div 8)\} & \leftarrow \text{Solve parentheses first} \\ = 65 + \{23 - (44 - 4)\} & \leftarrow \text{Dividing 32 by 5} \\ = 65 + \{23 - (40)\} & \leftarrow \text{Subtracting 4 from 44} \\ = 65 + \{23 - 40\} & \leftarrow \text{Solve curly brackets} \\ = 65 + (-17) & \leftarrow \text{Subtracting 23 and 40} \\ = 48 & \leftarrow \text{Subtracting 65 and 17} \end{aligned}$$

b) $78 - 45 \times 5$

Solution:

$$\begin{aligned} 78 - 45 \times 5 \\ = 78 - 225 & \leftarrow \text{Multiplying 45 by 5} \\ = -147 & \leftarrow \text{Subtracting 225 from 123} \end{aligned}$$

$$d) 91.5 - \left\{ \frac{4}{3} \times (9 - 3) \right\}$$

Solution:

$$91.5 - \left\{ \frac{4}{3} \times (9 - 3) \right\} \quad \leftarrow \text{Solve parentheses first}$$

$$= 91.5 - \left\{ \frac{4}{3} \times 6 \right\} \quad \leftarrow \text{Subtracting 9 by 3}$$

$$= 91.5 - \left\{ \frac{4}{\cancel{3}_1} \times \overset{2}{\cancel{6}} \right\} \quad \leftarrow \text{Multiplying } \frac{4}{3} \text{ from 6}$$

$$= 91.5 - \{8\}$$

$$= 91.5 - 8 \quad \leftarrow \text{Solve curly brackets}$$

$$= 83.5 \quad \leftarrow \text{Subtracting 91.5 by 8}$$

$$e) 324 + 192 \div 12 \times 6$$

Solution:

$$324 + 192 \div 12 \times 6$$

$$= 324 + 16 \times 6 \quad \leftarrow \text{Dividing 192 by 12}$$

$$= 324 + 96 \quad \leftarrow \text{Multiplying 16 and 6}$$

$$= 420 \quad \leftarrow \text{Adding 324 and 96}$$

$$f) 5 \times [16 \div 2\{16 + (4 + 5) \times 7\}]$$

Solution:

$$5 \times [16 \div 2\{16 + (4 + 5) \times 7\}] \quad \leftarrow \text{Solve parentheses first}$$

$$= 5 \times [16 \div 2\{16 + 9 \times 7\}] \quad \leftarrow \text{Add 4 and 5}$$

$$= 5 \times [16 \div 2\{16 + 63\}] \quad \leftarrow \text{Multiply 9 and 7}$$

$$= 5 \times [16 \div 2\{79\}] \quad \leftarrow \text{Solving curly brackets and adding 16 and 63}$$

$$= 5 \times [16 \div 2 \times 79]$$

$$= 5 \times [8 \times 79] \quad \leftarrow \text{Dividing 16 by 2 and solving square brackets}$$

$$= 5 \times [632]$$

$$= 5 \times 632 \quad \leftarrow \text{Multiply 5 and 632}$$

$$= 3160$$

$$g) 455 - \{76.2 \div 2(18 + 4 \times 6)\}$$

Solution:

$$455 - \{76.2 \div 2(18 + 4 \times 6)\}$$

$$= 455 - \{76.2 \div 2(18 + 24)\} \quad \leftarrow \text{solve parentheses and multiplying 4 and 6}$$

$$= 455 - \{76.2 \div 2(42)\} \quad \leftarrow \text{adding 18 and 24}$$

$$= 455 - \{76.2 \div 2 \times 42\} \quad \leftarrow \text{solving curly brackets}$$

$$= 455 - \{38.1 \times 42\} \quad \leftarrow \text{dividing 76.2 by 2}$$

$$= 455 - \{1600.2\} \quad \leftarrow \text{multiplying 38.1 and 42}$$

$$= 455 - 1600.2 = -1145.2$$

$$h) 76.48 + \{42.5 \times 7(6.14 - 1.2 \times 0.4)\}$$

Solution:

$$76.48 + \{42.5 \times 7(6.14 - 1.2 \times 0.4)\}$$

$$= 76.48 + \{42.5 \times 7(6.14 - 0.48)\} \quad \leftarrow \text{solve parentheses and multiplying 1.2 by 0.4}$$

$$= 76.48 + \{42.5 \times 7(5.66)\} \quad \leftarrow \text{subtracting 6.14 and 0.48}$$

$$= 76.48 + \{42.5 \times 7 \times 5.66\} \quad \leftarrow \text{solving curly brackets and multiplying}$$

$$= 76.48 + \{42.5 \times 39.62\}$$

$$= 76.48 + 1683.85 = 1760.33$$

$$d) \frac{8}{9} \times 3 \div 2 (0.9 \div 0.2) + 1.7 \times 3.1$$

Solution:

$$\frac{8}{9} \times 3 \div 2 (0.9 \div 0.2) + 1.7 \times 3.1$$

$$= \frac{8}{9} \times 3 \div 2 \times 4.5 + 1.7 \times 3.1 \quad \leftarrow \text{Solve parentheses and dividing 0.9 by 0.2}$$

$$= \frac{8}{9} \times 1.5 \times 4.5 + 1.7 \times 3.1 \quad \leftarrow \text{Dividing 3 by 2}$$

$$= \frac{8}{9} \times 1.5 \times 4.5 + 5.27 \quad \leftarrow \text{Multiplying 1.7 and 3.1}$$

$$= \frac{8}{9} \times 6.75 + 5.27 \quad \leftarrow \text{Multiplying 8 and 6.75 and dividing by 9}$$

$$= \frac{54}{9} + 5.27$$

$$= 6 + 5.27 = 11.27 \quad \leftarrow \text{Adding 5.27 and 6}$$

- 2. Hassan bought 2.5 kg of apples, 3.25 kg of mangoes, 1.25 kg of strawberries and 4.50 kg of melon from the fruit shop. If the cost of one kg of apples is Rs 120, mango is Rs 180, strawberry is Rs 145 and melon is Rs 50, what is the total amount of money he spends in all?**

Solution:

Given:

quantity of apples bought	= 2.5 kg
quantity of mangoes bought	= 3.25 kg
quantity of strawberries bought	= 1.25 kg
quantity of melons bought	= 4.50 kg

cost of apples per kg	= Rs 120
cost of mangoes per kg	= Rs 180
cost of strawberries per kg	= Rs 145
cost of melon per kg	= Rs 50

$$\text{total amount spent on 2.5 kg of apples} = \text{Rs } 120 \times 2.5$$

$$\text{total amount spent on 3.25 kg of mangoes} = \text{Rs } 180 \times 3.25$$

$$\text{total amount spent on 1.25 kg of strawberries} = \text{Rs } 145 \times 1.25$$

$$\text{total amount spent on 4.50 kg of melons} = \text{Rs } 50 \times 4.50$$

to find the total amount of money Hassan spent in all we add all the above values.

$$\begin{aligned} \text{Total amount spent} &= \text{amount spent on apples} + \text{amount spent on mangoes} + \\ &\quad \text{amount spent on strawberries} + \text{amount spent on melons.} \\ &= \text{Rs } 120 \times 2.5 + \text{Rs } 180 \times 3.25 + \text{Rs } 145 \times 1.25 + \\ &\quad \text{Rs } 50 \times 4.50 \end{aligned}$$

$$\begin{aligned} \text{or} \quad &= 120 \times 2.5 + 180 \times 3.25 + 145 \times 1.25 + 50 \times 4.50 \\ &= 300 + 585 + 181 + 225 = 1291.25 \end{aligned}$$

So, Hassan spent Rs 1291.25 on all the fruits in total.

3. In a cricket match series, team A wins 4 matches and gets 6.23 points for each match. Team B wins 6 matches and gets 5.12 points for each match.

a) find the total points team A gets.

Solution:

$$\begin{aligned}\text{Matches won by Team A} &= 4 \\ \text{Points won for each match} &= 6.23 \\ \text{Points won for 4 matches} &= 4 \times 6.23 \\ &= 24.92\end{aligned}$$

So, team A gets 24.92 points in total.

b) find the total points team B gets.

Solution:

$$\begin{aligned}\text{Matches won by Team B} &= 6 \\ \text{Points won for each match} &= 5.12 \\ \text{Points won for 6 matches} &= 6 \times 5.12 \\ &= 30.72\end{aligned}$$

So, team B gets 30.72 points in total.

c) find the total points both team gets.

Solution:

$$\begin{aligned}\text{Team A gets} &= 24.92 \text{ points} \\ \text{Team B gets} &= 30.72 \text{ points} \\ \text{Total points both teams get} &= 24.92 + 30.72 \\ &= 55.64\end{aligned}$$

So, both teams get 55.64 points in total.

d) find the difference between the points of both teams.

Solution:

$$\begin{aligned}\text{Team A gets} &= 24.92 \text{ points} \\ \text{Team B gets} &= 30.72 \text{ points} \\ \text{Difference} &= \text{Points of Team B} - \text{Points of Team A} \\ &= 30.72 - 24.92 \\ &= 5.8\end{aligned}$$

So, the difference between the points of both teams is 5.8.

Unit Evaluation

1. Choose the correct option.

- a) (i) b) (iii) c) (ii) d) (i) e) (iii)

2. Write all the integers between -12 to $+12$.

Solution:

The integers between -12 to $+12$ are:

$-12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0$
 $+1, +2, +3, +4, +5, +6, +7, +8, +9, +10, +11, +12.$

3. Write all the integers between -5 to $+15$.

Solution:

The integers between -5 to $+15$.

$-5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5, +6, +7, +8, +9, +10, +11, +12, +13, +14,$
 $+15.$

4. Write the absolute value of the following integers.

- a) $+23$

Solution:

absolute value of $-23 = |+23| = 23$

- b) -9

Solution:

absolute value of $-9 = |-9| = 9$

- c) -11

Solution:

absolute value of $-11 = |-11| = 11$

- d) $+131$

Solution:

absolute of $+131 = |+131| = 131$

- e) -19

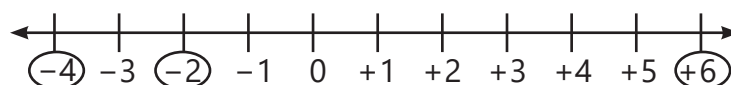
Solution:

absolute of $-19 = |-19| = 19$

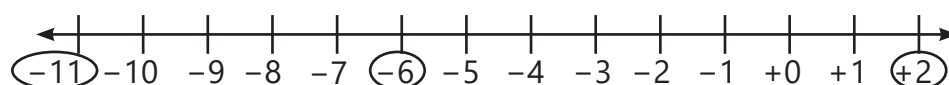
5. Show the given integers on a number line.

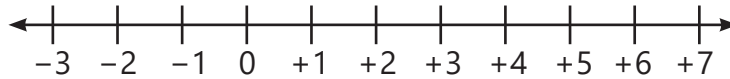
- a) $-4, -2, 6$

Solution:



- b) $-4, -6, +2$





6. Write the next five integers from the given sequence and show the integers on a number line.

a) $-14, -12, -10$

Solution:

We see that each integer is obtained when we add $+2$ to the previous integer, as

$$-14 + 2 = -12, -12 + 2 = -10$$

So, to obtain next five integers, we start by adding $+2$ to -10 .

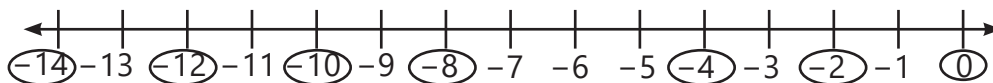
$$-10 + 2 = -8, -8 + 2 = -6, -6 + 2 = -4, -4 + 2 = -2$$

$$-2 + 2 = 0$$

So the sequence becomes:

$-14, -12, -10, -8, -6, -4, -2, 0$

On the number line:



b) $+3, +1, -1$

Solution:

We see that each integer of this sequence is obtained when we add -2 to the previous integer.

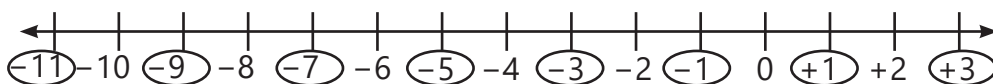
$$\text{i.e. } +3 + (-2) = +1, +1 + (-2) = -1, -1 + (-2) = -3, -3 + (-2) = -5$$

$$-5 + (-2) = -7, -7 + (-2) = -9, -9 + (-2) = -11$$

So the sequence becomes:

$+3, +1, -1, -3, -5, -7, -9, -11$

on the number line:



c) $-4, -8, -12$

Solution:

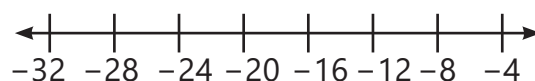
We see that each integer is obtained by adding -4 to the previous integer, i.e.

$$-4 + (-4) = -8, -8 + (-4) = -12, -12 + (-4) = -16, -16 + (-4) = -20$$

$$-20 + (-4) = -24, -24 + (-4) = -28, -28 + (-4) = -32$$

So the sequence becomes:

$-4, -8, -12, -16, -20, -24, -28, -32$ on the number line.



7. Write all the integers that come between the following.

a) -11 and -4

Solution:

Integers between -11 and -4 are: $-10, -9, -8, -7, -6, -5$

b) -2 and $+7$

Solution:

Integers between -2 and $+7$ are: $-1, 0, +1, 2, +3, +4, +5, +6$

c) -7 and $+7$

Solution:

Integers between -7 and $+7$ are:

$-6, -5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5, +6$

d) -1 and $+5$

Solution:

Integers and between -1 and $+5$ are:

$0, +1, +2, +3, +4$

e) -3 and $+6$

Solution:

Integers between -3 and $+6$ are:

$-2, -1, 0, +1, +2, +3, +4, +5$

f) -6 and $+8$

Solution:

Integers between -6 and $+8$ are:

$-5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5, +6, +7$

8. Compare and circle the greater one.

a) -44 and -4

Solution:

A greater integer with a negative sign is always smaller than a smaller integer with a negative sign. So -34 is greater than -44

$-44, \quad \textcircled{-34}$

b) -12 and $+15$

Solution:

A positive integer is always greater than a negative integer. So $+15$ is greater than -12 .

$-12, \quad \textcircled{+15}$

c) -19 and -7

Solution:

A greater integer with a negative sign is always smaller than a smaller integer with a negative sign. So, -7 is greater than -19 .

$-19, -7$

9. Arrange the following integers in ascending and descending order.

Solution:

Ascending order:

Starting from greater integer with a negative sign and ending at greater integer with a positive sign.

$-19, -8, -5, -2, +2, +4$

Descending order:

Starting from greater integer with a positive sign and ending at greater integer with a negative sign.

$+4, +2, -2, -5, -8, -19$

b) $-33, -12, -14, +14, +11, -17$

Solution:

Ascending order: Starting from the greater integer with a negative sign and ending at a greater integer with a positive sign.

$-33, -17, -14, -12, +11, +14$

Descending order: Starting from the greater integer with a positive sign and ending at a greater integer with a negative sign.

$+14, +11, -12, -14, -17, -33$

c) $-17, +14, -25, -20, +44$

Solution:

Ascending order: Starting from the greater integer with a negative sign and ending at a greater integer with a positive sign.

$-25, -20, -17, +14, +44$

Descending order: Starting from the greater integer with a positive sign and ending at a greater integer with a negative sign.

$+44, +14, -17, -20, -25$

d) $+44, -16, -23, +56, -13$

Solution:

Ascending order: Starting from the greater integer with a negative sign and ending at a greater integer with a positive sign.

$-3, -16, -13, +43, +56$

Descending order: Starting from the greater integer with a negative sign.

$+56, +43, -13, -16, -23$

10. Solve the following.

a) $+4 + (-6)$

Solution:

To add $+4$ and -6 , first take absolute values of both

$$|+4| = 4, |-6| = 6$$

Subtract these absolute values as both have opposite signs

$$6 - 4 = 2$$

Put the symbol of the greater integer.

$$+4 + (-6) = -2$$

b) $-7 - (-8)$

Solution:

To subtract first change the operation from subtraction to addition and change the sign of -8 to $+8$.

$$-7 + (+8)$$

Take absolute values of both.

$$|-7| = 7, |+8| = 8$$

Subtract these values of the greater integer

$$(-7) - (-8) = +1$$

c) $+5 - (+19)$

Solution:

To subtract first change the operation from subtraction to addition and change the sign of $+19$ to -19 .

$$+5 + (-19)$$

Take absolute values of both.

$$|+5| = 5, |-19| = 19$$

Subtract these values as both integers have opposite sign

$$19 - 5 = 14$$

put the sign of the greater integer

$$\text{so } +5 - (+19) = -14$$

d) -14×-19

Solution:

As both integers have same signs their product will be positive integer.

$$(-14) \times (-19)$$

$$\begin{array}{r} 14 \\ \times 19 \\ \hline 126 \\ 140 \\ \hline 266 \end{array}$$

so $(-14) \times (-19) = +266$

e) $-17 + (-67)$

Solution:

To add -17 and -67 first take their absolute values

$$|-17| = 17, |-67| = 67$$

Add these absolute values as both integers have same sign.

$$17 + 67 = 84$$

put $-$ sign with the answer as both integers have $-$ sign

$$-17 + (-67) = -84$$

f) $+12 \times (+23)$

Solution:

as both integers have the same sign so the product will also be a positive integer.

$$(+12) \times (+23)$$

$$\begin{array}{r} 12 \\ \times 23 \\ \hline 36 \\ 240 \\ \hline 276 \end{array}$$

So $+12 \times (+23) = +276$

g) $-24 \div (-12)$

Solution:

The result of division of integers having same sign is positive.

$$-24 \div (-12)$$

$$\begin{array}{r} 2 \\ 12 \overline{) 24} \\ \underline{-24} \\ 0 \end{array}$$

So $-24 \div (-12) = +2$

h) $+18 \div (-2)$

Solution:

The result of division of integers having opposite signs is negative.

$$+18 \div -2$$

$$\begin{array}{r} 9 \\ 2 \overline{) 18} \\ \underline{-18} \\ 0 \end{array}$$

so $+18 \div -2$

i) $-45 \times (-21)$

Solution:

The result of multiplication of two integers having the same sign is positive.

$(-45) \times (-21)$

$$\begin{array}{r} 45 \\ \times 21 \\ \hline 45 \\ 900 \\ \hline 945 \end{array}$$

So, $-45 \times (-21) = +945$

11. Verify the commutative property of addition and multiplication.

a) -4 and $+13$

Solution:

Commutative property of addition:

$$-4 + 13 = +9$$

changing the order of integers and adding

$$+13 \times (-4) = +9$$

So, changing the order of integers does not affect the result. Hence proved.

commutative property of multiplication.

first multiply -4 by $+13$

$$\begin{array}{r} (-4) \\ \times (+13) \\ \hline -52 \end{array}$$

Now multiply $+13$ by -4

$$\begin{array}{r} (+13) \\ \times (-4) \\ \hline -52 \end{array}$$

as $(-4) \times (+13) = -52$ and $(+13) \times (-4) = -52$.

So changing the order of integers does not change the result. Hence proved.

b) -6 and -11

commutative property of addition.

adding -6 and -11

$$-6 + (-11) = -17$$

changing the order of integers and adding.

$$-11 + (-6) = -17$$

so, changing the order of integers does not affect the result. Hence proved.

commutative property of multiplication

first multiply -6 by -11

$$\begin{array}{r} -6 \\ \times -11 \\ \hline + 66 \end{array}$$

Now multiply -11 by -6

$$\begin{array}{r} -11 \\ \times -6 \\ \hline + 66 \end{array}$$

As $-6 \times -11 = +66$ and $-11 \times -6 = +66$. so changing the order of integers does not change the result. Hence proved.

c) $+15$ and -19

Solution:

Commutative property of addition.

adding $+15$ and -19

$$+15 + (-19) = -4$$

changing the order and adding

$$(-19) + (+15) = -4$$

so, changing the order of integers does not affect the result. Hence proved.

commutative property of multiplication

first multiply $+15$ by -19

$$\begin{array}{r} +15 \\ \times -19 \\ \hline 135 \\ 150 \\ \hline - 285 \end{array}$$

Now multiply -19 by $+15$

$$\begin{array}{r} -19 \\ \times +15 \\ \hline 95 \\ 190 \\ \hline - 285 \end{array}$$

as $(+15) \times (-19) = -285$, and $(-19) \times (+15) = -285$. So changing the order of integers does not change the result. Hence proved.

d) $+19$ and $+31$

Solution:

commutative property of addition.

$$+19 \times (+31) = +50$$

changing the order of integers and adding.

$$(+31) \times (+19) = +50$$

so, changing the order of integers does not affect the result. Hence proved.

commutative property of multiplication

first multiply $+19$ by $+31$

$$\begin{array}{r} (+19) \\ \times (+31) \\ \hline 19 \\ 570 \\ \hline + 589 \end{array}$$

Now multiply $+31$ by $+19$

$$\begin{array}{r} (+31) \\ \times (+19) \\ \hline 279 \\ 310 \\ \hline + 589 \end{array}$$

as $(+19) \times (+31) = +589$ and $(31) \times (+19) = +589$. so changing the order of integers does not change the result. Hence proved.

12. Verify the associative property of addition and multiplication.

a) +5, +2 and -7

Solution:

Associative property of addition

According to associative law of addition.

$$\begin{array}{ll}
 [+5 + (+2)] + (-7) & = \quad +5 + [(+2) + (-7)] \\
 \text{LHS} & \text{RHS} \\
 = [+5 + (2)] + (-7) & = +5 + [(+2) + (-7)] \\
 = [+5 + 2] + (-7) & = +5 + [+2 - 7] \\
 = (+7) + (-7) & = +5 + (-5) \\
 = 0 & = 0
 \end{array}$$

LHS = RHS. Hence proved.

Associative property of multiplication

According to associative law of multiplication.

$$\begin{array}{ll}
 [+5 \times (+2)] \times (-7) & = \quad +5 \times [(+2) \times (-7)] \\
 \text{LHS} & \text{RHS} \\
 = [+5 \times (+2)] \times (-7) & = +5 \times [(+2) \times (-7)] \\
 = [+10] \times (-7) & = +5 \times (-14) \\
 = -70 & = -70
 \end{array}$$

LHS = RHS. Hence proved.

b) -25, -56 and -54

Solution:

Associative property of addition

$$\begin{array}{ll}
 [(-25) + (-56)] + (-54) & = \quad (-25) + (56) + (-54)] \\
 \text{LHS} & \text{RHS} \\
 = [(-25) + (-56)] + (-54) & = (-25) + [(-56) + (-54)] \\
 = (-81) + (-54) & = (-25) + (-110) \\
 = -135 & = -135 \\
 = 0 & = 0
 \end{array}$$

LHS = RHS. Hence proved.

Associative property of multiplication

$$\begin{array}{ll}
 [(-25) \times (-56)] \times (-54) & = \quad (-25) \times (56) \times (-54)] \\
 \text{LHS} & \text{RHS} \\
 = [(-25) \times (-56)] \times (-54) & = (-25) \times [(-56) \times (-54)] \\
 = (+1400) \times (-54) & = (-25) \times (+3024) \\
 = -75600 & = -75600
 \end{array}$$

LHS = RHS. Hence proved.

c) +14, +16 and -17

Solution:

Associative property of addition

$$\begin{array}{ll} [+14 + (+16)] + (-17) & = (+14) + [(+16) + (-17)] \\ \text{LHS} & \text{RHS} \\ = [+14 + (+16)] + (-17) & = (+14) + [(+16) + (-17)] \\ = (+30) + (-17) & = (+14) + (-1) \\ = +13 & = +13 \end{array}$$

LHS = RHS. Hence proved.

Associative property of multiplication

$$\begin{array}{ll} [+14 \times (+16)] \times (-17) & = (+14) \times [(+16) \times (-17)] \\ \text{LHS} & \text{RHS} \\ = [+14 \times (+16)] \times (-17) & = (+14) \times [(+16) \times (-17)] \\ = (+224) \times (-17) & = (+14) \times (-272) \\ = -3808 & = -3808 \end{array}$$

LHS = RHS. Hence proved.

13. Verify the distributive property of multiplication over addition and subtraction.

a) -11, -13 and +23

Solution:

we have to prove that:

$$\begin{array}{ll} \text{i) } -11 \times [(-13) + (+23)] & = [(-11 \times (-13)) + [(+11) \times (+23)]] \\ \text{ii) } -11 \times [(-13) - (+23)] & = [(-11 \times (-13)) - [(-11) \times (+23)]] \\ \text{i) } -11 \times [(-13) + (+23)] & = [(-11 \times (-13)) + [(+11) \times (+23)]] \\ \text{LHS} & \text{RHS} \\ = -11 \times [(-13) + (+23)] & = [(-11 \times (-13)) + [(-11) \times (+23)]] \\ = -11 \times (+10) & = (+143) + (-253) \\ = -110 & = -110 \end{array}$$

LHS = RHs. Hence proved.

$$\begin{array}{ll} \text{ii) } -11 \times [(-13) - (+23)] & = [(-11 \times (-13)) - [(-11) \times (+23)]] \\ \text{LHS} & \text{RHS} \\ = -11 \times [(-13) - (+23)] & = [(-11 \times (-13)) - [(-11) \times (+23)]] \\ = -11 \times (-36) & = 143 + (+253) \\ = +396 & = +396 \end{array}$$

LHS = RHs. Hence proved.

b) +44, +32 and +76

Solution:

we have to prove that:

$$\text{i) } +44 \times [(+32) + (+76)] = [(+44) \times (+32)] + [(+44) \times (+76)]$$

$$\begin{array}{ll}
 \text{ii) } +44 \times [(+32) - (+76)] & = [(+44) \times (+32)] + [(+44) \times (+76)] \\
 \text{i) } +44 \times [(+32) + (+76)] & = [(+44) \times (+32)] + [(+44) \times (+76)] \\
 \text{LHS} & \text{RHS} \\
 = +44 \times [(+32) + (+76)] & = [(+44) \times (+32)] + [(+44) \times (+76)] \\
 = +44 \times [+108] & = (+1408) + (+3344) \\
 = +4752 & = +4752 \\
 \text{ii) } +44 \times [(+32) - (+76)] & = [(+44) \times (+32)] + [(+44) \times (+76)] \\
 \text{LHS} & \text{RHS} \\
 = +44 \times [(+32) - (+76)] & = [(+44) \times (+32)] - [(+44) \times (+76)] \\
 = +44 \times [+32 + (-76)] & = +1408 - (+3344) \\
 = +44 \times (-44) & = +1408 + (-3344) \\
 = -1932 & = -1932
 \end{array}$$

LHS = RHS. Hence proved.

c) -1, +4 and -5

Solution:

we have to prove that:

$$\begin{array}{ll}
 \text{i) } (-1) \times [(+4) + (-5)] & = [(-1) \times (+4)] + [(-1) \times (-5)] \\
 \text{ii) } (-1) \times [(+4) - (-5)] & = [(-1) \times (+4)] - [(-1) \times (-5)] \\
 \text{i) } (-1) \times [(+4) + (-5)] & = [(-1) \times (+4)] \times [(-1) \times (-5)] \\
 \text{LHS} & \text{RHS} \\
 = (-1) \times [(+4) + (-5)] & = [(-1) \times (+4)] + [(-1) \times (-5)] \\
 = (-1) \times [-1] & = (-4) + (+5) \\
 = +1 & = +1
 \end{array}$$

LHS = RHS. Hence proved.

$$\begin{array}{ll}
 \text{ii) } (-1) \times [(+4) - (-5)] & = [(-1) \times (+4)] - [(-1) \times (-5)] \\
 \text{LHS} & \text{RHS} \\
 = (-1) \times [(+4) + (-5)] & = [(-1) \times (+4)] - [(-1) \times (-5)] \\
 = (-1) \times [(+4) + (+5)] & = (-4) - (+5) \\
 = (-1) \times (+9) & = (-4) + (-5) \\
 = -9 & = -9
 \end{array}$$

LHS = RHS. Hence proved.

14. Identify and write the name of the property.

a) $2 + (-3 + 5) = (2 + (-3)) + 5$

Solution:

Here we see that when we add two numbers at a time and then add the third number to it the order of grouping in addition does not affect the result. So this equation shows the “associative property w.r.t. addition”.

b) $-6 + 4 = 4 + (-6)$

Solution:

Here we see that when we add these two integers, changing the order of the integers does not affect the result. So this equation shows the “commutative property w.r.t. addition”.

c) $+5 \times (-3) = -3 \times (+5)$

Solution:

Here we see that changing the order of two integers when we are multiplying two integers does not affect the result. Hence it shows the “commutative property w.r.t. multiplication”.

$$d) 7 \times [(-2) + (-9)] = [(7 \times (-2))] + [(7 \times (-9))]$$

Solution:

Here we see that when we add two integers and then multiply with a third integer, it is the same as when we multiply the third integer by the first and second integer separately and then add their product. So this equation shows the “distributive property of multiplication over addition”.

15. Write the additive and multiplicative inverse of each.

a) -4

Solution:

Additive Inverse

$$\text{as } -4 + (+4) = 0$$

so +4 is the additive inverse of -4.

Multiplicative Inverse

$$\text{as } -4 \times \frac{1}{4} = 1, \text{ so } \frac{1}{4} \text{ is the multiplicative inverse of } -4.$$

b) +12

Solution:

Additive Inverse

$$\text{as } +12 + (-12) = 0$$

so -12 is the additive inverse of +12.

Multiplicative Inverse

$$\text{as } +12 \times \frac{1}{+12} = 1, \text{ so } \frac{1}{+12} \text{ is the multiplicative inverse of } +12.$$

c) -98

Solution:

Additive Inverse

$$\text{as } -98 + (+98) = 0$$

so +98 is the additive inverse of -98.

Multiplicative Inverse

$$\text{as } -8 \times \frac{1}{-98} = 1, \text{ so } \frac{1}{-98} \text{ is the multiplicative inverse of } -98.$$

d) 64

Solution:

Additive Inverse

$$\text{as } 64 + (-64) = 0$$

so -64 is the additive inverse of 64.

Multiplicative Inverse

$$\text{as } 64 \times \frac{1}{64} = 1, \text{ so } \frac{1}{64} \text{ is the multiplicative inverse of } 64.$$

16. Solve the following expressions.a) $709 + 54 - 66 \div 3 \times 2$ **Solution:**

$$709 + 54 - 66 \div 3 \times 2$$

$$= 709 + 54 - 22 \times 2 \quad \leftarrow \text{dividing 66 by 3}$$

$$= 709 + 54 - 44 \quad \leftarrow \text{multiplying 22 by 2}$$

$$= 763 - 44 \quad \leftarrow \text{adding 709 and 54}$$

$$= 719 \quad \leftarrow \text{subtracting 763 and 44}$$

b) $56.6 + 14.8 - (3.8 + 1.4 \times 0.2)$

Solution:

$$\begin{aligned}
 &56.6 + 14.8 - (3.8 + 1.4 \times 0.2) && \text{solve parentheses first} \\
 &= 56.6 + 14.8 - (4.08) && \text{multiplying 1.4 and 0.2} \\
 &= 56.6 + 14.8 + (-4.08) && \text{adding 3.8 and 0.28} \\
 &= 56.6 + 10.72 && \text{changing the operation and} \\
 &= 67.32 && \text{changing the sign of +4.08.} \\
 & && \text{adding 56.6 and 10.72}
 \end{aligned}$$

c) $\frac{2}{7} - [27 + \{32 - (6.2 + 3.4 \times 1.7)\}]$

Solution:

$$\begin{aligned}
 &\frac{2}{7} - [27 + \{32 - (6.2 + 3.4 \times 1.7)\}] && \text{solving parentheses and} \\
 &= \frac{2}{7} - [27 + \{32 - (6.2 + 5.78)\}] && \text{multiplying 3.4 and 1.7} \\
 &= \frac{2}{7} - [27 + \{32 - (11.98)\}] && \text{adding 6.2 and 5.78} \\
 &= \frac{2}{7} - [27 + \{32 + - 11.98\}] && \text{adding 6.2 and 5.78} \\
 &= \frac{2}{7} - [27 + \{20.02\}] && \text{solving curly brackets} \\
 &= \frac{2}{7} - [27 + (20.02)] && \text{adding 32 and 11.98} \\
 &= \frac{2}{7} - (47.02) && \text{adding 27 and 20.02} \\
 &= \frac{2}{7} + (- 47.02) \\
 &= 0.2857 - 47.02 && \text{dividing 2 by 7} \\
 &= -46.73 && \text{subtracting 0.2857 and 47.02}
 \end{aligned}$$

d) $177 - \{(34.2 + 4.5) \times (2.42 + 1.9)\}$

Solution:

$$\begin{aligned}
 &= 177 - \{38.7 \times 4.32\} \\
 &= 177 - \{167.184\} \\
 &= 9.816
 \end{aligned}$$

17. The temperature of Murree is -1°C at night and at noon, it is $+5^{\circ}\text{C}$. What is the difference in Murree's temperature between night and noon?

Solution:

Temperature of Muree at night = -1°C
 at noon = $+5^{\circ}\text{C}$

Difference between Muree's temp.

$$\begin{aligned}
 \text{at night and noon is} &= +5 - (-1) \\
 &= +5 + (+1) \\
 &= +6
 \end{aligned}$$

Therefore, the difference between Muree's temp. at night and noon is $+6^{\circ}\text{C}$.

Unit 3

Ratio, Rates and Percentages

3.1 Let's Practice

1. Find the percentage for the following.

a) 44 out of 100

Solution:

The percentage can be found by dividing the number by the total and multiplying by 100.

$$44 \text{ out of } 100 = \frac{44}{100}$$

$$\begin{aligned} \text{In percentage} &= \frac{44}{100} \times 100 \\ &= 44\% \end{aligned}$$

b) 50 out of 100

Solution:

$$50 \text{ out of } 100 = \frac{50}{100}$$

$$\begin{aligned} \text{In percentage} &= \frac{50}{100} \times 100 \\ &= 50\% \end{aligned}$$

c) 10 out of 15

Solution:

$$10 \text{ out of } 15 = \frac{10}{15}$$

$$\begin{aligned} \text{In percentage} &= \frac{10}{15} \times 100 \\ &= \frac{1000}{15} \\ &= 66.67\% \end{aligned}$$

d) 126 out of 255

Solution:

$$126 \text{ out of } 255 = \frac{126}{255}$$

$$\begin{aligned} \text{In percentage} &= \frac{126}{255} \times 100 \\ &= \frac{12600}{255} \\ &= 49.41\% \end{aligned}$$

e) 92 out of 300

Solution:

$$92 \text{ out of } 300 = \frac{92}{300}$$

$$\begin{aligned} \text{In percentage} &= \frac{92}{300} \times 100 \\ &= \frac{9200}{300} \\ &= 30.67\% \end{aligned}$$

2. Calculate:

a) 12% of 23

Solution:

$$\begin{aligned} 12\% \text{ of } 23 &= \frac{12}{100} \times 23 && \begin{array}{l} 12\% \text{ means } \frac{12}{100} \\ \text{or } 12 \text{ out of } 100 \end{array} \\ &= \frac{276}{100} \\ &= 2.76 \end{aligned}$$

b) 100% of 23

Solution:

$$\begin{aligned} 100\% \text{ of } 50 &= \frac{100}{100} \times 50 && \begin{array}{l} 100\% \text{ means } 100 \\ \text{out of } 100 \text{ or } \frac{100}{100} \end{array} \\ &= 1 \times 50 \\ &= 50 \end{aligned}$$

c) 72% of 35

Solution:

$$\begin{aligned} 72\% \text{ of } 35 &= \frac{72}{100} \times 35 && \begin{array}{l} \text{72\% means 72} \\ \text{out of 100 or } \frac{72}{100} \end{array} \\ &= \frac{2520}{100} \\ &= 25.2 \end{aligned}$$

d) 5% of 9

Solution:

$$\begin{aligned} 5\% \text{ of } 9 &= \frac{5}{100} \times 9 && \begin{array}{l} \text{5\% means 5 out of} \\ \text{100 or } \frac{5}{100} \end{array} \\ &= \frac{45}{100} \\ &= 2.76\% \end{aligned}$$

e) 1% of 15

Solution:

$$\begin{aligned} 1\% \text{ of } 15 &= \frac{1}{100} \times 15 && \begin{array}{l} \text{1\% means 1 out} \\ \text{of 100 or } \frac{1}{100} \end{array} \\ &= \frac{15}{100} \\ &= 0.15 \end{aligned}$$

3. There are 350 fruit trees in an orchard. Out of these, 176 trees are mango trees and remaining are apple trees. Find:

Solution:

Total number of a fruit trees = 350

Number of mango trees = 176

Number of apple trees = Total number of fruit trees – Number of mango trees
= 350 – 176
= 174

a) the percentage of mango trees

Solution:

$$\begin{aligned} \text{The percentage of mango trees} &= \frac{\text{Number of mango trees}}{\text{Total number of fruit trees}} \times 100 \\ &= \frac{176}{350} \times 100 \\ &= \frac{17600}{350} = 50.29\% \end{aligned}$$

So, the percentage of mango trees is 50.29%

b) the percentage of apple trees

Solution:

$$\begin{aligned} \text{The percentage of apple trees} &= \frac{\text{Number of apple trees}}{\text{Total number of fruit trees}} \times 100 \\ &= \frac{174}{350} \times 100 \\ &= \frac{17400}{350} = 49.7\% \end{aligned}$$

So, the percentage of apple trees is 49.7%

4. 18% of houses in a colony have solar panels. If there are 350 houses altogether, find the number of houses that have solar panels.

Solution:

Total number of houses in the colony = 350

Percentages of houses having solar panels = 18%

Number of houses having solar panels = 18%

As 18% of all houses in the colony have solar panels so we can write it as $\frac{18}{100}$ of 350 houses have solar panels.

So, the number of houses having.

$$\begin{aligned}\text{Solar panels} &= \frac{18}{100} \text{ of } 350 \\ &= \frac{18}{100} \times 350 \\ &= \frac{6300}{100} \times 350 \\ &= 63\end{aligned}$$

So, 63 houses have solar panels.

5. About 7 out of 10 parts of the Earth's surface is water. Find the percentage of water on Earth.

Solution:

Total parts of Earth's surface = 10

Parts of earth's surface that is water = 7

Percentage of water on Earth = ?

Percentage of water on Earth's surface can be found by dividing parts of earth's surface that is water by the total parts of earth surface and multiplying by 100.

$$\begin{aligned}\text{Percentage of water on Earth} &= \frac{7}{10} \times \frac{100}{100} \\ &= 70\%\end{aligned}$$

So, 70% percent of earth's space is water.

3.2 Let's Practice

1. Calculate and compare using percentage.

a) 43 out of 65 and 3 out of 10

Solution:

$$\begin{aligned}\text{Percentag of 43 out of 65} &= \frac{43}{65} \times 100 \\ &= \frac{4300}{65} = 66.15\%\end{aligned}$$

$$\text{Percentag of 3 out of 10} = \frac{3}{10} \times 100 = 30\%$$

So, $66.15\% > 30\%$ or 43 out of 65 is greater than 3 out of 10.

b) 70 out of 90 and 1 out of 2

Solution:

$$\begin{aligned}\text{Percentag of 70 out of 90} &= \frac{70}{90} \times 100 \\ &= \frac{700}{9} = 77.78\%\end{aligned}$$

$$\begin{aligned}\text{Percentag of 1 out of 2} &= \frac{1}{2} \times 100 \\ &= 1 \times 50 = 50\%\end{aligned}$$

So, $77.78\% > 50\%$. So 70 out of 90 is greater than 1 out of 2.

c) 20 out of 34 and 11 out of 15

Solution:

$$\begin{aligned}\text{Percentag of 20 out of 34} &= \frac{20}{34} \times 100 \\ &= \frac{2000}{34} = 58.82\%\end{aligned}$$

$$\begin{aligned}\text{Percentag of 11 out of 15} &= \frac{11}{15} \times 100 \\ &= \frac{1100}{15} = 73.33\%\end{aligned}$$

So, $73.33\% > 58.82\%$ or 11 out of 15 is greater than 20 out of 34.

d) 66 out of 70 and 66 out of 85

Solution:

$$\begin{aligned}\text{Percentag of 66 out of 70} &= \frac{66}{70} \times 100 \\ &= \frac{6600}{70} = 94.29\%\end{aligned}$$

$$\begin{aligned}\text{Percentag of 66 out of 85} &= \frac{66}{85} \times 100 \\ &= \frac{6600}{85} = 77.65\%\end{aligned}$$

So, $94.29\% > 77.65\%$ or percentage of 66 out of 70 is greater than 66 out of 85.

2. Calculate.

a) 65 increased by 80%

Solution:

First we find 80% of 65:

$$\begin{aligned}80\% \text{ of } 65 &= \frac{80}{100} \times 65 \\&= \frac{5200}{100} = 52\end{aligned}$$

So, 80% of 65 is 52.

c) 100 increased by 50%

Solution:

First we find 50% of 100:

$$\begin{aligned}50\% \text{ of } 100 &= \frac{50}{100} \times 100 \\&= 50\end{aligned}$$

So, 50% of 100 is 50.

e) 89 decreased by 89%

Solution:

First we find 89% of 89:

$$\begin{aligned}89\% \text{ of } 89 &= \frac{89}{100} \times 89 \\&= \frac{7921}{100} = 79.21\end{aligned}$$

So, 89% of 89 is 79.21.

$$\begin{aligned}\text{Now, } 89 \text{ decreased by } 89\% &= 89 - 79.21 \\&= 9.79\end{aligned}$$

So, 89 decrease by 89% is 9.79.

b) 198 increased by 1%

Solution:

First we find 1% of 198:

$$\begin{aligned}1\% \text{ of } 198 &= \frac{1}{100} \times 198 \\&= \frac{198}{100} = 1.98\end{aligned}$$

So, 1% of 198 is 1.98.

d) 244 decreased by 15%

Solution:

First we find 15% of 244:

$$\begin{aligned}15\% \text{ of } 244 &= \frac{15}{100} \times 244 \\&= \frac{3660}{100} = 36.6\end{aligned}$$

So, 15% of 244 is 36.6.

$$\begin{aligned}\text{Now, } 244 \text{ decreased by } 15\% &= 244 - 36.6 \\&= 207.4\end{aligned}$$

So, 244 decrease by 15% is 207.4

f) 900 decreased by 99%

Solution:

First we find 99% of 900:

$$\begin{aligned} 99\% \text{ of } 900 &= \frac{99}{100} \times 900 \\ &= 891 \end{aligned}$$

So, 99% of 900 is 891.

$$\begin{aligned} \text{Now, } 900 \text{ decreased by } 99\% &= 900 - 891 \\ &= 9 \end{aligned}$$

So, 99 decrease by 900% is 9.

- 3. A pre-school book has 44 pages. 15 pages have letters and 20 pages have pictures. Compare the number of pages with pictures to the number of pages with letters using percentages.**

Solution:

Number of pages in book = 44

Number of pages having letters = 15

Number of pages having pictures = 20

$$\begin{aligned} \text{Percentage of pages having letters} &= \frac{15}{44} \times 100 \\ &= \frac{1500}{44} = 34.09\% \end{aligned}$$

$$\begin{aligned} \text{Percentage of pages having pictures} &= \frac{20}{44} \times 100 \\ &= \frac{2000}{44} = 45.45\% \end{aligned}$$

So, number of pages having pictures is greater than number of pages having letters.

$$45.45\% > 34.09\%$$

- 4. Saad's salary is Rs 82,500 per month. He spends Rs 25,000 on groceries. Hamid's salary is Rs 94,000 per month. He spends Rs 40,000 on groceries. Compare the expenditure on groceries for both using percentages.**

Solution:

Saad's salary = Rs. 82,500

Saad spends on groceries = Rs. 25,000

Hamid's salary = Rs. 94,000

Hamid spends on groceries = Rs. 40,000

Percentage of Saad's income spent on groceries =

$$\begin{aligned}
&= \frac{\text{income spent on groceries}}{\text{Total income of Saad}} \times 100 \\
&= \frac{25000}{82500} \times 100 \\
&= \frac{2500000}{82500} \\
&= 30.30\%
\end{aligned}$$

Percentage of Hamid's income spent on groceries =

$$\begin{aligned}
&= \frac{\text{income spent on groceries}}{\text{Total income of Hamid}} \times 100 \\
&= \frac{40000}{94000} \times 100 \\
&= \frac{4000000}{94000} \\
&= \frac{4000}{94} \\
&= 42.55\%
\end{aligned}$$

as $42.55\% > 30.30\%$

So, Hamid spends more on groceries than Saad.

- 5. Mirha got 42 marks out of 50 marks in science and 72 marks out of 100 marks in mathematics. Compare her marks in science and mathematics by using percentages.**

Solution:

Mirha's marks in science = 42

Total marks of science = 50

Mirha's marks in mathematics = 72

Total marks of mathematics = 100

$$\begin{aligned}
\text{Percentage of Mirha's marks in science} &= \frac{\text{Mirha's marks in science}}{\text{Total marks}} \times 100 \\
&= \frac{42}{50} \times 100 \\
&= 84\%
\end{aligned}$$

Percentage of Mirha's marks in mathematics =

$$\begin{aligned}
&= \frac{\text{Mirha's marks in mathematics}}{\text{Total marks}} \times 100 \\
&= \frac{72}{100} \times 100 \\
&= 72\%
\end{aligned}$$

as $84\% > 72\%$

So Mirha got more marks in science than in mathematics.

6. The price of a laptop is Rs 98,075.

- a) If the price increases by 10%, find the increased price.
b) If the price decreases by 5%, find the decreased price.

Solution:

a) If the price....

original price of laptop = Rs. 98, 075

Percentage of increase in price = 10%

increased price = ?

To find increased price we need to find the increase in price.

Increase in price = 10% of Rs. 98, 075

$$\begin{aligned} &= \frac{10}{100} \times 98,075 \\ &= \text{Rs. } 9807.5 \end{aligned}$$

So, increased price = original price + increase in price

$$\begin{aligned} &= \text{Rs. } 98,075 + \text{Rs. } 9807.5 \\ &= \text{Rs. } 107, 882.5 \end{aligned}$$

b) If the price decreases.....

original price of laptop = Rs. 98, 075

Percentage of decrease in price = 5%

decreased price = ?

To find decreased price we need to find the decrease in price.

Decrease in price = 5% of Rs. 98, 075

$$\begin{aligned} &= \frac{5}{100} \times 98,075 \\ &= \text{Rs. } 4903.75 \end{aligned}$$

So, decreased price = original price – increase in price

$$\begin{aligned} &= \text{Rs. } 98,075 - \text{Rs. } 4903.75 \\ &= \text{Rs. } 93171.25 \end{aligned}$$

7. Anum got 428 marks out of 500 in the first term. In the second term, she got 17% more marks than in the first term. Find the increased marks.

Solution:

Anum's marks in first term = 428

Total marks = 500

Percentage of Anum's marks in first term = $\frac{428}{500} \times 100 = 85.6\%$

Percentage increase in marks in second term = 17%

So, percentage of marks in second term = $85.6\% + 17\% = 102.6\%$

3.3 Let's Practice

1. Find the ratios.

a) 30 to 110

b) 65 kg to 90 kg

c) 400 to 180

d) 58 to 220

e) 562 m to 880 m

f) 720 to 666

Solution:

a) 30 to 110

b) 65kg to 90kg

30 : 110

65 : 90

Dividing by their HCF to

Dividing by their HCF to

simplify this ratio

simplify this ratio

$$\frac{30}{10} : \frac{110}{10}$$

$$\frac{65}{5} : \frac{90}{5}$$

3 : 11

13 : 18

c) 400 to 180

d) 58 to 220

400 : 180

58 : 220

Dividing by their HCF to

Dividing by their HCF to

simplify this ratio

simplify this ratio

$$\frac{400}{20} : \frac{180}{20}$$

$$\frac{58}{2} : \frac{220}{2}$$

20 : 9

29 : 110

e) 562m to 880m

f) 720 to 666

562 : 880

720 : 666

Dividing by their HCF to

Dividing by their HCF to

simplify this ratio

simplify this ratio

$$\frac{562}{2} : \frac{880}{2}$$

$$\frac{720}{18} : \frac{666}{18}$$

281 : 440

40 : 37

2. Simplify the following ratios.

a) $20 : 14$

b) $32 : 64$

c) $112 : 98$

d) $530 : 680$

e) $90 : 33$

f) $140 : 360$

Solution:

a) $20:14$

Dividing these values by their

HCF to simplify.

$$\frac{20}{2} : \frac{14}{2}$$

$$10 : 7$$

c) $112:98$

Dividing these values by their

HCF to simplify.

$$\frac{112}{14} : \frac{98}{14}$$

$$8 : 7$$

e) $90:33$

Dividing these values by their

HCF to simplify.

$$\frac{90}{3} : \frac{33}{3}$$

$$30 : 11$$

b) $32:64$

Dividing these values by their

HCF to simplify.

$$\frac{32}{32} : \frac{64}{32}$$

$$1 : 2$$

d) $530:680$

Dividing these values by their

HCF to simplify.

$$\frac{530}{10} : \frac{680}{10}$$

$$53 : 68$$

f) $140:360$

Dividing these values by their

HCF to simplify.

$$\frac{140}{20} : \frac{360}{20}$$

$$7 : 18$$

3. Find the continued ratio $a : b : c$ for the following values.

a) If $a : b = 2 : 3$ and $b : c = 1 : 4$

b) If $a : b = 5 : 8$ and $b : c = 4 : 7$

c) If $a : b = 1 : 2$ and $b : c = 3 : 4$

d) If $a : b = 5 : 3$ and $b : c = 4 : 7$

e) If $a : b = 5 : 6$ and $b : c = 3 : 2$

Solution:

a) If $a : b = 2 : 3$ and $b : c = 1 : 4$

$$a : b = 2 : 3 \qquad b : c = 1 : 4$$

To find the continued ratio we write these values as follows:

$$\begin{array}{ccccc} a & : & b & : & c \\ 2 & : & 3 & : & \\ & \searrow & \downarrow & \searrow & \\ & & 1 & : & 4 \\ 2 \times 1 & : & 3 \times 1 & : & 3 \times 4 \\ 2 & : & 3 & : & 12 \end{array}$$

So the continued ratio for the given values is

$$2 : 3 : 12$$

b) If $a : b = 5 : 8$ and $b : c = 4 : 7$

$$a : b = 5 : 8 \quad b : c = 4 : 7$$

To find the continued ratio we write these values as follows:

$$\begin{array}{ccccc} a & : & b & : & c \\ 5 & : & 8 & : & \\ & \searrow & \downarrow & \searrow & \\ & & 4 & : & 7 \\ 5 \times 4 & : & 8 \times 4 & : & 8 \times 7 \\ 20 & : & 32 & : & 56 \end{array}$$

Dividing by the HCF of these numbers to simplify.

$$\frac{20}{4} : \frac{32}{4} : \frac{56}{4}$$

$$5 : 8 : 14$$

c) If $a : b = 1 : 2$ and $b : c = 3 : 4$

$$a : b = 1 : 2 \quad b : c = 3 : 4$$

To find the continued ratio we write these values as follows:

$$\begin{array}{ccccc} a & : & b & : & c \\ 1 & : & 2 & : & \\ & \searrow & \downarrow & \searrow & \\ & & 3 & : & 4 \\ 1 \times 3 & : & 2 \times 3 & : & 2 \times 4 \\ 3 & : & 6 & : & 8 \end{array}$$

So the continued ratio for the given values is

$$3 : 6 : 8$$

d) If $a : b = 5 : 3$ and $b : c = 4 : 7$

$$a : b = 5 : 3 \quad b : c = 4 : 7$$

To find the continued ratio we write these values as follows:

$$\begin{array}{ccccc} a & : & b & : & c \\ 5 & : & 3 & : & \\ & \searrow & \downarrow & \searrow & \\ & & 4 & : & 7 \\ 5 \times 4 & : & 3 \times 4 & : & 3 \times 7 \\ 20 & : & 12 & : & 21 \end{array}$$

So the continued ratio for the given values is

$$20 : 12 : 21$$

e) If $a : b = 5 : 6$ and $b : c = 3 : 2$

$$a : b = 5 : 6 \quad b : c = 3 : 2$$

To find the continued ratio we write these values as follows:

$$\begin{array}{ccc} a & : & b & : & c \\ 5 & : & 6 & : & \\ & \searrow & \downarrow & \searrow & \\ & & 3 & : & 2 \\ 5 \times 3 & : & 6 \times 3 & : & 6 \times 2 \\ 15 & : & 18 & : & 12 \end{array}$$

Dividing by the HCF of these numbers to simplify.

$$\frac{15}{3} : \frac{18}{3} : \frac{12}{3}$$

$$5 : 6 : 4$$

So the continued ratio for the given values is

$$5 : 6 : 4$$

- 4. The price of a book increased from Rs 560 to Rs 610. Find the ratio (in simplified form) of the increased price to the original price.**

Solution:

Original price = Rs. 560

increased price = Rs. 610

Ratio between prices:

$$\begin{array}{ccc} \text{Increased price} & : & \text{Original price} \\ 610 & : & 560 \end{array}$$

Dividing by their HCF to simplify this ratio.

$$\frac{610}{10} : \frac{560}{10}$$

$$61 : 56$$

So the ratio of the increased price to original price is 61 : 56.

- 5. Huma's weight decreased from 50 kg to 44 kg. Find the simplified ratio between her previous weight and her decreased weight.**

Solution:

Huma's previous weight = 50kg

Huma's decreased weight = 44 kg

Ratio between both weights:

$$\begin{array}{ccc} \text{Previous weight} & : & \text{decreased weight} \\ 50 & : & 44 \end{array}$$

Dividing by their HCF to simplify this ratio.

$$\frac{50}{2} : \frac{44}{2}$$

$$25 : 22$$

So, the ratio between Huma's previous weight and her decreased weight is 25:22.

6. The ratio of mango trees to apple trees is 4 : 5 and the ratio of apple trees to guava trees is 7 : 3. Find the continued ratio among them.

Solution:

Ratio of mango trees to apple trees = 4 : 5

Ratio of apple trees to guava trees = 7 : 3

Continued ratio

mango trees : apple trees : guava trees

$$\begin{array}{ccccc}
 4 & : & 5 & & \\
 & \searrow & \downarrow & \searrow & \\
 & & 7 & : & 3 \\
 4 \times 7 & : & 5 \times 7 & : & 5 \times 3 \\
 28 & : & 35 & : & 15
 \end{array}$$

So the continued ratio among mango trees, apple trees and guava trees is:

28 : 35 : 15

7. In a car showroom, the ratio of blue cars to red cars is 7 : 8 and the ratio of red cars to black cars is 3 : 5. Find the continued ratio among them.

Solution:

Ratio of blue cars to red cars = 7 : 8

Ratio of red cars to black cars = 3 : 5

Continued ratio

blue cars : red cars : black cars

$$\begin{array}{ccccc}
 7 & : & 8 & & \\
 & \searrow & \downarrow & \searrow & \\
 & & 3 & : & 5 \\
 7 \times 3 & : & 8 \times 3 & : & 8 \times 5 \\
 21 & : & 24 & : & 40
 \end{array}$$

So the continued ratio among blue cars, red cars and black cars is:

21 : 24 : 40

3.4 Let's Practice

1. Find the unit rate for each of the following.

a) Rs 120 for 3 packs of candies

Solution:

Write the given price and quantity as a ratio by removing units.

$$\frac{\text{Rs 120}}{3 \text{ packs}} = \frac{120}{3} = 120 : 3$$

To find unit rate convert second term of this ratio of 1 by dividing both terms by 3.

$$120 : 3 = \frac{120 \div 3}{3 \div 3} = \frac{40}{1}$$

The unit rate (rate per pack) of candies is Rs. 40 per pack.

b) 8 pages read in 40 minutes

Solution:

Write the given values as ratio by remaining units.

$$\frac{40 \text{ minutes}}{8 \text{ pages}} = \frac{40}{8} = 40 : 8$$

To find unit rate convert second term of this ratio to 1 by dividing both terms by 8.

$$40 : 8 = \frac{40 \div 8}{8 \div 8} = \frac{5}{1}$$

The unit rate is 5 minutes per page.

c) 4 kg bag of apples for Rs 920

Solution:

Write the given values as ratio by removing units.

$$\frac{\text{Rs 920}}{4 \text{ kg}} = \frac{920}{4} = 920 : 4$$

To find unit rate, convert the second term of this ratio to 1 by dividing both terms by 4.

$$920 : 4 = \frac{920 \div 4}{4 \div 4} = \frac{230}{1}$$

The unit rate is Rs 230 per kg bag of apples.

d) Rs 2560 for 8 books

Solution:

Write the given values as ratio by removing units.

$$\frac{\text{Rs 2560}}{8 \text{ books}} = \frac{2560}{8} = 2560 : 8$$

To find unit rate, convert second term of this ratio to 1, by dividing both terms by 8.

$$\frac{2560}{8} = \frac{2560 \div 8}{8 \div 8} = \frac{320}{1}$$

So, the unit rate is Rs 320 per book.

e) R1150 for 2.5 metres of fabric

Solution:

Write the given values as a ratio by removing units.

$$\frac{\text{Rs } 1150}{2.5 \text{ metres}} = \frac{1150}{2.5} = \frac{11500}{25} = 11500 : 25$$

To find unit rate convert second term of this ratio to 1 by dividing both terms by 25.

$$\frac{11500}{25} = \frac{11500 \div 25}{25 \div 25} = \frac{460}{1}$$

So, the unit rate is Rs 460 per meter of fabric.

- 2. The cost of a 21 metre long pipe is Rs 4305. What is the per metre rate for this pipe?**

Solution:

Length of pipe = 21 metre

Cost of 21 metre long pipe = Rs 4305

Per metre rate for the pipe = ?

Write the given price and quantity as a ratio by removing the units.

$$\frac{\text{Rs } 4305}{21 \text{ metre}} = \frac{4305}{21} = 4305 : 21$$

To find unit rate convert second term of this ratio to 1 by dividing both terms by 21.

$$\frac{4305}{21} = \frac{4305 \div 21}{21 \div 21} = \frac{204.8}{1} = \frac{205}{1} \text{ (rounding off)}$$

So, the unit rate is Rs 205 per metre of pipe.

- 3. The bus fare for 12 people is Rs 3792. What is the fare rate per person?**

Solution:

Number of people = 12

Bus fare of 12 people = Rs 3792

Fare rate per person = ?

Write the given fare and number of people as a ratio by removing the units.

$$\frac{\text{Rs } 3792}{12 \text{ people}} = \frac{3792}{12} = 3792 : 12$$

To find unit rate, convert the second term of this ratio to 1, by dividing both terms by 12.

$$\frac{3792}{12} = \frac{3792 \div 12}{12 \div 12} = \frac{316}{1}$$

So, the fare rate per person is Rs 316.

- 4. The cost of planting grass in a 2025 square metre playground is Rs 212625. What is the cost of planting grass per square metre?**

Solution:

Area of the ground = 2025 square metre

Cost of planting grass on it = Rs 212625

Cost of planting per square metre = ?

Write the given cost and area as a ratio by removing the units.

$$\frac{\text{Rs 212625}}{2025 \text{ square metre}} = \frac{212625}{2025}$$
$$= 212625 : 2025$$

To find the unit rate, convert the second term of this ratio to 1, by dividing both terms by 2025.

$$\frac{212625}{2025} = \frac{212625 \div 2025}{2025 \div 2025}$$
$$= \frac{105}{1}$$

So, the cost of planting grass per square metre is Rs 105.

- 5. A labourer earns Rs 22,550 for 8 hours of work in a construction company. What is his per hour rate?**

Solution:

Earnings of the labourer = Rs 22550

Duration of working = 8 hours

Per hour rate = ?

Write the given earnings and time duration as a ratio by removing the units.

$$\frac{\text{Rs 22550}}{8 \text{ hours}} = \frac{22550}{8}$$
$$= 22550 : 8$$

To find per hour rate, convert the second term of the fraction to 1, by dividing both terms by 8.

$$\frac{22550}{8} = \frac{22550 \div 8}{8 \div 8}$$
$$= \frac{2818.75}{1}$$

So, his per hour rate is Rs 2818.75.

Unit Evaluation

1. Choose the correct option.

- a) (iv) 71 % b) (ii) 15.3 c) (iv) 95.23 d) (iv) no unit
e) (iii) 288 f) (ii) Rs 2016.5

2. Find the percentage.

a) 78 out of 100

Solution:

$$78 \text{ out of } 100 = \frac{78}{100}$$

$$\begin{aligned} \text{In percentage} &= \frac{78}{100} \times 100 \\ &= 78\% \end{aligned}$$

c) 54 out of 80

Solution:

$$54 \text{ out of } 80 = \frac{54}{80}$$

$$\begin{aligned} \text{In percentage} &= \frac{54}{80} \times 100 \\ &= \frac{27}{40} \times 100 \\ &= \frac{27}{5} \times 2 = 67.5\% \end{aligned}$$

b) 21 out of 100

Solution:

$$21 \text{ out of } 100 = \frac{21}{100}$$

$$\begin{aligned} \text{In percentage} &= \frac{21}{100} \times 100 \\ &= 21\% \end{aligned}$$

d) 455 out of 600

Solution:

$$455 \text{ out of } 600 = \frac{455}{600}$$

$$\begin{aligned} \text{In percentage} &= \frac{455}{600} \times 100 \\ &= \frac{455}{6} = 75.83\% \end{aligned}$$

3. Calculate:

a) 54% of 45

Solution:

$$\begin{aligned} 54\% \text{ of } 45 &= \frac{54}{100} \times 45 \\ &= \frac{2430}{100} \\ &= 24.3 \end{aligned}$$

b) 100% of 90

Solution:

$$\begin{aligned} 100\% \text{ of } 90 &= \frac{100}{100} \times 90 \\ &= 90 \end{aligned}$$

c) 3% of 8

Solution:

$$\begin{aligned} 3\% \text{ of } 8 &= \frac{3}{100} \times 8 \\ &= \frac{24}{100} \\ &= 0.24 \end{aligned}$$

d) 4% of 2

Solution:

$$\begin{aligned} 4\% \text{ of } 2 &= \frac{4}{100} \times 2 \\ &= \frac{8}{100} \\ &= 0.08 \end{aligned}$$

4. A jar has 55 candies. 17 are chocolate candies. Find:

a) the percentage of chocolate candies

Solution:

Total number of candies in the jar = 55

Number of chocolate candies = 17

$$\begin{aligned}\text{Percentage of chocolate candies} &= \frac{\text{Number of chocolate candies}}{\text{Total number of candies}} \times 100 \\ &= \frac{17}{55} \times 100 \\ &= \frac{1700}{55} = 30.909\% \text{ or } 30.91\%\end{aligned}$$

So, the percentage of chocolate candies is 30.91%.

b) the percentage of candies other than chocolate candies

Solution:

Total number of candies in the jar = 55

Number of chocolate candies = 17

Number of candies other than chocolate = $55 - 17 = 38$

$$\begin{aligned}\text{Percentage of candies other than chocolate} &= \frac{\text{Number of candies other than chocolate}}{\text{Total number of candies}} \times 100 \\ &= \frac{38}{55} \times 100 \\ &= 69.09\%\end{aligned}$$

So, the percentage of candies other than chocolate candies is 69.09%.

5. A farm has 215 trees. 69% of trees are orange trees. How many other trees are there?

Solution:

Total number of trees = 215

Percentage of orange trees = 69%

Number of orange trees = 69% of 215

$$\begin{aligned}&= \frac{69}{100} \times 215 \\ &= 148.35 \\ &= 148\end{aligned}$$

$$\begin{aligned}\text{Number of other trees} &= \text{Total number trees} - \text{Number of orange trees} \\ &= 215 - 148 \\ &= 67\end{aligned}$$

So, there are 67 trees other than orange trees in the farm.

6. Calculate and compare using percentage.

a) 6 out of 8 and 3 out of 7

Solution:

$$\begin{aligned}\text{Percentage of 6 out of 8} &= \frac{6}{8} \times 100 \\ &= \frac{600}{8} = 75\%\end{aligned}$$

$$\begin{aligned}\text{Percentage of 3 out of 7} &= \frac{3}{7} \times 100 \\ &= \frac{300}{7} = 42.85\%\end{aligned}$$

So, $75\% > 42.85\%$ or 6 out of 8 is greater than 3 out of 7.

b) 20 out of 40 and 1 out of 2

Solution:

$$\begin{aligned}\text{Percentage of 20 out of 40} &= \frac{20}{40} \times 100 \\ &= \frac{2000}{40} = 50\%\end{aligned}$$

$$\begin{aligned}\text{Percentage of 1 out of 2} &= \frac{1}{2} \times 100 \\ &= \frac{100}{2} \\ &= 50\%\end{aligned}$$

as $50\% = 50\%$

So, 20 out of 40 is same as 1 out of 2.

c) 54 out of 77 and 52 out of 90

Solution:

$$\begin{aligned}\text{Percentage of 54 out of 77} &= \frac{54}{77} \times 100 \\ &= \frac{5400}{77} \\ &= 70.13\%\end{aligned}$$

$$\begin{aligned}\text{Percentage of 52 out of 90} &= \frac{52}{90} \times 100 \\ &= \frac{5200}{90} \\ &= 57.78\%\end{aligned}$$

as $70.13\% > 57.78\%$

So 54 out of 77 is greater than 52 out of 90.

7. Calculate:

a) 12 increased by 77%

Solution:

First we find 77% of 12

$$\begin{aligned} 77\% \text{ of } 12 &= \frac{77}{100} \times 12 \\ &= \frac{924}{100} = 9.24 \end{aligned}$$

So, 77% of 12 is 9.24.

Now, 12 increased by 77% = $12 + 9.24 = 21.24$

So, 12 increased by 77% is 21.24.

b) 23 increased by 89%

Solution:

First we find 89% of 23

$$\begin{aligned} 89\% \text{ of } 23 &= \frac{89}{100} \times 23 \\ &= \frac{2047}{100} = 20.47 \end{aligned}$$

So, 89% of 23 is 20.47.

Now, 23 increased by 89% = $23 + 20.47 = 43.47$

So, 23 increased by 89% is 43.47.

c) 1 increased by 50%

Solution:

First we find 50% of 1

$$\begin{aligned} 50\% \text{ of } 1 &= \frac{50}{100} \times 1 \\ &= 0.5 \end{aligned}$$

So, 50% of 1 is 0.5.

Now, 1 increased by 50% = $1 + 0.5 = 1.5$

So, 1 increased by 50% is 1.5.

d) 42 decreased by 42%

Solution:

First we find 42% of 42

$$\begin{aligned} 42\% \text{ of } 42 &= \frac{42}{100} \times 42 \\ &= \frac{1764}{100} = 17.64 \end{aligned}$$

So, 42% of 42 is 17.64.

Now, 42 decreased by 42% = $42 - 17.64 = 24.36$

So, 42 decreased by 42% is 24.36.

- 8. A farm has 345 animals. 98 are horses and 156 are buffaloes. Compare the number of horses with the number of buffaloes using percentage.**

Solution:

Total number of animals = 345

Number of horses = 98

$$\begin{aligned}\text{Percentage of horses on the farm} &= \frac{\text{Number of horses}}{\text{Total number of animals}} \times 100 \\ &= \frac{98}{345} \times 100 \\ &= \frac{9800}{345} = 28.405\% \text{ or } 28.41\%\end{aligned}$$

So, the percentage of horses on the farm is 28.41%

Number of buffaloes = 156

$$\begin{aligned}\text{Percentage of buffaloes on the farm} &= \frac{\text{Number of buffaloes}}{\text{Total number of animals}} \times 100 \\ &= \frac{156}{345} \times 100 \\ &= \frac{15600}{345} = 45.217\% \text{ or } 45.22\%\end{aligned}$$

So, the percentage of buffaloes on the farm is 45.22% as $28.41\% < 45.22\%$

So, the number of horses on the farm is less than the number of buffaloes.

- 9. Nida had Rs 15430. She bought a microwave oven for Rs 4350 and a blender for Rs 6435. Compare the price of both items using percentage.**

Solution:

Amount of money Nida had = Rs. 15430

Price of oven Nida bought = Rs. 4350

Price of blender Nida bought = Rs. 6435

$$\begin{aligned}\text{Percentage price of oven} &= \frac{\text{Price of oven}}{\text{Nida had amount of money}} \times 100 \\ &= \frac{4350}{15430} \times 100 \\ &= \frac{435000}{15430} = 28.19\%\end{aligned}$$

$$\begin{aligned}\text{Percentage price of blender} &= \frac{\text{Price of blender}}{\text{Nida had amount of money}} \times 100 \\ &= \frac{6435}{15430} \times 100 \\ &= \frac{643500}{15430} = 41.70\%\end{aligned}$$

So, the price of microwave oven is less than the price of blender.

10. The price of a car is Rs 1,256,755. Find the new price if:

Solution:

(a) Original price of car = Rs. 1,256,755

Percentage increase in price of car = 12%

Increase in the price of car = 12% of Rs. 1,256,755

$$= \frac{12}{100} \times 1,256,755$$

$$= \frac{15081060}{345} = \text{Rs. } 150810.6$$

Increased price = original price + increase in price

$$= \text{Rs. } 1,256,755 + \text{Rs. } 150,810.6$$

$$= \text{Rs. } 1,407,565.6$$

So, the price of car increased by 12% is Rs. 1,407,565.6

(b) Original price of car = Rs. 1,256,755

Percentage decrease in price of car = 8%

Decrease in the price of car = 8% of Rs. 1,256,755

$$= \frac{8}{100} \times 1,256,755$$

$$= \frac{10054040}{100} = \text{Rs. } 100,540.4$$

Decreased price = original price – decreased in price

$$= \text{Rs. } 1,256,755 - \text{Rs. } 100,540.4$$

$$= \text{Rs. } 1,156,214.6$$

So, the price of the car decreased by 8% is Rs. 1,156,214.6

11. Find the ratios.

(a) 45 to 60

Solution:

45 to 60

45 : 60

Dividing the values by their HCF
to simplify the ratio.

$$\frac{45}{15} : \frac{60}{15}$$
$$3 : 4$$

(c) Rs. 1245 to Rs. 5600

Solution:

Rs. 1245 to Rs. 5600

1245 : 5600

Dividing the values by their HCF to simplify the ratio.

$$\frac{1245}{5} : \frac{5600}{5}$$
$$249 : 1120$$

(b) 120 m to 60 m

Solution:

120 m + 60 m

120 : 60

Dividing the values by their HCF
to simplify the ratio.

$$\frac{120}{60} : \frac{60}{60}$$
$$2 : 1$$

12. Simplify the following ratios.

(a) 76 : 44

Solution:

76 : 44

Dividing these values by their HCF to simplify.

$$\frac{76}{4} : \frac{44}{4}$$

$$19 : 11$$

(c) 88 : 122

Solution:

88 : 122

Dividing these values by their HCF to simplify.

$$\frac{88}{2} : \frac{122}{2}$$

$$44 : 61$$

(b) 40 : 2

Solution:

40 : 2

Dividing these values by their HCF to simplify.

$$\frac{40}{2} : \frac{2}{2}$$

$$20 : 1$$

13. Find the continued ratio a : b : c for the following values.

(a) If a : b = 4 : 9 and b : c = 5 : 7

Solution:

a : b = 4 : 9 , b : c = 5 : 7

$$\begin{array}{ccccc} a & : & b & : & c \\ 4 & : & 9 & : & \\ & \searrow & \downarrow & \searrow & \\ & 5 & : & 7 & \end{array}$$

$$4 \times 5 : 9 \times 5 : 9 \times 7$$

$$20 : 45 : 63$$

(b) If a : b = 3 : 1 and b : c = 8 : 9

Solution:

a : b = 3 : 1 , b : c = 8 : 9

$$\begin{array}{ccccc} a & : & b & : & c \\ 3 & : & 1 & : & \\ & \searrow & \downarrow & \searrow & \\ & 8 & : & 9 & \end{array}$$

$$3 \times 8 : 1 \times 8 : 1 \times 9$$

$$24 : 8 : 9$$

(c) If a : b = 2 : 5 and b : c = 6 : 7

Solution:

a : b = 2 : 5 , b : c = 6 : 7

$$\begin{array}{ccccc} a & : & b & : & c \\ 2 & : & 5 & : & \\ & \searrow & \downarrow & \searrow & \\ & 6 & : & 7 & \end{array}$$

$$2 \times 6 : 5 \times 6 : 5 \times 7$$

$$12 : 30 : 35$$

14. The ratio between the heights of Ali and Sara is 3 : 1 and the ratio between the heights of Sara and Saad is 4 : 5. Find the continued ratio among their heights.

Solution:

Ratio between the heights of Ali and Sara = 3 : 1

Ratio between the heights of Sara and Saad = 4 : 5

Ali's height : Sara's height : Saad's height

$$\begin{array}{ccc} 3 & : & 1 \\ & \searrow & \downarrow \quad \nearrow \\ & 4 & : & 5 \end{array}$$

$$3 \times 4 : 1 \times 4 : 1 \times 5$$

$$12 : 4 : 5$$

So, the continued ratio among the heights of Ali, Sara and Saad is:

Ali : Sara : Saad

$$12 : 4 : 5$$

15. Find the unit rate for each of the following.

(a) Rs. 655 for 5 packs of erasers.

Solution:

Write the given values as a ratio by removing units.

$$\frac{\text{Rs. 655}}{5 \text{ packs}} = \frac{655}{5} = 655 : 5$$

To find unit rate, convert second term of this ratio to 1 by dividing both terms by 5.

$$655 : 5 = \frac{655 \div 5}{5 \div 5} = \frac{131}{1}$$

So, the unit rate is Rs. 131 per pack of erasers.

(b) 4290 km covered in 4 hours.

Solution:

Write the given values as a ratio by removing units.

$$\frac{420 \text{ km}}{4 \text{ hour}} = \frac{420}{5} = 420 : 4$$

To find unit rate, convert second term of this ratio to 1 by dividing both terms by 4.

$$420 : 4 = \frac{420 \div 4}{4 \div 4} = \frac{105}{1}$$

So, the unit rate is Rs. 105 km per hour.

(c) 7 litres oil for Rs. 3850.

Solution:

Write the given values as a ratio by removing units.

$$\frac{\text{Rs. 3850}}{7 \text{ litres}} = \frac{3850}{7} = 3850 : 7$$

To find unit rate, convert second term of this ratio to 1 by dividing both terms by 7.

$$3850 : 7 = \frac{3850 \div 7}{7 \div 7} = \frac{550}{1}$$

So, the unit rate is Rs. 550 per litre oil.

16. The cost of 200 kg of cement is Rs 310,000. What is the per kg rate for the cement?

Solution:

Quantity of cement = 200 kg

Price of 200 kg cement = Rs. 4440

Per kg rate of cement = ?

Write the given cost and quantity as a ratio by removing units.

$$\frac{\text{Rs. 4440}}{200 \text{ kg}} = \frac{4440}{200} = 4440 : 200$$

To find unit rate, convert second term of this ratio to 1 by dividing both terms by 200.

$$4440 : 200 = \frac{4440 \div 200}{200 \div 200} = \frac{22.2}{1}$$

So, the unit rate for cement is Rs. 22.2 per kg of cement.

4.1 Let's Practice**1. Identify if the collection of given objects is a set.**

a) Collection of the first ten prime numbers

Answer:

It is a set because it has distinct elements and no element is repeated.

b) Collection of funny students in grade 6

Answer:

It is not a set because the elements of this set are not well defined as "funny" means different for different people.

c) Collection of the multiples of 2

Answer:

It is a set because it is well defined it has distinct elements and no elements is repeated.

d) $\{2, 3, 5, 8, 2\}$

Answer:

This collection of numbers has distinct elements, they are well defined but the elements 2 is repeated so it is not a set.

e) Collection of even numbers

Answer:

it is a set because it is well defined, it has distinct elements and no element is repeated.

f) Collection of all students in a class

Answer:

it is a set because it is well defined, the members of this set are distinct and also not repeated.

g) Collection of stationery items

Answer:

it is a set because it has distinct elements, they are well defined and no element is repeated.

h) $\{\text{parrot, pigeon, sparrow, crow}\}$

it is a set because it has distinct elements, they are well defined and no element is repeated.

i) $\{a, e, i, o, u\}$

Answer:

it is a set because it has well defined elements, the elements are distinct and no element is repeated.

j) {P, A, K, I, S, T, A, N}

Answer: This is not a set although it has distinct and well defined elements but the element A is repeated, therefore it is not a set

2. Write the number of elements in each set.

a) Set of odd numbers less than 20

Solution:

Let's represent the set of odd numbers by O, so set of odd numbers less than 20 is,

$O = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$

so it has 10 elements

b) Set of provinces of Pakistan

Solution:

Let's represent the set of provinces of Pakistan by P so,

$P = \{\text{Sindh, Balochistan, KPK, Punjab}\}$

So, it has 4 elements.

c) Set of alphabets between K and T

Solution:

Let's represent the set of alphabets between K and T by A so

$A = \{L, M, N, O, P, Q, R, S\}$

so it has 8 elements.

d) $\{3, 6, 9, \dots, 27\}$

Solution:

This set represents the multiples of 3 up to 27. The elements of this set are

$3, 6, 9, 12, 15, 18, 21, 24, 27$.

so the number of elements of this set is 9.

e) $\{\text{June, July, August}\}$

Solution:

the number of elements of this set is 3.

f) Set of prime numbers between 7 and 30

Solution:

Let's represent this set by letter P. The set P has the following elements.

$P = \{11, 13, 17, 19, 23, 29\}$

3. Write true or false for the given statements.

a) $8 \in$ the set of even numbers $\rightarrow$ True, as 8 is a member of set of even numbers.

b) $4 \in \{2, 4, 6, 8\} \rightarrow$ True

c) $9 \in$ the set of prime numbers $\rightarrow$ False, as 9 is not a prime number.

d) $24 \in$ the set of multiples of 8 $\rightarrow$ True, as 24 is the third multiple of 8

e) Sunday $\in$ {months of a year} $\rightarrow$ False, as Sunday is not the name of any month.

f) $7 \in$ the set of factors of 56 $\rightarrow$ True, as 7 is included in the factors of 56.

4.2 Let's Practice

1. Identify and separate out the finite sets and the infinite sets.

a) Set of whole numbers less than 15.

Answer:

Finite set. The set of whole numbers less than 15 is $W = \{1, 3, 5\}$. The members of this set are countable. So, it is a finite set.

b) Set of all even numbers.

Answer:

Infinite set. The set of all even numbers is $E = \{2, 4, 6, \dots\}$. The members of this set are uncountable and unlimited. So, it is an infinite set.

c) Set of factors of 25.

Answer:

Finite set. The set of factors of 25 is, $A = \{5\}$. The members of this set are countable and limited. So, it is a finite set.

d) $\{4, 8, 12, 16, 20, \dots\}$

Answer:

Infinite set. This is a set of multiples of 4. The members of this set are unlimited and uncountable. So, it is an infinite set.

e) set of multiples of 4.

Answer:

Infinite set. The set of multiples of 4 has unlimited and uncountable members. So, it is an infinite set.

f) set of factors of 4.

Answer:

Finite set. The set of factors of 4 is, $B = \{1, 2, 4\}$. It has limited and countable members. So, it is a finite set.

g) Set of all positive integers.

Answer:

Infinite set. The set of all positive integers has unlimited and uncountable members. So, it is an infinite set.

h) set of first 15 multiples of 10.

Answer:

Finite set. The set of first 15 multiples of 10 is $A = \{15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180, 195, 210, 225\}$. It has limited and countable members. So it is a finite set.

2. Which of these are empty sets?

a) The set of the days of the week that start with the letter N.

Answer:

This is an empty set because no day of the week starts with letter N.

b) The set of odd numbers less than 20 but greater than 19.

Answer:

This is an empty set because there is no odd number less than 20 or greater than 19.

c) The set of letters in the word Pakistan.

Answer:

This is not an empty set. The set of letters in the word Pakistan is, $L = \{P, a, k, i, s, t, n\}$

d) The set of odd numbers divisible by 10.

Answer:

This is an empty set, because there is no odd number which is divisible by 10.

3. Which of these are singleton set?

a) The set of neutral integers.

Answer: This is a singleton set. The only neutral integer in the set of integers is 0. The set of neutral integer is $A = \{0\}$.

b) The set of all students of grade 6.

Answer:

It is not a singleton set. There must be more than one student in grade 6.

c) The set of all even prime numbers.

Answer:

This is a singleton set. The only even prime number is 2. The set of all even prime numbers is, $L = \{2\}$

d) The set of factors of 1.

Answer:

This is a singleton set. The only factor of 1 is 1 itself. The set of factors of 1 is, $F = \{1\}$.

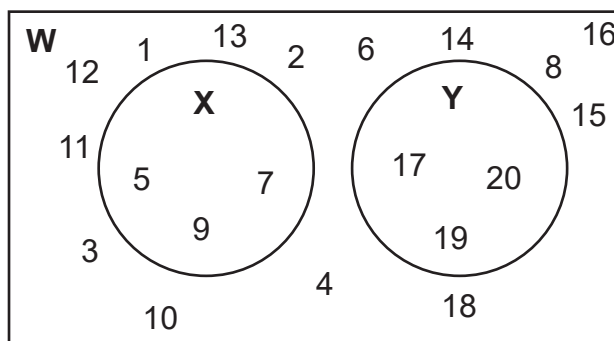
4. Identify the universal set from the given sets and then show it by using a Venn diagram.

a) $W = \{1, 2, 3, \dots, 20\}$ $X = \{5, 7, 9\}$ $Y = \{17, 19, 20\}$

Solution:

Here the universal set is W as it contains all the members included in the other two sets. Using venn diagram we can show this as follows.

Universal set = W

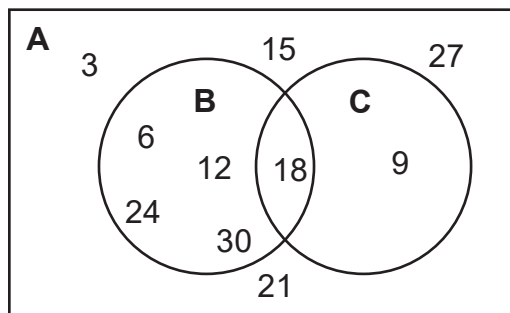


$$b) A = \{3, 6, 9, 12, \dots, 30\} \quad B = \{6, 12, \dots, 30\} \quad C = \{9, 18\}$$

Solution:

Here, the set A is the universal set as it contains all the members included in the other two sets. We can show this using a venn diagram as follows.

Universal set = A



Unit Evaluation

1. Choose the correct option.

- a) (ii) set b) (ii) $\{2, 1, 1\}$ c) (iii) Set of the first common multiple of 3 and 6
d) (ii) Set of the first 6 multiples of 2
e) (ii) Set of the multiples of 6 and (iv) Set of the first five alphabets of English.

2. List five singleton sets

Answer:

- 1) The set of even prime numbers, $A = \{2\}$
- 2) The set of odd numbers between 1 and 4, $B = \{3\}$
- 3) The set of neutral integers, $N = \{0\}$
- 4) The set of factors of 1. $B = \{1\}$
- 5) The set of days of the week that start with the letter F, $W = \{\text{Friday}\}$

3. Separate out empty, finite, infinite and singleton sets from the following.

- a) Set of the days of the week.

Answer:

This is a finite set. The days of the week are countable and limited.

$W = \{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$

- b) Set of integers.

Answer:

This is an infinite set. The set of integers is uncountable and unlimited.

$Z = \{0, \pm 1, \pm 2, \pm 3, \dots\}$

- c) Set of stationery items

Answer:

This is a finite set. The set of stationery items is countable and limited.

- d) Set of common factors of 3 and 4.

Answer:

This is a singleton set. 1 is the only number which is the common factor of 3 and 4, or which divides both 3 and 4.

e) Set of months of year that have 32 days.

Answer:

This is an empty set. No month of year has 32 days.

f) Set of all animals.

Answer:

This is an infinite set. There is unlimited number of animals in the whole world.

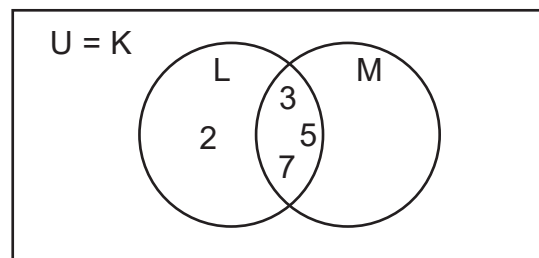
4. Identify and separate out the universal set from the given sets and then show the sets through a Venn diagram.

a) $K = \{2, 3, 5, 7, 11, \dots, 41\}$ $L = \{2, 3, 5, 7\}$ $M = \{3, 5, 7\}$

Solution:

Here K is the universal set as it has all the elements included in the other two sets.

We can show it using venn diagram as follows:



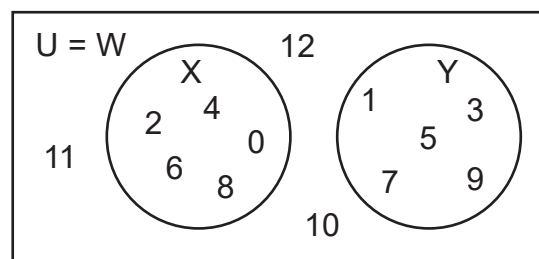
b) $W =$ Set of whole numbers up to 12

$X =$ Set of first 5 even numbers

$Y =$ Set of first 5 odd numbers

Solution:

Here W is the universal set. We can show it using venn diagram as follows:



5.1 : Let's Practice

1. By using the term-to-term rule, find the next three terms of the given sequences.

- a) 2, 8, 14, 20, _____, _____, _____
b) 45, 41, 37, _____, _____, _____
c) 3200, 1600, 800, _____, _____, _____
d) 6, 12, 24, 48, _____, _____, _____
e) 5, 7, 9, 11, _____, _____, _____

Solution:

- a) Observe the sequence. The term-to-term rule helps us find the next terms in the sequence:

2, 8, 14, 20, _____, _____, _____

First, find the difference between the two terms.

$$8 - 2 = 6$$

The difference between the two terms is 6, so the next terms of the sequence can be obtained by adding 6 to each previous term.

The next three terms of the sequence are:

$$20 + 6 = 26, 26 + 6 = 32 \text{ and } 32 + 6 = 38$$

The sequence is:

2, 8, 14, 20, 26, 32, 38

- b) Observe the sequence. The term-to-term rule helps us find the next terms in the sequence:

45, 41, 37, _____, _____, _____

First, find the difference between the two terms.

$$41 - 45 = -4$$

The difference between the two terms is -4 , so the next terms of the sequence can be obtained by subtracting 4 to each previous term.

The next three terms of the sequence are:

$$37 - 4 = 33$$

$$33 - 4 = 29$$

$$29 - 4 = 25$$

The sequence is:

45, 41, 37, 33, 29, 25

- c) Observe the sequence. The term-to-term rule helps us find the next terms in the sequence:

3200, 1600, 800, _____, _____, _____

First, find the ratio between the two terms.

$$1600 : 3200 = \frac{1600}{3200} = \frac{1}{2}$$

The ratio between the two terms is $\frac{1}{2}$, so the next terms of the sequence can be obtained by dividing 2 to each previous term.

The next three terms of the sequence are:

$$\frac{800}{2} = 400, \frac{400}{2} = 200 \text{ and } \frac{200}{2} = 100$$

The sequence is:

3200, 1600, 800, 400, 200, 100

- d) Observe the sequence. The term-to-term rule helps us find the next terms in the sequence:

6, 12, 24, 48, _____, _____, _____

First, find the ratio between the two terms.

$$12 : 6 = \frac{12}{6} = 2$$

The ratio between the two terms is 2, so the next terms of the sequence can be obtained by multiplying to each previous term.

The next three terms of the sequence are:

$$48 \times 2 = 96, 96 \times 2 = 192 \text{ and } 192 \times 2 = 384$$

The sequence is:

6, 12, 24, 48, 96, 192, 384

- e) Observe the sequence. The term-to-term rule helps us find the next terms in the sequence:

5, 7, 9, 11, _____, _____, _____

First, find the difference between the two terms.

$$7 - 5 = 2$$

The difference between the two terms is 2, so the next terms of the sequence can be obtained by adding 2 to each previous term.

The next three terms of the sequence are:

$$11 + 2 = 13, 13 + 2 = 15 \text{ and } 15 + 2 = 17$$

The sequence is:

5, 7, 9, 11, 13, 15, 17

2. By using the position-to-term rule, find the 12th, 16th and 21th numbers in the following.

a) 3, 4, 5, 6, 7, ...

b) 8, 16, 24, 32, 40, ...

c) 15, 16, 17, 18, 19, ...

d) 14, 28, 42, 56, ...

e) 10, 20, 30, 40, 50, ...

Solution:

a) 3, 4, 5, 6, 7, ...

In the sequence,

3, 4, 5, 6, 7, ...

If we look at the pattern closely, we can see that the terms are obtained by adding 2 to each position of the terms.

Since, 3 is the first number in the sequence, then:

$$3 = 1 + 2$$

Similarly,

$$4 = 2 + 2$$

$$5 = 3 + 2$$

$$6 = 4 + 2$$

So, we can find the 12th, 16th and 21th term by adding 2 to it.

$$12^{\text{th}} \text{ term} = 12 + 2 = 14$$

$$16^{\text{th}} \text{ term} = 16 + 2 = 18$$

$$21^{\text{th}} \text{ term} = 21 + 2 = 23$$

b) 8, 16, 24, 32, 40, ...

In the sequence,

8, 16, 24, 32, 40, ...

If we look at the pattern closely, we can see that the terms are obtained by multiplying 8 to each position of the terms.

Since, 8 is the first number in the sequence, then:

$$8 = 1 \times 8$$

Similarly,

$$16 = 2 \times 8$$

$$24 = 3 \times 8$$

$$32 = 4 \times 8$$

So, we can find the 12th, 16th and 21th term by multiplying 8 to it.

$$12^{\text{th}} \text{ term} = 12 \times 8 = 96$$

$$16^{\text{th}} \text{ term} = 16 \times 8 = 128$$

$$21^{\text{th}} \text{ term} = 21 \times 8 = 168$$

c) 15, 16, 17, 18, 19, ...

In the sequence,

15, 16, 17, 18, 19, ...

If we look at the pattern closely, we can see that the terms are obtained by adding 14 to each position of the terms.

Since, 15 is the first number in the sequence, then:

$$15 = 1 + 14$$

Similarly,

$$16 = 2 + 14$$

$$17 = 3 + 14$$

$$18 = 4 + 14$$

So, we can find the 12th, 16th and 21th term by adding 14 to it.

$$12^{\text{th}} \text{ term} = 12 + 14 = 26$$

$$16^{\text{th}} \text{ term} = 16 + 14 = 30$$

$$21^{\text{th}} \text{ term} = 21 + 14 = 35$$

d) 14, 28, 42, 56, ...

In the sequence,

14, 28, 42, 56, ...

If we look at the pattern closely, we can see that the terms are obtained by multiplying 14 to each position of the terms.

Since, 14 is the first number in the sequence, then:

$$14 = 1 \times 14$$

Similarly,

$$28 = 2 \times 14$$

$$42 = 3 \times 14$$

$$56 = 4 \times 14$$

So, we can find the 12th, 16th and 21th term by multiplying 14 to it.

$$12^{\text{th}} \text{ term} = 12 \times 14 = 168$$

$$16^{\text{th}} \text{ term} = 16 \times 14 = 224$$

$$21^{\text{th}} \text{ term} = 21 \times 14 = 294$$

e) 10, 20, 30, 40, 50, ...

In the sequence,

10, 20, 30, 40, 50, ...

If we look at the pattern closely, we can see that the terms are obtained by multiplying 10 to each position of the terms.

Since, 10 is the first number in the sequence, then:

$$10 = 1 \times 10$$

Similarly,

$$20 = 2 \times 10$$

$$30 = 3 \times 10$$

$$40 = 4 \times 10$$

$$50 = 5 \times 10$$

So, we can find the 12th, 16th and 21th term by multiplying 10 to it.

$$12^{\text{th}} \text{ term} = 12 \times 10 = 120$$

$$16^{\text{th}} \text{ term} = 16 \times 10 = 160$$

$$21^{\text{th}} \text{ term} = 21 \times 10 = 210$$

3. **Qasim started walking in the park. On the first day, he took 2 rounds of the walking track; on the second day, he took 3 rounds; and on the third day, he took 4 rounds around the track. How many rounds will he takes on the 7th and 15th days?**

Solution:

$$1^{\text{st}} \text{ day round} = 2$$

$$2^{\text{nd}} \text{ day round} = 3$$

$$3^{\text{rd}} \text{ day round} = 4$$

If we look at the pattern closely, we can see that the terms are obtained by adding 1 to each position of the terms.

Since, 2 is the first number in the sequence, then:

$$2 = 1 + 1$$

Similarly,

$$3 = 2 + 1$$

$$4 = 3 + 1$$

So, we can find the 7th and 15th term by adding 1 to it.

$$7^{\text{th}} \text{ term} = 7 + 1 = 8$$

$$15^{\text{th}} \text{ term} = 15 + 1 = 16$$

Therefore, Qasim will take 8 rounds on the 7th day and 16 rounds on the 15th day.

4. **Hira went to embroidery school to learn. On the first day, she made 2 flowers on a piece of cloth; on the second day, she made 4 flowers; and on the third day, she made 6 flowers. How many flowers will she make during the next three days?**

Solution:

Here, sequence is:

2, 4, 6, ...

$$\text{Difference} = 4 - 2 = 2$$

The difference between the two terms is 2, so the next terms of the sequence can be obtained by adding to each previous term.

The next three terms of the sequence are:

$$6 + 2 = 8, 8 + 2 = 10 \text{ and } 10 + 2 = 12$$

The sequence is:

2, 4, 6, 8, 10, 12

5.2 : Let's Practice

1. **Separate the variable and constant terms in the following algebraic expressions.**

a) $2a + 3b - 5$

b) $a^3 + a^2 - a + 17$

c) $5m - 3n - 7$

d) $6p + 5q + 25$

e) $12r - 4s + 3$

f) $4x + 17y - 1$

Solution:

a) $2a + 3b - 5$

Variable terms: $2a, 3b$

Constant term: -5

b) $a^3 + a^2 - a + 17$

Variable terms: $a^3, a^2, -a$

Constant term: 17

- c) $5m - 3n - 7$
Variable terms: $5m, -3n$
Constant term: -7
- d) $6p + 5q + 25$
Variable terms: $6p, 5q$
Constant term: 25
- e) $12r - 4s + 3$
Variable terms: $12r, -4s$
Constant term: 3
- f) $4x + 17y - 1$
Variable terms: $4x, 17y$
Constant term: -1

2. Write down the coefficients of the variable terms in the following algebraic expressions.

- a) $10a - 25b$ b) $75x - 33y$ c) $16p + 13q + 24$
- d) $5a^4 - 4a^3 + 3a^2 - 2a + 20$ e) $15x^3 + 4x^2 - 60$
- f) $4a^2b + 3ab^2 - 12ab - 15$ g) $7x^3y + 4x^2y^2 - 5xy^3$

Solution:

- a) $10a - 25b$
Coefficient of the variable term with $a = 10$
Coefficient of the variable term with $b = -25$
- b) $75x - 33y$
Coefficient of the variable term with $x = 75$
Coefficient of the variable term with $y = -33$
- c) $16p + 13q + 24$
Coefficient of the variable term with $p = 16$
Coefficient of the variable term with $q = 13$
- d) $5a^4 - 4a^3 + 3a^2 - 2a + 20$
Coefficient of the variable term with $a^4 = 5$
Coefficient of the variable term with $a^3 = -4$
Coefficient of the variable term with $a^2 = 3$
Coefficient of the variable term with $a = -2$
- e) $15x^3 + 4x^2 - 60$
Coefficient of the variable term with $x^3 = 15$
Coefficient of the variable term with $x^2 = 4$
- f) $4a^2b + 3ab^2 - 12ab - 15$
Coefficient of the variable term with $a^2b = 4$
Coefficient of the variable term with $ab^2 = 3$
Coefficient of the variable term with $ab = -12$
- g) $7x^3y + 4x^2y^2 - 5xy^3$
Coefficient of the variable term with $x^3y = 7$
Coefficient of the variable term with $x^2y^2 = 4$
Coefficient of the variable term with $xy^3 = -5$

3. Write down the exponents of the variables in the following:

- a) $7a^5$ b) $-3b^2$ c) $-25p^9$ d) $30g^{12}$ e) x^4 f) $-6y^3$

Solution:

a) $7a^5$

Exponent of the variable $a = 5$

b) $-3b^2$

Exponent of the variable $b = 2$

c) $-25p^9$

Exponent of the variable $p = 9$

d) $30g^{12}$

Exponent of the variable $g = 12$

e) x^4

Exponent of the variable $x = 4$

f) $-6y^3$

Exponent of the variable $y = 3$ **4. Evaluate the following algebraic expressions if $p = 4$, $q = 5$ and $r = -6$.**

a) $2p + 3q - 4r$

b) $3p^2 + 4q^2 + 10r$

c) $p^3 + r^3$

d) $p^2 + q^2 - r^2$

e) $pq + qr + pqr$

f) $7p^2 - 4pq + 5r^2$

Solution:

a) $2p + 3q - 4r$

Substituting $p = 4$, $q = 5$, and $r = -6$

$$\begin{aligned} 2p + 3q - 4r &= 2(4) + 3(5) - 4(-6) \\ &= 8 + 15 + 24 \\ &= 47 \end{aligned}$$

b) $3p^2 + 4q^2 + 10r$

Substituting $p = 4$, $q = 5$, and $r = -6$

$$\begin{aligned} 3p^2 + 4q^2 + 10r &= 3(4)^2 + 4(5)^2 + 10(-6) \\ &= 3(16) + 4(25) - 60 \\ &= 48 + 100 - 60 \\ &= 88 \end{aligned}$$

c) $p^3 + r^3$

Substituting $p = 4$ and $r = -6$

$$\begin{aligned} p^3 + r^3 &= 4^3 + (-6)^3 \\ &= 64 + (-216) \\ &= 64 - 216 \\ &= -152 \end{aligned}$$

d) $p^2 + q^2 - r^2$

Substituting $p = 4$, $q = 5$, and $r = -6$

$$\begin{aligned} p^2 + q^2 - r^2 &= 4^2 + 5^2 - (-6)^2 \\ &= 16 + 25 - 36 \\ &= 5 \end{aligned}$$

e) $pq + qr + pqr$

Substituting $p = 4$, $q = 5$, and $r = -6$

$$\begin{aligned} pq + qr + pqr &= 4(5) + 5(-6) + 4(5)(-6) \\ &= 20 - 30 - 120 \\ &= -130 \end{aligned}$$

f) $7p^2 - 4pq + 5r^2$

Substituting $p = 4$, $q = 5$, and $r = -6$

$$\begin{aligned} 7p^2 - 4pq + 5r^2 &= 7(4)^2 - 4(4)(5) + 5(-6)^2 \\ &= 7(16) - 4(20) + 5(36) \\ &= 112 - 80 + 180 \\ &= 212 \end{aligned}$$

5. If the values of $x = 4$, $y = -3$ and $z = 9$, find the values of the given expressions.

a) $\frac{x}{y}$

b) $\frac{y}{x}$

c) $\frac{x^3}{y}$

d) $\frac{y^2}{x}$

e) $\frac{x+y}{z}$

f) $\frac{x^2 - y^2}{z^2}$

g) $\frac{x^2 + xy + y^2}{2x - 3y + 2z}$

h) $\frac{7x^2 + 5xy + z^2}{(x-y)(x+y)}$

Solution:

a) $\frac{x}{y}$

Substituting $x = 4$ and $y = -3$

$$\frac{x}{y} = \frac{4}{-3} = -\frac{4}{3}$$

b) $\frac{y}{x}$

Substituting $x = 4$ and $y = -3$

$$\frac{y}{x} = \frac{-3}{4} = -\frac{3}{4}$$

c) $\frac{x^3}{y}$

Substituting $x = 4$ and $y = -3$

$$\begin{aligned}\frac{x^3}{y} &= \frac{4^3}{-3} \\ &= \frac{64}{-3} \\ &= -\frac{64}{3}\end{aligned}$$

d) $\frac{y^2}{x}$

Substituting $x = 4$ and $y = -3$

$$\frac{y^2}{x} = \frac{(-3)^2}{4} = \frac{9}{4}$$

e) $\frac{x+y}{z}$

Substituting $x = 4$, $y = -3$, and $z = 9$

$$\begin{aligned}\frac{x+y}{z} &= \frac{4 + (-3)}{9} \\ &= \frac{4 - 3}{9} \\ &= \frac{1}{9}\end{aligned}$$

f) $\frac{x^2 - y^2}{z^2}$

Substituting $x = 4$, $y = -3$, and $z = 9$

$$\begin{aligned}\frac{x^2 - y^2}{z^2} &= \frac{4^2 - (-3)^2}{9^2} \\ &= \frac{(16 - 9)}{81} \\ &= \frac{7}{81}\end{aligned}$$

g) $\frac{x^2 + xy + y^2}{2x - 3y + 2z}$

Substituting $x = 4$, $y = -3$, and $z = 9$

$$\begin{aligned}\frac{x^2 + xy + y^2}{2x - 3y + 2z} &= \frac{4^2 + 4(-3) + (-3)^2}{2(4) - 3(-3) + 2(9)} \\ &= \frac{16 - 12 + 9}{8 + 9 + 18} \\ &= \frac{13}{35}\end{aligned}$$

$$\begin{aligned}
 \text{h) } & \frac{7x^2 + 5xy + z^2}{(x-y)(x+y)} \\
 & \text{Substituting } x = 4, y = -3, \text{ and } z = 9 \\
 & \frac{7x^2 + 5xy + z^2}{(x-y)(x+y)} = \frac{7(4)^2 + 5(4)(-3) + 9^2}{(4 - (-3))(4 + (-3))} \\
 & = \frac{7(16) + 5(4)(-3) + (9)^2}{(4+3)(4-3)} \\
 & = \frac{112 - 60 + 81}{(7)(1)} \\
 & = \frac{133}{7} \\
 & = 19
 \end{aligned}$$

5.3 : Let's Practice

1. Find the sum of the following expressions.

- $2a + 3b + 4c, 5a + 4b + 2c$
- $8a - 5b + 2c, -10a + 5b + 15c$
- $5m + 4n - 2p, 4m - 10n + 11p$
- $3x + 4y - 15, -10x + 11y - 10$
- $5a + 12b + 20c, 10a - 20b - 12c, 6a + 8b - 7c$
- $20l + 8m - 15n, -18l - 5m - 10n, 6l - 3m + 7n$
- $p^4 - 15p^3 - 18p^2, 20p^3 + 22p^2 + 13p, 8p^2 + 2p - 30$

Solution:

- $2a + 3b + 4c, 5a + 4b + 2c$
 $(2a + 3b + 4c) + (5a + 4b + 2c)$
 $= 2a + 3b + 4c + 5a + 4b + 2c$
 $= 2a + 5a + 3b + 4b + 4c + 2c$
 $= 7a + 7b + 6c$
- $8a - 5b + 2c, -10a + 5b + 15c$
 $(8a - 5b + 2c) + (-10a + 5b + 15c)$
 $= 8a - 5b + 2c - 10a + 5b + 15c$
 $= 8a - 10a - 5b + 5b + 2c + 15c$
 $= -2a + 0b + 17c$
 $= -2a + 17c$
- $5m + 4n - 2p, 4m - 10n + 11p$
 $(5m + 4n - 2p) + (4m - 10n + 11p)$
 $= 5m + 4n - 2p + 4m - 10n + 11p$
 $= 5m + 4m + 4n - 10n - 2p + 11p$
 $= 9m - 6n + 9p$
- $3x + 4y - 15, -10x + 11y - 10$
 $(3x + 4y - 15) + (-10x + 11y - 10)$
 $= 3x + 4y - 15 - 10x + 11y - 10$
 $= 3x - 10x + 4y + 11y - 15 - 10$
 $= -7x + 15y - 25$
- $5a + 12b + 20c, 10a - 20b - 12c, 6a + 8b - 7c$
 $(5a + 12b + 20c) + (10a - 20b - 12c) + (6a + 8b - 7c)$
 $= 5a + 12b + 20c + 10a - 20b - 12c + 6a + 8b - 7c$
 $= 5a + 10a + 6a + 12b - 20b + 8b + 20c - 12c - 7c$
 $= 21a + 0b + 1c$
 $= 21a + c$
- $20l + 8m - 15n, -18l - 5m - 10n, 6l - 3m + 7n$

$$\begin{aligned}
& (20l + 8m - 15n) + (-18l - 5m - 10n) + (6l - 3m + 7n) \\
& = 20l + 8m - 15n - 18l - 5m - 10n + 6l - 3m + 7n \\
& = 20l - 18l + 6l + 8m - 5m - 3m - 15n - 10n + 7n \\
& = 8l + 0m - 18n \\
& = 8l - 18n
\end{aligned}$$

$$\begin{aligned}
\text{g) } & p^4 - 15p^3 - 18p^2, 20p^3 + 22p^2 + 13p, 8p^2 + 2p - 30 \\
& (p^4 - 15p^3 - 18p^2) + (20p^3 + 22p^2 + 13p) + (8p^2 + 2p - 30) \\
& = p^4 - 15p^3 - 18p^2 + 20p^3 + 22p^2 + 13p + 8p^2 + 2p - 30 \\
& = p^4 + 20p^3 - 15p^3 + 22p^2 - 18p^2 + 8p^2 + 13p + 2p - 30 \\
& = p^4 + 5p^3 + 12p^2 + 15p - 30
\end{aligned}$$

2. Subtract the first expression from the second in each of the following.

- a) $5a + 6b + 4c, 10a + 9b + 11c$
- b) $6p - 3q - 2r, 11p - 7q + 4r$
- c) $25x + 40y - 70z, 35x + 10y - 100z$
- d) $16a^4 + 9a^3 - 18a^2, 20a^4 - 10a^3 + 10a^2 + 21a$
- e) $2b^3 + 7b^2 - 12b - 12, 8b^4 - 12b + 15$
- f) $81x^6 - 125y^6, 100x^6 + 25x^5y + 45x^4y^2 - 125y^6$
- g) $6x^3 - 13x^2 + 17x - 30, 14x^4 + 20x - 18$

Solution:

- a) $5a + 6b + 4c, 10a + 9b + 11c$
 $(10a + 9b + 11c) - (5a + 6b + 4c)$
 $= 10a + 9b + 11c - 5a - 6b - 4c$
 $= (10a - 5a) + (9b - 6b) + (11c - 4c)$
 $= 5a + 3b + 7c$
- b) $6p - 3q - 2r, 11p - 7q + 4r$
 $(11p - 7q + 4r) - (6p - 3q - 2r)$
 $= 11p - 7q + 4r - 6p + 3q + 2r$
 $= (11p - 6p) + (-7q + 3q) + (4r + 2r)$
 $= 5p - 4q + 6r$
- c) $25x + 40y - 70z, 35x + 10y - 100z$
 $(35x + 10y - 100z) - (25x + 40y - 70z)$
 $= 35x + 10y - 100z - 25x - 40y + 70z$
 $= (35x - 25x) + (10y - 40y) + (-100z + 70z)$
 $= 10x - 30y - 30z$
- d) $16a^4 + 9a^3 - 18a^2, 20a^4 - 10a^3 + 10a^2 + 21a$
 $(20a^4 - 10a^3 + 10a^2 + 21a) - (16a^4 + 9a^3 - 18a^2)$
 $= 20a^4 - 10a^3 + 10a^2 + 21a - 16a^4 - 9a^3 + 18a^2$
 $= (20a^4 - 16a^4) + (-10a^3 - 9a^3) + (10a^2 + 18a^2) + 21a$
 $= 4a^4 - 19a^3 + 28a^2 + 21a$
- e) $2b^3 + 7b^2 - 12b - 12, 8b^4 - 12b + 15$
 $(8b^4 - 12b + 15) - (2b^3 + 7b^2 - 12b - 12)$
 $= 8b^4 - 12b + 15 - 2b^3 - 7b^2 + 12b + 12$
 $= 8b^4 - 2b^3 - 7b^2 + (-12b + 12b) + (15 + 12)$
 $= 8b^4 - 2b^3 - 7b^2 + 0 + 27$
 $= 8b^4 - 2b^3 - 7b^2 + 27$
- f) $81x^6 - 125y^6, 100x^6 + 25x^5y + 45x^4y^2 - 125y^6$
 $(100x^6 + 25x^5y + 45x^4y^2 - 125y^6) - (81x^6 - 125y^6)$
 $= 100x^6 + 25x^5y + 45x^4y^2 - 125y^6 - 81x^6 + 125y^6$
 $= (100x^6 - 81x^6) + 25x^5y + 45x^4y^2 + (-125y^6 + 125y^6)$
 $= 19x^6 + 25x^5y + 45x^4y^2 + 0$

$$\begin{aligned}
&= 19x^6 + 25x^5y + 45x^4y^2 \\
\text{g) } &6x^3 - 13x^2 + 17x - 30, 14x^4 + 20x - 18 \\
&(14x^4 + 20x - 18) - (6x^3 - 13x^2 + 17x - 30) \\
&= 14x^4 + 20x - 18 - 6x^3 + 13x^2 - 17x + 30 \\
&= 14x^4 - 6x^3 + 13x^2 + (20x - 17x) + (-18 + 30) \\
&= 14x^4 - 6x^3 + 13x^2 + 3x + 12
\end{aligned}$$

5.4 : Let's Practice

1. Simplify the given expressions.

- $2(4a - 3b) + 5b$
- $5(3m + n) - 10n$
- $7(2a - 3b) - 8(a - 4b)$
- $3(5p - 2q + 3r) + 7(3p + 2q - 2r)$
- $-2(4x + 3y + 15) - 5(2x - 4y - 10)$
- $5(x - y) - 2(2x - 3y + 2z) + (4x + 9y + 5z)$
- $-7(2u - v) + 2(v - 2w) - 6(3w - 4u)$
- $5[6a - 4\{5a - 2(4a + 5)\}]$

Solution:

- $$\begin{aligned}
2(4a - 3b) + 5b \\
2(4a - 3b) + 5b &= 8a - 6b + 5b \\
&= 8a - b
\end{aligned}$$
- $$\begin{aligned}
5(3m + n) - 10n \\
5(3m + n) - 10n &= 15m + 5n - 10n \\
&= 15m - 5n
\end{aligned}$$
- $$\begin{aligned}
7(2a - 3b) - 8(a - 4b) \\
7(2a - 3b) - 8(a - 4b) &= 14a - 21b - 8a + 32b \\
&= 14a - 8a - 21b + 32b \\
&= 6a + 11b
\end{aligned}$$
- $$\begin{aligned}
3(5p - 2q + 3r) + 7(3p + 2q - 2r) \\
3(5p - 2q + 3r) + 7(3p + 2q - 2r) &= 15p - 6q + 9r + 21p + 14q - 14r \\
&= 15p + 21p - 6q + 14q + 9r - 14r \\
&= 36p + 10q - 5r
\end{aligned}$$
- $$\begin{aligned}
-2(4x + 3y + 15) - 5(2x - 4y - 10) \\
-2(4x + 3y + 15) - 5(2x - 4y - 10) &= -8x - 6y - 30 - 10x + 20y + 50 \\
&= -8x - 10x - 6y + 20y - 30 + 50 \\
&= -18x + 14y + 20
\end{aligned}$$
- $$\begin{aligned}
5(x - y) - 2(2x - 3y + 2z) + (4x + 9y + 5z) \\
&= 5x - 5y - 4x + 6y - 4z + 4x + 9y + 5z \\
&= 5x - 4x + 4x - 5y + 6y + 9y - 4z + 5z \\
&= 5x + 10y + z
\end{aligned}$$
- $$\begin{aligned}
-7(2u - v) + 2(v - 2w) - 6(3w - 4u) \\
&= -14u + 7v + 2v - 4w - 18w + 24u \\
&= -14u + 24u + 7v + 2v - 4w - 18w \\
&= 10u + 9v - 22w
\end{aligned}$$
- $$\begin{aligned}
5[6a - 4\{5a - 2(4a + 5)\}] \\
5[6a - 4\{5a - 2(4a + 5)\}] &= 5[6a - 4\{5a - 8a - 10\}] \\
&= 5[6a - 4\{-3a - 10\}] \\
&= 5[6a + 12a + 40] \\
&= 5[18a + 40] \\
&= 90a + 200
\end{aligned}$$

Unit Evaluation 5

1. Choose the correct option.

- a) Which rule helps us find any specific term if its position in the pattern and the rule of the pattern is known?
 i) Pattern-to-term rule ii) Term-to-term rule
 iii) Position-to-pattern rule **iv) Position-to-term rule**
- b) The 16th term of the given pattern 9, 10, 11, 12, ... is:
 i) 23 ii) 25 iii) 29 **iv) 24**
- c) Which of these has a coefficient of 3?
 i) $2x^3$ **ii) $3x^2$** iii) x^3 iv) x
- d) In an expression, the terms are separated by:
 i) + or $\times$ **ii) - or +** iii) $\times$ or $\div$ iv) + or $\div$
- e) If $x = 5$, $y = 9$ and $z = 14$, then the value of the expression $4x^3 - 3y^2 + 12z$ is:
 i) 349 ii) 389 **iii) 280** iv) 264

2. Evaluate the following expressions if $x = 9$, $y = -3$ and $z = -4$.

- a) x^2y b) $x^2 + y^2 + z^2$ c) $2x^3 - 3x^2 + 5x$
 d) $5y^3 + 10y^2 - 15y$ e) $7z^3 - 8z^2 + 10z - 25$ f) $5x^3 + 3x^2 + 10x - 30$

Solution:

- a) x^2y

Substituting $x = 9$ and $y = -3$

$$x^2y = 9^2(-3) = 81(-3) = -243$$

- b) $x^2 + y^2 + z^2$

Substituting $x = 9$, $y = -3$, and $z = -4$

$$\begin{aligned} x^2 + y^2 + z^2 &= 9^2 + (-3)^2 + (-4)^2 \\ &= 81 + 9 + 16 \\ &= 106 \end{aligned}$$

- c) $2x^3 - 3x^2 + 5x$

Substituting $x = 9$

$$\begin{aligned} 2x^3 - 3x^2 + 5x &= 2(9)^3 - 3(9)^2 + 5(9) \\ &= 2(729) - 3(81) + 45 \\ &= 1458 - 243 + 45 \\ &= 1260 \end{aligned}$$

- d) $5y^3 + 10y^2 - 15y$

Substituting $y = -3$

$$\begin{aligned} 5y^3 + 10y^2 - 15y &= 5(-3)^3 + 10(-3)^2 - 15(-3) \\ &= 5(-27) + 10(9) + 45 \\ &= -135 + 90 + 45 \\ &= 0 \end{aligned}$$

- e) $7z^3 - 8z^2 + 10z - 25$

Substituting $z = -4$

$$\begin{aligned} 7z^3 - 8z^2 + 10z - 25 &= 7(-4)^3 - 8(-4)^2 + 10(-4) - 25 \\ &= 7(-64) - 8(16) - 40 - 25 \\ &= -448 - 128 - 40 - 25 \\ &= -641 \end{aligned}$$

- f) $5x^3 + 3x^2 + 10x - 30$

Substituting $x = 9$

$$5x^3 + 3x^2 + 10x - 30 = 5(9)^3 + 3(9)^2 + 10(9) - 30$$

$$\begin{aligned}
 &= 5(729) + 3(81) + 90 - 30 \\
 &= 3645 + 243 + 90 - 30 \\
 &= 3948
 \end{aligned}$$

3. Find the sum of the following expressions.

- a) $9a + 15b + 14$, $10a + 10b + 20$
 b) $15b - 12c - 17d$, $12b + 12c - 3d$
 c) $25p + 30q - 45r$, $20p - 30q + 45r$
 d) $7a^3 - 4a^2 + 5a - 11$, $4a^2 - 4a + 10$

Solution:

- a) $9a + 15b + 14$, $10a + 10b + 20$
 $(9a + 15b + 14) + (10a + 10b + 20)$
 $= 9a + 15b + 14 + 10a + 10b + 20$
 $= (9a + 10a) + (15b + 10b) + (14 + 20)$
 $= 19a + 25b + 34$
 b) $15b - 12c - 17d$, $12b + 12c - 3d$
 $(15b - 12c - 17d) + (12b + 12c - 3d)$
 $= 15b - 12c - 17d + 12b + 12c - 3d$
 $= (15b + 12b) + (-12c + 12c) + (-17d - 3d)$
 $= 27b + 0c - 20d$
 $= 27b - 20d$
 c) $25p + 30q - 45r$, $20p - 30q + 45r$
 $(25p + 30q - 45r) + (20p - 30q + 45r)$
 $= 25p + 30q - 45r + 20p - 30q + 45r$
 $= (25p + 20p) + (30q - 30q) + (-45r + 45r)$
 $= 45p + 0q + 0r$
 $= 45p$
 d) $7a^3 - 4a^2 + 5a - 11$, $4a^2 - 4a + 10$
 $(7a^3 - 4a^2 + 5a - 11) + (4a^2 - 4a + 10)$
 $= 7a^3 - 4a^2 + 5a - 11 + 4a^2 - 4a + 10$
 $= 7a^3 + (-4a^2 + 4a^2) + (5a - 4a) + (-11 + 10)$
 $= 7a^3 + 0a^2 + a - 1$
 $= 7a^3 + a - 1$

4. Subtract the first expression from the second in each of the following.

- a) $17a + 25b - 31c$, $12a + 25b + 9c$
 b) $4c - 6d + 9e$, $11c - 16d + 8e$
 c) $5m + 21n - 34p + 15$, $25n - 40p - 20$
 d) $26p + 31q - 42r$, $25p + 31q - 50r - 4$

Solution:

- a) $17a + 25b - 31c$, $12a + 25b + 9c$
 $(12a + 25b + 9c) - (17a + 25b - 31c)$
 $= 12a + 25b + 9c - 17a - 25b + 31c$
 $= (12a - 17a) + (25b - 25b) + (9c + 31c)$
 $= -5a + 0b + 40c$
 $= -5a + 40c$
 b) $4c - 6d + 9e$, $11c - 16d + 8e$
 $(11c - 16d + 8e) - (4c - 6d + 9e)$
 $= 11c - 16d + 8e - 4c + 6d - 9e$
 $= (11c - 4c) + (-16d + 6d) + (8e - 9e)$
 $= 7c - 10d - e$
 c) $5m + 21n - 34p + 15$, $25n - 40p - 20$

$$\begin{aligned}
& (25n - 40p - 20) - (5m + 21n - 34p + 15) \\
& = 25n - 40p - 20 - 5m - 21n + 34p - 15 \\
& = -5m + (25n - 21n) + (-40p + 34p) + (-20 - 15) \\
& = -5m + 4n - 6p - 35
\end{aligned}$$

$$\begin{aligned}
\text{d) } & 26p + 31q - 42r, 25p + 31q - 50r - 4 \\
& (25p + 31q - 50r - 4) - (26p + 31q - 42r) \\
& = 25p + 31q - 50r - 4 - 26p - 31q + 42r \\
& = (25p - 26p) + (31q - 31q) + (-50r + 42r) - 4 \\
& = -p + 0q - 8r - 4 \\
& = -p - 8r - 4
\end{aligned}$$

5. Simplify the following algebraic expressions.

$$\begin{aligned}
\text{a) } & 12[7p + 3\{8p - (7p - 75)\}] \\
& 12[7p + 3\{8p - (7p - 75)\}] = 12[7p + 3\{8p - 7p + 75\}] \\
& = 12[7p + 3\{p + 75\}] \\
& = 12[7p + 3p + 225] \\
& = 12[10p + 225] \\
& = 120p + 2700
\end{aligned}$$

$$\begin{aligned}
\text{b) } & 20[3(2x + 2y) - 4\{4x - 3(x - y)\}] \\
& = 20[3(2x + 2y) - 4\{4x - 3(x - y)\}] \\
& = 20[6x + 6y - 4\{4x - 3x + 3y\}] \\
& = 20[6x + 6y - 4\{x + 3y\}] \\
& = 20[6x + 6y - 4x - 12y] \\
& = 20[2x - 6y] \\
& = 40x - 120y
\end{aligned}$$

$$\begin{aligned}
\text{c) } & 17[15x - 3\{16x - (18x - 10 + 6x)\}] \\
& = 17[15x - 3\{16x - (24x - 10)\}] \\
& = 17[15x - 3\{16x - 24x + 10\}] \\
& = 17[15x - 3\{-8x + 10\}] \\
& = 17[15x + 24x - 30] \\
& = 17[39x - 30] \\
& = 663x - 510
\end{aligned}$$

$$\begin{aligned}
\text{d) } & -9[12a + \{3a - 5(8a - 10)\}] - 3(5b - 4c) \\
& = -9[12a + \{3a - 40a + 50\}] - 15b + 12c \\
& = -9[12a + \{-37a + 50\}] - 15b + 12c \\
& = -9[12a - 37a + 50] - 15b + 12c \\
& = -108a + 333a - 450 - 15b + 12c \\
& = 225a - 15b + 12c - 450
\end{aligned}$$

6.1 : Let's Practice

1. Identify and separate out the equations.

a) $3 + 2 = 5 \times 1$

b) $7 \times 4 = 30 - 2$

c) $5 + 9 = 5 - 4$

d) $8 \div 4 = 3 + 1$

e) $18 \times 2 = 9 \times 4$

f) $17 - 2 = 3 \times 5$

g) $1 \times 9 = 7 + 3$

h) $11 + 5 = 12 + 4$

Solution:

a) $3 + 2 = 5 \times 1$

LHS = $3 + 2 = 5$

RHS = $5 \times 1 = 5$

LHS = RHS

So, it is an equation.

b) $7 \times 4 = 30 - 2$

LHS = $7 \times 4 = 28$

RHS = $30 - 2 = 28$

LHS = RHS

So, it is an equation.

c) $5 + 9 = 5 - 4$

LHS = $5 + 9 = 14$

RHS = $5 - 4 = 1$

LHS $\neq$ RHS

So, it is not an equation.

d) $8 \div 4 = 3 + 1$

LHS = $8 \div 4 = 2$

RHS = $3 + 1 = 4$

LHS $\neq$ RHS

So, it is not an equation.

e) $18 \times 2 = 9 \times 4$

LHS = $18 \times 2 = 36$

RHS = $9 \times 4 = 36$

LHS = RHS

So, it is an equation.

f) $17 - 2 = 3 \times 5$

LHS = $17 - 2 = 15$

RHS = $3 \times 5 = 15$

LHS = RHS

So, it is an equation.

g) $1 \times 9 = 7 + 3$

LHS = $1 \times 9 = 9$

RHS = $7 + 3 = 10$

LHS $\neq$ RHS

So, it is not an equation.

h) $11 + 5 = 12 + 4$

LHS = $11 + 5 = 16$

RHS = $12 + 4 = 16$

LHS = RHS

So, it is an equation.

2. Identify and separate out the algebraic equations.

- | | | |
|------------------|--------------------------|------------------|
| a) $4x + 3y$ | b) $2x^3 + 5x = 9$ | c) $8 + 4x + 3y$ |
| d) $5t - 3v = 7$ | e) $7x + 6y^2 + t^3 = 1$ | f) $1 + y = z$ |
| g) $5d + 4e = 5$ | h) $7 + h^2 + 5g^2$ | |

Solution:

- a) $4x + 3y$

This is not an equation because it has no equal sign.

- b) $2x^3 + 5x = 9$

It is an equation because there is equal sign between two expressions.

- c) $8 + 4x + 3y$

This is not an equation because it has no equal sign.

- d) $5t - 3v = 7$

It is an equation because there is equal sign between two expressions.

- e) $7x + 6y^2 + t^3 = 1$

It is an equation because there is equal sign between two expressions.

- f) $1 + y = z$

It is an equation because there is equal sign between two expressions.

- g) $5d + 4e = 5$

It is an equation because there is equal sign between two expressions.

- h) $7 + h^2 + 5g^2$

This is not an equation because it has no equal sign.

3. Identify the linear equations from the given algebraic equations.

- | | | |
|---------------------|-------------------|-------------------|
| a) $6x + 9y = 2$ | b) $5y^3 + s = 6$ | c) $2t + 5s = 11$ |
| d) $c^3 + 5n^4 = 7$ | e) $5r + 3s = 9$ | f) $3 - 5k = 11j$ |
| g) $f^2 + g = 4$ | h) $s + t = 2$ | |

Solution:

- a) $6x + 9y = 2$

This is a linear equation. The variables x and y are raised to the power of 1, and there are no other terms involving higher powers of x and y .

- b) $5y^3 + s = 6$

This is not a linear equation. The variable y is raised to the power of 3. Therefore, it is not a linear equation.

- c) $2t + 5s = 11$

This is a linear equation. The variables t and s are raised to the power of 1, and there are no other terms involving higher powers of t and s .

- d) $c^3 + 5n^4 = 7$

This is not a linear equation. Both the variables c and n are raised to powers 3 and 4, respectively. Therefore, it is not a linear equation.

- e) $5r + 3s = 9$

This is a linear equation. The variables r and s are raised to the power of 1, and there are no other terms involving higher powers of r and s .

- f) $3 - 5k = 11j$

This is a linear equation. The variables k and j are raised to the power of 1, and there are no other terms involving higher powers of k and j .

- g) $f^2 + g = 4$

This is not a linear equation. The variable f is raised to the power of 2. Therefore, it is not a linear equation.

h) $s + t = 2$

This is a linear equation. The variables s and t are raised to the power of 1, and there are no other terms involving higher powers of s and t .

4. Write 5 linear equations in one variable.

Solution:

a) $3x + 5 = 12$

This equation represents a linear relationship between x and the constant terms 5 and 12.

b) $2x - 8 = 10$

This equation represents a linear relationship between x and the constant terms -8 and 10.

c) $6x + 2 = 4x - 3$

This equation represents a linear relationship between x and the constant terms 2 and -3 on both sides of the equation.

d) $-4x + 7 = 3$

This equation represents a linear relationship between x and the constant terms 7 and 3.

e) $9x - 6 = 2x + 10$

This equation represents a linear relationship between x and the constant terms -6 and 10 on both sides of the equation.

6.2 : Let's Practice

1. Write the linear equation in one variable for each of the following.

- a) If 7 is added to a number, the sum is 19.
- b) If 12 is added to twice a number, the sum is 28.
- c) If twice a number is added to three times itself, the sum is 35.
- d) If thrice a number is subtracted from four times itself, the difference is 12.

Solution:

- a) If 7 is added to a number, the sum is 19.

Suppose the number is x .

"7 is added to a number x " means " $x + 7$ ".

So, $x + 7 = 19$ is the required linear equation.

- b) If 12 is added to twice a number, the sum is 28.

Suppose the number = x .

Twice a number = $2x$

12 added to twice a number = $2x + 12$

So, $2x + 12 = 28$ is the required linear equation.

- c) If twice a number is added to three times itself, the sum is 35.

Suppose the number = x .

Twice a number = $2x$

Three times a number = $3x$

twice a number is added to three times itself = $2x + 3x$

So, $2x + 3x = 35$ is the required linear equation.

- d) If thrice a number is subtracted from four times itself, the difference is 12.

Suppose the number = x .

Three times a number = $3x$

Four times a number = $4x$

thrice a number is subtracted from four times itself = $4x - 3x$

So, $4x - 3x = 12$ is the required linear equation.

2. Write the sentence for the given equations.

a) $5x = 9$

b) $\frac{3}{t} = 4$

c) $4m + 6 = 15$

d) $x + 2 = 7$

e) $x - 5 = 24$

f) $7 - x = 3$

Solution:

a) $5x = 9$

5 times a number equal 9

b) $3t = 4$

3 divided by a number equals 4

c) $4m + 6 = 15$

6 added to 4 times a number equals 15

d) $x + 2 = 7$

2 added to a number equals 7

e) $x - 5 = 24$

5 subtracted from a number equals 24

f) $7 - x = 3$

A number subtracted from 7 equals 3

3. Solve the following equations.

a) $\frac{2}{3}a = 8$

b) $\frac{5}{7}b = 15$

c) $\frac{1}{2}(a - 4) + 8 = 20$

d) $4(\frac{1}{2}p - \frac{5}{2}) = 5$

e) $\frac{2}{3}(q + 3) - 4 = \frac{4}{5}$

f) $\frac{3}{5}(5m - 10) - 5 = \frac{3}{5}$

g) $0.5y - 1.3 = 0.3(0.1y + 1.2)$

h) $3(3.4d + 0.2) - 2.3 = 6.2d + 3.3$

i) $0.7r - 4.0 = 5.0 - 1.3r$

j) $2.4(x + 2) = 3.3 + 0.4x$

Solution:

a) $\frac{2}{3}a = 8$

Multiplying $\frac{3}{2}$ on both sides

$$\frac{3}{2} \times \frac{2}{3}a = 8 \times \frac{3}{2}$$

$$a = 4 \times 3$$

$$a = 12$$

b) $\frac{5}{7}b = 15$

Multiplying $\frac{7}{5}$ on both sides

$$\frac{7}{5} \times \frac{5}{7}b = 15 \times \frac{7}{5}$$

$$b = 3 \times 7$$

$$b = 21$$

c) $\frac{1}{2}(a - 4) + 8 = 20$

Subtracting 8 on both sides

$$\frac{1}{2}(a - 4) + 8 - 8 = 20 - 8$$

$$\frac{1}{2}(a - 4) = 12$$

Multiplying 2 on both sides

$$2 \times \frac{1}{2}(a - 4) = 12 \times 2$$

$$a - 4 = 24$$

Adding 4 on both sides

$$a - 4 + 4 = 24 + 4$$

$$a = 28$$

d) $4\left(\frac{1}{2}p - \frac{5}{2}\right) = 5$

$$4 \times \frac{1}{2}p - 4 \times \frac{5}{2} = 5$$

$$2p - 10 = 5$$

Adding 10 on both sides

$$2p - 10 + 10 = 5 + 10$$

$$2p = 15$$

Dividing 2 on both sides

$$\frac{2p}{2} = \frac{15}{2}$$

$$p = 7\frac{1}{2}$$

e) $\frac{2}{3}(q + 3) - 4 = \frac{4}{5}$

Multiplying LCM 15 on both sides

$$15 \times \frac{2}{3}(q + 3) - 15 \times 4 = 15 \times \frac{4}{5}$$

$$10(q + 3) - 60 = 12$$

Adding 60 on both sides

$$10(q + 3) - 60 + 60 = 12 + 60$$

$$10(q + 3) = 72$$

Dividing 10 on both sides

$$\frac{10(q + 3)}{10} = \frac{72}{10}$$

$$q + 3 = \frac{36}{5}$$

Subtracting 3 on both sides

$$q + 3 - 3 = \frac{36}{5} - 3$$

$$q = \frac{36 - 15}{5}$$

$$q = \frac{21}{5}$$

$$q = 4\frac{1}{5}$$

f) $\frac{3}{5}(5m - 10) - 5 = \frac{3}{5}$

Multiplying 5 on both sides

$$5 \times \frac{3}{5}(5m - 10) - 5 \times 5 = 5 \times \frac{3}{5}$$

$$3(5m - 10) - 25 = 3$$

$$15m - 30 - 25 = 3$$

$$15m - 55 = 3$$

Adding 55 on both sides

$$15m - 15 + 15 = 3 + 55$$

$$15m = 58$$

Dividing 15 on both sides

$$\frac{15m}{15} = \frac{58}{15}$$

$$m = 3\frac{13}{15}$$

g) $0.5y - 1.3 = 0.3(0.1y + 1.2)$

$$0.5y - 1.3 = 0.03y + 0.36$$

Taking variable on LHS

$$0.5y - 0.03y = 0.36 + 1.3$$

$$0.47y = 1.66$$

Dividing 0.47 on both sides

$$\frac{0.47y}{0.47} = \frac{1.66}{0.47}$$

$$y = 3.53$$

h) $3(3.4d + 0.2) - 2.3 = 6.2d + 3.3$

$$10.2d + 0.6 - 2.3 = 6.2d + 3.3$$

Taking variable on LHS

$$10.2d - 6.2d = 3.3 - 0.6 + 2.3$$

$$4d = 5$$

Dividing 4 on both sides

$$\frac{4d}{4} = \frac{5}{4}$$

$$d = 1.25$$

i) $0.7r - 4.0 = 5.0 - 1.3r$

$$0.7r - 4.0 = 5.0 - 1.3r$$

Taking variable on LHS

$$0.7r + 1.3r = 5.0 + 4.0$$

$$2r = 9$$

Dividing 2 on both sides

$$\frac{2r}{2} = \frac{9}{2}$$

$$r = 4.5$$

j) $2.4(x + 2) = 3.3 + 0.4x$

$$2.4x + 4.8 = 3.3 + 0.4x$$

Taking variable on LHS

$$2.4x - 0.4x = 3.3 - 4.8$$

$$2x = -1.5$$

Dividing 2 on both sides

$$\frac{2x}{2} = -\frac{1.5}{2}$$

$$x = -0.75$$

4. When four times a number is added to three times itself, the sum obtained is 28. Find the number.

Solution:

Suppose the number = x .

Three times a number = $3x$

Four times a number = $4x$

Four times a number is added to three times = $4x + 3x$

So, $4x + 3x = 28$

$$7x = 28$$

Dividing 7 on both sides

$$\frac{7x}{7} = \frac{28}{7}$$

$$x = 4$$

Therefore, the number is 4.

5. When two-thirds of a number is added to four-fifths of itself, the sum obtained

is 44. Find the number.

Solution:

Suppose the number = x .

Two-third of a number = $(\frac{2}{3})x$

Four-fifth of a number = $(\frac{4}{5})x$

Two-thirds of a number is added to four-fifths = $(\frac{2}{3})x + (\frac{4}{5})x$

So, $(\frac{2}{3})x + (\frac{4}{5})x = 44$

$15 \times (\frac{2}{3})x + 15 \times (\frac{4}{5})x = 15 \times 44$ (Multiplying by LCM 15 on both sides)

$5 \times 2x + 3 \times 4x = 660$

$10x + 12x = 660$

$22x = 660$

$\frac{22x}{22} = \frac{660}{22}$

(Dividing 22 on both sides)

$x = 30$

Therefore, the number is 30.

6. The perimeter of an equilateral triangle is 12 cm. Find the length of each of its

sides.

Solution:

Let's x represent the length of each side in equilateral triangle.

Perimeter of equilateral triangle = 12 cm

Side + side + side = perimeter of triangle

$x + x + x = 12$

$3x = 12$

Dividing 3 on both sides

$\frac{3x}{3} = \frac{12}{3}$

$x = 4$

Therefore, each side of the equilateral triangle has a length of 4 cm.

7. The length of a blue wire is 16 metres longer than the length of a yellow wire.

Find the length of the blue wire if the sum of their lengths is 58 metres.

Solution:

Let x be the length of yellow wire.

Length of blue wire = $x + 16$

Sum of both wires = 58

$x + (x + 16) = 58$

$x + x + 16 = 58$

$2x + 16 = 58$

Subtracting 16 on both sides

$2x + 16 - 16 = 58 - 16$

$2x = 42$

Dividing 2 on both sides

$\frac{2x}{2} = \frac{42}{2}$

$x = 21$

Hence,

Length of yellow wire = $x = 21$ m

Length of blue wire = $x + 16 = 21 + 16 = 37$ m

8. Nida's age is one-third the age of her father. If the difference between their ages is 32 years, find their present ages.

Solution:

Let x be the Nida's age.

Her father's age = $3x$

Difference between their ages = 32 years

$$3x - x = 32$$

$$2x = 32$$

Dividing 2 on both sides

$$\frac{2x}{2} = \frac{32}{2}$$

$$x = 16$$

Therefore,

Nida's age = $x = 16$ years

Her father's age = $3x = 3(16) = 48$ years

9. The cost of a raincoat is half the cost of an umbrella. Find the price of the raincoat if the total cost of an umbrella and a raincoat is Rs 1740.

Solution:

Let x be the cost of raincoat.

The cost of umbrella = $2x$

Total cost = Rs.1, 740

$$x + 2x = 1740$$

$$3x = 1740$$

Dividing 3 on both sides

$$\frac{3x}{3} = \frac{1740}{3}$$

$$x = 580$$

Therefore, the price of the raincoat is Rs 580.

Unit Evaluation 6

1. Choose the correct option.

a) Which of these is an algebraic equation?

i) $2x + 4$

ii) $3 - t$

iii) $4d + 9 = 3$

iv) $4 + 2(5 - r)$

b) Which of the following is not an equation in one variable?

i) $4x + 2 = 10$

ii) $6x = 54$

iii) $3x^2 = 27$

iv) $3x + 8y = 22$

c) The value of x in $7x - 4 = 24$ is:

i) 20

ii) 24

iii) 28

iv) 4

d) All the variables in a linear equation have the power of:

i) 1

ii) 2

iii) 3

iv) 0

e) Twice a number is equal to seven:

i) $7 = 2z$

ii) $7z = 2$

iii) $Z - 2 = 7$

iv) $z = 7 - 2$

2. Identify the linear equations from the given algebraic equations.

a) $2x + 3 = 9$

b) $7t^4 + 4 = 1$

c) $4y + 7 = 15$

d) $x^4 + 5y^4 = 13$

e) $2 + 5r = 72$

f) $6h^2 + 2g^2 = 12$

g) $4y^2 + 7u^3 = 1$

h) $2s + 5t = 10$

Solution:

a) $2x + 3 = 9$

This is a linear equation. The variable x is raised to the power of 1, and there are no other terms involving higher powers of x .

- b) $7t^4 + 4 = 1$
This is not a linear equation. The variable t is raised to the power of 4. Therefore, it is not a linear equation.
- c) $4y + 7 = 15$
This is a linear equation. The variable y is raised to the power of 1, and there are no other terms involving higher powers of y .
- d) $x^4 + 5y^4 = 13$
This is not a linear equation. Both the variables x and y are raised to powers 4. Therefore, it is not a linear equation.
- e) $2 + 5r = 72$
This is a linear equation. The variable r is raised to the power of 1, and there are no other terms involving higher powers of r .
- f) $6h^2 + 2g^2 = 12$
This is not a linear equation. The variables h and g are raised to the power of 2. Therefore, it is not a linear equation.
- g) $4y^2 + 7u^3 = 1$
This is not a linear equation. The variable y is raised to the power of 2 and the variable u is raised to the power of 3. Therefore, it is not a linear equation.
- h) $2s + 5t = 10$
This is a linear equation. The variables s and t are raised to the power of 1, and there are no other terms involving higher powers of s and t .

3. Write a linear equation for each of the following.

- a) A number divided by 2 is 12.
Suppose the number is x .
"A number is divided by 2" means " $\frac{x}{2}$ ".
So, $\frac{x}{2} = 12$ is the required linear equation.
- b) The difference of a number from 7 when divided by 4 is 10.
Suppose the number is x .
"The difference of a number from 7" means " $7 - x$ ".
Divide the difference by 4 means " $\frac{7 - x}{4}$ ".
So, $\frac{7 - x}{4} = 10$ is the required linear equation.
- c) 46 added to a number is the same as 3 times the number.
Suppose the number is x .
"46 added to a number" means " $x + 46$ ".
Three times a number = $3x$
So, $x + 46 = 3x$ is the required linear equation.
- e) 8 more than a number is equal to 3 less than twice the number.
Suppose the number is x .
"8 more than a number" means " $x + 8$ ".
Twice a number = $2x$
"3 less than twice the number" means " $2x - 3$ ".
So, $x + 8 = 2x - 3$ is the required linear equation.
- f) 5 added to twice a number is the same as thrice the number.
Suppose the number is x .
Twice a number = $2x$
"5 added to twice a number" means " $2x + 5$ ".
Thrice a number = $3x$
So, $2x + 5 = 3x$ is the required linear equation.

4. The perimeter of a rectangular masjid is 404 metres. If its length is 22 metres more than its width, find the dimensions of the masjid.

Solution:

Let

Width of rectangular Masjid = w

Length of rectangular Masjid = $w + 22$

Perimeter of rectangle = 404 m

$$2(\text{length} + \text{width}) = 404$$

$$2(w + 22 + w) = 404$$

$$2w + 44 + 2w = 404$$

$$4w + 44 = 404$$

$$4w + 44 - 44 = 404 - 44 \quad (\text{Subtracting 44 on both sides})$$

$$4w = 360$$

$$\frac{4w}{4} = \frac{360}{4} \quad (\text{Dividing 4 on both sides})$$

$$w = 90$$

Hence, width of rectangular Masjid = $w = 90$ m

And length of rectangular Masjid = $w + 22 = 90 + 22 = 112$ m

5. The sum of Umar and Hassan's ages is 30 years. If Umar's age is one-fourth of

Hassan's age, find their present ages.

Solution:

Let

Hassan's age = x

Umar's age = $(\frac{1}{4})x$

Sum of their ages = 30 years

$$x + (\frac{1}{4})x = 30$$

$$4x + x = 120 \quad (\text{Multiplying 4 on both sides})$$

$$5x = 120$$

$$\frac{5x}{5} = \frac{120}{5} \quad (\text{Dividing 5 on both side})$$

$$x = 24$$

Hence,

Hassan's age = $x = 24$ years

Umar's age = $(\frac{1}{4})x = (\frac{1}{4})(24) = 6$ years

6. A cat is four times as heavy as a hen. If the sum of their masses is 10 kg, find the mass of the cat.

Solution:

Let

The weight of hen = x kg

The weight of cat = $4x$

According to given condition

$$x + 4x = 10$$

$$5x = 10$$

$$\frac{5x}{5} = \frac{10}{5}$$

$$x = 2$$

Therefore,

Weight of hen = $x = 2$ kg

Weight of cat = $4x = 4(2) = 8$ kg

7.1 : Let's Practice

1. Find the area of a square-shaped ground if its length is 12 m.

Solution;

Length of a square-shaped ground = $L = 12\text{m}$

Area of a square-shaped ground = L^2

Area of a square-shaped ground = $(12)^2$
 $= 144\text{ m}^2$

Therefore, the area of the square-shaped ground is 144 m^2 .

2. Find the perimeter of a masjid if its length is 88 m and its width is 65 m.

Solution:

Length = $L = 88\text{ m}$

Width = $W = 65\text{ m}$

Perimeter = ?

We know that

Perimeter of a rectangular Masjid = $2(L + W)$
 $= 2(88 + 65)$
 $= 2(153)$
 $= 306\text{ m}$

Therefore, the perimeter of the Masjid is 306 meters.

3. If the length of a square clock is 8 cm, what will be the area of its surface?

Solution:

Length of a square clock = 8 cm

Area of a square clock = ?

We know that

Area of a square clock = $(\text{length})^2$
 $= (8\text{ cm})^2$
 $= 64\text{ cm}^2$

Therefore, the area of the surface of the square clock is 64 cm^2 .

4. A hall is 30 m long and 25 m wide. Find the cost of tiling its floor at the rate of Rs 180 per square metre.

Solution:

Length of the hall = 30 m

Width of the hall = 25 m

Cost per square meter = Rs 180

Area = Length $\times$ Width

Area = $30\text{ m} \times 25\text{ m}$

Area = 750 m^2

Cost = Area $\times$ Cost per square meter Cost

$= 750\text{ square meters} \times \text{Rs } 180/\text{square meter Cost}$

$= \text{Rs } 135,000$

Therefore, the cost of tiling the floor of the hall at the rate of Rs 180 per m^2 would be Rs 135,000.

5. Find the area of a rectangle if its length is 20 cm and its width is 15 cm.

Solution:

Length of rectangle = 20 cm

Width of rectangle = 15 cm

Area of rectangle = ?

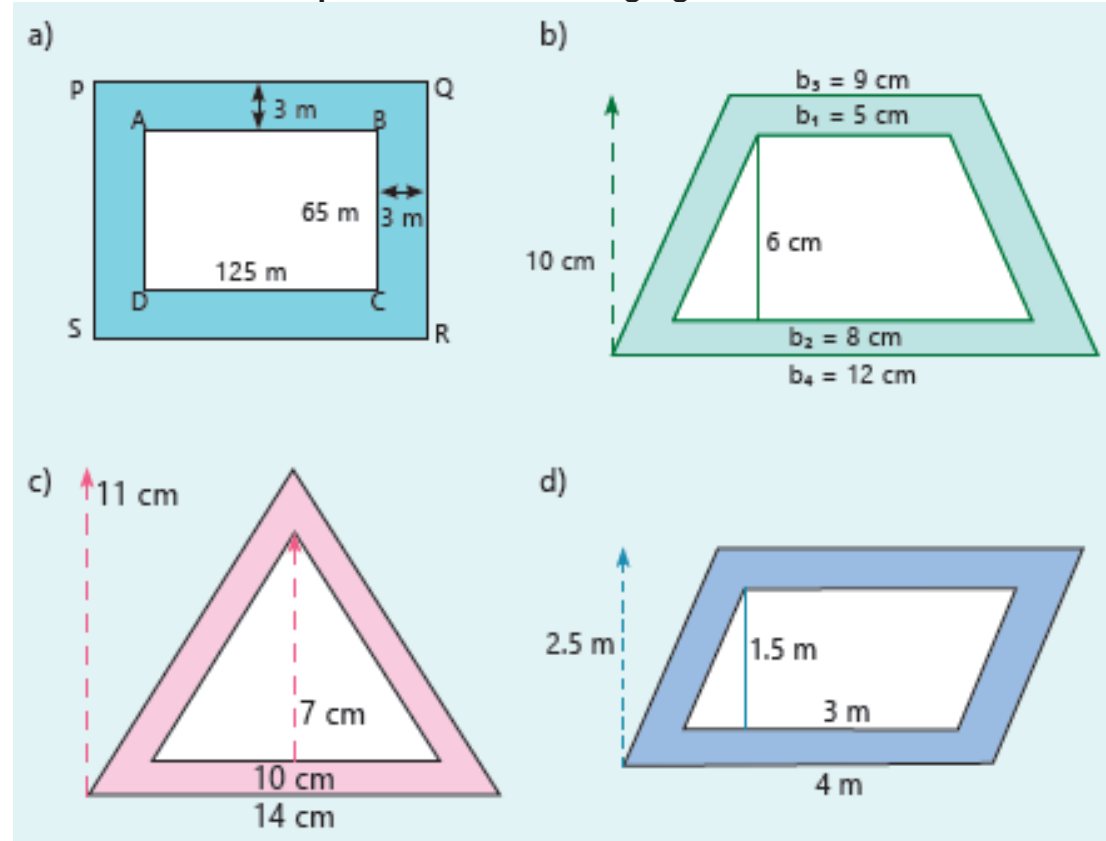
We know that

$$\begin{aligned}\text{Area of rectangle} &= \text{length} \times \text{width} \\ &= 20 \text{ cm} \times 15 \text{ cm} \\ &= 300 \text{ cm}^2\end{aligned}$$

Therefore, the area of the rectangle is 300 cm^2 .

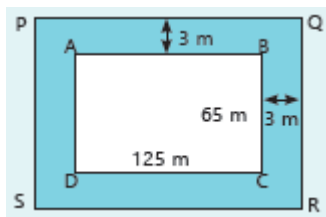
7.2 : Let's Practice

1. Find the area of the path for the following figures.



Solution:

a)



Suppose ABCD is the given rectangular park with a length of 125 m and a width of 65 m. The outer rectangle is shown by PQRS. We need to find the area of the shaded region.

Area of the shaded region = Area of the rectangle PQRS – Area of the rectangle ABCD

$$\begin{aligned}\text{Area of the inner rectangle ABCD} &= \text{length} \times \text{width} \\ &= 125 \text{ m} \times 65 \text{ m} = 8125 \text{ m}^2\end{aligned}$$

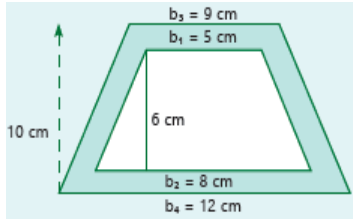
$$\begin{aligned}\text{Length of the outer rectangle PQRS} &= CD + (2 \times 3 \text{ m}) \\ &= 125 \text{ m} + 6 \text{ m} = 131 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Width of the outer rectangle PQRS} &= BC + (2 \times 3 \text{ m}) \\ &= 65 \text{ m} + 6 \text{ m} = 71 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Area of the rectangle PQRS} &= 131 \text{ m} \times 71 \text{ m} \\ &= 9301 \text{ m}^2\end{aligned}$$

$$\text{Area of the shaded region} = 9301 \text{ m}^2 - 8125 \text{ m}^2 = 1176 \text{ m}^2$$

b)



First, we find the area of the inner trapezium that is:

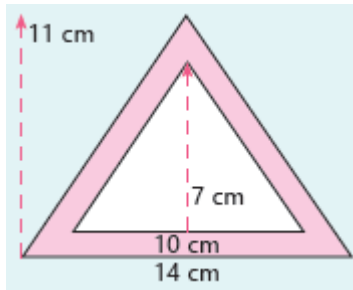
$$\begin{aligned}\text{Area of the inner trapezium} &= \text{altitude} \times \left(\frac{b_1 + b_2}{2} \right) \\ &= 6 \text{ cm} \times \frac{5+8}{2} \text{ cm} \\ &= 3 \text{ cm} \times 13 \text{ cm} \\ &= 39 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of the outer trapezium} &= \text{altitude} \times \left(\frac{b_3 + b_4}{2} \right) \\ &= 10 \text{ cm} \times \frac{9+12}{2} \text{ cm} \\ &= 5 \text{ cm} \times 21 \text{ cm} \\ &= 105 \text{ cm}^2\end{aligned}$$

Area of the shaded region = Area of the outer trapezium – Area of the inner trapezium

$$\text{Area of the shaded region} = 105 \text{ cm}^2 - 39 \text{ cm}^2 = 66 \text{ cm}^2$$

c)



First, we find the area of the inner triangle that is:

$$\begin{aligned}\text{Area of the inner triangle} &= \frac{1}{2} (\text{altitude} \times \text{base}) \\ &= \frac{1}{2} (7 \text{ cm} \times 10 \text{ cm}) \\ &= \frac{1}{2} \times 70 \text{ cm}^2 \\ &= 35 \text{ cm}^2\end{aligned}$$

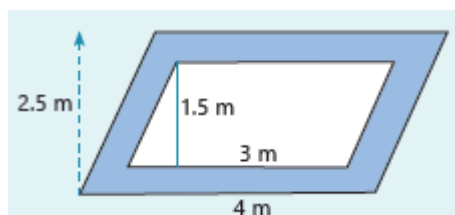
$$\begin{aligned}\text{Area of the outer triangle} &= \frac{1}{2} (\text{altitude} \times \text{base}) \\ &= \frac{1}{2} (11 \text{ cm} \times 14 \text{ cm}) \\ &= 11 \times 7 \text{ cm}^2 \\ &= 77 \text{ cm}^2\end{aligned}$$

Area of a path around the triangle = Area of outer triangle – Area of inner triangle

$$\text{Area of the path around the triangle} = 77 \text{ cm}^2 - 35 \text{ cm}^2 = 42 \text{ cm}^2$$

So, the area of the path around the triangle is 42 cm².

d)



$$\begin{aligned}\text{Area of the inner parallelogram} &= \text{base} \times \text{altitude} \\ &= 3 \text{ m} \times 1.5 \text{ m} \\ &= 4.5 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Area of the outer parallelogram} &= \text{base} \times \text{altitude} \\ &= 4 \text{ m} \times 2.5 \text{ m} \\ &= 10 \text{ m}^2\end{aligned}$$

Area of the path of the parallelogram = Area of the outer parallelogram – Area of the inner parallelogram

$$\begin{aligned}\text{Area of the path of the parallelogram} &= 10 \text{ m}^2 - 4.5 \text{ m}^2 \\ &= 5.5 \text{ m}^2\end{aligned}$$

2. Find the area of a triangular-shaped pool with a base of 25 m and a height of 11 m.

Solution:

Base = 25 m

Height = 11 m

Area of triangle = ?

We know that

$$\begin{aligned}\text{Area of triangle} &= \frac{1}{2} (\text{height} \times \text{base}) \\ &= \frac{1}{2} (11 \text{ cm} \times 25 \text{ cm}) \\ &= \frac{1}{2} (275 \text{ cm}^2) \\ &= 137.5 \text{ cm}^2\end{aligned}$$

3. A triangular wall clock covers a 424 cm² area of the wall. If the height of the clock is 28 cm, find the base of the clock.

Solution:

Area of triangle = 424 cm²

Height = 28 cm

Base = ?

We know that

$$\text{Area of triangle} = \frac{1}{2} (\text{height} \times \text{base})$$

$$424 \text{ cm}^2 = \frac{1}{2} (28 \text{ cm} \times \text{base})$$

$$424 \text{ cm}^2 = 14 \text{ cm} \times \text{base}$$

Dividing 14 cm on both sides

$$\frac{424 \text{ cm}^2}{14 \text{ cm}} = \frac{14 \text{ cm}}{14 \text{ cm}} \times \text{base}$$

$$30.29 \text{ cm} = \text{base}$$

Therefore, the base of the triangular wall clock is approximately 30.29 cm.

4. Find the area of a parallelogram-shaped field with a base of 250 m and a height of 200 m. Also, find the cost of ploughing it at the rate of Rs 250 per square metre.

Solution:

Base = 250 m

Height = 200 m

Area of parallelogram = ?

We know that

$$\begin{aligned}\text{Area of the parallelogram} &= \text{base} \times \text{height} \\ &= 250 \text{ m} \times 200 \text{ m} \\ &= 50000 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Cost of ploughing @ Rs.250 per m}^2 &= 50000 \times 250 \\ &= \text{Rs. } 12,500,000\end{aligned}$$

Therefore, the cost of ploughing the field at the rate of Rs 250 per square meter is Rs 12,500,000.

5. Find the height of a parallelogram-shaped room with a base of 15 m and an area of 615 m².

Solution:

Base = 15 m

Area of parallelogram = 615 m²

Height of parallelogram = ?

We know that

Area of the parallelogram = base $\times$ height

$$615 \text{ m}^2 = 15 \text{ m} \times \text{height}$$

Dividing 15 m on both sides

$$\frac{615 \text{ m}^2}{15 \text{ m}} = \frac{15 \text{ m}}{15 \text{ m}} \times \text{height}$$

$$41 = \text{height}$$

Therefore, the height of the parallelogram-shaped room is 41 meters.

6. Find the area of a trapezium-shaped masjid if the length of its two parallel sides is 7 m and 9 m and the distance between them is 6 m.

Solution:

Length of base 1 = b₁ = 7m

Length of base 2 = b₂ = 9 m

Length of altitude = 6 m

Area of trapezium = ?

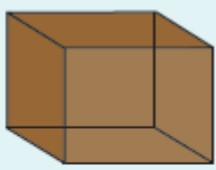
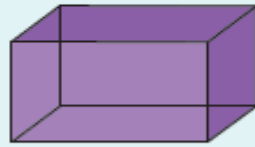
We know that

$$\begin{aligned}\text{Area of the inner trapezium} &= \text{altitude} \times \left(\frac{b_1 + b_2}{2} \right) \\ &= 6 \text{ m} \times \frac{7+9}{2} \text{ m} \\ &= 6 \text{ m} \times \frac{16}{2} \text{ m} \\ &= 6 \text{ m} \times 8 \text{ m} \\ &= 48 \text{ m}^2\end{aligned}$$

Therefore, the area of the trapezium-shaped masjid is 48 square meters.

7.3 : Let's Practice

1. Name the following shapes and write their number of faces, edges and vertices.

a)  Faces Edges Vertices Name: _____	b)  Faces Edges Vertices Name: _____
c)  Faces Edges Vertices Name: _____	d)  Faces Edges Vertices Name: _____
e)  Faces Edges Vertices Name: _____	f)  Faces Edges Vertices Name: _____

Solution:

- a) Cone
Faces = 2, Edges = 1, Vertices = 1
- b) Cube
Faces = 6, Edges = 12, Vertices = 8
- c) Sphere
Faces = 1, Edges = 0, Vertices = 0
- d) Cylinder
Faces = 3, Edges = 2, Vertices = 0
- e) Cuboid
Faces = 6, Edges = 12, Vertices = 8
- f) Hemi-sphere
Faces = 2, Edges = 1, Vertices = 0

7.4 : Let's Practice

1. Substitute the given values for the length of a cube in the formula to calculate the volume and surface area.

- a) 9 m b) 14 cm c) 23 cm d) 11 m e) 7.5 m

Solution:

a) 9 m

$$\text{Length} = l = 9 \text{ m}$$

We know that

$$\text{Volume of a cube} = \text{length} \times \text{width} \times \text{height}$$

$$= l \times l \times l$$

$$= l^3$$

$$= (9\text{m})^3$$

$$= 729 \text{ m}^3$$

Also,

$$\text{Surface area of a cube} = 6 \times l^2$$

$$= 6 \times (9\text{m})^2$$

$$= 6 \times 81 \text{ m}^2$$

$$= 486 \text{ m}^2$$

b) 14 m

$$\text{Length} = l = 14 \text{ m}$$

We know that

$$\text{Volume of a cube} = \text{length} \times \text{width} \times \text{height}$$

$$= l \times l \times l$$

$$= l^3$$

$$= (14\text{m})^3$$

$$= 2744 \text{ m}^3$$

Also,

$$\text{Surface area of a cube} = 6 \times l^2$$

$$= 6 \times (14\text{m})^2$$

$$= 6 \times 196 \text{ m}^2$$

$$= 1176 \text{ m}^2$$

c) 23 cm

$$\text{Length} = l = 23 \text{ cm}$$

We know that

$$\text{Volume of a cube} = \text{length} \times \text{width} \times \text{height}$$

$$= l \times l \times l$$

$$= l^3$$

$$= (23 \text{ cm})^3$$

$$= 12167 \text{ cm}^3$$

Also,

$$\text{Surface area of a cube} = 6 \times l^2$$

$$= 6 \times (23 \text{ cm})^2$$

$$= 6 \times 529 \text{ cm}^2$$

$$= 3174 \text{ cm}^2$$

d) 11 m

$$\text{Length} = l = 11 \text{ m}$$

We know that

$$\text{Volume of a cube} = \text{length} \times \text{width} \times \text{height}$$

$$= l \times l \times l$$

$$\begin{aligned}
 &= l^3 \\
 &= (11 \text{ m})^3 \\
 &= 1331 \text{ m}^3
 \end{aligned}$$

Also,

$$\begin{aligned}
 \text{Surface area of a cube} &= 6 \times l^2 \\
 &= 6 \times (11 \text{ m})^2 \\
 &= 6 \times 121 \text{ m}^2 \\
 &= 726 \text{ m}^2
 \end{aligned}$$

e) 7.5 m

$$\text{Length} = l = 7.5 \text{ m}$$

We know that

$$\begin{aligned}
 \text{Volume of a cube} &= \text{length} \times \text{width} \times \text{height} \\
 &= l \times l \times l \\
 &= l^3 \\
 &= (7.5 \text{ m})^3 \\
 &= 421.875 \text{ m}^3
 \end{aligned}$$

Also,

$$\begin{aligned}
 \text{Surface area of a cube} &= 6 \times l^2 \\
 &= 6 \times (7.5 \text{ m})^2 \\
 &= 6 \times 56.25 \text{ m}^2 \\
 &= 337.5 \text{ m}^2
 \end{aligned}$$

2. Substitute the given values in the formula to calculate the volume and surface area of the cuboids.

- | | | |
|-------------------------|----------------------|----------------------|
| a) $l = 5.1 \text{ cm}$ | $w = 1.4 \text{ cm}$ | $h = 2.8 \text{ cm}$ |
| b) $l = 7.8 \text{ m}$ | $w = 5.5 \text{ m}$ | $h = 5 \text{ m}$ |
| c) $l = 25 \text{ cm}$ | $w = 20 \text{ cm}$ | $h = 14 \text{ cm}$ |
| d) $l = 16 \text{ m}$ | $w = 11 \text{ m}$ | $h = 12 \text{ m}$ |

Solution:

a) Given that

$$l = 5.1 \text{ cm}$$

$$w = 1.4 \text{ cm}$$

$$h = 2.8 \text{ cm}$$

We know that

$$\begin{aligned}
 \text{Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\
 &= 5.1 \text{ cm} \times 1.4 \text{ cm} \times 2.8 \text{ cm} \\
 &= 19.99 \text{ cm}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\
 &= 2 \times (5.1 \text{ cm} \times 1.4 \text{ cm}) + 2 \times (5.1 \text{ cm} \times 2.8 \text{ cm}) + 2 \times (1.4 \text{ cm} \times 2.8 \text{ cm})
 \end{aligned}$$

$$\begin{aligned}
 &= 2 \times (7.14 \text{ cm}^2) + 2 \times (14.28 \text{ cm}^2) + 2 \times (3.92 \text{ cm}^2) \\
 &= 14.28 \text{ cm}^2 + 28.56 \text{ cm}^2 + 7.84 \text{ cm}^2 \\
 &= 50.68 \text{ cm}^2
 \end{aligned}$$

b) Given that

$$l = 7.8 \text{ m}$$

$$w = 5.5 \text{ m}$$

$$h = 5 \text{ m}$$

We know that

$$\begin{aligned}
 \text{Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\
 &= 7.8 \text{ m} \times 5.5 \text{ m} \times 5 \text{ m} \\
 &= 214.5 \text{ m}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\
 &= 2 \times (7.8 \text{ m} \times 5.5 \text{ m}) + 2 \times (7.8 \text{ m} \times 5 \text{ m}) + 2 \times (5.5 \text{ m} \times 5 \text{ m}) \\
 &= 2 \times (42.9 \text{ m}^2) + 2 \times (39 \text{ m}^2) + 2 \times (27.5 \text{ m}^2) \\
 &= 85.8 \text{ m}^2 + 78 \text{ m}^2 + 55 \text{ m}^2 \\
 &= 218.8 \text{ m}^2
 \end{aligned}$$

c) Given that

$$l = 25 \text{ cm}$$

$$w = 20 \text{ cm}$$

$$h = 14 \text{ cm}$$

We know that

$$\begin{aligned}
 \text{Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\
 &= 25 \text{ cm} \times 20 \text{ cm} \times 14 \text{ cm} \\
 &= 7000 \text{ cm}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\
 &= 2 \times (25 \text{ cm} \times 20 \text{ cm}) + 2 \times (25 \text{ cm} \times 14 \text{ cm}) + 2 \times (20 \text{ cm} \times 14 \text{ cm})
 \end{aligned}$$

$$\begin{aligned}
 &= 2 \times (500 \text{ cm}^2) + 2 \times (350 \text{ cm}^2) + 2 \times (280 \text{ cm}^2) \\
 &= 1000 \text{ cm}^2 + 700 \text{ cm}^2 + 560 \text{ cm}^2 \\
 &= 2260 \text{ cm}^2
 \end{aligned}$$

d) Given that

$$l = 16 \text{ m}$$

$$w = 11 \text{ m}$$

$$h = 12 \text{ m}$$

We know that

$$\begin{aligned}
 \text{Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\
 &= 16 \text{ m} \times 11 \text{ m} \times 12 \text{ m} \\
 &= 2112 \text{ m}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\
 &= 2 \times (16 \text{ m} \times 11 \text{ m}) + 2 \times (16 \text{ m} \times 12 \text{ m}) + 2 \times (11 \text{ m} \times 12 \text{ m})
 \end{aligned}$$

$$\begin{aligned}
 &= 2 \times (176 \text{ m}^2) + 2 \times (192 \text{ m}^2) + 2 \times (132 \text{ m}^2) \\
 &= 352 \text{ m}^2 + 384 \text{ m}^2 + 264 \text{ m}^2 \\
 &= 1000 \text{ m}^2
 \end{aligned}$$

3. Find the capacity of an oil container in litres whose length is 600 cm, width is 450 cm and height is 300 cm.

Solution:

$$l = 600 \text{ cm}$$

$$w = 450 \text{ cm}$$

$$h = 300 \text{ cm}$$

We know that

$$\begin{aligned}
 \text{Volume of a cube} &= \text{length} \times \text{width} \times \text{height} \\
 &= 600 \text{ cm} \times 450 \text{ cm} \times 300 \text{ cm} \\
 &= 81000000 \text{ cm}^3
 \end{aligned}$$

$$\text{Capacity} = \frac{\text{volume}}{1000}$$

$$\text{Capacity} = \frac{81000000 \text{ cm}^3}{1000}$$

$$\text{Capacity} = 81000 \text{ liters}$$

Therefore, the capacity of the oil container is 81000 liters.

4. A wooden box is 25 cm long, 14 cm wide and 22 cm high. Calculate the volume of the wooden box.

Solution:

$$l = 25 \text{ cm}$$

$$w = 14 \text{ cm}$$

$$h = 22 \text{ cm}$$

We know that

$$\begin{aligned}\text{Volume of a cube} &= \text{length} \times \text{width} \times \text{height} \\ &= 25 \text{ cm} \times 14 \text{ cm} \times 22 \text{ cm} \\ &= 7700 \text{ cm}^3\end{aligned}$$

Therefore, the volume of the wooden box is 7700 cm^3 .

5. The dimensions of a cuboid box are 6 m × 400 m × 1.5 m. Find:

a) total volume

b) cost of painting its entire outer surface at the rate of Rs 22 per m^2 .

Solution:

$$l = 6 \text{ m}$$

$$w = 400 \text{ m}$$

$$h = 1.5 \text{ m}$$

We know that

$$\begin{aligned}\text{a) Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\ &= 6 \text{ m} \times 400 \text{ m} \times 1.5 \text{ m} \\ &= 3600 \text{ m}^3\end{aligned}$$

Therefore, the total volume of the cuboid box is 3600 m^3 .

$$\begin{aligned}\text{b) Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\ &= 2 \times (6 \text{ m} \times 400 \text{ m}) + 2 \times (6 \text{ m} \times 1.5 \text{ m}) + 2 \times (400 \text{ m} \times 1.5 \text{ m}) \\ &= 2 \times (2400 \text{ m}^2) + 2 \times (9 \text{ m}^2) + 2 \times (600 \text{ m}^2) \\ &= 4800 \text{ m}^2 + 18 \text{ m}^2 + 1200 \text{ m}^2 \\ &= 6018 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{cost of painting its entire surface @ Rs 22 per m}^2 &= 22 \times 6018 \\ &= \text{Rs. } 132,396\end{aligned}$$

6. A hall is 21 m long, 15 m wide and 30 m high. Find:

a) total surface area

b) area of four walls

c) area of the roof

roof

Solution:

$$l = 21 \text{ m}$$

$$w = 15 \text{ m}$$

$$h = 30 \text{ m}$$

We know that

$$\begin{aligned}\text{a) Total surface area} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\ &= 2 \times (21 \text{ m} \times 15 \text{ m}) + 2 \times (21 \text{ m} \times 30 \text{ m}) + 2 \times (15 \text{ m} \times 30 \text{ m}) \\ &= 2 \times (315 \text{ m}^2) + 2 \times (630 \text{ m}^2) + 2 \times (450 \text{ m}^2) \\ &= 630 \text{ m}^2 + 1260 \text{ m}^2 + 900 \text{ m}^2 \\ &= 2790 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{b) Area of four walls} &= 2 \times (l \times h) + 2 \times (w \times h) \\ &= 2 \times (21 \text{ m} \times 30 \text{ m}) + 2 \times (15 \text{ m} \times 30 \text{ m}) \\ &= 2 \times (630 \text{ m}^2) + 2 \times (450 \text{ m}^2) \\ &= 1260 \text{ m}^2 + 900 \text{ m}^2 \\ &= 2160 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{c) Area of the roof} &= l \times w \\ &= 21 \text{ m} \times 15 \text{ m}\end{aligned}$$

$$= 315 \text{ m}^2$$

7. A room is 3.2 m long, 2.5 m wide and 4.1 m high. Find:

- total surface area of the room including the roof and floor.
- cost of painting its four walls at the rate of Rs 250 per m^2 .
- cost of painting the ceiling at the rate of Rs 290 per m^2 .

Solution:

$$l = 3.2 \text{ m}$$

$$w = 2.5 \text{ m}$$

$$h = 4.1 \text{ m}$$

We know that

$$\begin{aligned} \text{a) Total surface area} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\ &= 2 \times (3.2 \text{ m} \times 2.5 \text{ m}) + 2 \times (3.2 \text{ m} \times 4.1 \text{ m}) + 2 \times (2.5 \text{ m} \times 4.1 \text{ m}) \end{aligned}$$

m)

$$\begin{aligned} &= 2 \times (8 \text{ m}^2) + 2 \times (13.12 \text{ m}^2) + 2 \times (10.25 \text{ m}^2) \\ &= 16 \text{ m}^2 + 26.24 \text{ m}^2 + 20.5 \text{ m}^2 \\ &= 62.74 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{b) Area of four walls} &= 2 \times (l \times h) + 2 \times (w \times h) \\ &= 2 \times (3.2 \text{ m} \times 4.1 \text{ m}) + 2 \times (2.5 \text{ m} \times 4.1 \text{ m}) \\ &= 2 \times (13.12 \text{ m}^2) + 2 \times (10.25 \text{ m}^2) \\ &= 26.24 \text{ m}^2 + 20.5 \text{ m}^2 \\ &= 46.74 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Cost of painting four walls @ Rs. 250 per m}^2 &= 250 \times 46.74 \\ &= \text{Rs. 11,685} \end{aligned}$$

$$\begin{aligned} \text{c) Area of ceiling} &= l \times w \\ &= 3.2 \text{ m} \times 2.5 \text{ m} \\ &= 8 \text{ m}^2 \end{aligned}$$

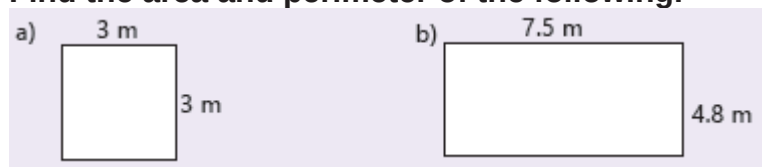
$$\begin{aligned} \text{Cost of painting the ceiling @ Rs. 290 per m}^2 &= 8 \times 290 \\ &= \text{Rs. 2,320} \end{aligned}$$

Unit Evaluation 7

1. Choose the correct option.

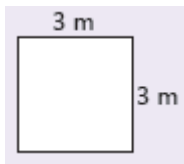
- The formula to find the area of a rectangle is:
i) $2(l + w)$ ii) $2(l \times w)$ iii) $4l$ **iv) $l \times w$**
- The shortest distance between the top of a shape to its opposite side is called:
i) perimeter ii) area iii) surface area **iv) altitude**
- If the lengths of the bases of a trapezium are 20 cm and 25 cm and the length of its altitude is 22 cm, the area of the trapezium is:
i) 990 cm^2 ii) 990 cm iii) 495 cm **iv) 495 cm^2**
- The number of flat faces a cube has:
i) 8 ii) 12 iii) 4 **iv) 6**
- The surface area of a cube is 384 m^2 . The length of the cube is:
i) 9 m^2 ii) 8 m^2 **iii) 8 m** iv) 9 m

2. Find the area and perimeter of the following.



Solution:

a)

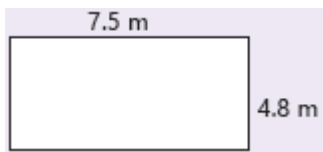


Length = $l = 3 \text{ m}$

$$\begin{aligned}\text{Area of the square} &= (l)^2 \\ &= (3 \text{ m})^2 \\ &= 9 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Perimeter of the square} &= 4 \times \text{length} \\ &= 4 \times 3 \text{ m} \\ &= 12 \text{ m}\end{aligned}$$

b)



Length = $l = 7.5 \text{ m}$

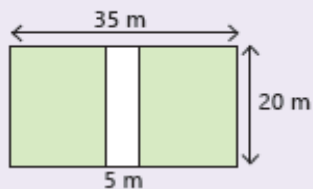
Width = $w = 4.8 \text{ m}$

$$\begin{aligned}\text{Area of the rectangle} &= l \times w \\ &= 7.5 \text{ m} \times 4.8 \text{ m} \\ &= 36 \text{ m}^2\end{aligned}$$

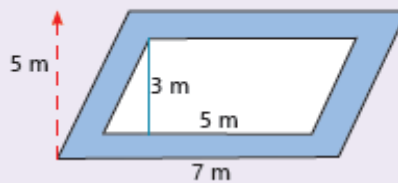
$$\begin{aligned}\text{Perimeter of the rectangle} &= 2(l + w) \\ &= 2(7.5 \text{ m} + 4.8 \text{ m}) \\ &= 2(12.3 \text{ m}) \\ &= 24.6 \text{ m}\end{aligned}$$

3. Find the area of the shaded region.

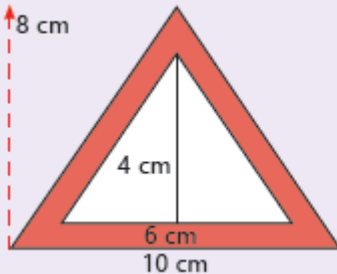
a)



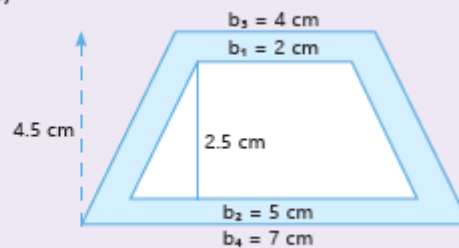
b)



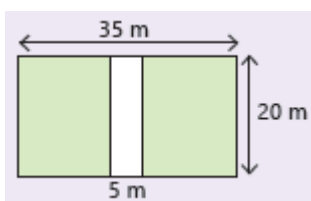
c)



d)



a)

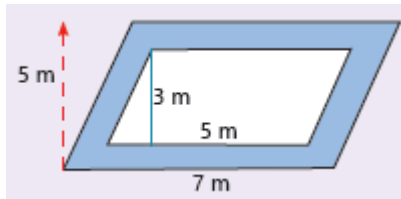


$$\begin{aligned}
 \text{Area of rectangle} &= l \times w \\
 &= 35 \text{ m} \times 20 \text{ m} \\
 &= 700 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of unshaded rectangle} &= l \times w \\
 &= 5 \text{ m} \times 20 \text{ m} \\
 &= 100 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of shaded region} &= \text{area of rectangle} - \text{area of unshaded rectangle} \\
 &= 700 \text{ m}^2 - 100 \text{ m}^2 \\
 &= 600 \text{ m}^2
 \end{aligned}$$

b)



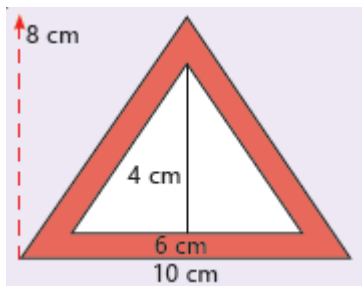
$$\begin{aligned}
 \text{Area of the inner parallelogram} &= \text{base} \times \text{altitude} \\
 &= 5 \text{ m} \times 3 \text{ m} \\
 &= 15 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the outer parallelogram} &= \text{base} \times \text{altitude} \\
 &= 7 \text{ m} \times 5 \text{ m} \\
 &= 35 \text{ m}^2
 \end{aligned}$$

Area of the path of the parallelogram = Area of the outer parallelogram – Area of the inner parallelogram

$$\begin{aligned}
 \text{Area of the path of the parallelogram} &= 35 \text{ m}^2 - 15 \text{ m}^2 \\
 &= 20 \text{ m}^2
 \end{aligned}$$

c)



First, we find the area of the inner triangle that is:

$$\begin{aligned}
 \text{Area of the inner triangle} &= \frac{1}{2} (\text{altitude} \times \text{base}) \\
 &= \frac{1}{2} (4 \text{ cm} \times 6 \text{ cm}) \\
 &= \frac{1}{2} \times 24 \text{ cm}^2 \\
 &= 12 \text{ cm}^2
 \end{aligned}$$

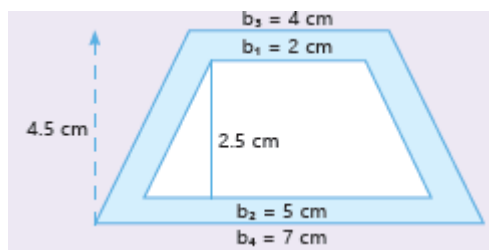
$$\begin{aligned}
 \text{Area of the outer triangle} &= \frac{1}{2} (\text{altitude} \times \text{base}) \\
 &= \frac{1}{2} (8 \text{ cm} \times 10 \text{ cm}) \\
 &= 4 \text{ cm} \times 10 \text{ cm} \\
 &= 40 \text{ cm}^2
 \end{aligned}$$

Area of a path around the triangle = Area of outer triangle – Area of inner triangle

$$\text{Area of the path around the triangle} = 40 \text{ cm}^2 - 12 \text{ cm}^2 = 28 \text{ cm}^2$$

So, the area of the path around the triangle is 28 cm².

d)



First, we find the area of the inner trapezium that is:

$$\begin{aligned}\text{Area of the inner trapezium} &= \text{altitude} \times \left(\frac{b_1 + b_2}{2} \right) \\ &= 2.5 \text{ cm} \times \frac{2+5}{2} \text{ cm} \\ &= 2.5 \text{ cm} \times \frac{7}{2} \text{ cm} \\ &= 2.5 \text{ cm} \times 3.5 \text{ cm} \\ &= 8.75 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of the outer trapezium} &= \text{altitude} \times \left(\frac{b_3 + b_4}{2} \right) \\ &= 4.5 \text{ cm} \times \frac{4+7}{2} \text{ cm} \\ &= 4.5 \text{ cm} \times \frac{11}{2} \text{ cm} \\ &= 4.5 \text{ cm} \times 5.5 \text{ cm} \\ &= 24.75 \text{ cm}^2\end{aligned}$$

Area of the shaded region = Area of the outer trapezium – Area of the inner trapezium

$$\text{Area of the shaded region} = 24.75 \text{ cm}^2 - 8.75 \text{ cm}^2 = 16 \text{ cm}^2$$

4. Draw the following solids and then write their number of edges, vertices and faces.

a) cuboid

b) cube

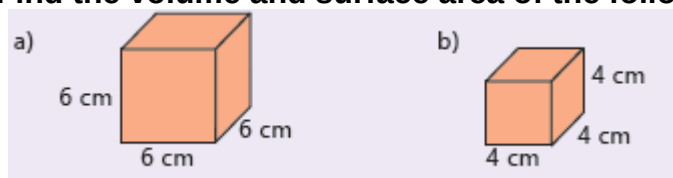
c) cone

d) cylinder

e) sphere

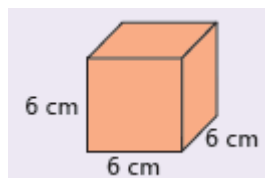
f) hemisphere

5. Find the volume and surface area of the following cubes.



Solution:

a)



Length = l = 6 cm

We know that

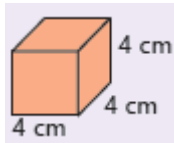
$$\begin{aligned}\text{Volume of a cube} &= \text{length} \times \text{width} \times \text{height} \\ &= l \times l \times l \\ &= l^3 \\ &= (6 \text{ cm})^3 \\ &= 216 \text{ cm}^3\end{aligned}$$

Also,

$$\text{Surface area of a cube} = 6 \times l^2$$

$$\begin{aligned}
 &= 6 \times (6 \text{ cm})^2 \\
 &= 6 \times 36 \text{ cm}^2 \\
 &= 216 \text{ cm}^3
 \end{aligned}$$

b)



Length = l = 4 cm

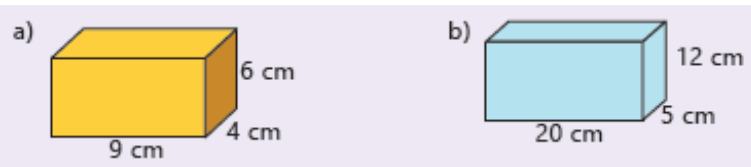
We know that

$$\begin{aligned}
 \text{Volume of a cube} &= \text{length} \times \text{width} \times \text{height} \\
 &= l \times l \times l \\
 &= l^3 \\
 &= (4 \text{ cm})^3 \\
 &= 64 \text{ cm}^3
 \end{aligned}$$

Also,

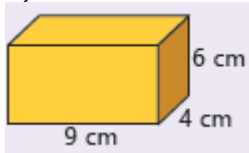
$$\begin{aligned}
 \text{Surface area of a cube} &= 6 \times l^2 \\
 &= 6 \times (4 \text{ cm})^2 \\
 &= 6 \times 16 \text{ cm}^2 \\
 &= 96 \text{ cm}^2
 \end{aligned}$$

6. Find the volume and surface area of the following cuboids.



Solution:

a)



$$l = 9 \text{ cm}$$

$$w = 4 \text{ cm}$$

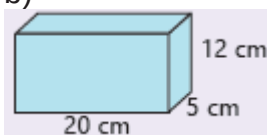
$$h = 6 \text{ cm}$$

We know that

$$\begin{aligned}
 \text{Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\
 &= 9 \text{ cm} \times 4 \text{ cm} \times 6 \text{ cm} \\
 &= 216 \text{ cm}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\
 &= 2 (9 \text{ cm} \times 4 \text{ cm}) + 2 (9 \text{ cm} \times 6 \text{ cm}) + 2 (4 \text{ cm} \times 6 \text{ cm}) \\
 &= 2 \times (36 \text{ cm}^2) + 2 \times (54 \text{ cm}^2) + 2 \times (24 \text{ cm}^2) \\
 &= 72 \text{ cm}^2 + 108 \text{ cm}^2 + 48 \text{ cm}^2 \\
 &= 228 \text{ cm}^2
 \end{aligned}$$

b)



$$l = 20 \text{ cm}$$

$$w = 5 \text{ cm}$$

$$h = 12 \text{ cm}$$

We know that

$$\begin{aligned} \text{Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\ &= 20 \text{ cm} \times 5 \text{ cm} \times 12 \text{ cm} \\ &= 1200 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\ &= 2(20 \text{ cm} \times 5 \text{ cm}) + 2(20 \text{ cm} \times 12 \text{ cm}) + 2(5 \text{ cm} \times 12 \text{ cm}) \end{aligned}$$

$$\begin{aligned} &= 2 \times (100 \text{ cm}^2) + 2 \times (240 \text{ cm}^2) + 2 \times (60 \text{ cm}^2) \\ &= 200 \text{ cm}^2 + 480 \text{ cm}^2 + 120 \text{ cm}^2 \\ &= 800 \text{ cm}^2 \end{aligned}$$

7. Find the perimeter of a square-shaped chessboard whose length is 18 cm.

Solution:

$$\text{Length} = l = 18 \text{ cm}$$

$$\begin{aligned} \text{Perimeter of the square-shaped chessboard} &= 4 \times l \\ &= 4 \times 18 \text{ cm} \\ &= 72 \text{ cm} \end{aligned}$$

8. Find the length of a rectangular park if its perimeter is 2470 m and its width is

580 m. Find the cost of fencing around the park at a rate of Rs 40 per metre.

Solution:

$$\text{Perimeter of rectangular park} = 2470 \text{ m}$$

$$\text{Width} = 580 \text{ m}$$

$$\text{Length} = l = ?$$

$$\text{Cost of fencing @ Rs. 40 per metre} = ?$$

We know that

$$\text{Perimeter of rectangle} = 2470 \text{ m}$$

$$2(l + w) = 2470$$

Dividing 2 on both sides

$$\frac{2(l+w)}{2} = \frac{2470}{2}$$

$$l + w = 1235$$

$$l + 580 = 1235$$

$$l = 1235 - 580$$

$$l = 655$$

Therefore, length of the rectangular park is 655 m.

$$\begin{aligned} \text{Cost of fencing @ Rs. 40 per metre} &= 40 \times 2470 \\ &= \text{Rs. } 98,800 \end{aligned}$$

9. A square-shaped parking area is 120 m long. There is a 2 m wide tiled border around the parking area. Find the area of the border.

Solution:

$$\text{Length of the small square} = 120 \text{ m}$$

$$\text{Width of the border} = 2 \text{ m}$$

$$\text{Length of the large square} = \text{length} + 2 (\text{width of the border})$$

$$= 120 \text{ m} + 2 (2\text{m})$$

$$= 120 \text{ m} + 4 \text{ m} = 124 \text{ m}$$

Area of the border = area of the larger square – area of the small square

$$= (124 \text{ m})^2 - (120 \text{ m})^2$$

$$= 15376 \text{ m}^2 - 14400 \text{ m}^2$$

$$= 976 \text{ m}^2$$

Therefore, the area of the border is 976 square meters.

- 10. A cornfield is 240 m long and 176 m wide. A 2.5 m wide path is made all around the inside of the path. Find the area of the path.**

Solution:

Length = 240 m

Width = 176 m

Width of the path = 2.5 m

Length of the larger rectangle = length of the small rectangle + 2 (width of the path)

$$= 240 \text{ m} + 2 (2.5 \text{ m})$$

$$= 240 \text{ m} + 5 \text{ m} = 245 \text{ m}$$

Width of the larger rectangle = width of small rectangle + 2 (width of the path)

$$= 176 \text{ m} + 2 (2.5 \text{ m})$$

$$= 176 \text{ m} + 5 \text{ m} = 181 \text{ m}$$

Area of the path = Area of the larger rectangle - Area of the cornfield

Area of the path = (245 m) × (181 m) - (240 m) × (176 m)

Area of the path = 44345 m² - 42240 m²

Area of the path = 2105 m²

Therefore, the area of the path is 2105 square meters.

- 11. The length of the two bases of a trapezium is 10.5 cm and 8.9 cm, respectively. If the length of the altitude is 7.5 cm, find the area of the trapezium.**

Solution:

Length of base 1 = b_1 = 10.5 cm

Length of base 2 = b_2 = 8.9 cm

Altitude = 7.5 cm

Area of trapezium = ?

We know that

Area of the trapezium = altitude × $\left(\frac{b_1 + b_2}{2}\right)$

$$= 7.5 \text{ cm} \times \frac{10.5 + 8.9}{2} \text{ cm}$$

$$= 7.5 \text{ cm} \times \frac{19.4}{2} \text{ cm}$$

$$= 7.5 \text{ cm} \times 9.7 \text{ cm}$$

$$= 72.75 \text{ cm}^2$$

- 12. The front wall of an auditorium is triangular in shape. Find the base of the triangle if its area is 240 cm² and its altitude is 24 cm.**

solution:

Area of the triangle = 240 cm²

Altitude = 24 cm

Base = ?

We know that

Area of the triangle = $\frac{1}{2}$ (altitude × base)

$$240 \text{ cm}^2 = \frac{1}{2} (24 \text{ cm} \times \text{base})$$

$$240 \text{ cm}^2 = 12 \text{ cm} \times \text{base}$$

Dividing 12 cm on both sides

$$\frac{240 \text{ cm}^2}{12 \text{ cm}} = \frac{12 \text{ cm} \times \text{base}}{12 \text{ cm}}$$

$$\Rightarrow \text{base} = 20 \text{ cm}$$

Therefore, the base of the triangle is 20 cm.

13. Find the total surface area of a cuboid ice block whose length is 12 cm, width

is 3.8 cm and height is 5.9 cm.

Solution:

$$l = 12 \text{ cm}$$

$$w = 3.8 \text{ cm}$$

$$h = 5.9 \text{ cm}$$

We know that

$$\begin{aligned} \text{Surface area of a cuboid} &= 2 \times (l \times w) + 2 \times (l \times h) + 2 \times (w \times h) \\ &= 2(12\text{cm} \times 3.8\text{cm}) + 2(12\text{cm} \times 5.9\text{cm}) + 2(3.8\text{cm} \times \\ &5.9\text{cm}) \end{aligned}$$

$$\begin{aligned} &= 2 \times (45.6 \text{ cm}^2) + 2 \times (70.8 \text{ cm}^2) + 2 \times (22.42 \text{ cm}^2) \\ &= 91.2 \text{ cm}^2 + 141.6 \text{ cm}^2 + 44.84 \text{ cm}^2 \\ &= 277.64 \text{ cm}^2 \end{aligned}$$

Therefore, the total surface area of the cuboid ice block is 277.64 square centimeters.

14. Find the capacity of a water tank whose length, width and height are 4 m, 6 m and 5 m, respectively.

Solution:

$$l = 4 \text{ m}$$

$$w = 6 \text{ m}$$

$$h = 5 \text{ m}$$

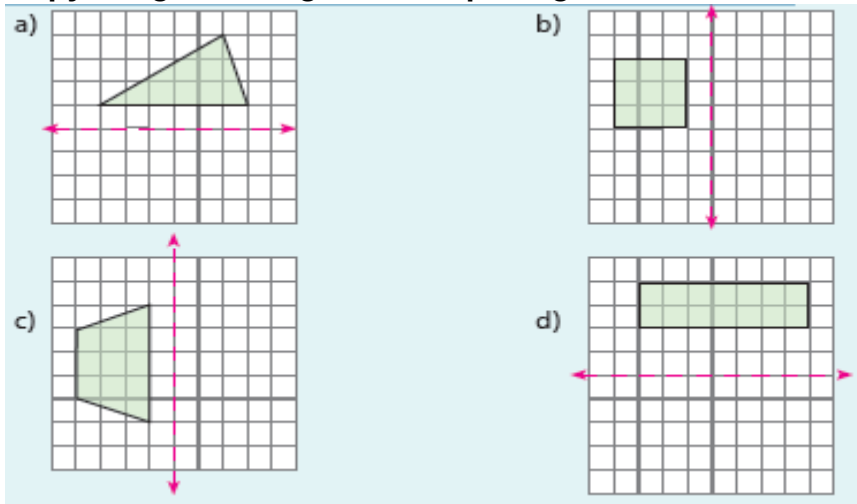
We know that

$$\begin{aligned} \text{Volume of a cuboid} &= \text{length} \times \text{width} \times \text{height} \\ &= 4 \text{ m} \times 6 \text{ m} \times 5 \text{ m} \\ &= 120 \text{ m}^3 \end{aligned}$$

Therefore, the capacity of the water tank is 120 cubic meters.

8.1: Let's Practice

1. Copy the given images on a square grid and reflect them.



Solution:

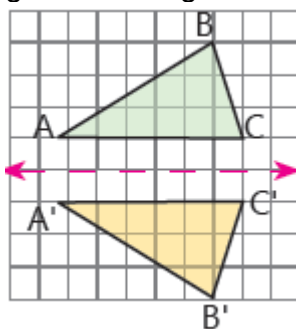
a)

Name the points (vertices) of the given shape for convenience. Choose the closest point (from the line of reflection) of the given image and mark it on the opposite side of the line at exactly the same distance as the original image.

For A, mark A' which is the same distance from the line of reflection.

Similarly, mark all the other points on the other side.

Join A' to B' , B' to C' , C' to A' . This is the required reflected image of the given rectangle.



b)

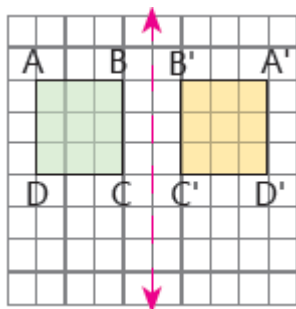
Name the points (vertices) of the given shape for convenience.

Choose the closest point (from the line of reflection) of the given image and mark it on the opposite side of the line at exactly the same distance as the original image.

For A, mark A' which is the same distance from the line of reflection.

Similarly, mark all the other points on the other side.

Join A' to B' , B' to C' , C' to D' and D' to A' . This is the required reflected image of the given square.



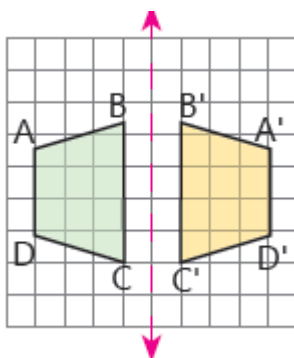
c)

Name the points (vertices) of the given shape for convenience. Choose the closest point (from the line of reflection) of the given image and mark it on the opposite side of the line at exactly the same distance as the original image.

For A, mark A' which is the same distance from the line of reflection.

Similarly, mark all the other points on the other side.

Join A' to B' , B' to C' , C' to D' and D' to A' . This is the required reflected image of the given trapezium.



d)

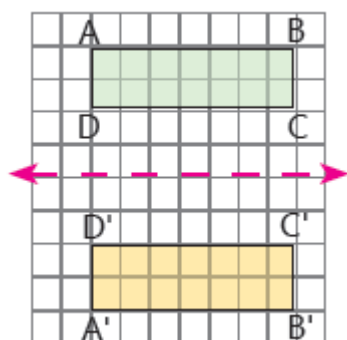
Name the points (vertices) of the given shape for convenience.

Choose the closest point (from the line of reflection) of the given image and mark it on the opposite side of the line at exactly the same distance as the original image.

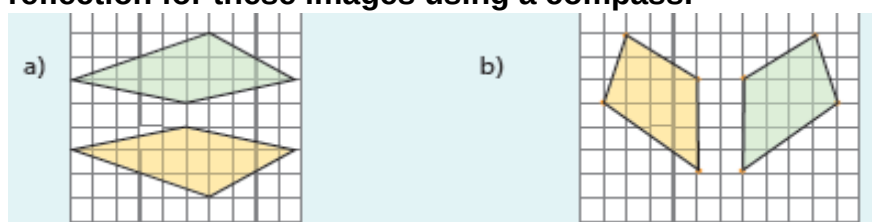
For A, mark A' which is the same distance from the line of reflection.

Similarly, mark all the other points on the other side.

Join A' to B' , B' to C' , C' to D' and D' to A' . This is the required reflected image of the given rectangle.



2. Copy the given figures on a square grid. Then draw the lines of reflection for these images using a compass.



Solution:

a)

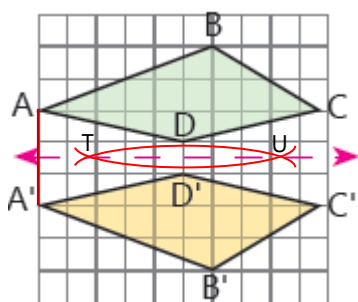
Follow the steps to construct the line of reflection.

Step I: Choose any vertex of the given rhombus $ABCD$ and join it with the corresponding vertex of the reflected image $A'B'C'D'$ using a straight line. Here, we join A and A' .

Step II: Open the compass with more than half the measure of the line segment AA' and draw an arc on the top and bottom of AA' , placing the compass pointer at A .

Step III: Similarly, draw an arc on the top and bottom of AA' , placing the compass pointer at A' which intersects the previous arcs at points T and U , respectively.

Step IV: Join T to U . The perpendicular bisector TU is the required line of reflection.



b)

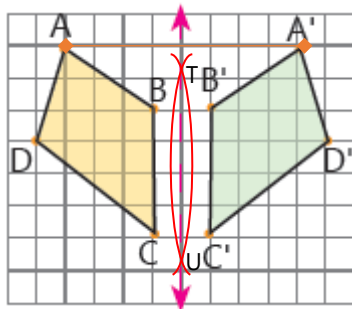
Follow the steps to construct the line of reflection.

Step I: Choose any vertex of the given rhombus $ABCD$ and join it with the corresponding vertex of the reflected image $A'B'C'D'$ using a straight line. Here, we join A and A' .

Step II: Open the compass with more than half the measure of the line segment AA' and draw an arc on the top and bottom of AA' , placing the compass pointer at A .

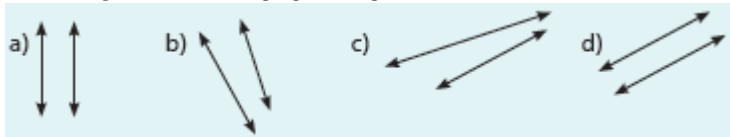
Step III: Similarly, draw an arc on the top and bottom of AA' , placing the compass pointer at A' which intersects the previous arcs at points T and U , respectively.

Step IV: Join T to U . The perpendicular bisector TU is the required line of reflection.



8.2 : Let's Practice

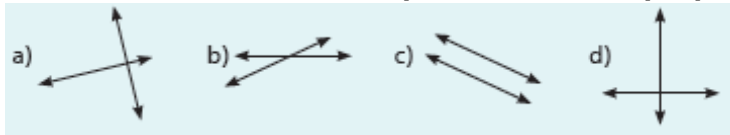
1. Identify and tick (✓) the parallel lines.



Solution:

- a) Parallel lines because parallel lines never intersect at any point.
- b) Non-parallel lines because non-parallel lines at a single point.
- c) Non-parallel lines because non-parallel lines at a single point.
- d) Parallel lines because parallel lines never intersect at any point.

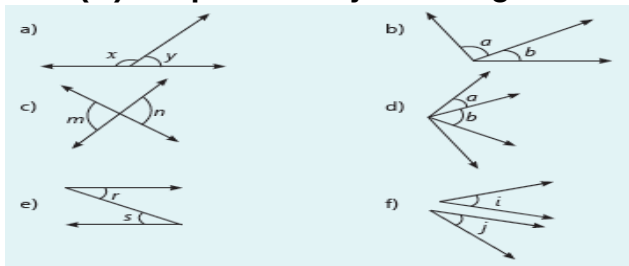
2. Determine which of these pairs of lines is perpendicular.



Solution:

- a) These lines are perpendicular.
- b) These lines are not perpendicular.
- c) These lines are not perpendicular.
- d) These lines are perpendicular.

3. Tick (✓) the pairs of adjacent angles.



Solution:

- These are pairs of adjacent angles because non-parallel lines meet at a common point.
- These are pairs of adjacent angles because non-parallel lines meet at a common point.
- These are not pairs of adjacent angles because non-parallel lines do not meet at a common point.
- These are not pairs of adjacent angles because non-parallel lines do not meet at a common point.
- These are not pairs of adjacent angles because non-parallel lines do not meet at a common point.
- These are not pairs of adjacent angles because non-parallel lines do not meet at a common point.

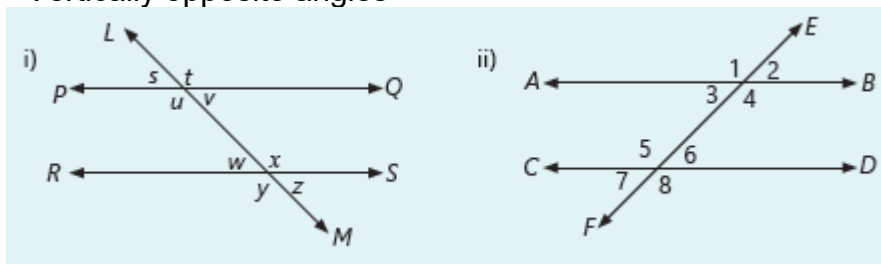
8.3 : Let's Practice

- 1. Two parallel lines are intersected by a transversal, as shown in the given figures. Observe these figures and write answers to the following questions:**

a) Write the names of the parallel lines and the transversal.

b) Write the names of the pairs of:

- Corresponding angles
- Alternate interior and exterior angles
- Vertically opposite angles



i) a) $\overleftrightarrow{PQ} \parallel \overleftrightarrow{RS}$ and $\overleftrightarrow{LM}$ is a transversal line

b) Corresponding angles:

In the given figure, the following corresponding angles are equal:

$\angle s$ and $\angle w$, $\angle t$ and $\angle x$, $\angle u$ and $\angle y$, $\angle v$ and $\angle z$

Alternate interior angles:

In the given figure, the following alternate interior angles are equal:

$\angle u = \angle x$ and $\angle v = \angle z$

Alternate exterior angles:

In the given figure, the following alternate exterior angles are equal:

$\angle s = \angle z$ and $\angle t = \angle y$

ii) a) $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ and $\overleftrightarrow{EF}$ is a transversal line

b) Corresponding angles:

$\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 3$ and $\angle 7$, $\angle 4$ and $\angle 8$

In the given figure, the following corresponding angles are equal:

Alternate interior angles:

In the given figure, the following alternate interior angles are equal:

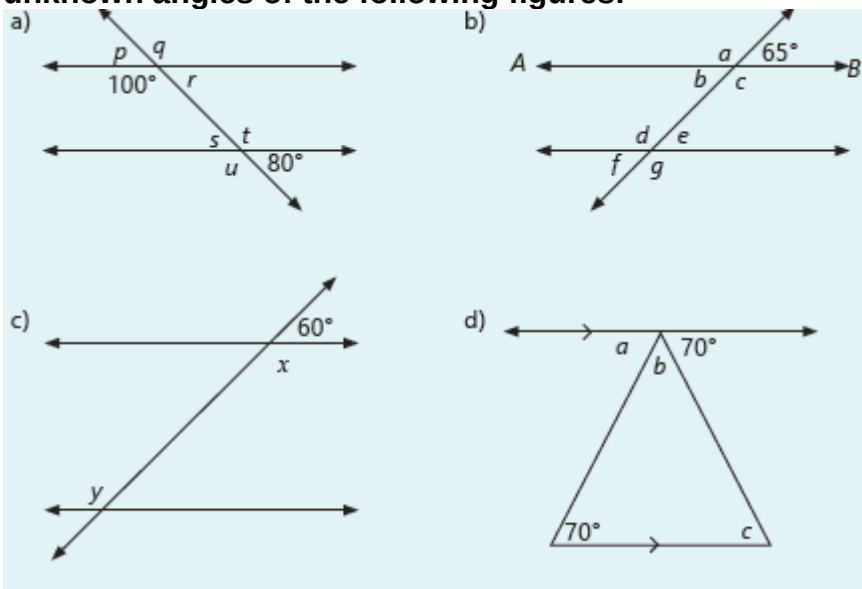
$$\angle 3 = \angle 6 \text{ and } \angle 4 = \angle 5$$

Alternate exterior angles:

In the given figure, the following alternate exterior angles are equal:

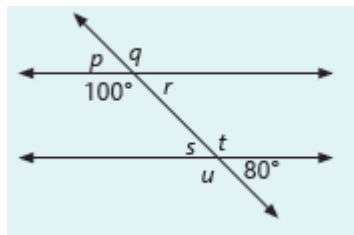
$$\angle 1 = \angle 8 \text{ and } \angle 2 = \angle 7$$

2. Using the properties of parallel lines and a transversal, find the unknown angles of the following figures:



Solution:

a)



Since, vertical opposite angles are equal. So,

$$\angle q = 60^\circ \text{ and } \angle s = 80^\circ$$

As angles in a straight line are supplementary. Thus

$$\angle p + \angle q = 180^\circ$$

$$\angle p + 60^\circ = 180^\circ$$

$$\angle p = 180^\circ - 60^\circ$$

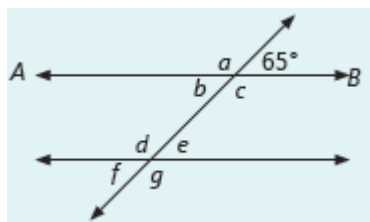
$$\angle p = 120^\circ$$

As corresponding angles are equal. So,

$$\angle q = \angle t \Rightarrow \angle t = 60^\circ$$

$$\angle r = 80^\circ \text{ and } \angle u = 100^\circ$$

b)



As angles in a straight line are equal. So,

$$\angle a + 65^\circ = 180^\circ$$

$$\angle a = 180^\circ - 65^\circ$$

$$\angle a = 115^\circ$$

As corresponding angles are equal. So

$$\angle a = \angle d \Rightarrow \angle d = 115^\circ$$

$$\angle e = 65^\circ$$

Also, vertical opposite angles are equal. So,

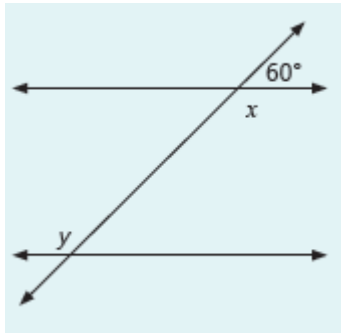
$$\angle b = 65^\circ$$

$$\angle c = \angle a \Rightarrow \angle c = 115^\circ$$

$$\angle d = \angle g \Rightarrow \angle g = 115^\circ$$

$$\angle e = \angle f \Rightarrow \angle f = 65^\circ$$

c)



As angles in a same straight line are supplementary. So,

$$\angle x + 60^\circ = 180^\circ$$

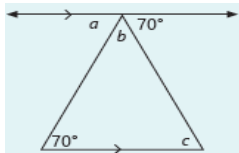
$$\angle x = 180^\circ - 60^\circ$$

$$\angle x = 120^\circ$$

As alternate interior angles are equal. So,

$$\angle x = \angle y \Rightarrow \angle y = 120^\circ$$

d)



In parallel lines, alternate angles are equal. So,

$$\angle a = 70^\circ \text{ and } \angle c = 70^\circ$$

Since,

Sum of angles of triangle = 180°

$$\angle b + \angle c + 70^\circ = 180^\circ$$

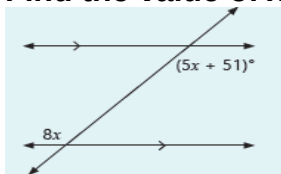
$$\angle b + 70^\circ + 70^\circ = 180^\circ$$

$$\angle b + 140^\circ = 180^\circ$$

$$\angle b = 180^\circ - 140^\circ$$

$$\angle b = 40^\circ$$

3. Find the value of x.



Solution:

Since, alternate interior angles are equal. So,

$$(5x + 51)^\circ = 8x$$

$$5^0x + 51^0 = 8^0x$$

$$51^0 = 8^0x - 5^0x$$

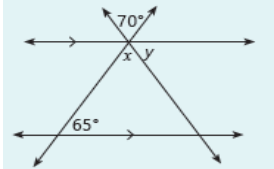
$$\Rightarrow 3^0x = 51^0$$

Dividing 3^0 on both sides

$$\frac{3^0x}{3^0} = \frac{51^0}{3^0}$$

$$x = 17^0$$

4. Find the value of x and y .



Solution:

Since, vertical opposite angles are equal. So,

$$\angle x = 70^0$$

As angles in a straight line are supplementary. So,

$$\angle x + \angle y + 65^0 = 180^0$$

$$70^0 + \angle y + 65^0 = 180^0$$

$$\angle y + 135^0 = 180^0$$

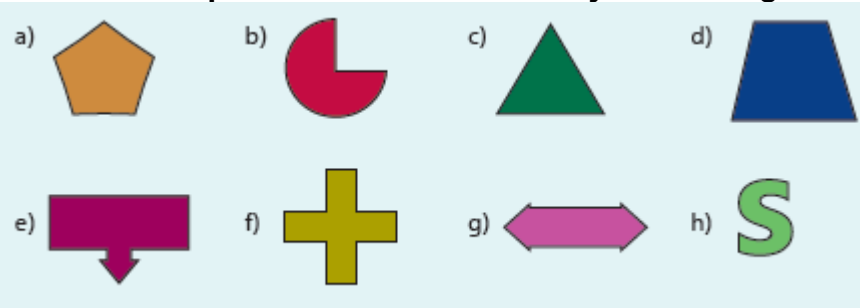
$$\angle y = 180^0 - 135^0$$

$$\angle y = 45^0$$

8.4 : Let's Practice

1. Tick (✓) the figures that have rotational symmetry. Write the order of rotation

and mark the point of rotation for the symmetric figures.



Solution:

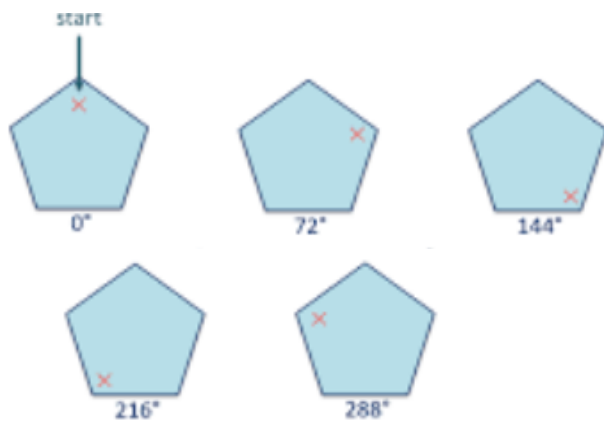
a)



We can see that the shape looks exactly the same five times in full rotation.

This

shows that it has a rotational symmetry of order 5 because they can be rotated 72^0 around their centre to match their original shapes. The point of rotation is the centre of the pentagon. Now observe the figures below.



b)



Here, the shape looks exactly the same only once in full rotation. This shows that it doesn't have any rotational symmetry.

Look at the image. Let's rotate it around its centre point.

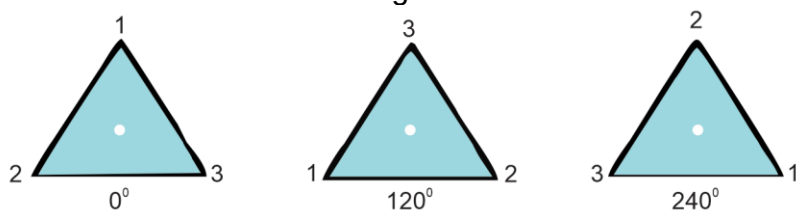


c)



We can see that the shape looks exactly the same three times in full rotation. This

shows that it has a rotational symmetry of order 3 because it can be rotated 120° around its centroid to match its original shape. The point of rotation is the centroid. Now observe the figures below.



d)



Here, the shape looks exactly the same only once in full rotation. This shows that it doesn't have any rotational symmetry.

Look at the image. Let's rotate it around its centre point.

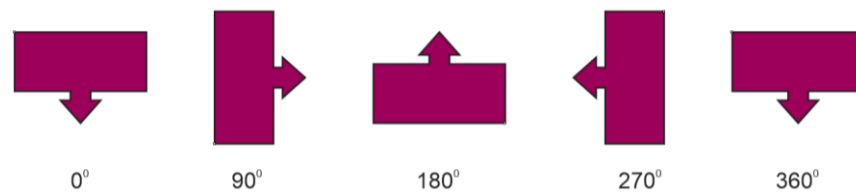


e)



Here, the shape looks exactly the same only once in full rotation. This shows that it doesn't have any rotational symmetry.

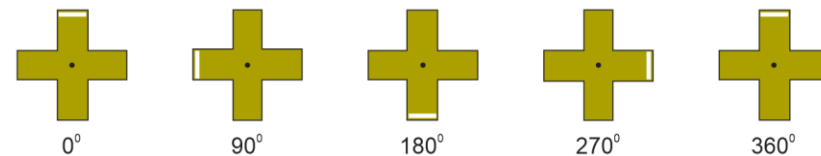
Look at the image. Let's rotate it around its centre point.



f)



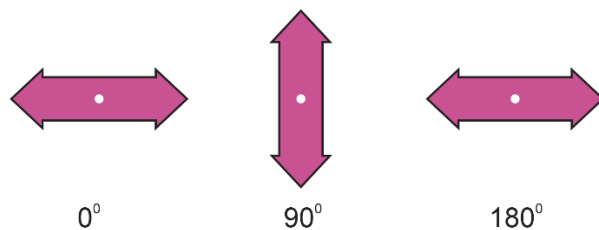
We can see that the shape looks exactly the same four times in full rotation. This shows that it has a rotational symmetry of order 4 because it can be rotated 90° around its centre to match its original shape. Now observe the figures below.



g)



We can see that the shape looks exactly the same two times in full rotation. This shows that it has a rotational symmetry of order 2 because it can be rotated 90° around its centre to match its original shape. Now observe the figures below.



h)



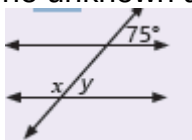
We can see that the shape looks exactly the same two times in full rotation. This shows that it has a rotational symmetry of order 2 because it can be rotated 90° around its centre to match its original shape. Now observe the figures below.



Unit Evaluation 8

1. Choose the correct option.

- The lines that intersect each other at 90° are called:
 - parallel lines
 - adjacent lines**
 - perpendicular lines
 - horizontal lines
- The sum of angles on a straight line is:
 - 90°
 - 180°**
 - 270°
 - 360°
- The angles having a common arm, a common vertex and lying on the opposite side of the common arm are called:
 - Vertically opposite
 - Adjacent**
 - Alternate
 - Corresponding
- When two lines cross each other, how many pairs of vertically opposite angles are formed?
 - 1
 - 2**
 - 3
 - 4
- The unknown angles (x and y) in the given figure are:



- 75° and 75°
- 75° and 105°**
- 105° and 105°
- 75° and 95°

2. What is the difference between parallel lines, perpendicular lines and a transversal? Give examples.

Parallel lines:

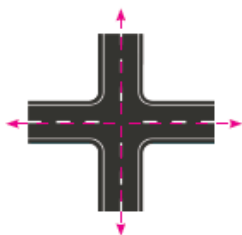
Two lines are said to be parallel if they do not intersect when extended in both directions.

Look at the railway track. We can see that the distance between the two lines of the track remains the same.



Perpendicular lines:

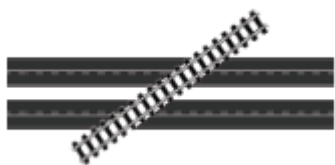
The lines intersecting each other at 90° are known as perpendicular lines. Look at the crossroads. The two straight roads are meeting at 90° . These lines represent perpendicular lines. Perpendicular lines cross each other at 90° .



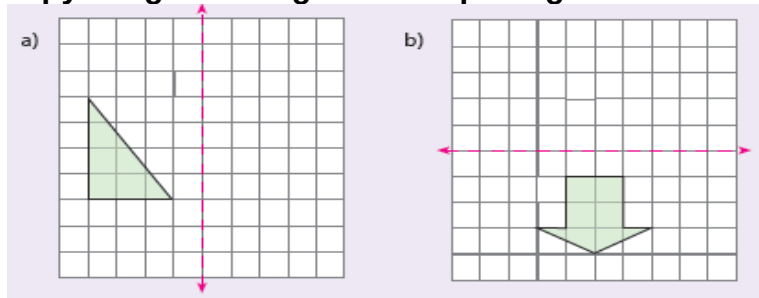
Transversal:

A transversal is a line that passes through two or more given lines, rays or segments at distinct points.

Look at the picture showing two parallel roads and a railway track. The railway track is crossing the parallel roads. This track is an example of a transversal. A transversal crosses two or more lines.



3. Copy the given images on a square grid and reflect them.



Solution:

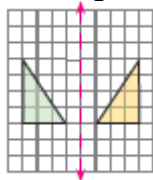
a)

Name the points (vertices) of the given shape for convenience.

Choose the closest point (from the line of reflection) of the given image and mark it on the opposite side of the line at exactly the same distance as the original image.

For A, mark A' which is the same distance from the line of reflection. Similarly, mark all the other points on the other side.

Join A' to B' , B' to C' , C' to A' . This is the required reflected image of the given rectangle.



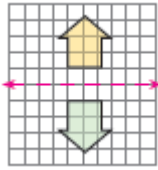
b)

Name the points (vertices) of the given shape for convenience.

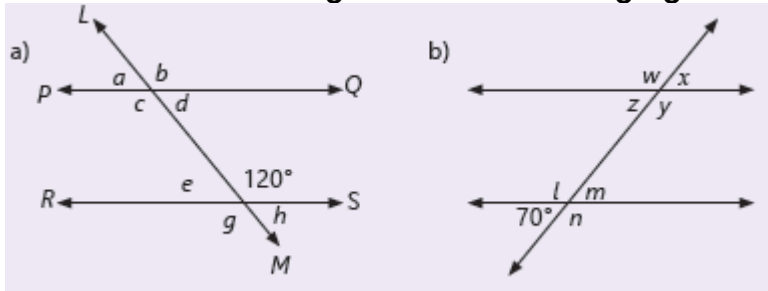
Choose the closest point (from the line of reflection) of the given image and mark it on the opposite side of the line at exactly the same distance as the original image.

For A, mark A' which is the same distance from the line of reflection. Similarly, mark all the other points on the other side.

Join A' to B' , B' to C' , C' to D' , D' to E' and E' to A' . This is the required reflected image of the given rectangle.

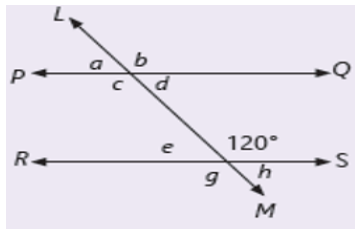


4. Find the unknown angles of the following figures.



Solution:

a)



As angles in a straight line are supplementary. Thus

$$\angle e + 120^\circ = 180^\circ$$

$$\angle e = 180^\circ - 120^\circ$$

$$\angle e = 60^\circ$$

Since, vertical opposite angles are equal. So,

$$\angle g = 120^\circ \text{ and } \angle e = \angle h \Rightarrow \angle h = 60^\circ$$

As corresponding angles are equal. So,

$$\angle a = \angle e \Rightarrow \angle a = 60^\circ$$

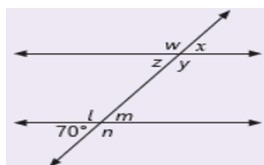
$$\angle b = 120^\circ$$

$$\angle c = \angle g \Rightarrow \angle c = 120^\circ$$

and

$$\angle d = \angle h \Rightarrow \angle d = 60^\circ$$

b)



As angles in a straight line are supplementary. Thus

$$\angle n + 70^\circ = 180^\circ$$

$$\angle n = 180^\circ - 70^\circ$$

$$\angle n = 110^\circ$$

Since, vertical opposite angles are equal. So,

$$\angle m = 70^\circ \text{ and } \angle l = \angle n \Rightarrow \angle l = 110^\circ$$

As corresponding angles are equal. So,

$$\angle w = \angle l \Rightarrow \angle w = 110^\circ$$

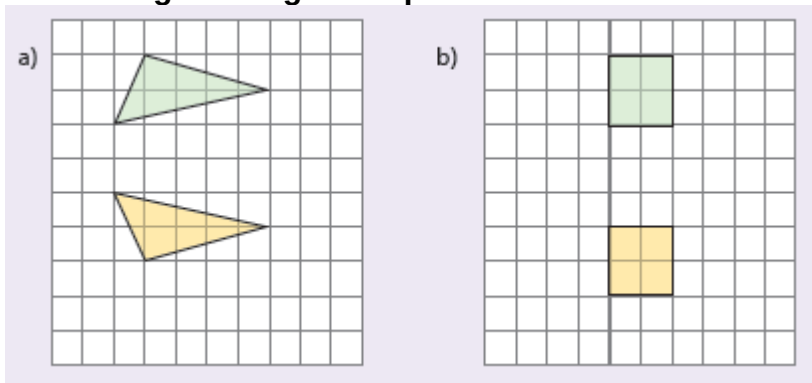
$$\angle x = \angle m \Rightarrow \angle x = 70^\circ$$

$$\angle y = \angle n \Rightarrow \angle y = 110^\circ$$

And

$$\angle z = 70^\circ$$

5. Copy the given figures on a square grid. Then draw the lines of reflection for these images using a compass.



Solution:

a)

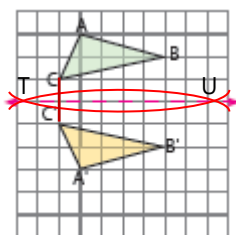
Follow the steps to construct the line of reflection.

Step I: Choose any vertex of the given rhombus ABC and join it with the corresponding vertex of the reflected image $A'B'C'$ using a straight line. Here, we join C and C' .

Step II: Open the compass with more than half the measure of the line segment AA' and draw an arc on the top and bottom of CC' , placing the compass pointer at A .

Step III: Similarly, draw an arc on the top and bottom of CC' , placing the compass pointer at C' which intersects the previous arcs at points T and U , respectively.

Step IV: Join T to U . The perpendicular bisector TU is the required line of reflection.



b)

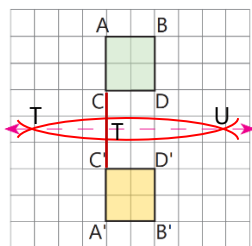
Follow the steps to construct the line of reflection.

Step I: Choose any vertex of the given rhombus $ABCD$ and join it with the corresponding vertex of the reflected image $A'B'C'D'$ using a straight line. Here, we join C and C' .

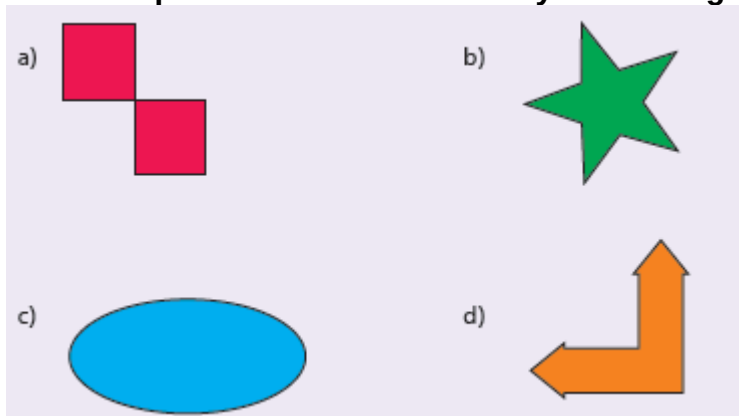
Step II: Open the compass with more than half the measure of the line segment AA' and draw an arc on the top and bottom of CC' , placing the compass pointer at A .

Step III: Similarly, draw an arc on the top and bottom of CC' , placing the compass pointer at C' which intersects the previous arcs at points T and U , respectively.

Step IV: Join T to U . The perpendicular bisector TU is the required line of reflection.



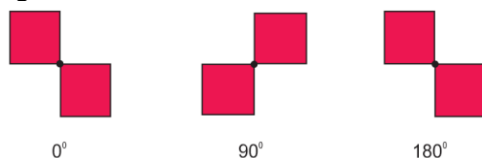
6. Tick (✓) the figures that have rotational symmetry. Write the order of rotation and mark the point of rotation for the symmetric figures.



Solution:

We can see that the shape looks exactly the same two times in full rotation. This

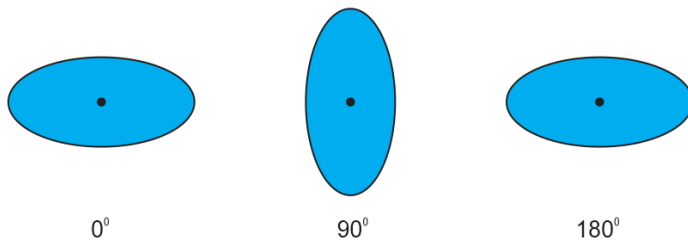
a) shows that it has a rotational symmetry of order 2 because it can be rotated 90° around its centre to match its original shape. Now observe the figures below.



b) We can see that the shape looks exactly the same five times in full rotation. This shows that it has a rotational symmetry of order 5. Now observe the figures below.

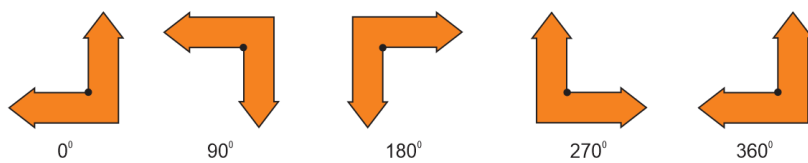


c) We can see that the shape looks exactly the same two times in full rotation. This shows that it has a rotational symmetry of order 2. Now observe the figures below.



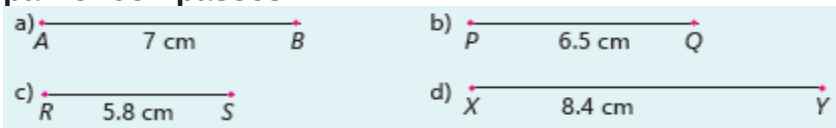
d) Here, the shape looks exactly the same only once in full rotation. This shows that it doesn't have any rotational symmetry.

Look at the image. Let's rotate it around its centre point.



9.1: Let's Practice

1. Draw the perpendicular bisectors of the following line segments using a pair of compasses.



Solution:

a) $\overline{AB} = 7\text{ cm}$

Steps of Construction:

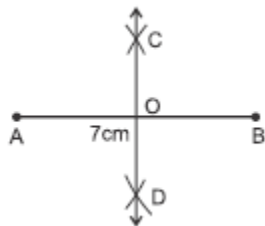
Step I: With the pointer of the compass at point A , open the compass (more than half the measure of the line segment $\overline{AB} = 7\text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step III: Repeat step I and II by placing the pointer of the compass at point B . The arcs will intersect the previous arcs at point C and D , respectively.

Step IV: Join points C to D with a line drawn using ruler that bisects the line segment $\overline{AB}$ at point O .

$\overleftrightarrow{CD}$ is the required perpendicular bisector of $\overline{AB}$.



b) $\overline{PQ} = 6.5\text{ cm}$

Steps of Construction:

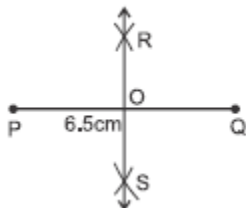
Step I: With the pointer of the compass at point P , open the compass (more than half the measure of the line segment $\overline{PQ} = 6.5\text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step III: Repeat step I and II by placing the pointer of the compass at point Q . The arcs will intersect the previous arcs at point R and S , respectively.

Step IV: Join points R to S with a line drawn using ruler that bisects the line segment $\overline{PQ}$ at point O .

$\overleftrightarrow{RS}$ is the required perpendicular bisector of $\overline{PQ}$.



c) $\overline{RS} = 5.8\text{ cm}$

Steps of Construction:

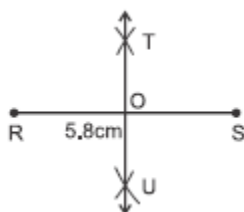
Step I: With the pointer of the compass at point R , open the compass (more than half the measure of the line segment $\overline{RS} = 5.8 \text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step III: Repeat step I and II by placing the pointer of the compass at point S . The arcs will intersect the previous arcs at point T and U , respectively.

Step IV: Join points T to U with a line drawn using ruler that bisects the line segment $\overline{RS}$ at point O .

$\overleftrightarrow{TU}$ is the required perpendicular bisector of $\overline{RS}$.



d) $\overline{XY} = 8.4 \text{ cm}$

Steps of Construction:

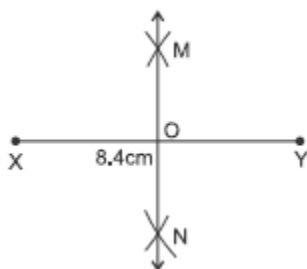
Step I: With the pointer of the compass at point X , open the compass (more than half the measure of the line segment $\overline{XY} = 8.4 \text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step III: Repeat step I and II by placing the pointer of the compass at point Y . The arcs will intersect the previous arcs at point M and N , respectively.

Step IV: Join points M to N with a line drawn using ruler that bisects the line segment $\overline{XY}$ at point O .

$\overleftrightarrow{MN}$ is the required perpendicular bisector of $\overline{XY}$.



2. Draw the following line segments and bisect them.

a) 8 cm

b) 6.8 cm

c) 7.5 cm

d) 5.6 cm

Solution:

a) 8 cm

Steps of Construction:

Step I: Draw a line segment $\overline{PQ} = 8 \text{ cm}$.

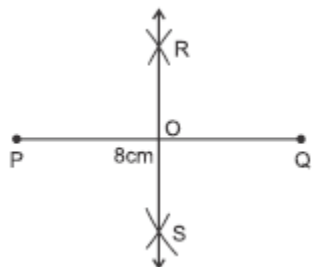
Step II: With the pointer of the compass at point P , open the compass (more than half the measure of the line segment $\overline{PQ}$).

Step III: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step IV: Repeat step I and II by placing the pointer of the compass at point Q . The arcs will intersect the previous arcs at point R and S , respectively.

Step V: Join points R to S with a line drawn using ruler that bisects the line segment $\overline{PQ}$ at point O .

$\overleftrightarrow{RS}$ is the required perpendicular bisector of $\overline{PQ}$.



b) 6.8 cm

Steps of Construction:

Step I: Draw a line segment $\overline{PQ} = 6.8$ cm.

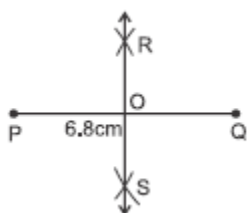
Step II: With the pointer of the compass at point P , open the compass (more than half the measure of the line segment $\overline{PQ}$).

Step III: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step IV: Repeat step I and II by placing the pointer of the compass at point Q . The arcs will intersect the previous arcs at point R and S , respectively.

Step V: Join points R to S with a line drawn using ruler that bisects the line segment $\overline{PQ}$ at point O .

$\overleftrightarrow{RS}$ is the required perpendicular bisector of $\overline{PQ}$.



c) 7.5 cm

Steps of Construction:

Step I: Draw a line segment $\overline{PQ} = 7.5$ cm.

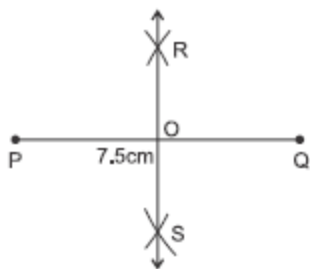
Step II: With the pointer of the compass at point P , open the compass (more than half the measure of the line segment $\overline{PQ}$).

Step III: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step IV: Repeat step I and II by placing the pointer of the compass at point Q . The arcs will intersect the previous arcs at point R and S , respectively.

Step V: Join points R to S with a line drawn using ruler that bisects the line segment $\overline{PQ}$ at point O .

$\overleftrightarrow{RS}$ is the required perpendicular bisector of $\overline{PQ}$.



d) 5.6 cm

Steps of Construction:

Step I: Draw a line segment $\overline{PQ} = 5.6$ cm.

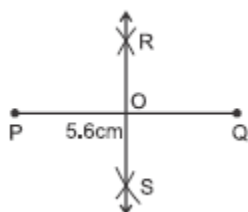
Step II: With the pointer of the compass at point P , open the compass (more than half the measure of the line segment $\overline{PQ}$).

Step III: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step IV: Repeat step I and II by placing the pointer of the compass at point Q . The arcs will intersect the previous arcs at point R and S , respectively.

Step V: Join points R to S with a line drawn using ruler that bisects the line segment $\overline{PQ}$ at point O .

$\overleftrightarrow{RS}$ is the required perpendicular bisector of $\overline{PQ}$.



3. Draw perpendiculars to the lines from the given points on them by using a compass.



Solution:

a)

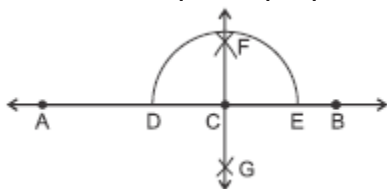
Steps of Construction:

Step I: With C as the centre, draw an arc that cuts AB at D and E .

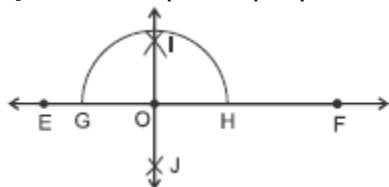
Step II: With D and E as centres, draw arcs with equal radii that cut each other at F .

Step III: Join F to C and extend it to point G .

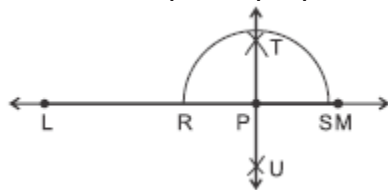
$\overleftrightarrow{FG}$ is the required perpendicular line on AB at point C .



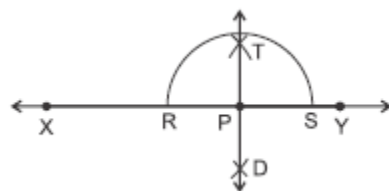
b)

Steps of Construction:**Step I:** With O as the centre, draw an arc that cuts EF at G and H .**Step II:** With G and H as centres, draw arcs with equal radii that cut each other at I .**Step III:** Join I to O and extend it to point J . $\overleftrightarrow{IJ}$ is the required perpendicular line on EF at point O .

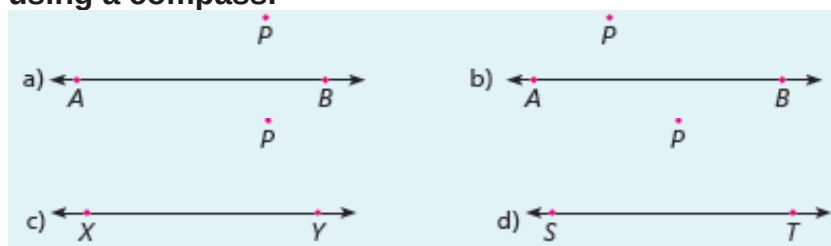
c)

Steps of Construction:**Step I:** With P as the centre, draw an arc that cuts LM at R and S .**Step II:** With R and S as centres, draw arcs with equal radii that cut each other at T .**Step III:** Join T to P and extend it to point U . $\overleftrightarrow{TU}$ is the required perpendicular line on LM at point P .

d)

Steps of Construction:**Step I:** With P as the centre, draw an arc that cuts XY at R and S .**Step II:** With R and S as centres, draw arcs with equal radii that cut each other at T .**Step III:** Join T to P and extend it to point U . $\overleftrightarrow{TU}$ is the required perpendicular line on XY at point P .

4. Draw perpendiculars to the given lines from the given point not on them using a compass.

**Solution:**

a)

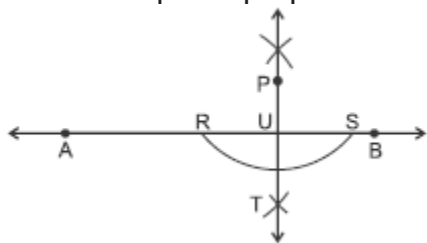
Steps of Construction:

Step I: With P as the centre, draw an arc that cuts $\overleftrightarrow{AB}$ at points R and S , as shown in the figure.

Step II: With R and S as centres, draw arcs with equal radii that cut each other at point T .

Step III: Join P to T , which cuts $\overleftrightarrow{AB}$ at U .

$\overleftrightarrow{PT}$ is the required perpendicular from point P at $\overleftrightarrow{AB}$.



b)

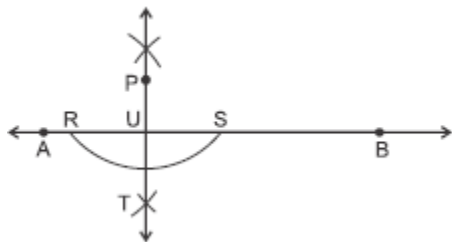
Steps of Construction:

Step I: With P as the centre, draw an arc that cuts $\overleftrightarrow{AB}$ at points R and S , as shown in the figure.

Step II: With R and S as centres, draw arcs with equal radii that cut each other at point T .

Step III: Join P to T , which cuts $\overleftrightarrow{AB}$ at U .

$\overleftrightarrow{PT}$ is the required perpendicular from point P at $\overleftrightarrow{AB}$.



c)

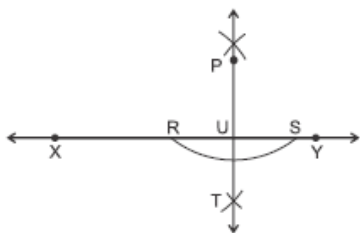
Steps of Construction:

Step I: With P as the centre, draw an arc that cuts $\overleftrightarrow{XY}$ at points R and S , as shown in the figure.

Step II: With R and S as centres, draw arcs with equal radii that cut each other at point T .

Step III: Join P to T , which cuts $\overleftrightarrow{XY}$ at U .

$\overleftrightarrow{PT}$ is the required perpendicular from point P at $\overleftrightarrow{XY}$.



d)

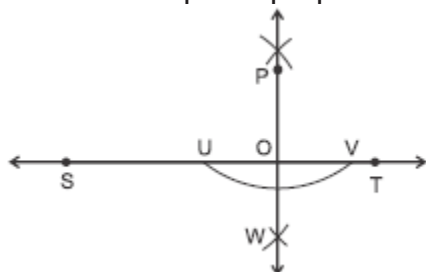
Steps of Construction:

Step I: With P as the centre, draw an arc that cuts $\overleftrightarrow{ST}$ at points U and V , as shown in the figure.

Step II: With U and V as centres, draw arcs with equal radii that cut each other at point W .

Step III: Join P to W , which cuts $\overleftrightarrow{ST}$ at O .

$\overleftrightarrow{PW}$ is the required perpendicular from point P at $\overleftrightarrow{ST}$.



5. Draw a line $\overleftrightarrow{RS}$ and draw a perpendicular on it from a point P not on it.

Solution:

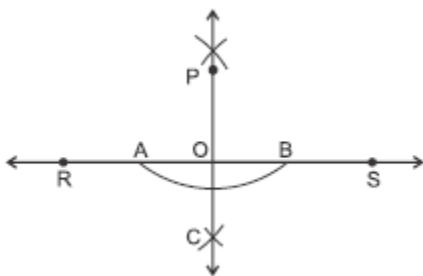
Steps of Construction:

Step I: With P as the centre, draw an arc that cuts $\overleftrightarrow{RS}$ at points A and B , as shown in the figure.

Step II: With A and B as centres, draw arcs with equal radii that cut each other at point C .

Step III: Join P to C , which cuts $\overleftrightarrow{RS}$ at O .

$\overleftrightarrow{PC}$ is the required perpendicular from point P at $\overleftrightarrow{RS}$.



6. Draw a line $\overleftrightarrow{ST}$ and draw a perpendicular from point M on it.

Solution:

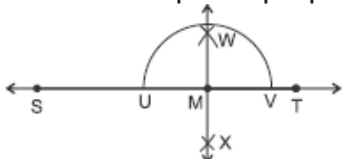
Steps of Construction:

Step I: With M as the centre, draw an arc that cuts $\overleftrightarrow{ST}$ at U and V .

Step II: With U and V as centres, draw arcs with equal radii that cut each other at W .

Step III: Join W to M and extend it to point X .

$\overleftrightarrow{WX}$ is the required perpendicular line on $\overleftrightarrow{ST}$ at point M .



7. Draw a line segment $\overline{AB} = 6.8$ cm. Draw a perpendicular from point P not on it.

Solution:

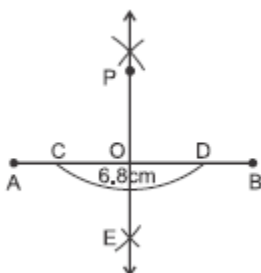
Steps of Construction:

Step I: With P as the centre, draw an arc that cuts $\overline{AB} = 6.8$ at points C and D , as shown in the figure.

Step II: With C and D as centres, draw arcs with equal radii that cut each other at point E .

Step III: Join P to E , which cuts $\overleftrightarrow{AB}$ at O .

$\overleftrightarrow{PE}$ is the required perpendicular from point P at $\overleftrightarrow{AB}$.



9.2: Let's Practice

1. Construct the following angles by using a protractor and bisect them by using a pair of compasses.

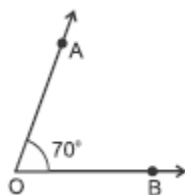
- a) 70° b) 85° c) 110° d) 130° e) 160°

Solution:

- a) 70°

Steps of Construction:

- Draw a ray $\overrightarrow{OB}$.
- Place the protractor such that the center of the protractor is at point O and the baseline of the protractor is aligned with ray $\overrightarrow{OB}$.
- Mark a point A at the 70° mark on the protractor scale.
- Use a compass to bisect the angle $\angle AOB$ by drawing an arc.
- Label the angle $\angle AOB = 70^\circ$.

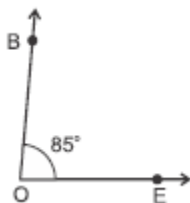


- b) 85°

Steps of Construction:

- Draw a ray $\overrightarrow{OE}$.
- Place the protractor such that the center of the protractor is at point O and the baseline of the protractor is aligned with ray $\overrightarrow{OE}$.
- Mark a point B at the 85° mark on the protractor scale.

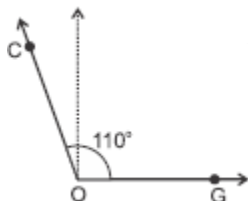
- Use a compass to bisect the angle $\angle BOF$ by drawing an arc.
- Label the angle $\angle BOE = 85^\circ$.



c) 110°

Steps of Construction:

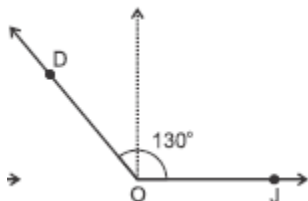
- Draw a ray $\overrightarrow{OG}$.
- Place the protractor such that the center of the protractor is at point O and the baseline of the protractor is aligned with ray $\overrightarrow{OG}$.
- Mark a point C at the 110° mark on the protractor scale.
- Use a compass to bisect the angle $\angle COH$ by drawing an arc.
- Label the angle $\angle COG = 110^\circ$.



d) 130°

Steps of Construction:

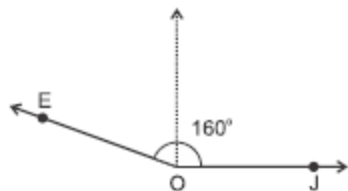
- Draw a ray $\overrightarrow{OH}$.
- Place the protractor such that the center of the protractor is at point O and the baseline of the protractor is aligned with ray $\overrightarrow{OH}$.
- Mark a point D at the 130° mark on the protractor scale.
- Use a compass to bisect the angle $\angle DOJ$ by drawing an arc.
- Label the angle $\angle DOG = 85^\circ$.



e) 160°

Steps of Construction:

- Draw a ray $\overrightarrow{OJ}$.
- Place the protractor such that the center of the protractor is at point O and the baseline of the protractor is aligned with ray $\overrightarrow{OJ}$.
- Mark a point E at the 160° mark on the protractor scale.
- Use a compass to bisect the angle $\angle EOJ$ by drawing an arc from points E and L.
- Label the angle $\angle EOJ = 160^\circ$.



2. Construct the following angles by using a pair of compasses. Then bisect them.

- a) 30° b) 45° c) 60° d) 75° e) 90°
 f) 105° g) 120°

Solution:

a) 30°

Steps of Construction:

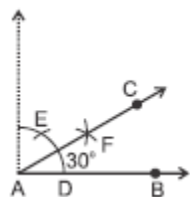
Step I: Draw a ray $\overrightarrow{AB}$ and construct an angle of 90° at A.

Step II: With A as the centre, draw an arc of a suitable radius that cuts $\overrightarrow{AB}$ at D.

Step III: Using the same opening of compasses, with D and E as the centre, Draw an arc that cuts the previous arc at point F.

Step IV: Draw $\overrightarrow{AC}$ passing through F.

$m\angle BAC = 30^\circ$ which is the required angle.



b) 45°

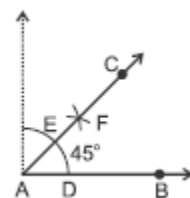
Steps of Construction:

Step I: Draw a ray $\overrightarrow{AB}$ and construct an angle of 90° at A.

Step II: Using the same opening of compasses, with D and E as centres, draw arcs that cut each other at point F.

Step III: Draw $\overrightarrow{AC}$ passing through F.

$m\angle BAC = 45^\circ$ which is the required angle.



c) 60°

Steps of Construction:

Steps of Construction:

Step I: Draw a ray $\overrightarrow{AB}$.

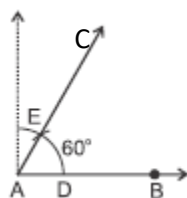
Step II: With A as the centre, draw an arc of a suitable radius that cuts $\overrightarrow{AB}$ at D.

Step III: Using the same opening of compasses, with D as the centre, draw an arc

that cuts the previous arc at point E .

Step IV: Draw $\overrightarrow{AB}$ passing through E .

$m\angle BAC = 60^\circ$ which is the required angle.



d) 75°

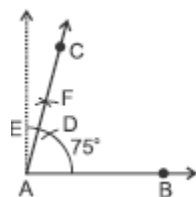
Steps of Construction:

Step I: Draw a ray $\overrightarrow{AB}$ and construct an angle of 90° at A .

Step II: Using the same opening of compasses, with D and E as centres, draw arcs that cut each other at point F .

Step III: Draw $\overrightarrow{AB}$ passing through F .

$m\angle BAC = 75^\circ$ which is the required angle.



e) 90°

Steps of Construction:

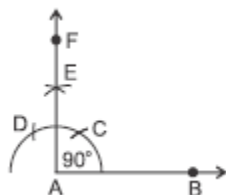
Step I: Draw $\overrightarrow{AB}$.

Step II: With A as the centre, draw an arc that cuts AB at point C .

Step III: Using the same opening of compasses, with C as the centre, draw an arc that cuts the previous arc at point D . Similarly, with D as the centre, draw an arc that cuts the same arc at point E .

Step IV: Draw $\overrightarrow{AF}$ passing through point E .

$m\angle BAF = 90^\circ$ which is the required angle.



f) 105°

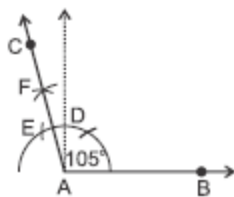
Steps of Construction:

Step I: Draw a ray $\overrightarrow{AB}$ and construct an angle of 90° at A .

Step II: Using the same opening of compasses, with D and E as centres, draw two arcs that cut each other at point F .

Step III: Draw $\overrightarrow{AC}$ passing through F .

$m\angle BAC = 105^\circ$ which is the required angle.

g) 120° **Steps of Construction:****Step I:** Draw $\overrightarrow{AB}$.**Step II:** With A as the centre, draw an arc of a suitable radius that cuts $\overrightarrow{AB}$ at D.**Step III:** Using the same opening of compasses, with D as the centre, draw an arc that cuts the previous arc at E. Similarly, with E as the centre, draw an arc that cuts the arc at F.**Step IV:** Draw ray $\overrightarrow{AC}$ passing through F. $m\angle BAC = 120^\circ$ which is the required angle.**Unit Evaluation 9****1. Choose the correct Option.**

- a) A perpendicular bisector is a line that divides any given line segment into two equal parts making an angle of:
 i) 60° **ii) 90°** iii) 110° iv) 180°
- b) Bisecting of an angle means to divide the angle into _____.
 i) one equal part **ii) two equal parts**
 iii) three equal parts iv) four equal parts
- c) A perpendicular bisector passes through the:
 i) end point of a line segment **ii) midpoint of a line segment**
 iii) starting point of a line segment iv) outside of a line segment
- d) If we bisect a 60° angle, we get two _____ angles.
 i) 15° **ii) 30°** iii) 45° iv) 60°
- e) Which of these angles cannot be constructed using a compass?
 i) 60° ii) 75° **iii) 95°** iv) 135°

2. Draw the perpendicular bisectors of the following line segments using a pair of compasses.

- | | |
|----|----|
| a) | b) |
| c) | d) |

Solution:

e) a) $\overline{FG} = 4 \text{ cm}$

Steps of Construction:

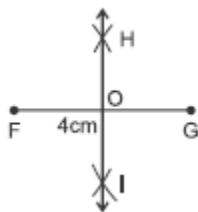
Step I: With the pointer of the compass at point F , open the compass (more than half the measure of the line segment $\overline{FG} = 4 \text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step III: Repeat step I and II by placing the pointer of the compass at point G . The arcs will intersect the previous arcs at point H and I , respectively.

Step IV: Join points H to I with a line drawn using ruler that bisects the line segment $\overline{FG}$ at point O .

$\overleftrightarrow{HI}$ is the required perpendicular bisector of $\overline{FG}$.



b) $\overline{WX} = 6.9 \text{ cm}$

Steps of Construction:

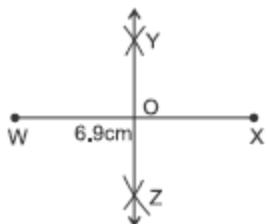
Step I: With the pointer of the compass at point W , open the compass (more than half the measure of the line segment $\overline{WX} = 6.9 \text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step III: Repeat step I and II by placing the pointer of the compass at point X . The arcs will intersect the previous arcs at point Y and Z , respectively.

Step IV: Join points Y to Z with a line drawn using ruler that bisects the line segment $\overline{WX}$ at point O .

$\overleftrightarrow{YZ}$ is the required perpendicular bisector of $\overline{WX}$.



c) $\overline{RS} = 8.6 \text{ cm}$

Steps of Construction:

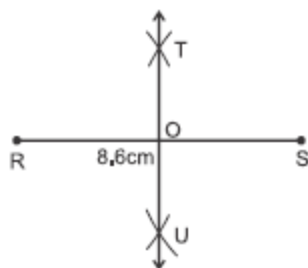
Step I: With the pointer of the compass at point R , open the compass (more than half the measure of the line segment $\overline{RS} = 8.6 \text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

Step III: Repeat step I and II by placing the pointer of the compass at point S . The arcs will intersect the previous arcs at point T and U , respectively.

Step IV: Join points T to U with a line drawn using ruler that bisects the line segment $\overline{RS}$ at point O .

$\overleftrightarrow{TU}$ is the required perpendicular bisector of $\overline{RS}$.



d) $\overline{XY} = 3.5 \text{ cm}$

Steps of Construction:

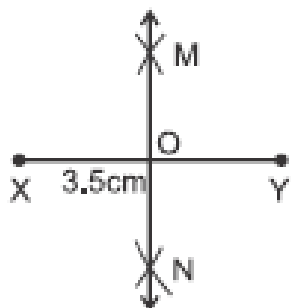
Step I: With the pointer of the compass at point X, open the compass (more than half the measure of the line segment $\overline{XY} = 3.5 \text{ cm}$).

Step II: Draw two arcs on the top and bottom, respectively, as shown in the figure.

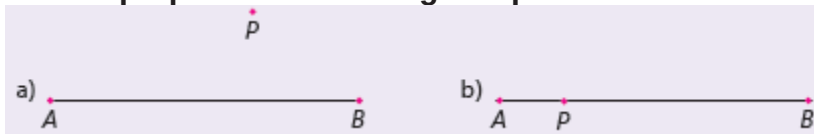
Step III: Repeat step I and II by placing the pointer of the compass at point Y. The arcs will intersect the previous arcs at point M and N, respectively.

Step IV: Join points M to N with a line drawn using ruler that bisects the line segment $\overline{XY}$ at point O.

$\overleftrightarrow{MN}$ is the required perpendicular bisector of $\overline{XY}$.



3. Draw a perpendicular from given point P to the line segment AB.



a)

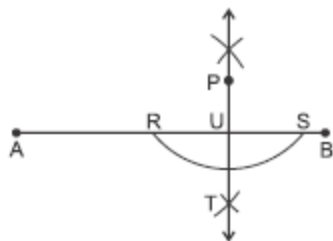
Steps of Construction:

Step I: With P as the centre, draw an arc that cuts $\overline{AB}$ at points R and S, as shown in the figure.

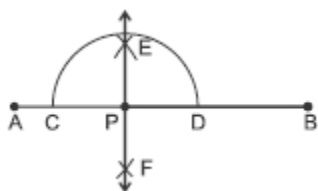
Step II: With R and S as centres, draw arcs with equal radii that cut each other at point T.

Step III: Join P to T, which cuts $\overline{AB}$ at U.

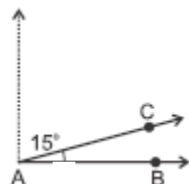
$\overleftrightarrow{PT}$ is the required perpendicular from point P at $\overline{AB}$.



b)

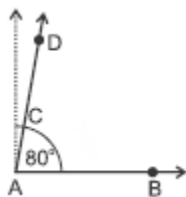
Steps of Construction:**Step I:** With P as the centre, draw an arc that cuts $\overline{AB}$ at C and D.**Step II:** With C and D as centres, draw arcs with equal radii that cut each other at E.**Step III:** Join E to P and extend it to point F. $\overleftrightarrow{EF}$ is the required perpendicular line on $\overline{AB}$ at point P.**4. Construct the following angles by using a protractor and bisect them by using a compass.**a) 15° b) 80° c) 125° **Solution:**a) 15°

- Draw a ray $\overrightarrow{AB}$.
- Place the protractor with its center at point A and its baseline along ray $\overrightarrow{AB}$.
- Mark a point C on the protractor where the 15° mark intersects the arc.
- Without changing the protractor's position, place the compass's tip at point A and set its width to point C.
- Draw an arc intersecting ray AB at point C.
- Draw a ray $\overrightarrow{AC}$ starting from point A. This ray makes a 15° angle with ray $\overrightarrow{AB}$.

b) 80°

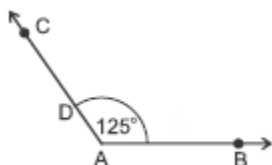
- Draw a ray $\overrightarrow{AB}$.
- Place the protractor with its center at point A and its baseline along ray $\overrightarrow{AB}$.
- Mark a point C on the protractor where the 80° mark intersects the arc.
- Without changing the protractor's position, place the compass's tip at point A and set its width to point C.
- Draw an arc intersecting ray $\overrightarrow{AB}$ at point C.

- Draw a ray $\overrightarrow{CD}$ starting from point C and passing through point A. This ray makes an 80° angle with ray $\overrightarrow{AB}$.



c) 125°

- Draw a ray $\overrightarrow{AB}$.
- Place the protractor with its center at point A and its baseline along ray $\overrightarrow{AB}$.
- Mark a point C on the protractor where the 125° mark intersects the arc.
- Without changing the protractor's position, place the compass's tip at point A and set its width to point C.
- Draw an arc intersecting ray $\overrightarrow{AB}$ at point D.
- Draw a ray $\overrightarrow{CD}$ starting from point C and passing through point A. This ray makes a 125° angle with ray $\overrightarrow{AB}$.



5. Construct the following angles by using a pair of compasses. Then bisect them.

a) 60°

b) 120°

c) 75°

Solution:

a) 60°

Steps of Construction:

Steps of Construction:

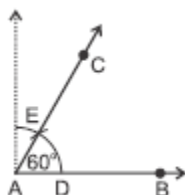
Step I: Draw a ray $\overrightarrow{AB}$.

Step II: With A as the centre, draw an arc of a suitable radius that cuts $\overrightarrow{AB}$ at D.

Step III: Using the same opening of compasses, with D as the centre, draw an arc that cuts the previous arc at point E.

Step IV: Draw $\overrightarrow{AE}$ passing through E.

$m\angle BAC = 60^\circ$ which is the required angle.



b) 120°

Steps of Construction:

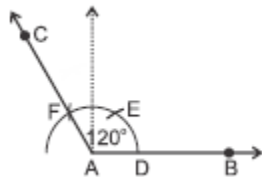
Step I: Draw $\overrightarrow{AB}$.

Step II: With A as the centre, draw an arc of a suitable radius that cuts $\overrightarrow{AB}$ at D .

Step III: Using the same opening of compasses, with D as the centre, draw an arc that cuts the previous arc at E . Similarly, with E as the centre, draw an arc that cuts the arc at F .

Step IV: Draw ray $\overrightarrow{AC}$ passing through F .

$m\angle BAC = 120^\circ$ which is the required angle.



c) 75°

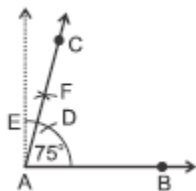
Steps of Construction:

Step I: Draw a ray $\overrightarrow{AB}$ and construct an angle of 90° at A .

Step II: Using the same opening of compasses, with D and E as centres, draw arcs that cut each other at point F .

Step III: Draw $\overrightarrow{AC}$ passing through F .

$m\angle BAC = 75^\circ$ which is the required angle.



10.1: Let's Practice

1. The following data shows the populations in cities X, Y and Z for three years. Draw a vertical multiple bar graph for this data using an appropriate scale.

City	Population in Thousands		
	Year 1	Year 2	Year 3
X	50	85	120
Y	60	80	100
Z	65	90	90

Solution:

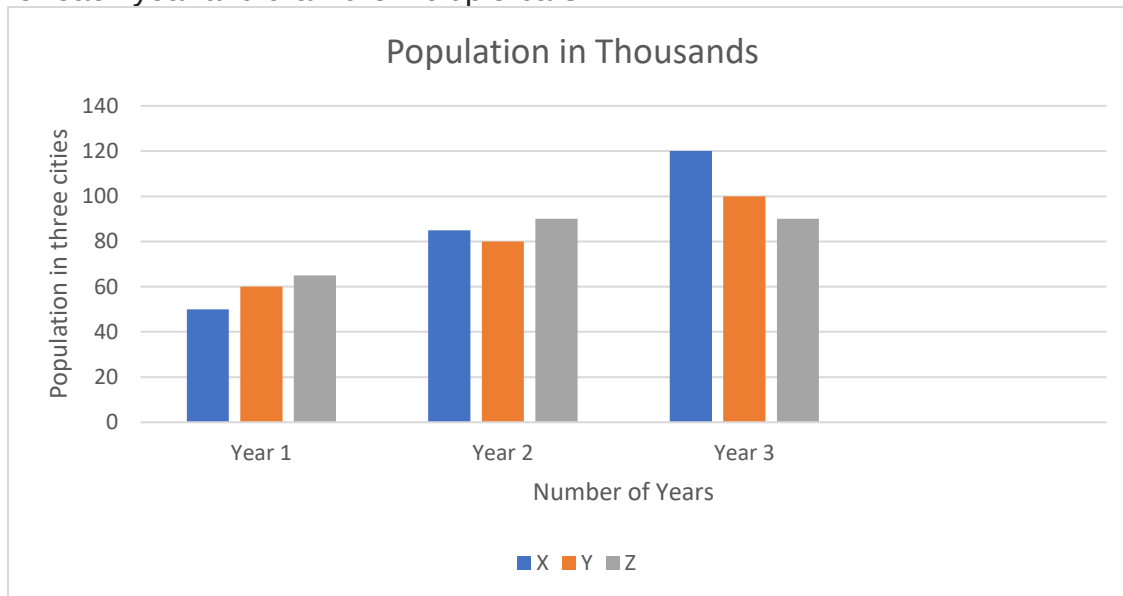
We use the following steps to draw the vertical multiple bar graph.

Step I: Draw the x-axis (horizontal line) and the y-axis (vertical line) perpendicular to each other and write the number of years along the x-axis and the populations in three cities along the y-axis.

Step II: Choose cities for each category (X, Y, Z) and set an appropriate scale. Here, one mark represents 1 city along the x-axis.

Step III: For year 1, the population in city X is 50, in city Y it is 60, and in city Z it is 65. So, we will color up to the marks of 50, 60, and 65 for X, Y, and Z, respectively.

Step IV: Similarly, keep following the steps, colour the correct numbers of marks for each year and draw the multiple bars.



2. The following data shows the survey of diabetes in males and females in the rural areas of Pakistan for 4 years. Draw a horizontal multiple bar graph for this data using an appropriate scale.

Patients	Patients in percentage			
	2019	2020	2021	2022
Males	16	23	12	18
Females	12	28	25	17

Solution:

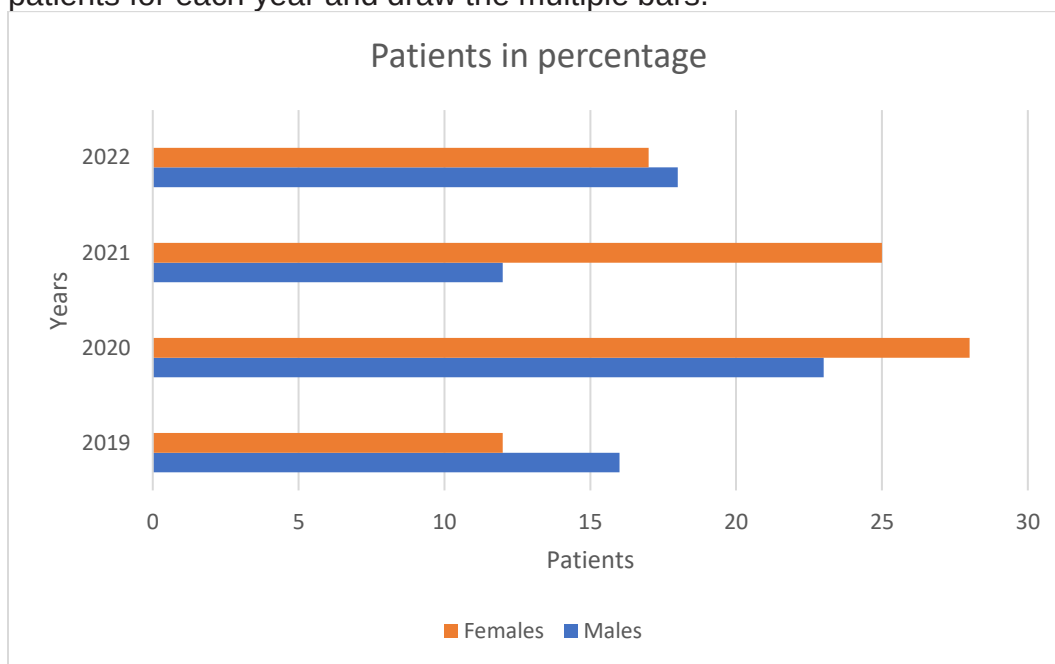
We use the following steps to draw the multiple bar graph.

Step I: Draw the x-axis (horizontal line) and the y-axis (vertical line) perpendicular to each other and write the number of patients along the x-axis and the years along the y-axis.

Step II: Choose a colour for each category (male and female) and set an appropriate scale. Here, one mark represents 1 patient along the x-axis.

Step III: For 2019, male patients are 16 and female patients are 12. So, we will colour upto the mark of 16 and 12 for males and Females respectively.

Step IV: Similarly, keep following the steps, colour the correct numbers of patients for each year and draw the multiple bars.



3. The following data shows the marks of five pupils who took exams in math and science. Draw a vertical multiple bar graph for this data using an appropriate scale.

Subject	Zainab	Yasmeen	Nida	Wajiha	Maria
Math	23	17	21	12	18
Science	20	15	17	19	14

Solution:

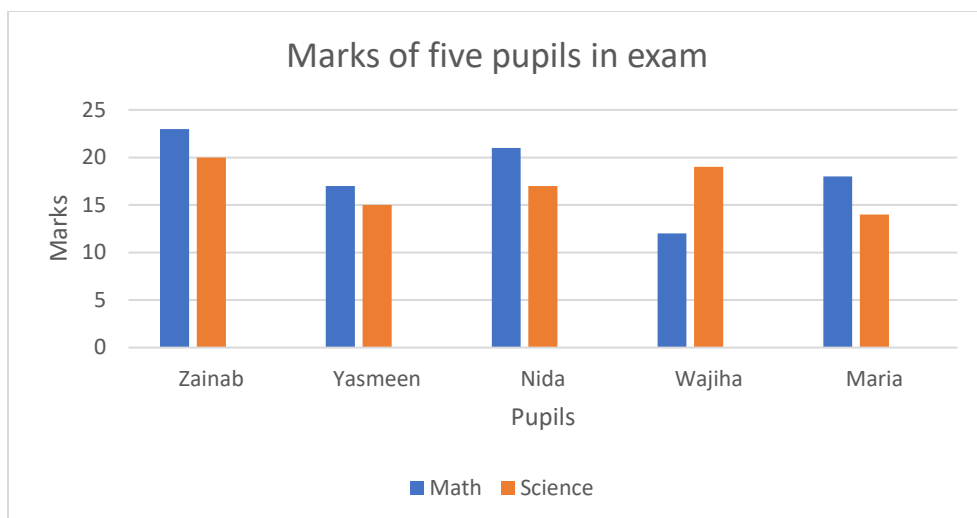
We use the following steps to draw the vertical multiple bar graph.

Step I: Draw the x-axis (horizontal line) and the y-axis (vertical line) perpendicular to each other and write the pupils along the x-axis and the marks along the y-axis.

Step II: Choose a colour for each category (math and science) and set an appropriate scale. Here, one mark represents 1 pupil along the x-axis.

Step III: For Zainab, the the marks in math is 23 and the marks in science is 20. So, we will color up to the marks of 23 and 20 for math and science, respectively.

Step IV: Similarly, keep following the steps, colour the correct marks for each pupil and draw the multiple bars.



4. The following data shows the price (per litre) of petrol and diesel for six months. Draw a horizontal multiple bar graph for this data using an appropriate scale.

Month	April	May	June	July	August	September
Petrol	110	108	110	112	119	123
Diesel	113	110	112	113	116	120

Solution:

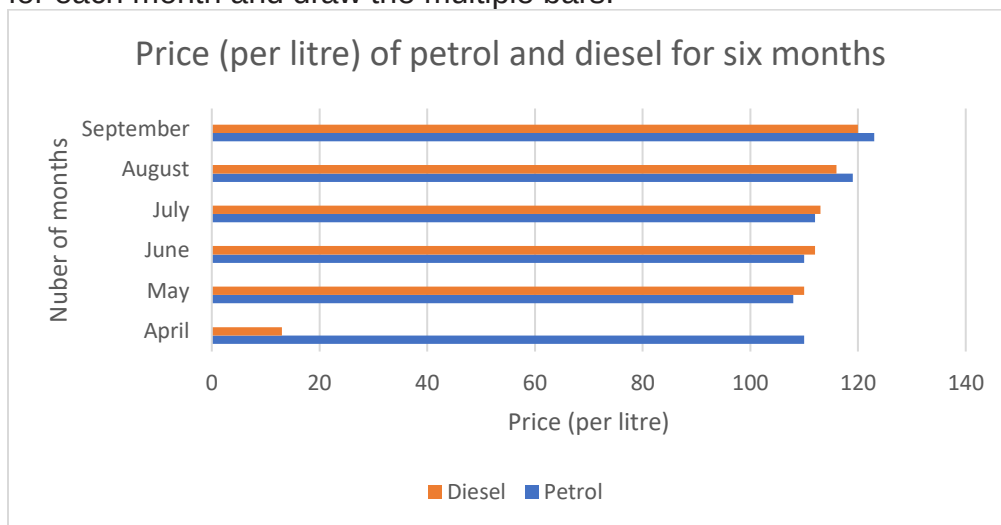
We use the following steps to draw the multiple bar graph.

Step I: Draw the x-axis (horizontal line) and the y-axis (vertical line) perpendicular to each other and write the price (per litre) along the x-axis and the months along the y-axis.

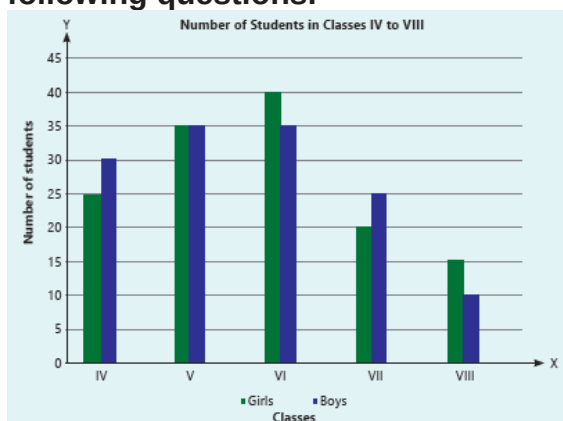
Step II: Choose a colour for each category (petrol and diesel) and set an appropriate scale. Here, one mark represents 1 litre along the x-axis.

Step III: On April, price of petrol per litre is Rs.110 and price of diesel per litre is Rs.113. So, we will colour upto the price of 110 and 113 per litre for petrol and diesel, respectively.

Step IV: Similarly, keep following the steps, colour the price of petrol and diesel for each month and draw the multiple bars.



5. Observe the given multiple bar graph in vertical form and answer the following questions.

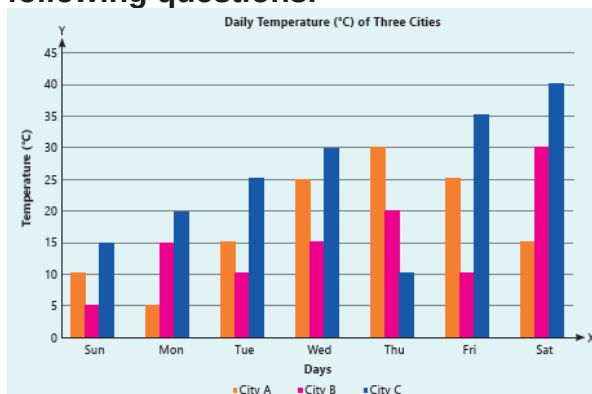


- What is the title of the graph?
- What information is shown on the x-axis?
- What information is shown on the y-axis?
- How many students are in class VI?
- In which class are the least number of students present?
- In which class are the greatest number of students present?
- In which class is the same number of boys and girls present?
- What is the total number of students in classes VII and VIII?
- What is the difference between the number of students in class IV and class VI?

Solution:

- The title of the graph is the number of students in classes IV and VIII.
- The information shown on the x-axis is boys and girls.
- The information shown on the y-axis is the number of students.
- The number of students in class VI is 75.
- The least number of students are present in class VIII.
- The greatest number of students are present in class VI.
- The same number of boys and girls are present in class V.
- The total number of students in classes VII and VIII is 70.
- The difference between the number of students in class IV and class VI is 20.

6. Observe the given multiple bar graph in vertical form and answer the following questions.



- What is the title of the graph?
- What information is shown on the x-axis?
- What information is shown on the y-axis?
- What is the temperature of City A on Tuesday?
- What is the temperature of City C on Wednesday?
- On which day is the temperature of City B very low?
- On which day is the temperature of City A high?
- What is the total temperature of City C for Tuesday and Wednesday?
- What is the difference between the temperature of City C on Monday and Friday?

Solution:

- The title of the graph is the daily temperature ($^{\circ}\text{C}$) of three cities.
- The information shown on the x-axis is the days.
- The information shown on the y-axis is the city temperature ($^{\circ}\text{C}$).
- The temperature of City A on Tuesday is 15°C .
- The temperature of City C on Wednesday is 30°C .
- The temperature of City B is very low on Sunday.
- The temperature of City A is very high on Thursday.
- The total temperature of City C for Tuesday and Wednesday is 55°C .
- The difference between the temperature of City C on Monday and Friday is 15°C .

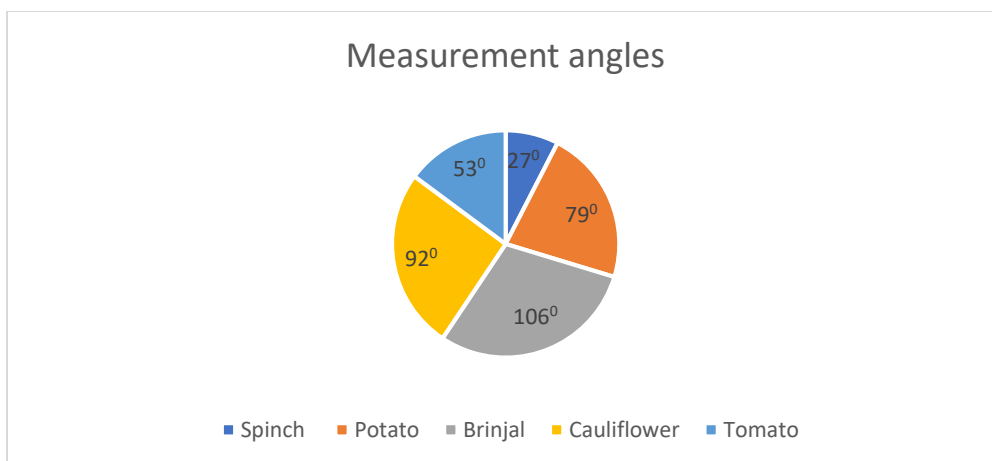
10.2: Let's Practice

- The following data shows the amount Sadia paid for purchasing different vegetables. Draw a pie graph for the following data.

Vegetable	Price
Spinach	105
Potato	300
Brinjal	400
Cauliflower	350
Tomato	200

Solution:

Vegetables	Price	Measurement of the central angle
Spinach	105	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^{\circ} = \frac{105}{1355} \times 360^{\circ} = 27^{\circ}$
Potato	300	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^{\circ} = \frac{300}{1355} \times 360^{\circ} = 79^{\circ}$
Brinjal	400	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^{\circ} = \frac{400}{1355} \times 360^{\circ} = 106^{\circ}$
Cauliflower	350	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^{\circ} = \frac{350}{1355} \times 360^{\circ} = 92^{\circ}$
Tomato	200	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^{\circ} = \frac{200}{1355} \times 360^{\circ} = 53^{\circ}$
Total	1355	

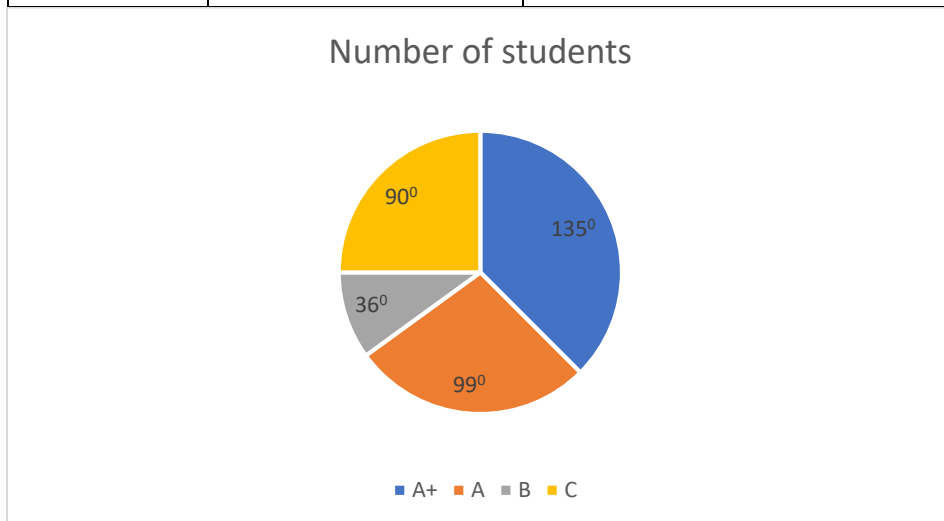


2. The given table shows the results of a class of 40 students. Draw a pie graph for the data.

Result	Number of Students
A+	15
A	11
B	4
C	10

Solution:

Result	Number of students	Measurement of the central angle
A ⁺	15	$\frac{A^+ \text{ grade students}}{\text{Total students}} \times 360^\circ = \frac{15}{40} \times 360^\circ = 135^\circ$
A	11	$\frac{A \text{ grade students}}{\text{Total students}} \times 360^\circ = \frac{11}{40} \times 360^\circ = 99^\circ$
B	4	$\frac{B \text{ grade students}}{\text{Total students}} \times 360^\circ = \frac{4}{40} \times 360^\circ = 36^\circ$
C	10	$\frac{C \text{ grade students}}{\text{Total students}} \times 360^\circ = \frac{10}{40} \times 360^\circ = 90^\circ$
Total	40	

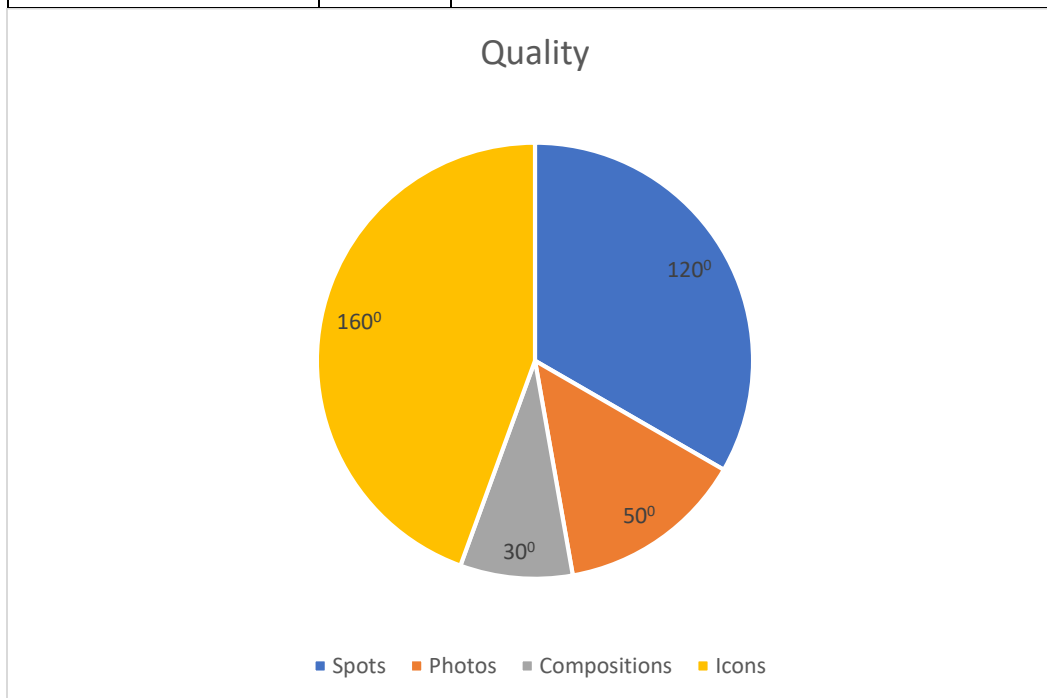


3. The number of different types of illustrations in a book is given below. Draw a pie graph for this data.

Type of Illustration	Quantity
spots	60
photos	25
compositions	15
icons	80

Solution:

Types of illustration	Quality	Measurement of the central angle
Spots	60	$\frac{\text{Number of spots}}{\text{Total illustrations}} \times 360^\circ = \frac{60}{180} \times 360^\circ = 120^\circ$
Photos	25	$\frac{\text{Number of spots}}{\text{Total illustrations}} \times 360^\circ = \frac{25}{180} \times 360^\circ = 50^\circ$
Compositions	15	$\frac{\text{Number of spots}}{\text{Total illustrations}} \times 360^\circ = \frac{15}{180} \times 360^\circ = 30^\circ$
Icons	80	$\frac{\text{Number of spots}}{\text{Total illustrations}} \times 360^\circ = \frac{80}{180} \times 360^\circ = 160^\circ$
Total	180	

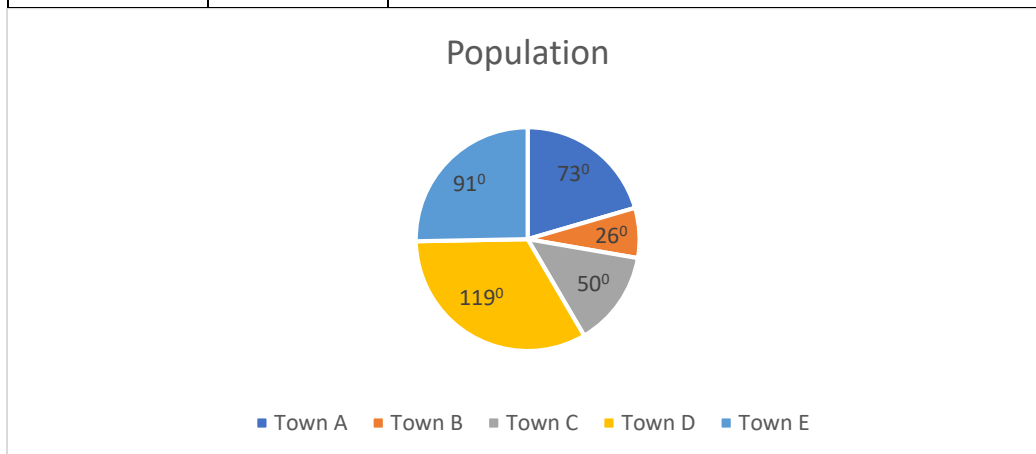


4. The population of five towns in a certain city is given below. Draw a pie graph for this data.

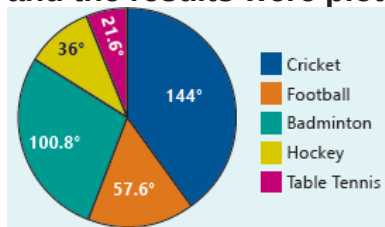
Town	Population
Town A	34000
Town B	12000
Town C	23000
Town D	55000
Town E	42000

Solution:

Town	Population	Measurement of the central angle
Town A	34000	$\frac{\text{Town A population}}{\text{Total population}} \times 360^\circ = \frac{34000}{166000} \times 360^\circ = 73^\circ$
Town B	12000	$\frac{\text{Town A population}}{\text{Total population}} \times 360^\circ = \frac{12000}{166000} \times 360^\circ = 26^\circ$
Town C	23000	$\frac{\text{Town A population}}{\text{Total population}} \times 360^\circ = \frac{23000}{166000} \times 360^\circ = 50^\circ$
Town D	55000	$\frac{\text{Town A population}}{\text{Total population}} \times 360^\circ = \frac{55000}{166000} \times 360^\circ = 119^\circ$
Town E	42000	$\frac{\text{Town A population}}{\text{Total population}} \times 360^\circ = \frac{42000}{166000} \times 360^\circ = 91^\circ$
Total	166000	



5. 100 Students of a school were asked which sport was their favourite to play and the results were plotted on the pie graph as follows.



Read the pie graph and answer the following questions:

- How many students like to play hockey?
- Which sport was the least favourite among students?
- Which sport was the most popular among students?
- How many students like table tennis and cricket altogether?

Solution:

- There are 10 students who like to play hockey. As
 Students who like to play hockey = $\frac{\text{Angle of the sector}}{360^\circ} \times \text{total students}$
 $= \frac{36^\circ}{360^\circ} \times 100 = 10 \text{ students}$

- Least favourite sport among students is table tennis.

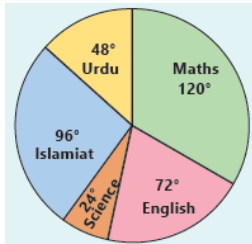
$$\begin{aligned}\text{Students who like to play table tennis} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{21.6^\circ}{360^\circ} \times 100 = 6 \text{ students}\end{aligned}$$

c) Most favourite sport among students is cricket.

$$\begin{aligned}\text{Students who like to play cricket} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{144^\circ}{360^\circ} \times 100 = 40 \text{ students}\end{aligned}$$

d) There are 46 students who like table tennis and cricket together.

6. The pie graph below shows the data of the favourite subjects 300 students of a school. Read the graph and answer the following questions.



- What is the ratio of students that liked Math to students that liked Islamiyat?
- What is the ratio of students that liked English to students that liked Urdu?
- What percentage of students liked English the most?
- Which subject was the least favourite among students?
- Which subject was the most popular among students?
- How many students liked Urdu and Islamiyat together?

Solution:

$$\begin{aligned}\text{a) Number of Math students} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{120^\circ}{360^\circ} \times 300 = 100\end{aligned}$$

$$\begin{aligned}\text{Number of Islamiyat students} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{96^\circ}{360^\circ} \times 300 = 80\end{aligned}$$

$$\text{Ratio} = 100 : 80 = \frac{100}{80} = \frac{5}{4} = 5 : 4$$

$$\begin{aligned}\text{b) Number of English students} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{72^\circ}{360^\circ} \times 300 = 60\end{aligned}$$

$$\begin{aligned}\text{Number of Urdu students} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{48^\circ}{360^\circ} \times 300 = 40\end{aligned}$$

$$\text{Ratio} = 60 : 40 = \frac{60}{40} = \frac{3}{2} = 3 : 2$$

$$\text{c) Percentage of English students} = \frac{60}{300} \times 100 = 20\%$$

d) Least favourite subject among students is science.

$$\begin{aligned}\text{Least favourite subject among students} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{24^\circ}{360^\circ} \times 300 = 20\end{aligned}$$

e) Most popular subject among students is mathematics.

$$\begin{aligned}\text{Most popular subject among students} &= \frac{\text{Angle of the sector}}{360^\circ} \times \text{total students} \\ &= \frac{120^\circ}{360^\circ} \times 300 = 100\end{aligned}$$

f) There are 120 students who like Urdu and Islamiyat together.

10.3: Let's Practice

1. The following typos errors were found after reviewing the first 30 pages of a book. Show this in the form of grouped data.

3, 2, 5, 3, 2, 1, 0, 5, 2, 2, 0, 2, 0, 3, 0, 2, 3, 3, 0, 4, 3, 4, 0, 1, 1, 1, 4, 3, 1, 4

Solution:

Arranging the given data in ascending order:

0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5

Minimum error = 0

Maximum error = 5

Errors	Tally marks	Frequency
0	I	6
1		5
2	I	6
3	II	7
4		4
5		2
Total		30

2. The given data shows the price of 20 stationery items. Show it in the form of grouped data.

Rs 40	Rs 45	Rs 10	Rs 15
Rs 40	Rs 15	Rs 20	Rs 20
Rs 25	Rs 10	Rs 35	Rs 20
Rs 20	Rs 40	Rs 30	Rs 30
Rs 30	Rs 35	Rs 45	Rs 25

Solution:

Arranging the given data in ascending order:

10, 10, 15, 15, 20, 20, 20, 20, 25, 25, 30, 30, 30, 35, 35, 40, 40, 40, 45, 45

Next, divide the weight of students into 4 groups having equal interval.

The groups are: 10 – 19, 20 – 29, 30 – 39, 40 – 49.

Now draw a table with four columns and complete it as given below.

Groups/Classes	Price of stationary items	Tally marks	Frequency
10 – 19	10, 10, 15, 15		4
20 – 29	20, 20, 20, 20, 25, 25	I	6
30 – 39	30, 30, 30, 35, 35		5
40 – 49	40, 40, 40, 45, 45		5
Total			20

3. The given data is the weight of class 6 students:

20 kg	24 kg	21 kg	18 kg	20 kg	24 kg
21 kg	18 kg	19 kg	24 kg	19 kg	21 kg
24 kg	20 kg	18 kg	18 kg	20 kg	24 kg
18 kg	24 kg	26 kg	21 kg	19 kg	26 kg

Show it in the form of grouped data. Also:

- Tell the number of students that have a weight less than 20 kg.
- Tell the number of students that have a weight greater than 21 kg.

Solution:

Arranged data in ascending order:

18 kg, 18 kg, 18 kg, 18 kg, 18 kg, 19 kg, 19 kg, 19 kg, 20 kg, 20 kg, 20 kg, 20 kg, 21 kg, 21 kg, 21 kg, 21 kg, 24 kg, 24 kg, 24 kg, 24 kg, 24 kg, 24 kg, 26 kg, 26 kg

Next, divide the weight of students into 5 groups having equal interval.

The groups are: 18 – 19, 20 – 21, 22 – 23, 24 – 25, 26 – 27.

Now draw a table with four columns and complete it as given below.

Groups/Classes	Price of stationary items	Tally marks	Frequency
18 – 19	18, 18, 18, 18, 18, 19, 19, 19	HHH III	8
20 – 21	20, 20, 20, 20, 21, 21, 21, 21	HHH III	8
22 – 23			0
24 – 25	24, 24, 24, 24, 24, 24	HHH I	6
26 – 27	26, 26	II	2
Total			24

- The number of students who have a weight less than 20 kg is 8.
- The number of students that have a weight greater than 21 kg is 8.

10.4: Let's Practice

1. Find the arithmetic mean of the following data.

- 52, 36, 58, 23, 84, 71
- 36, 67, 58, 79, 60, 97, 73, 27, 25
- 128, 146, 155, 135, 186

Solution:

- 52, 36, 58, 23, 84, 71

$$\text{Arithmetic mean} = \bar{X} = \frac{\sum x}{n}, \quad \text{Here, } n = 6$$

$$\bar{X} = \frac{52 + 36 + 58 + 23 + 84 + 71}{6} = \frac{324}{6} = 54$$

- 36, 67, 58, 79, 60, 97, 73, 27, 25

$$\text{Arithmetic mean} = \bar{X} = \frac{\sum x}{n}, \quad \text{Here, } n = 9$$

$$\bar{X} = \frac{36 + 67 + 58 + 79 + 60 + 97 + 73 + 27 + 25}{9} = \frac{522}{9} = 58$$

- 128, 146, 155, 135, 186

$$\text{Arithmetic mean} = \bar{X} = \frac{\sum x}{n}, \quad \text{Here, } n = 5$$

$$\bar{X} = \frac{128 + 146 + 155 + 135 + 186}{5} = \frac{750}{5} = 150$$

2. The heights of 5 students are 135 cm, 155 cm, 144 cm, 159 cm and 167 cm. Calculate the arithmetic mean of the heights of the students.

Solution:

135, 155, 144, 159, 167

Arithmetic mean = $\bar{X} = \frac{\sum x}{n}$, Here, $n = 5$

$$\bar{X} = \frac{135 + 155 + 144 + 159 + 167}{5} = \frac{760}{5} = 152$$

So, the arithmetic mean height of the 5 students is 152 cm.

3. The ages of 8 teachers of a school are 45, 23, 35, 59, 43, 52, 40, and 31. Calculate the arithmetic mean and median of the data.

Solution:

45, 23, 35, 59, 43, 52, 40, 31

For Arithmetic mean:Arithmetic mean = $\bar{X} = \frac{\sum x}{n}$, Here, $n = 8$

$$\bar{X} = \frac{45 + 23 + 35 + 59 + 43 + 52 + 40 + 31}{8} = \frac{328}{8} = 41$$

So, the arithmetic mean age of the teachers is 41.

For Median:

Arranging the ages in ascending order:

23, 31, 35, 40, 43, 45, 52, 59

Number of values = $n = 10$ (even)Median = arithmetic mean of $\left(\frac{n}{2}\right)$ th and $\left[\left(\frac{n}{2}\right) + 1\right]$ th observation= arithmetic mean of $\left(\frac{10}{2}\right)$ th and $\left[\left(\frac{10}{2}\right) + 1\right]$ th observation= arithmetic mean of the 4th and 5th observation

23, 31, 35, 40, 43, 45, 52, 59

4th observation = 40 and 5th observation = 43Arithmetic mean of 4th and 5th observation = $\frac{40+43}{2} = \frac{83}{2} = 41.5$

So, the median of the given data values is 41.5

4. Find the median of the following sets of observations.

a) 13, 6, 11, 8, 17

b) 167, 120, 148, 114, 180, 123, 137

c) 96, 81, 57, 68, 73, 27, 39, 56, 44

Solution:

a) 13, 6, 11, 8, 17

Arrange the given data in ascending order:

6, 8, 11, 13, 17

Number of values = $n = 5$ (odd)Median = $\left(\frac{n+1}{2}\right)$ th observation= $\left(\frac{5+1}{2}\right)$ th observation = $\left(\frac{6}{2}\right)$ th observation = 3rd observation

Median = 11

So, the median of the first data set is 11.

b) 167, 120, 148, 114, 180, 123, 137

Arrange the given data in ascending order:

114, 120, 123, 137, 148, 167, 180

Number of values = $n = 7$ (odd)Median = $\left(\frac{n+1}{2}\right)$ th observation

$$= \left(\frac{7+1}{2}\right)\text{th observation} = \left(\frac{8}{2}\right)\text{th observation} = 4^{\text{th}} \text{ observation}$$

Median = 137

So, the median of the first data set is 137.

- c) 96, 81, 57, 68, 73, 27, 39, 56, 44

Arrange the given data in ascending order:

27, 39, 44, 56, 57, 68, 73, 81, 96

Number of values = $n = 9$ (odd)

$$\text{Median} = \left(\frac{n+1}{2}\right)\text{th observation}$$

$$= \left(\frac{9+1}{2}\right)\text{th observation} = \left(\frac{10}{2}\right)\text{th observation} = 5^{\text{th}} \text{ observation}$$

Median = 57

So, the median of the first data set is 57.

5. Identify the mode of each of the following sets of data.

- a) 2, 8, 3, 5 2, 7, 9

- b) 158, 110, 116, 110, 132, 110, 113

- c) 357, 410, 548, 362, 630

Solution:

- a) 2, 8, 3, 5 2, 7, 9

The frequency of 2 is 2, which is the greatest frequency.

Hence, the mode = $\hat{X} = 2$.

- b) 158, 110, 116, 110, 132, 110, 113

The frequency of 10 is 3, which is the greatest frequency.

Hence, the mode = $\hat{X} = 110$.

- c) 357, 410, 548, 362, 630

Each value appears only once in this set, so there is no mode.

Hence, the mode = $\hat{X} = 0$.

10.5: Let's Practice

- 1. A box contains 10 pencils of different colours, including 3 red, 2 green, 2 blue, 1 yellow, and 2 black pencils. If a pencil is chosen at random, what is the probability of getting a green pencil?**

Solution:

Sample space = 3 red, 2 green, 2 blue, 1 yellow, and 2 black pencils

Total number of possible outcomes = $3 + 2 + 2 + 1 + 2 = 10$

Let G be the event that getting a green pencil.

$$\text{Probability of an event} = P(G) = \frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$$

$$\text{Probability of an event} = P(G) = \frac{2}{10} = \frac{1}{5}$$

So, the probability of getting a green pencil is $\frac{1}{5}$.

- 2. A 6-sided dice is rolled once. What is the probability of getting a multiple of 3?**

Solution:

Sample space is: 1, 2, 3, 4, 5, 6

Total number of possible outcomes = 6

Favourable outcomes are: 3, 6

Number of favourable outcomes = 2

Probability of an event = $P(E) = \frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

Probability of getting a multiple of 3 = $P(E) = \frac{2}{6} = \frac{1}{3}$

So, the probability of rolling a multiple of 3 on a 6-sided dice is $\frac{1}{3}$.

3. Ibrahim spun a spinner. What is the probability of each of the following outcomes?

- a) Stopping at yellow
c) Stopping at purple

- b) Stopping at red
d) Stopping at blue



Solution:

Total possible outcomes = 2 (yellow) + 2 (red) + 3 (purple) + 1 (blue) = 8

- a) Stopping at yellow

Number of favourable outcomes = 2

Probability of an event = $P(\text{yellow}) = \frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

Probability of stopping at yellow = $\frac{2}{8} = \frac{1}{4}$

So, probability of stopping at yellow is $\frac{1}{4}$

- b) Stopping at red

Number of favourable outcomes = 2

Probability of an event = $P(\text{red}) = \frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

Probability of stopping at red = $\frac{2}{8} = \frac{1}{4}$

So, probability of stopping at red is $\frac{1}{4}$

- c) Stopping at purple

Number of favourable outcomes = 3

Probability of an event = $P(\text{purple}) = \frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

Probability of stopping at purple = $\frac{3}{8}$

So, probability of stopping at purple is $\frac{3}{8}$

- d) Stopping at blue

Number of favourable outcomes = 1

Probability of an event = $P(\text{blue}) = \frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

Probability of stopping at blue = $\frac{1}{8}$

So, probability of stopping at blue is $\frac{1}{8}$

4. A bag contains 3 grey marbles, 5 black marbles, 3 blue marbles, 5 white marbles and 4 red marbles. A student picked one marble without looking. Find the probability of the following events

- a) P (blue marble) b) P (white marble) c) P (black marble)
d) P (grey marble) e) P (red marble)

Solution:

Total possible outcomes = 3 (grey) + 5 (black) + 3 (blue) + 5 (white) + 4 (red) = 20

- a) P (blue marble)

Number of favourable outcomes = 3

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{blue marble}) = \frac{3}{20}$$

So, probability of blue marble is $\frac{3}{20}$

b) P (white marble)

Number of favourable outcomes = 5

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{white marble}) = \frac{5}{20} = \frac{1}{4}$$

So, probability of white marble is $\frac{1}{4}$

c) P (black marble)

Number of favourable outcomes = 5

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{black marble}) = \frac{5}{20} = \frac{1}{4}$$

So, probability of black marble is $\frac{1}{4}$

d) P (grey marble)

Number of favourable outcomes = 3

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{grey marble}) = \frac{3}{20}$$

So, probability of grey marble is $\frac{3}{20}$

e) P (red marble)

Number of favourable outcomes = 4

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{red marble}) = \frac{4}{20} = \frac{1}{5}$$

So, probability of red marble is $\frac{1}{5}$

5. Look at the different coloured balls and find the probability:

a) P (blue balls)

b) P (green balls)

c) P (yellow balls)

d) P (black balls)



Solution:

Total number of possible outcomes = 3(blue) + 7(green) + 5(yellow) + 3(black) = 18

a) P (blue balls)

Number of favourable outcomes = 3

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{blue balls}) = \frac{3}{18} = \frac{1}{6}$$

So, probability of blue balls is $\frac{1}{6}$

b) P (green balls)

Number of favourable outcomes = 7

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{green balls}) = \frac{7}{18}$$

So, probability of green balls is $\frac{7}{18}$

c) P (yellow balls)

Number of favourable outcomes = 5

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{yellow balls}) = \frac{5}{18}$$

So, probability of yellow balls is $\frac{5}{18}$

d) P (black balls)

Number of favourable outcomes = 3

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$P(\text{black balls}) = \frac{3}{18} = \frac{1}{6}$$

So, probability of black balls is $\frac{1}{6}$

6. Tasbeeha picks a number disk randomly.



a) What is the sample space?

b) What is the probability that a randomly selected number disk will have a 7?

c) What is the probability that a randomly selected number disk will have a 3?

Solution:

a) sample space = 4, 7, 7, 8, 3, 8 = 6

b) Probability that a randomly selected number disk will have a 7.

Number of favourable outcomes = 2

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$\text{Probability of getting number disk 7} = \frac{2}{6} = \frac{1}{3}$$

So, Probability of getting number disk 7 is $\frac{1}{3}$.

d) Probability that a randomly selected number disk will have a 3?

Number of favourable outcomes = 2

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

$$\text{Probability of getting number disk 3} = \frac{1}{6}$$

So, Probability of getting number disk 3 is $\frac{1}{6}$.

Unit Evaluation 10

1. Choose the correct option.

a) The most frequent observation in a data set is called:

i) mean

ii) median

iii) **mode**

iv) chart

b) In a data set with an even number of values, the median is:

i) **The average of the two middle values**

ii) The middle value

iii) The value that occurs most frequently in the data set

- iv) The repeated value
- c) Which of these shows discrete data?
- i) **The number of computers in a room** ii) The weight of a glass
- iii) The length of a rope iv) The capacity of oil in a tank
- d) The mode of 35, 35, 56, 59, 67, 75, 87, 75, 89, 75 is:
- i) 35 ii) 59 **iii) 75** iv) 87
- e) The median of 8, 29, 18, 14, 6, 22, 12, 24, 36 is:
- i) 14 ii) 17 **iii) 18** iv) 22
- f) Any potential result of a random experiment is considered to be an:
- i) outcome** ii) event iii) trial iv) sample space

2. The following data shows the daily load-shedding duration in hours for 30 regions of the city. Show this in the form of grouped data and answer the following questions:

10, 2, 12, 5, 8, 7, 7, 6, 3, 8 10, 1, 7, 6, 11, 10, 12, 7, 6, 7, 9, 11, 7, 6, 9, 2, 7

- a) Find the least load-shedding hours.
- b) Find the most frequent load-shedding hours.

Solution:

Arranging the given data in ascending order:

1, 2, 2, 3, 5, 6, 6, 6, 6, 6, 7, 7, 7, 7, 7, 7, 8, 8, 9, 9, 10, 10, 10, 11, 11, 12, 12

Next, divide the weight of students into 6 groups having equal interval.

The groups are: 1 – 2, 3 – 4, 5 – 6, 7 – 8, 9 – 10, 11 – 12.

Now draw a table with four columns and complete it as given below.

Load-shedding duration	Hours	Tally marks	Frequency
1 – 2	1, 2, 2	III	3
3 – 4	3	I	1
5 – 6	5, 6, 6, 6, 6, 6	HHH I	6
7 – 8	7, 7, 7, 7, 7, 7, 8, 8	HHH III	8
9 – 10	9, 9, 10, 10, 10	HHH	5
11 – 12	11, 11, 12, 12	IIII	4
Total			27

- a) Least load-shedding hours = 3 – 4
- b) Most load-shedding hours = 7 – 8

3. The following table shows the weekly sales of notebooks in March and April. Draw a multiple vertical bar graph for this data.

Weeks	March	April
Week 1	7000	5000
Week 2	6000	3000
Week 3	5000	1000
Week 4	4000	1000

Solution:

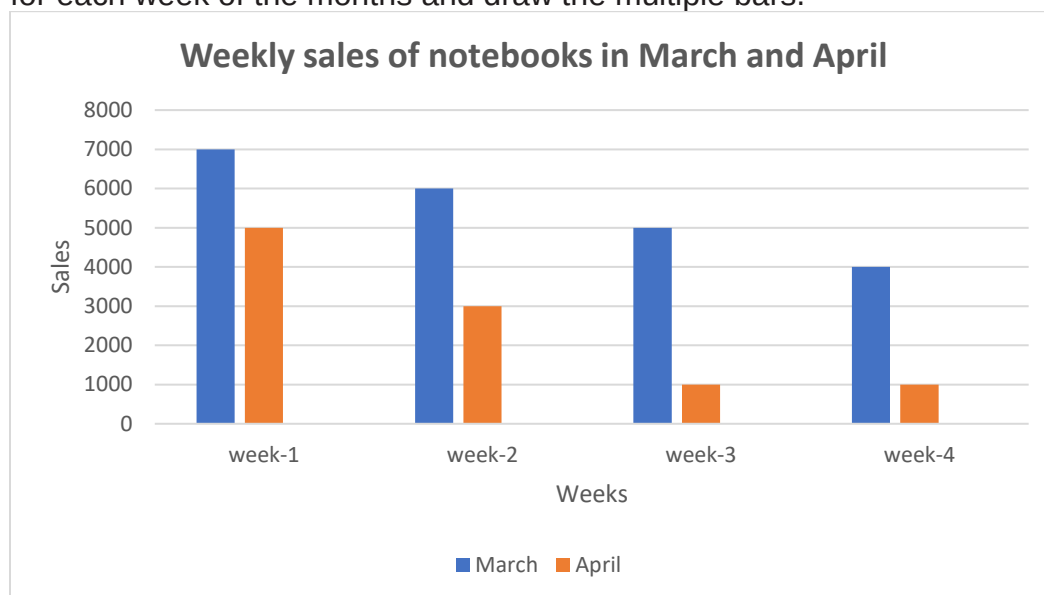
We use the following steps to draw the vertical multiple bar graph.

Step I: Draw the x-axis (horizontal line) and the y-axis (vertical line) perpendicular to each other and write the number of weeks along the x-axis and the weekly sales of notebooks along the y-axis.

Step II: Choose weeks for each month (March and April) and set an appropriate scale. Here, one mark represents 1 week along the x-axis.

Step III: For week 1, the sale in March is Rs.7,000, and in April is Rs.5,000. So, we will color up to the sales of 7000 and 5000 for March and April, respectively.

Step IV: Similarly, keep following the steps, colour the correct numbers of sales for each week of the months and draw the multiple bars.



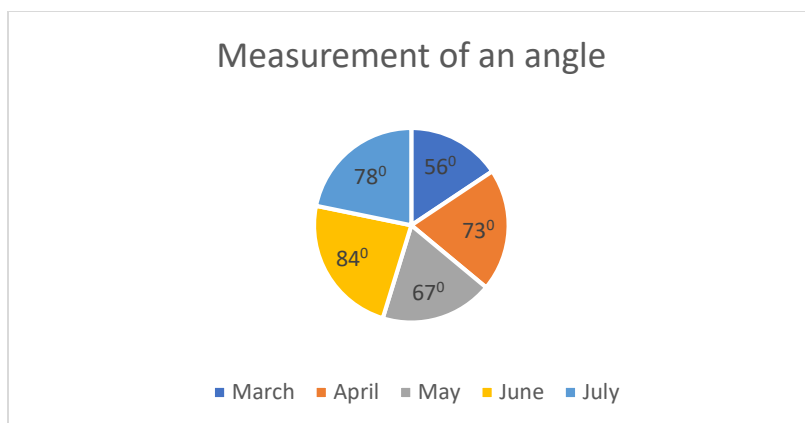
4. The following table shows how much Ahad spends on food in the first 5 months of the year. Draw a pie graph for the following data.

Months	March	April	May	June	July
Amount spent on food (in Rs)	10000	13000	12000	15000	14000

Solution:

Months	Amount spent on food (in Rs)	Measurement of the central angle
March	10000	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^\circ = \frac{10000}{64000} \times 360^\circ = 56^\circ$
April	13000	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^\circ = \frac{13000}{64000} \times 360^\circ = 73^\circ$
May	12000	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^\circ = \frac{12000}{64000} \times 360^\circ = 67^\circ$
June	15000	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^\circ = \frac{15000}{64000} \times 360^\circ = 84^\circ$
July	14000	$\frac{\text{Amount spent}}{\text{Total amount}} \times 360^\circ = \frac{14000}{64000} \times 360^\circ = 78^\circ$
Total	64000	

Pie Graph:



5. Find the mean, median and mode for the following data.

a) 35, 42, 53, 30, 32

b) 520, 356, 450, 180, 256, 368

Solution:

a) 35, 42, 53, 30, 32

For arithmetic mean:

Arithmetic mean = $\bar{X} = \frac{\sum x}{n}$, Here, $n = 5$

$$\bar{X} = \frac{35 + 42 + 53 + 30 + 32}{5} = \frac{192}{5} = 38.4$$

So, the arithmetic mean is 38.4.

For median:

Arranging the data in ascending order:

30, 32, 35, 42, 53

Number of values = $n = 5$ (odd)

Median = $\left(\frac{n+1}{2}\right)$ th observation

$$= \left(\frac{5+1}{2}\right)\text{th observation} = \left(\frac{6}{2}\right)\text{th observation} = 3^{\text{rd}} \text{ observation}$$

Median = 35

So, the median of the given data set is 35.

For mode:

Each value appears only once in this set, so there is no mode.

Hence, the mode = $\hat{X} = 0$.

b) 520, 356, 450, 180, 256, 368

For arithmetic mean:

Arithmetic mean = $\bar{X} = \frac{\sum x}{n}$, Here, $n = 6$

$$\bar{X} = \frac{520 + 356 + 450 + 180 + 256 + 368}{6} = \frac{2130}{6} = 355$$

So, the arithmetic mean is 355.

For median:

Arranging the data in ascending order:

180, 256, 356, 368, 450, 520

Number of values = $n = 6$ (even)

Median = arithmetic mean of $\left(\frac{n}{2}\right)$ th and $\left[\left(\frac{n}{2}\right) + 1\right]$ th observation

$$= \text{arithmetic mean of } \left(\frac{6}{2}\right)\text{th and } \left[\left(\frac{6}{2}\right) + 1\right]\text{th observation}$$

$$= \text{arithmetic mean of the } 3^{\text{rd}} \text{ and } 4^{\text{th}} \text{ observation}$$

180, 256, 356, 368, 450, 520

3rd observation = 356 and 4th observation = 368

Arithmetic mean of 4th and 5th observation = $\frac{356+368}{2} = \frac{724}{2} = 362$

So, the median of the given data set is 362.

For mode:

Each value appears only once in this set, so there is no mode.

Hence, the mode = $\hat{X} = 0$.

6. A box contains slips of numbers from 1 to 10. Arham takes one slip randomly without looking at the slips. Find:

- probability of picking the number 7
- probability of picking a number from a multiple of 2
- probability of picking a number from a multiple of 3
- probability of picking a number from a multiple of 5

Solution:

Sample space is: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Total number of possible outcomes = 10

- probability of picking the number 7

Number of favourable outcomes = 1

Probability of an event = $P(7) = \frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

probability of picking the number 7 = $\frac{1}{10}$

So, probability of picking the number 7 is $\frac{1}{10}$

- probability of picking a number from a multiple of 2

Number of favourable outcomes = 5

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

probability of picking a number from a multiple of 2 = $\frac{5}{10} = \frac{1}{2}$

So, probability of picking a number from a multiple of 2 is $\frac{1}{2}$

- probability of picking a number from a multiple of 3

Number of favourable outcomes = 3

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

probability of picking a number from a multiple of 3 = $\frac{3}{10}$

So, probability of picking a number from a multiple of 3 is $\frac{3}{10}$

- probability of picking a number from a multiple of 5

Number of favourable outcomes = 2

Probability of an event = $\frac{\text{Number of favourable (desired) outcomes}}{\text{Total number of possible outcomes}}$

probability of picking a number from a multiple of 5 = $\frac{2}{10} = \frac{1}{5}$

So, probability of picking a number from a multiple of 5 is $\frac{1}{5}$

7. Draw shapes (as sample space) for each of these according to the mentioned probabilities.

a) $P(\text{triangle}) = \frac{2}{5}$

b) $P(\text{square}) = \frac{2}{10}$

c) $P(\text{circle}) = \frac{4}{10}$

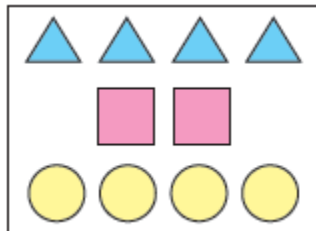
Solution:

Number of triangular figures = 4

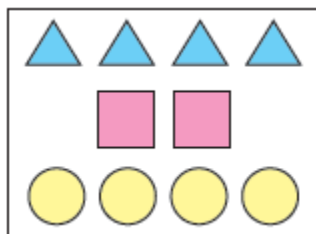
Number of square figures = 2

Number of circular figures = 4

a) $P(\text{triangle}) = \frac{4}{10}$



b) $P(\text{square}) = \frac{2}{10}$



c) $P(\text{circle}) = \frac{4}{10}$

